

Exposé du problème

Un ordonnancement des tâches consiste à placer dans un certain ordre chronologique des opérations qui font intervenir des dépenses, des ressources, des performances et du temps.

Alors, il consiste à organiser l'exécution des tâches en vue de réaliser le projet dans les meilleurs délais en utilisant de manière optimale les ressources disponibles et en contrôlant de manière permanente, au cours de la réalisation, l'état d'avancement des travaux..

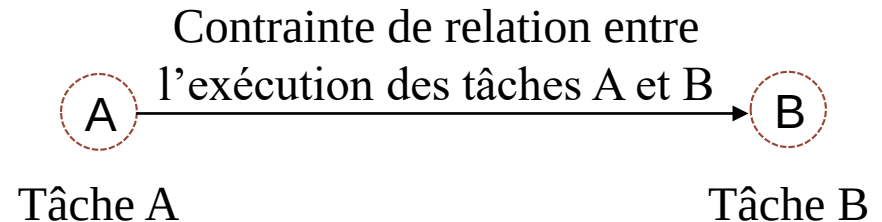
Méthodes

Pour la mise au point du planning ou calendrier d'un projet, on peut utiliser les méthodes utilisant :

- ☐ le concept de graphes ;
- ☐ les calculs dans des tableaux ;
- ☐ le diagramme de GANTT.

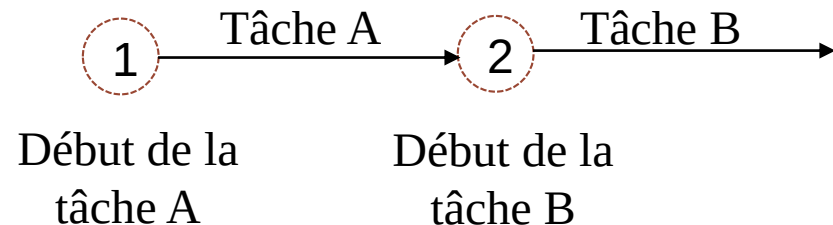
Méthode MPM (Méthode des Potentiels METRA)

Dans le concept de graphes :
sommet → tâche
arc → contrainte (durée)



Méthode CPM (Critical Path Method)

Dans le concept de graphes :
sommet → événement
arc → tâche



Cette méthode a une variante, incluant des données aléatoires, appelée P.E.R.T. (Program Evaluation and Review Task ou Program Evaluation and Research Task).

Tâche	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
Durée	7	7	15	30	45	15	45	60	20	30	30	15	30	15	30	15	15	15	30	7	4	2	7
T.ant.	-	a	b	c	d	e	d	d	h	i	f	k	g j l	m	n	m	o	q	q	p	r t	s t	r s

Tâche	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
Durée	7	7	15	30	45	15	45	60	20	30	30	15	30	15	30	15	15	15	30	7	4	2	7
T.ant.	-	a	b	c	d	e	d	d	h	i	f	k	g j l	m	n	m	o	q	q	p	r t	s t	r s
T.suc.																							

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Tâche	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
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Tâche	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
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Tâche	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
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Graphe MPM

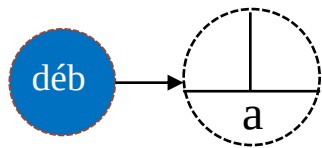
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Graphe MPM

déb

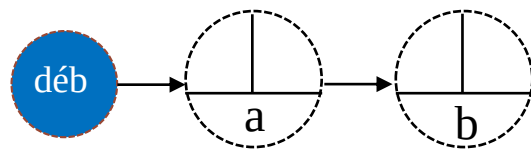
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Graphe MPM



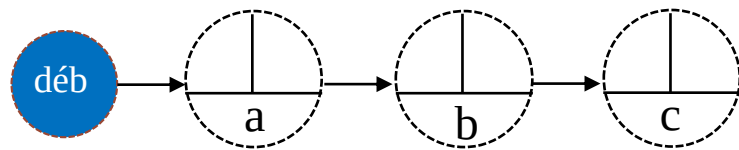
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Graphe MPM



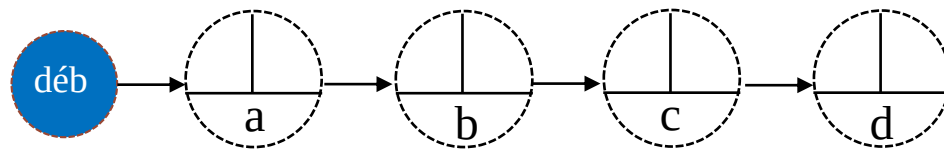
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Graphe MPM



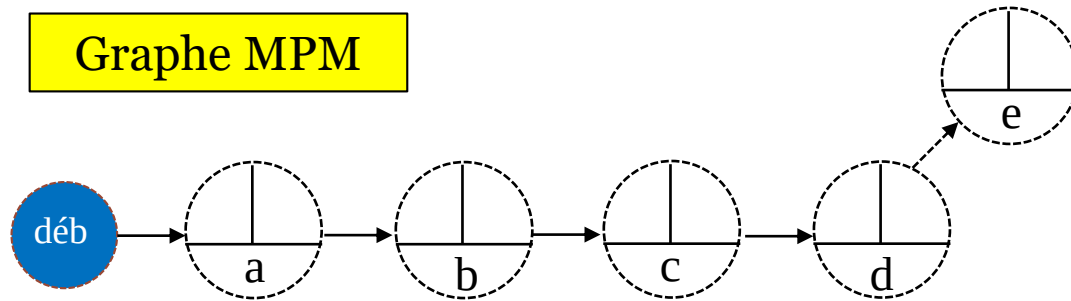
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Graphe MPM



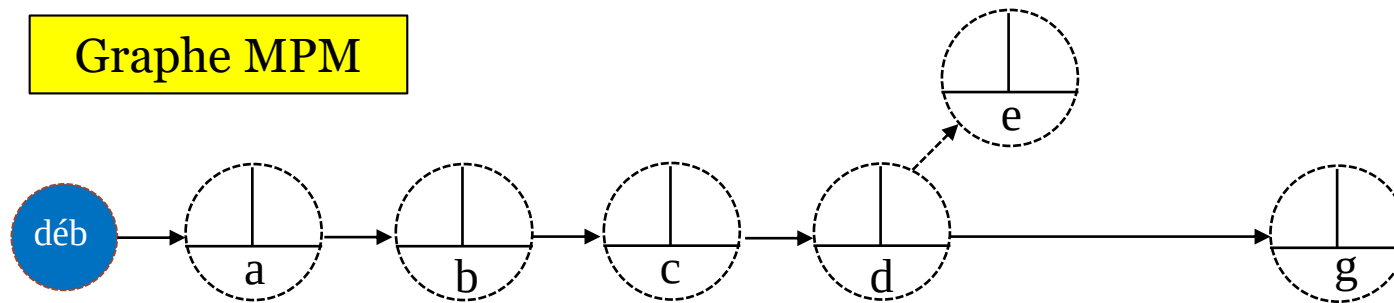
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Graphe MPM



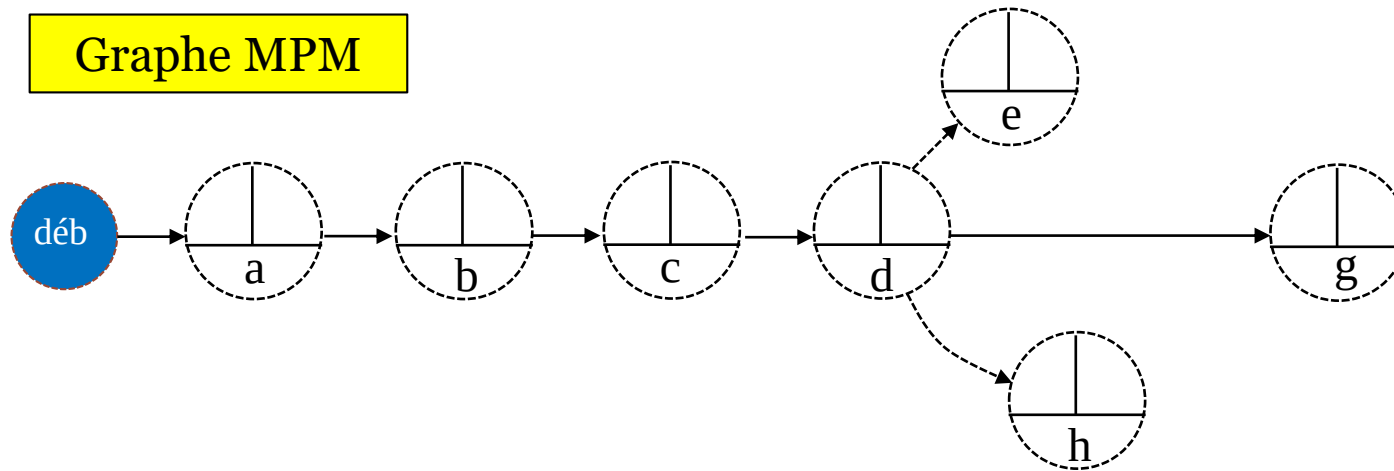
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Graphe MPM



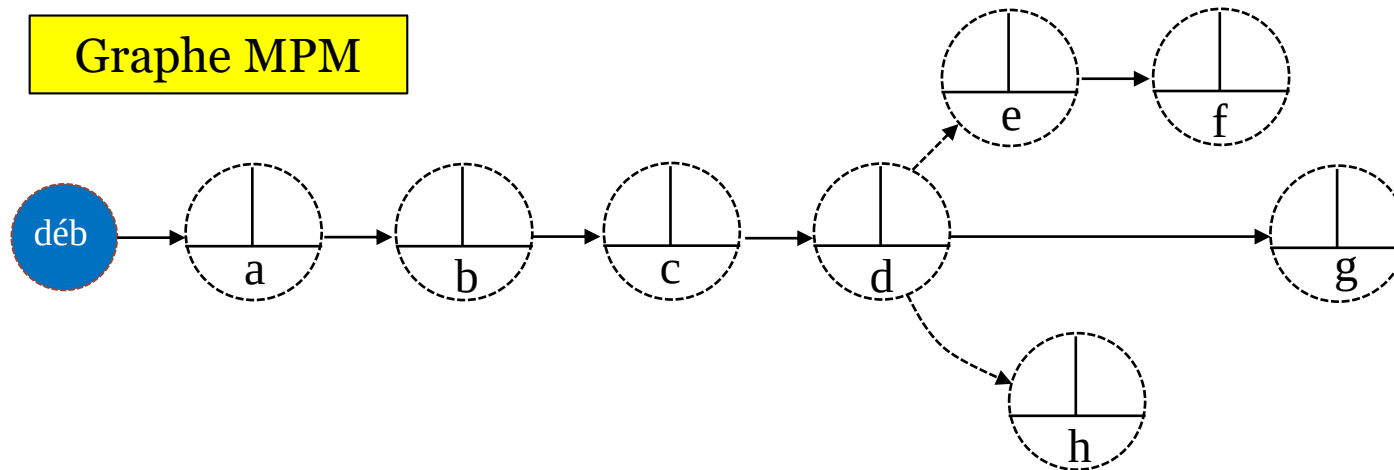
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Graphe MPM



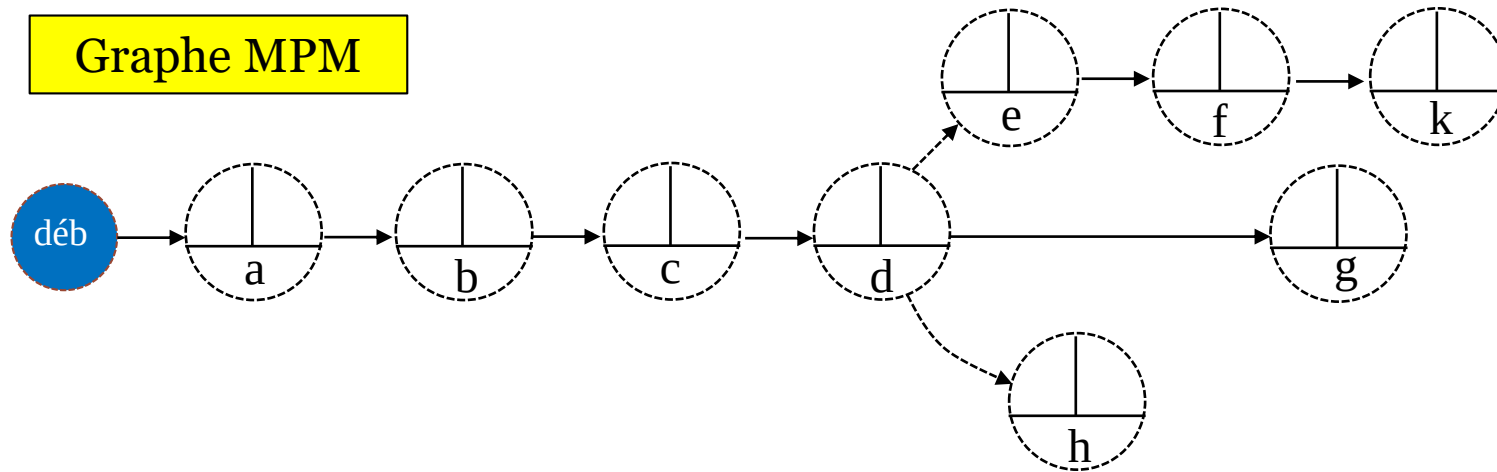
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Graphe MPM



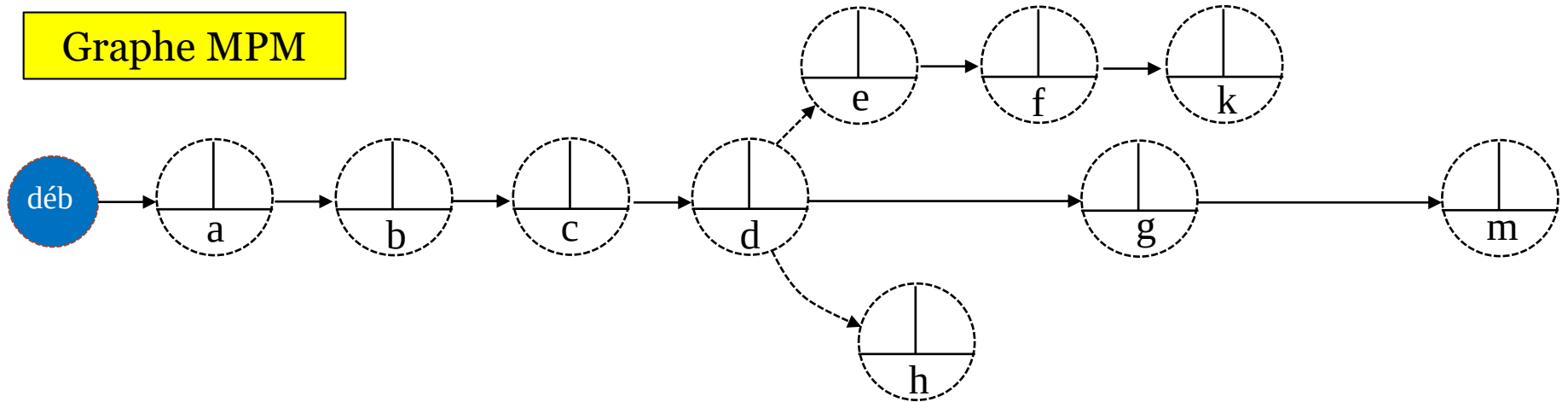
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Graphe MPM



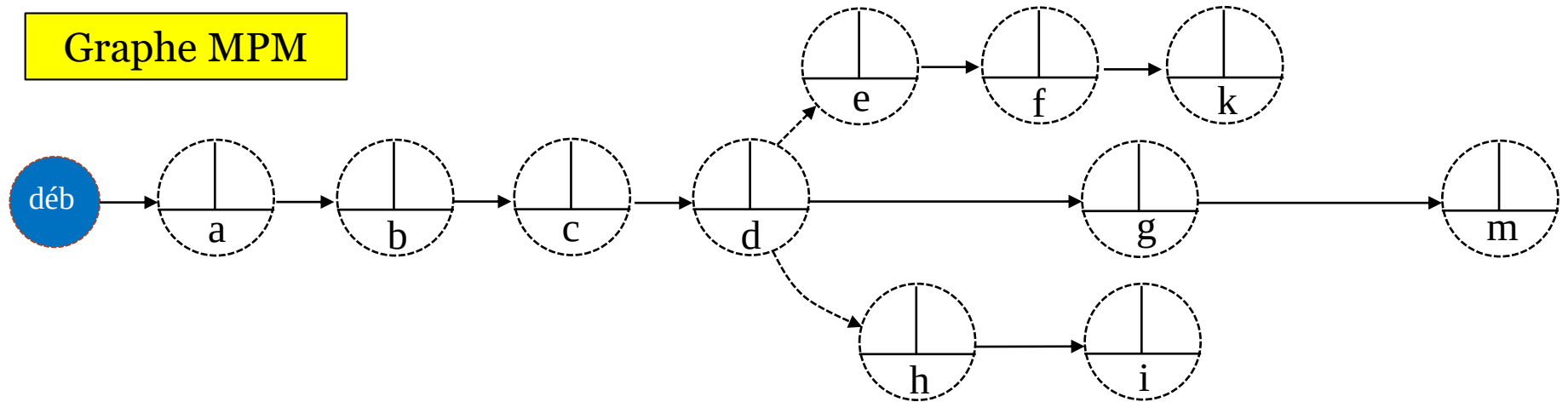
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T.suc.	b	c	d	e g h	f	k	m	i	j	m	l	m	n p	o	q	t	r s	u w	v w	u v	fin	fin	fin

Graphe MPM



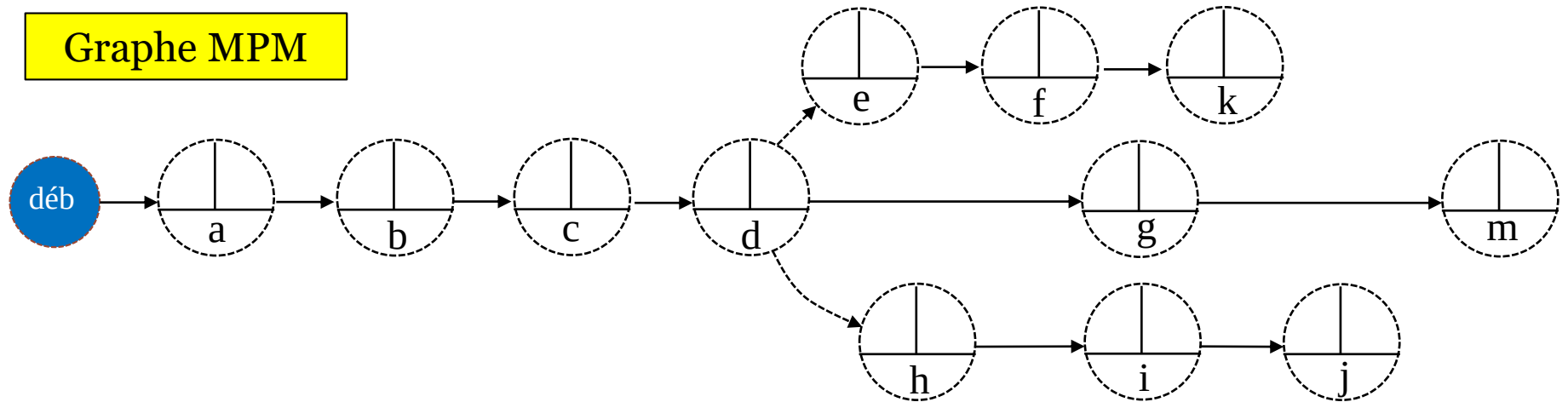
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Graphe MPM



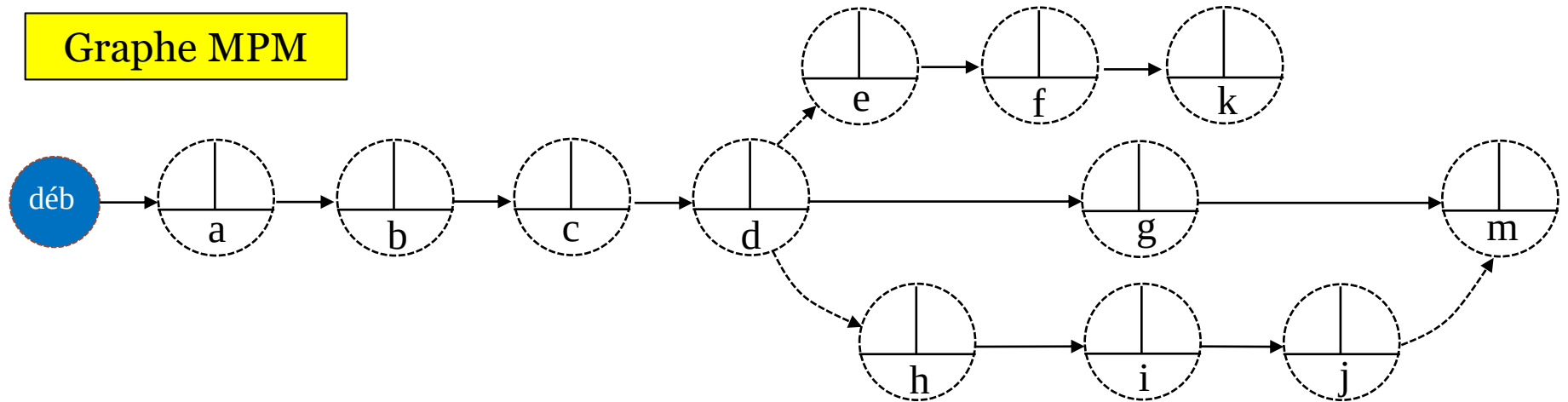
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Graphe MPM



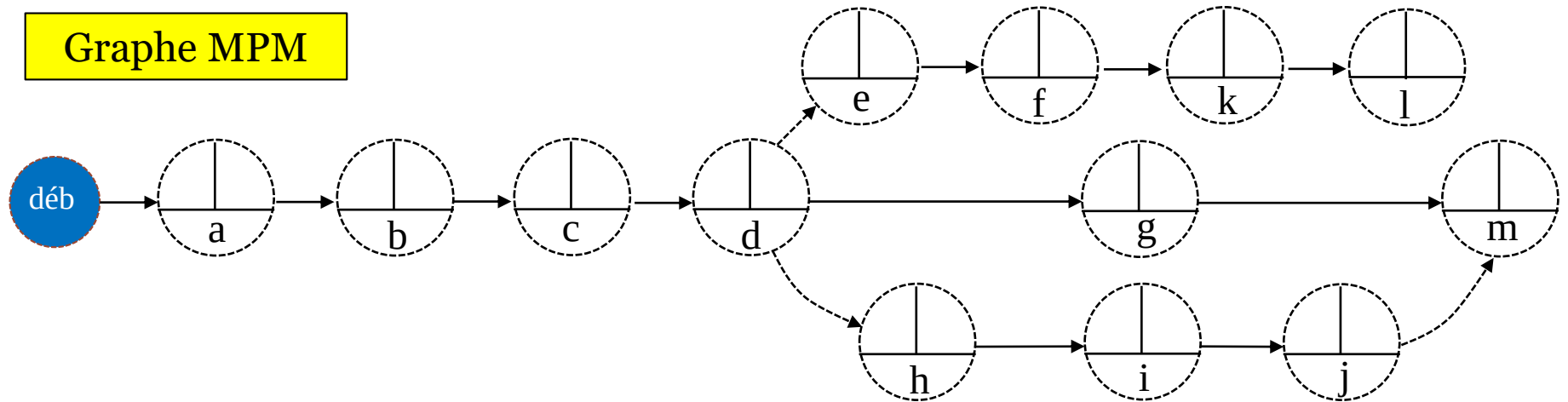
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Graphe MPM



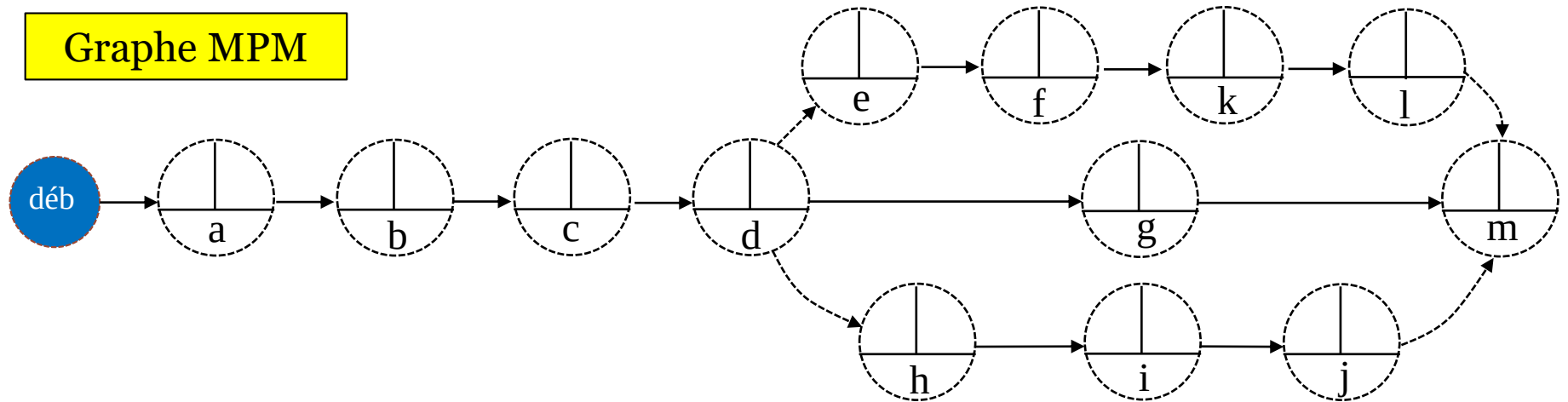
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Graphe MPM



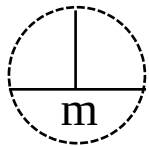
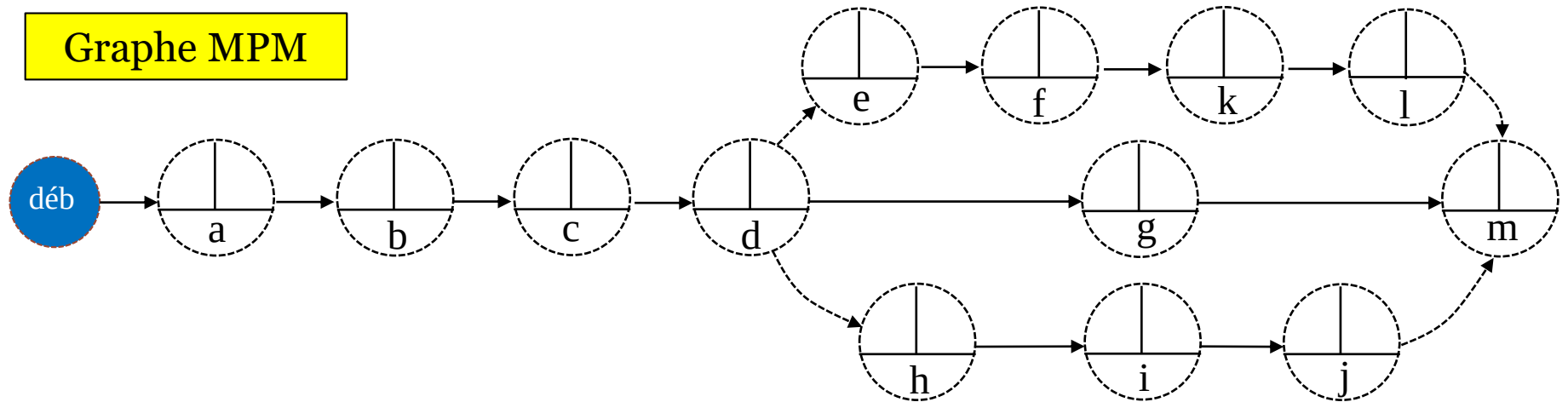
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T.suc.	b	c	d	e g h	f	k	m	i	j	m	l	m	n p	o	q	t	r s	u w	v w	u v	fin	fin	fin

Graphe MPM



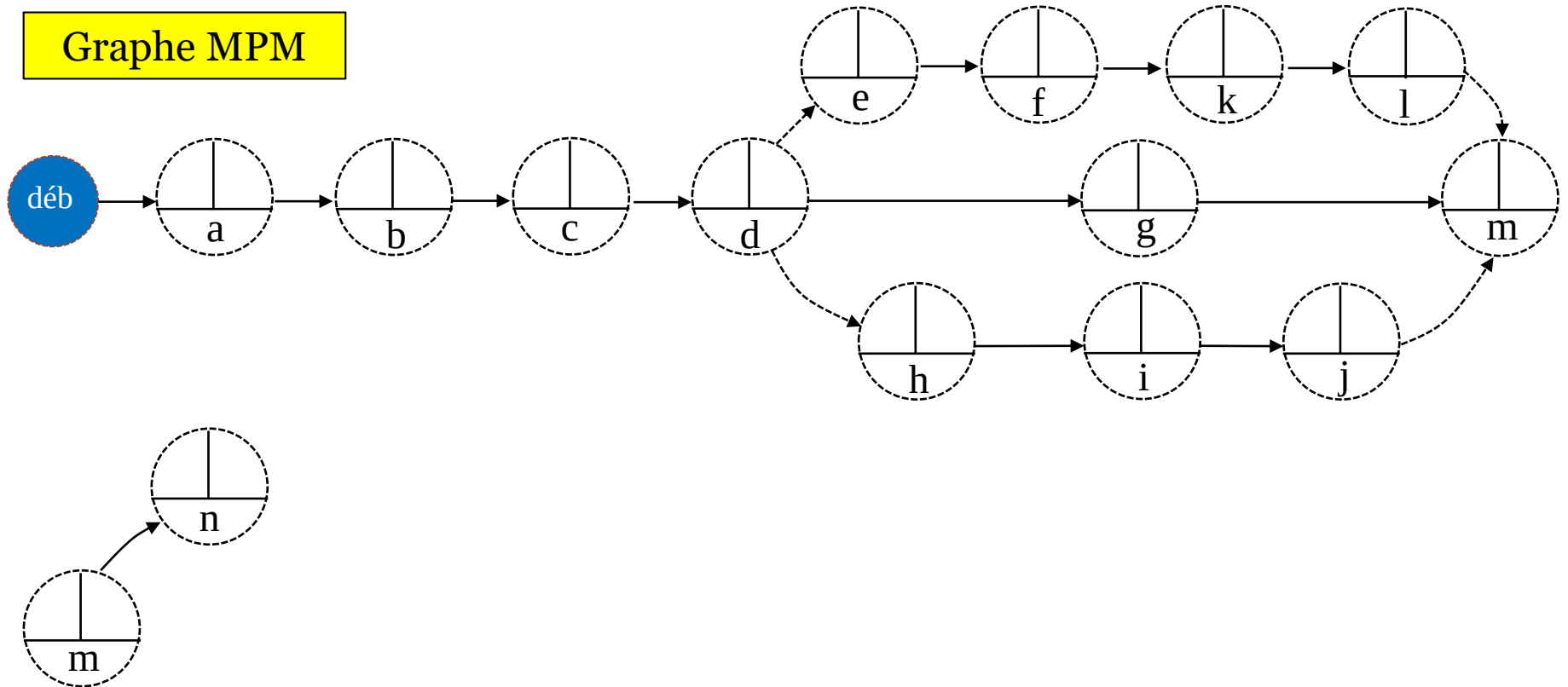
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T.suc.	b	c	d	e g h	f	k	m	i	j	m	l	m	n p	o	q	t	r s	u w	v w	u v	fin	fin	fin

Graphe MPM



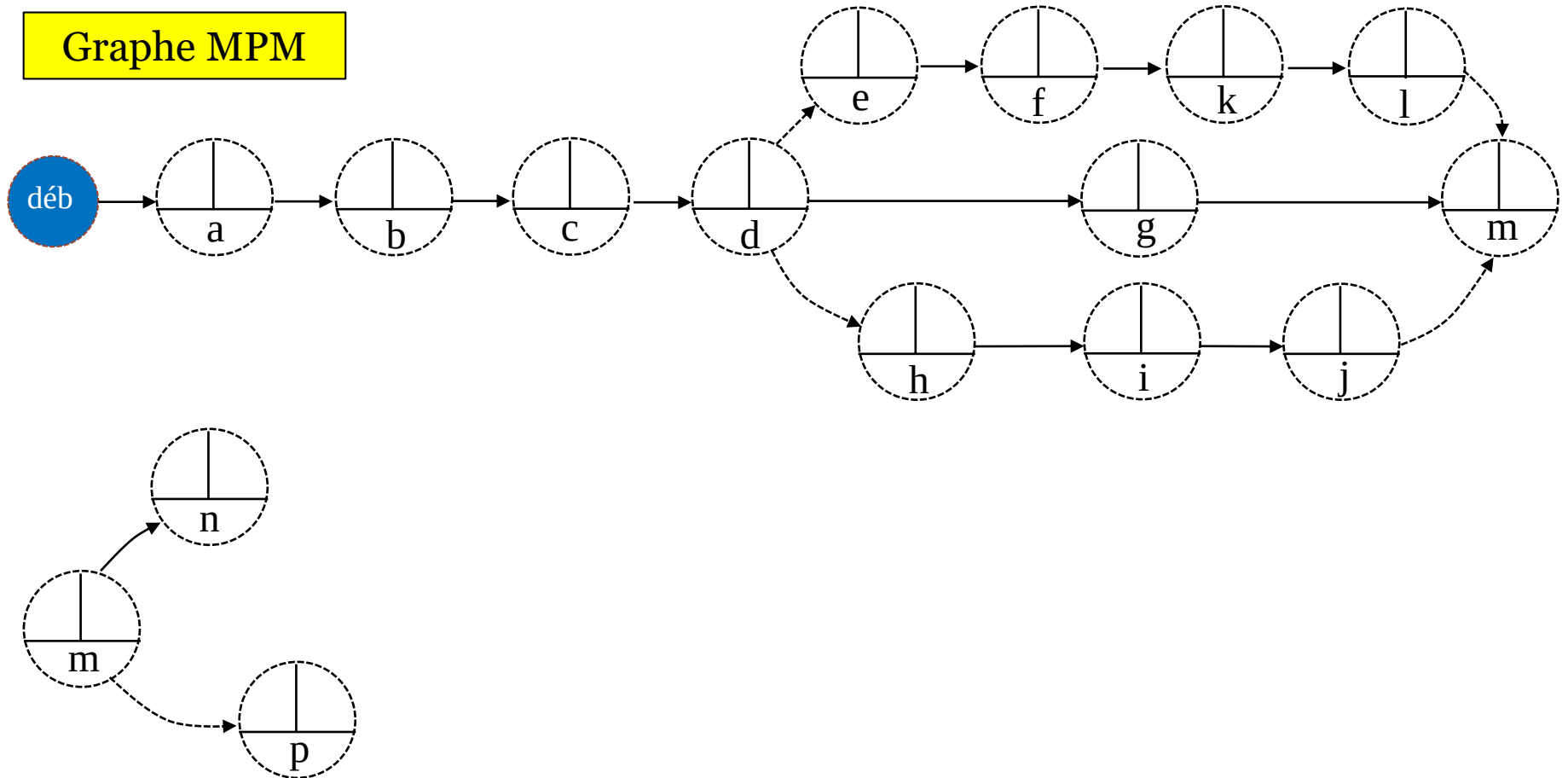
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Graphe MPM



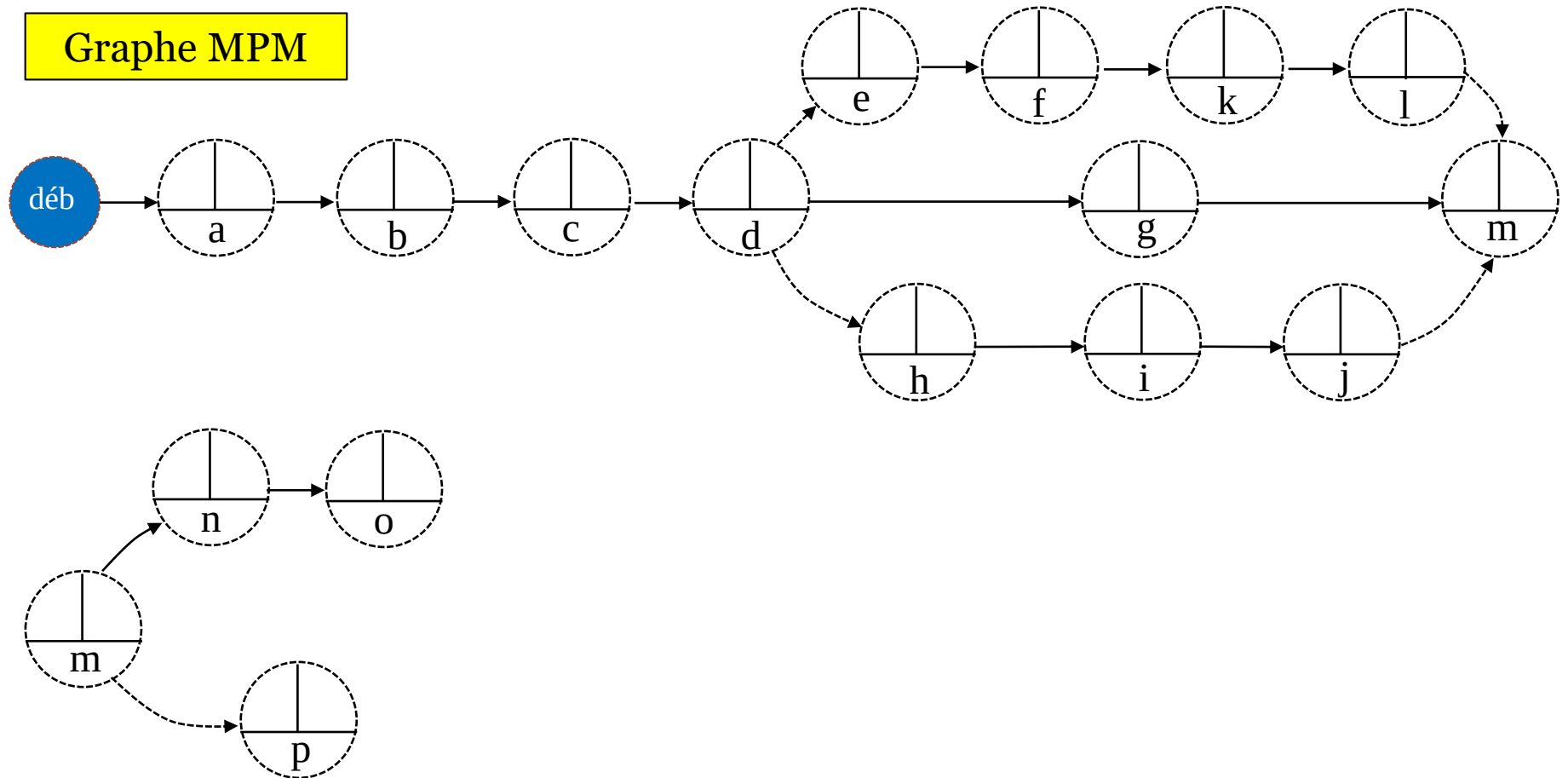
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Graphe MPM



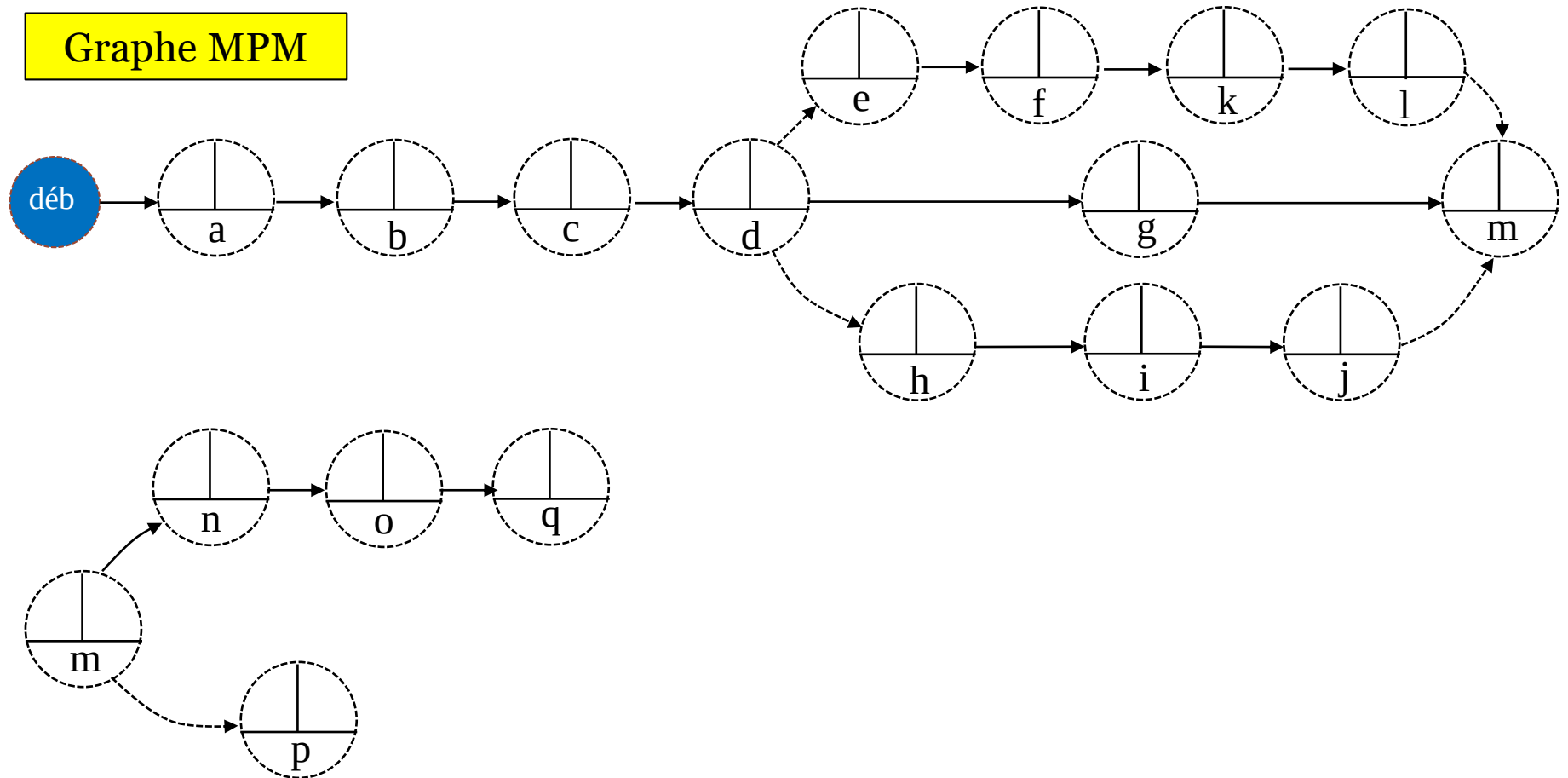
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T.suc.	b	c	d	e g h	f	k	m	i	j	m	l	m	n p	o	q	t	r s	u w	v w	u v	fin	fin	fin

Graphe MPM



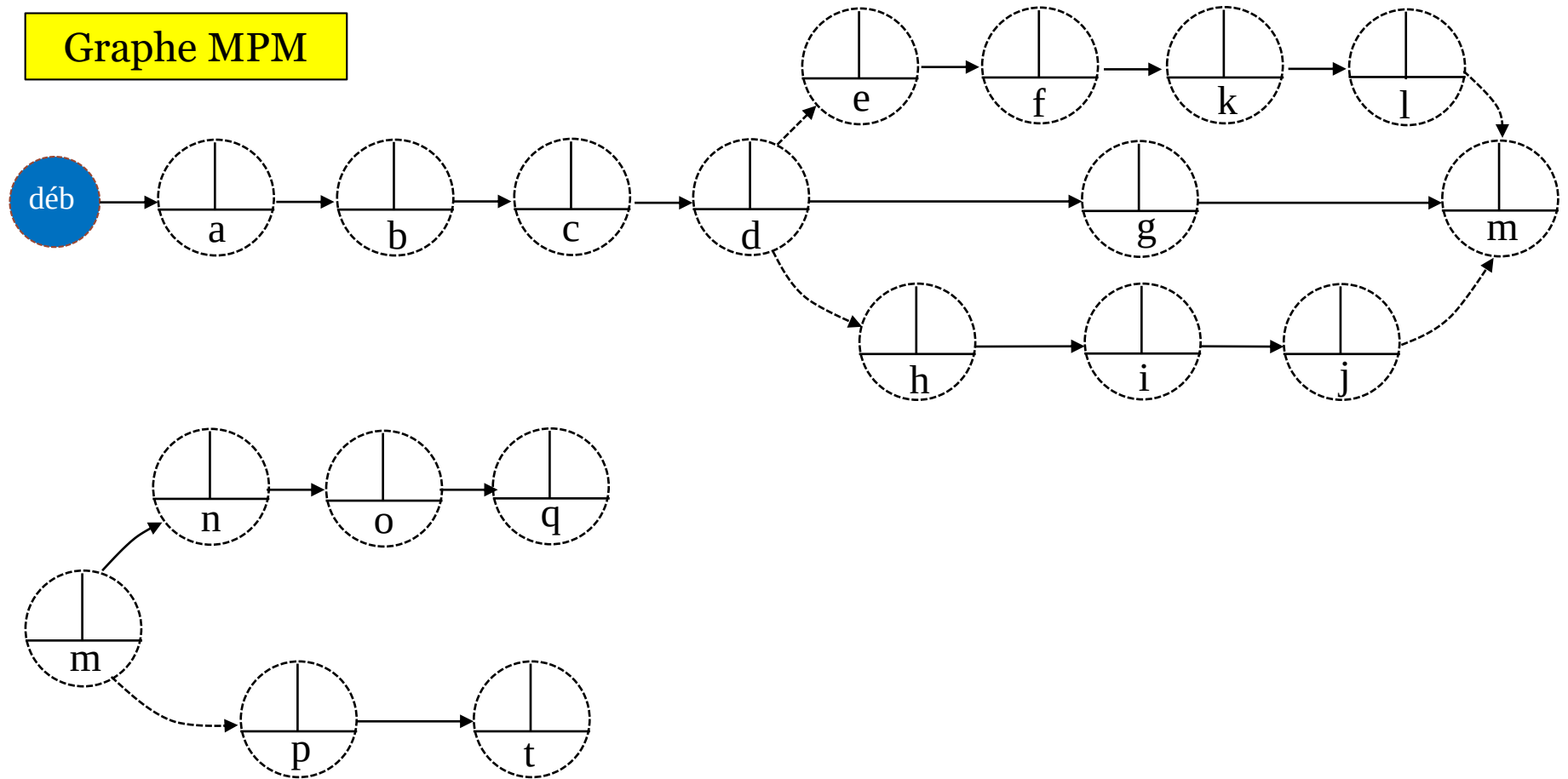
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Graphe MPM



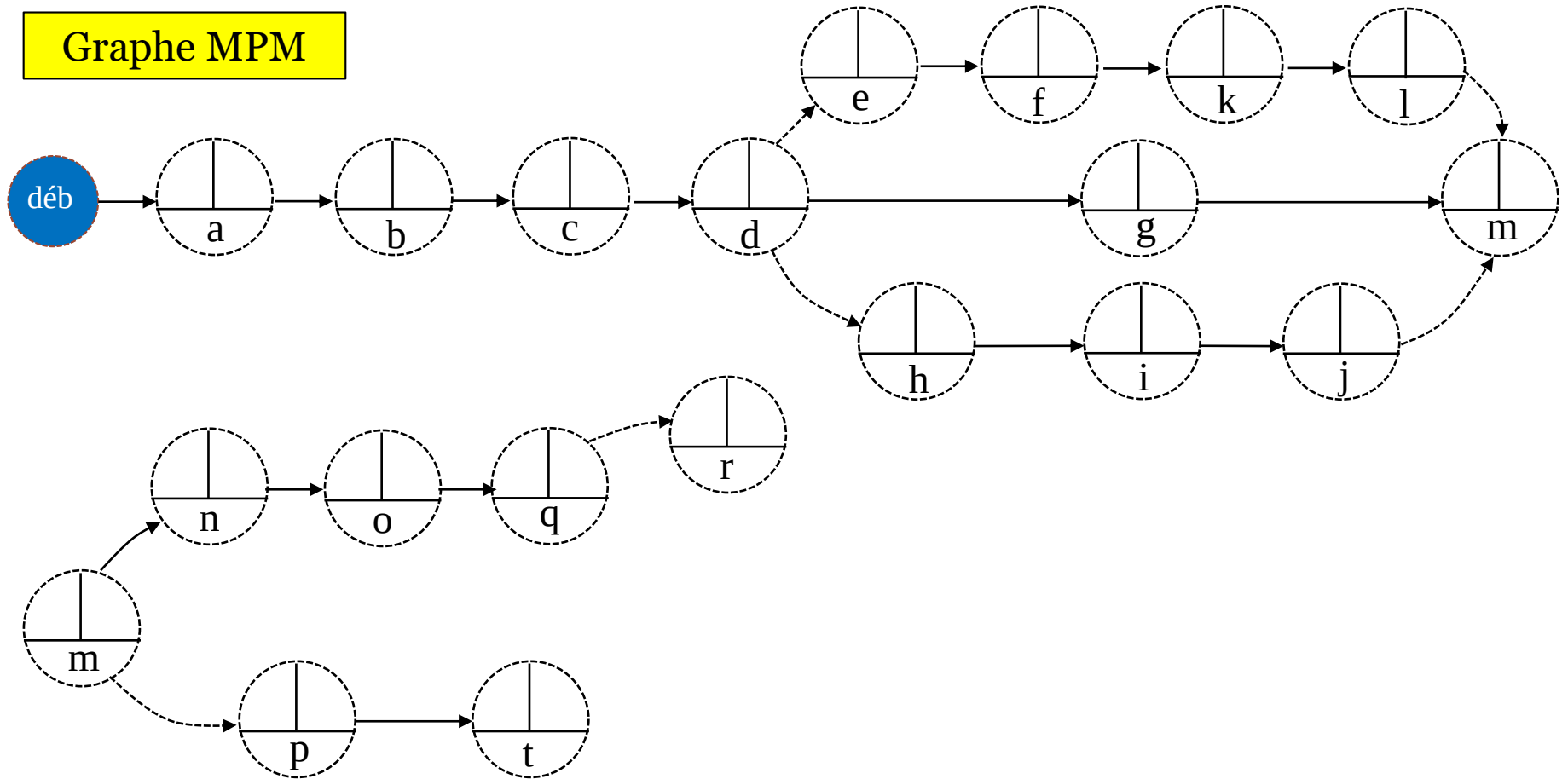
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Graphe MPM



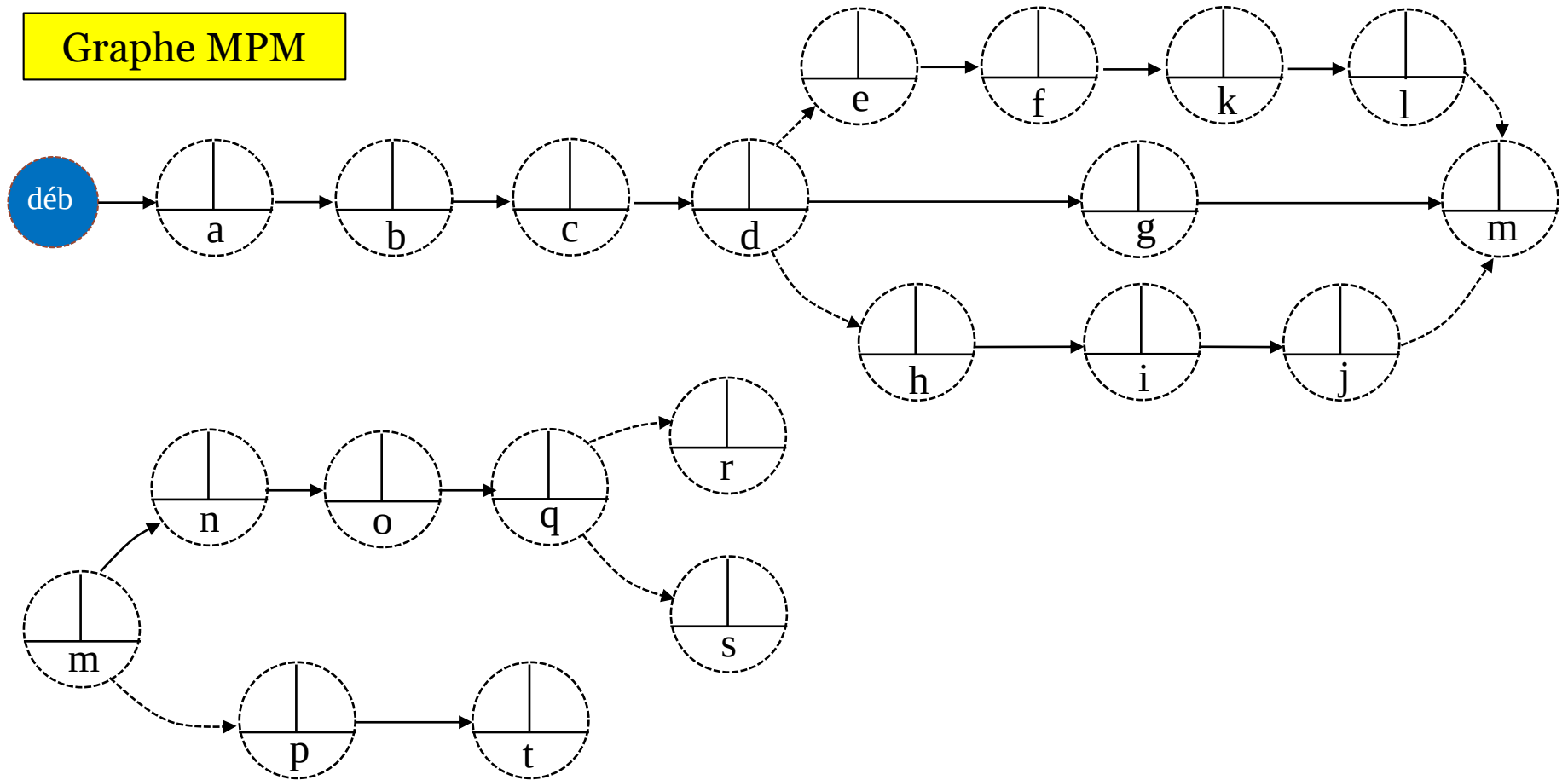
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Graphe MPM



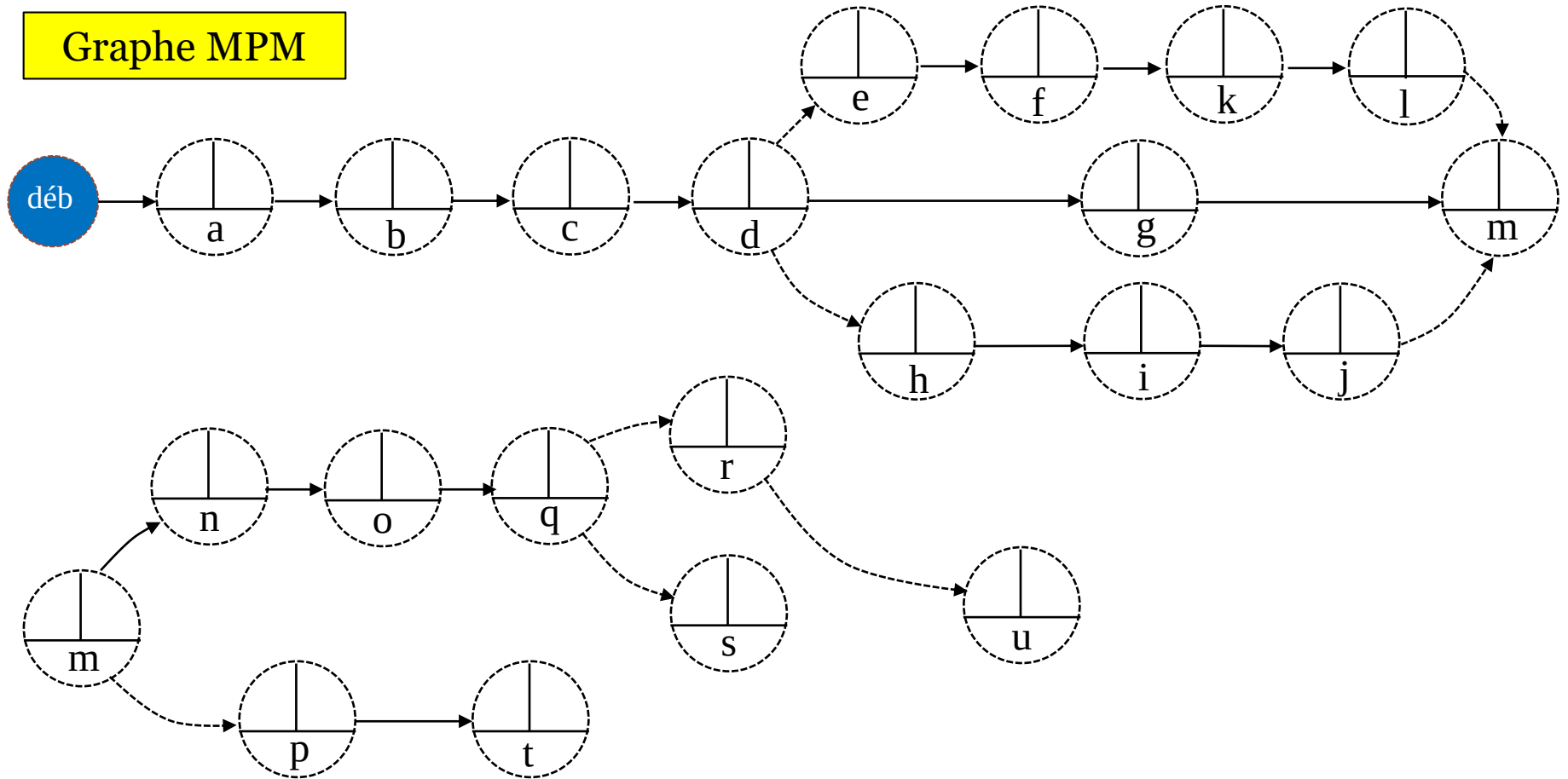
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Graphe MPM



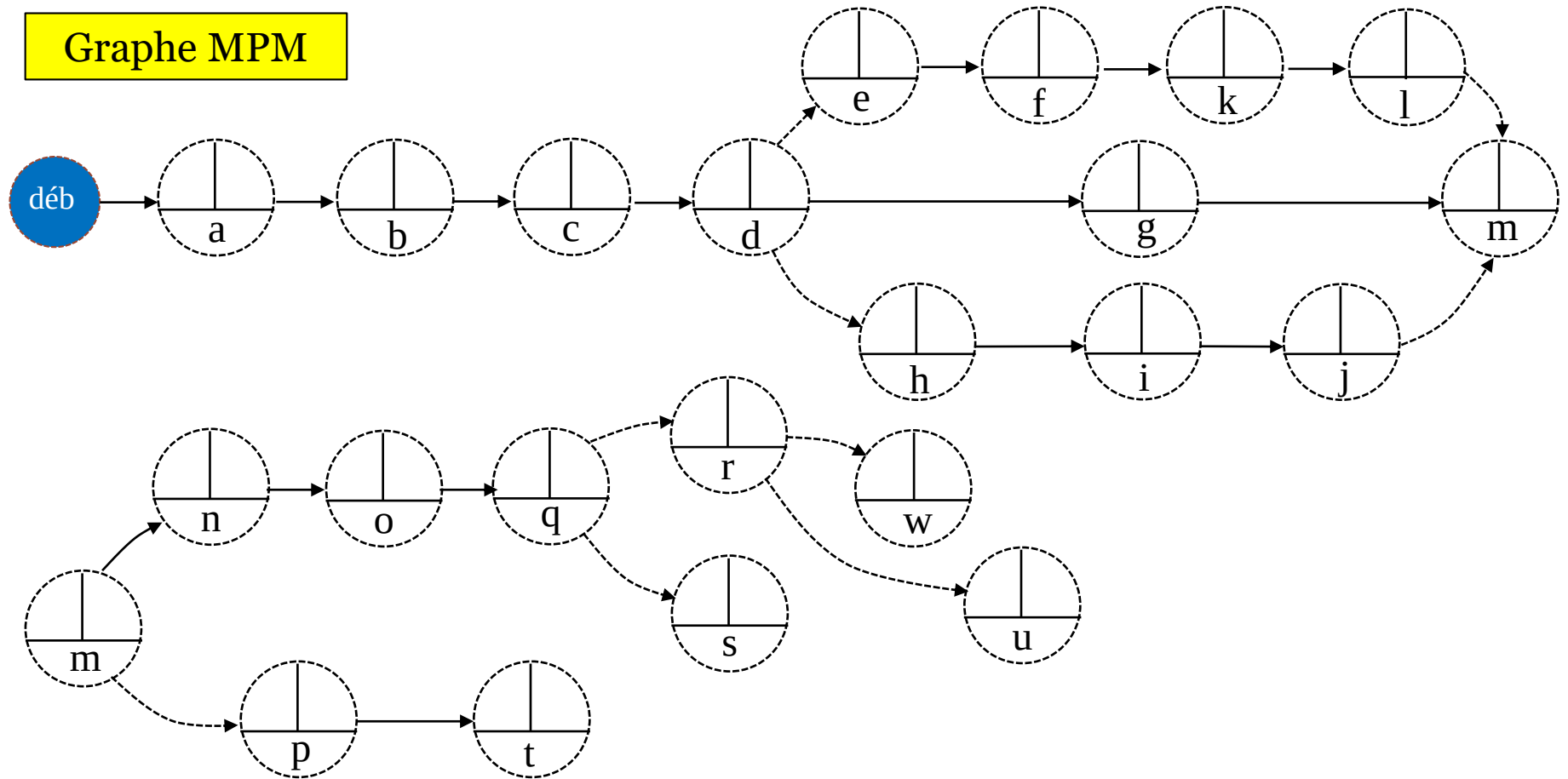
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Graphe MPM



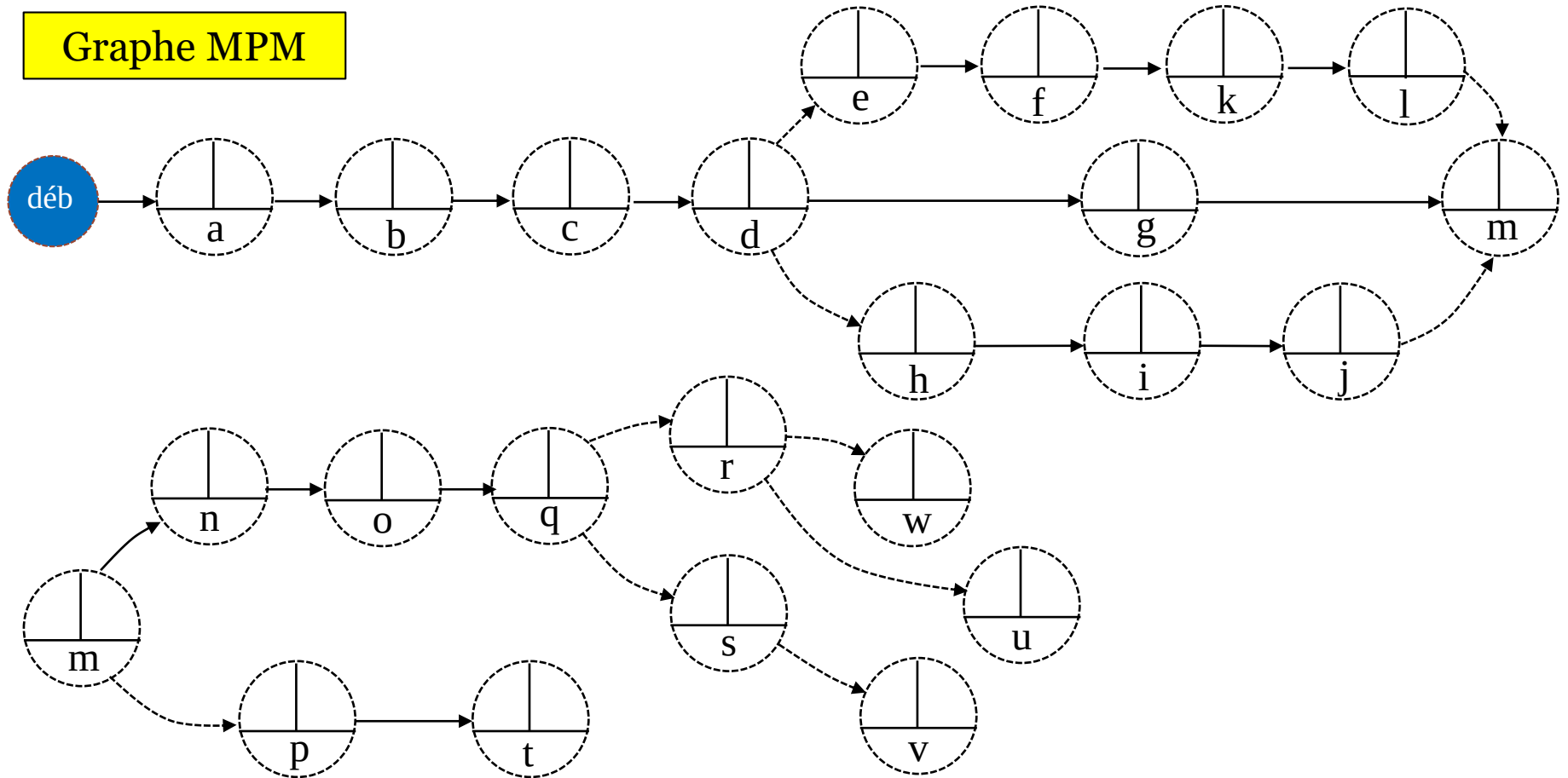
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Graphe MPM



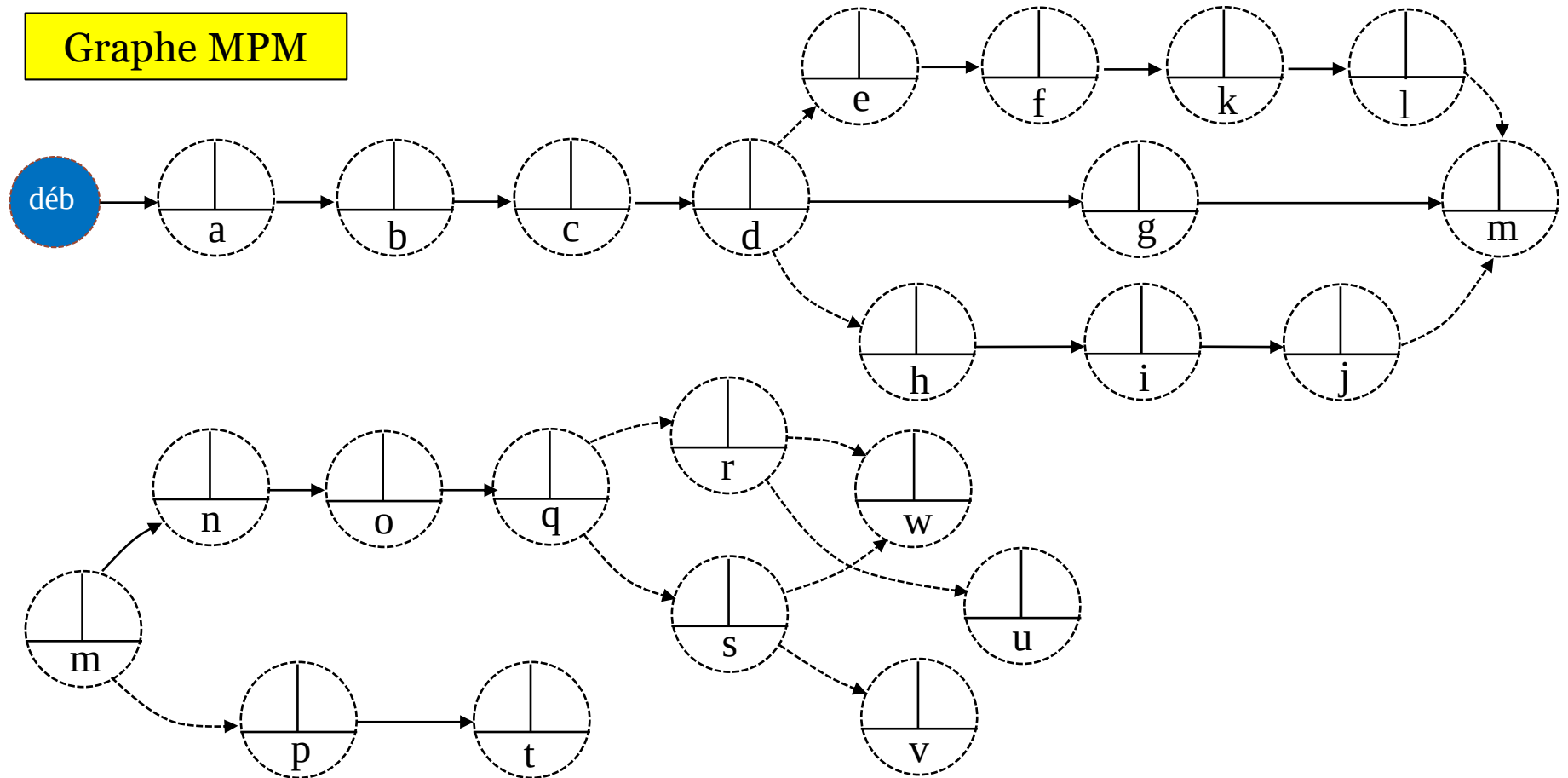
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Graphe MPM



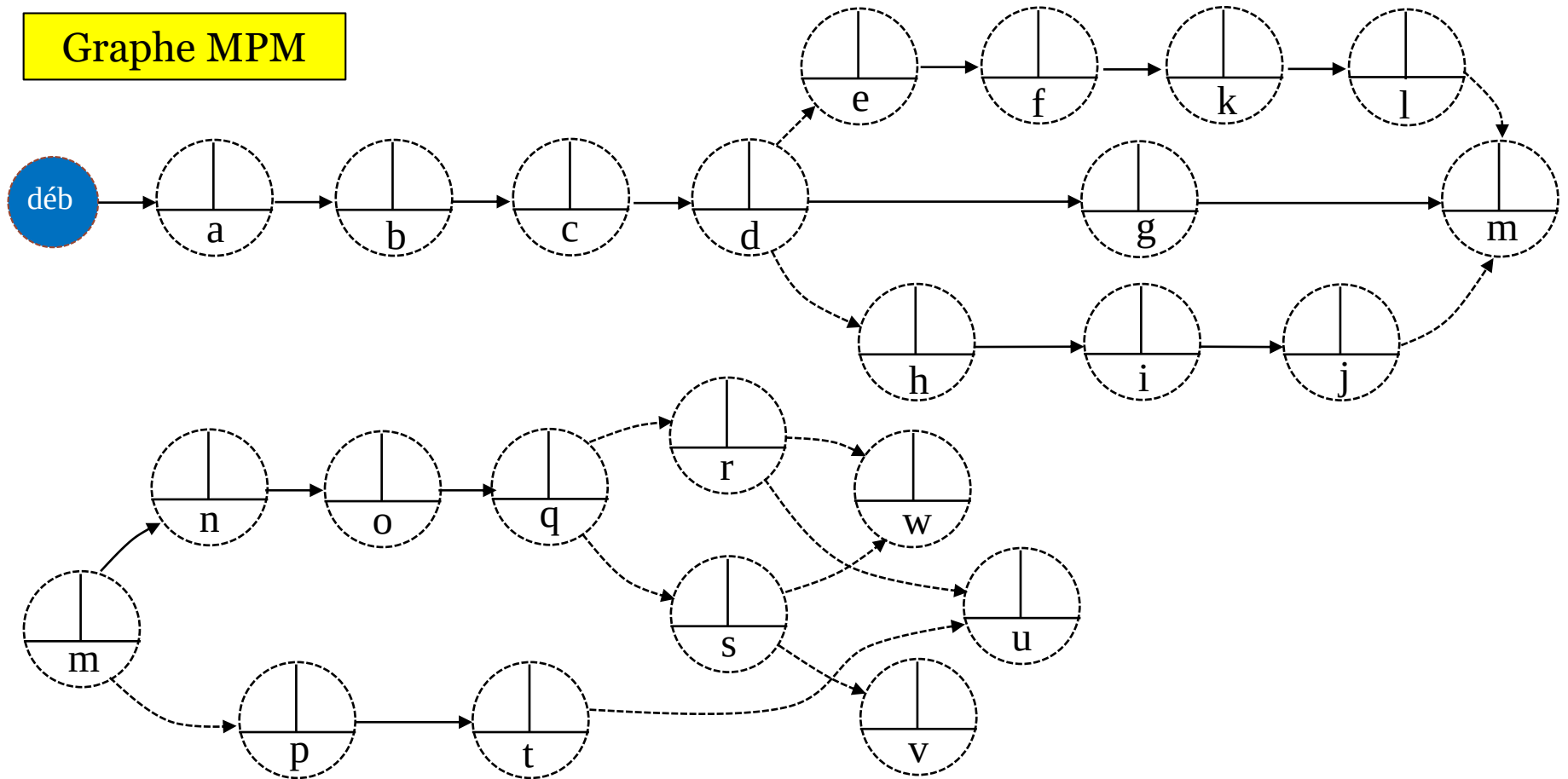
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Graphe MPM



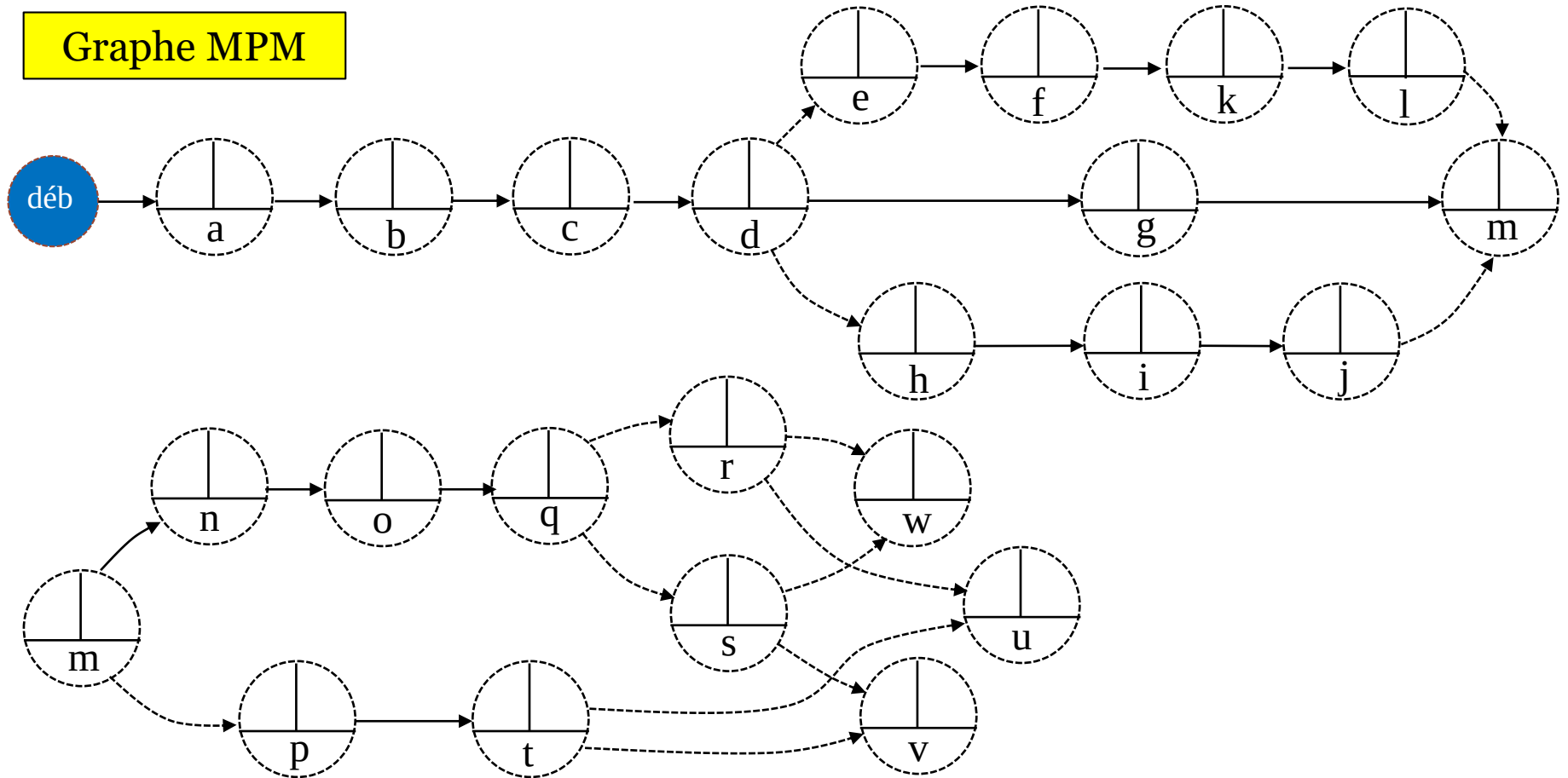
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Graphe MPM



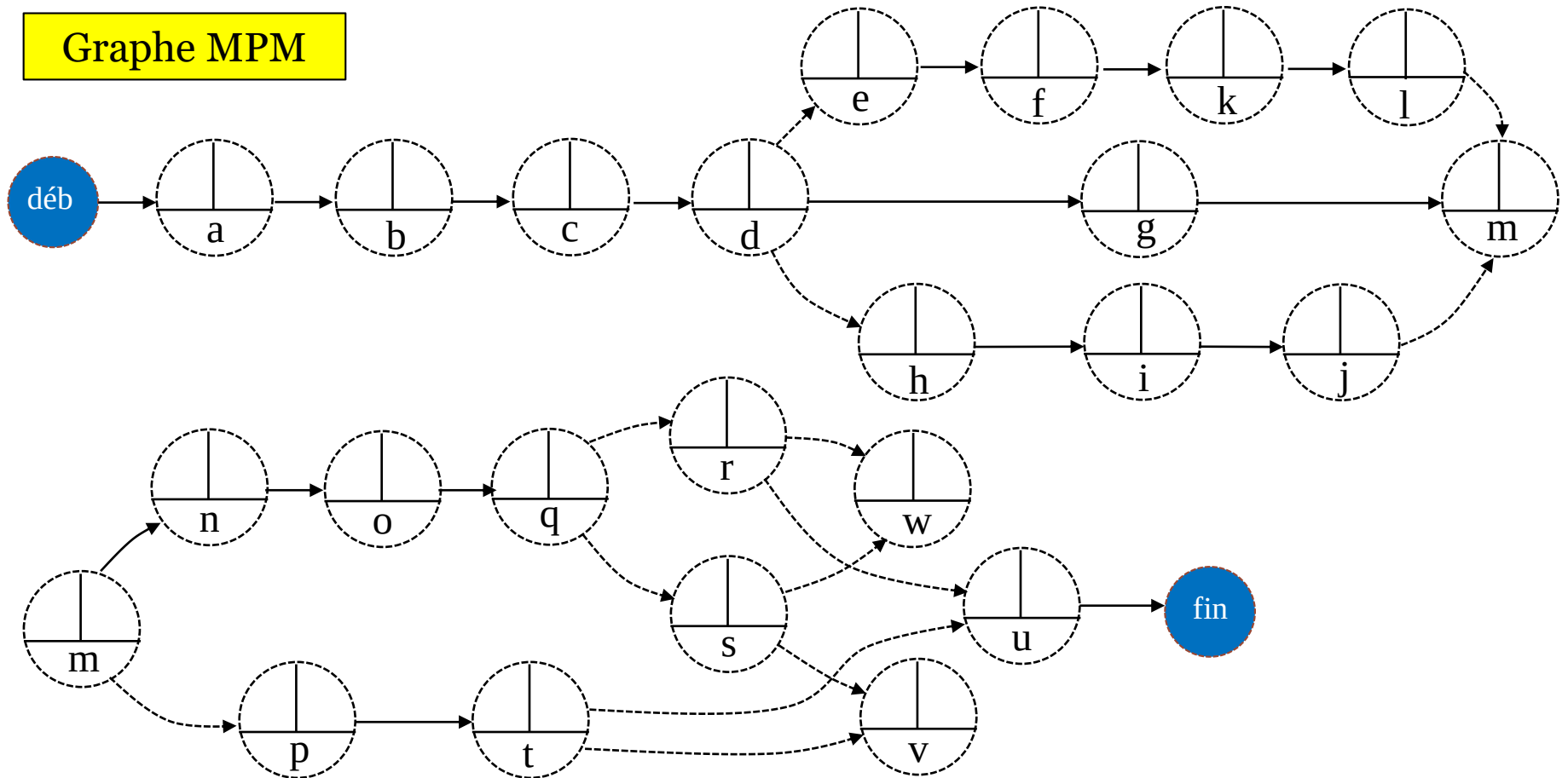
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Graphe MPM



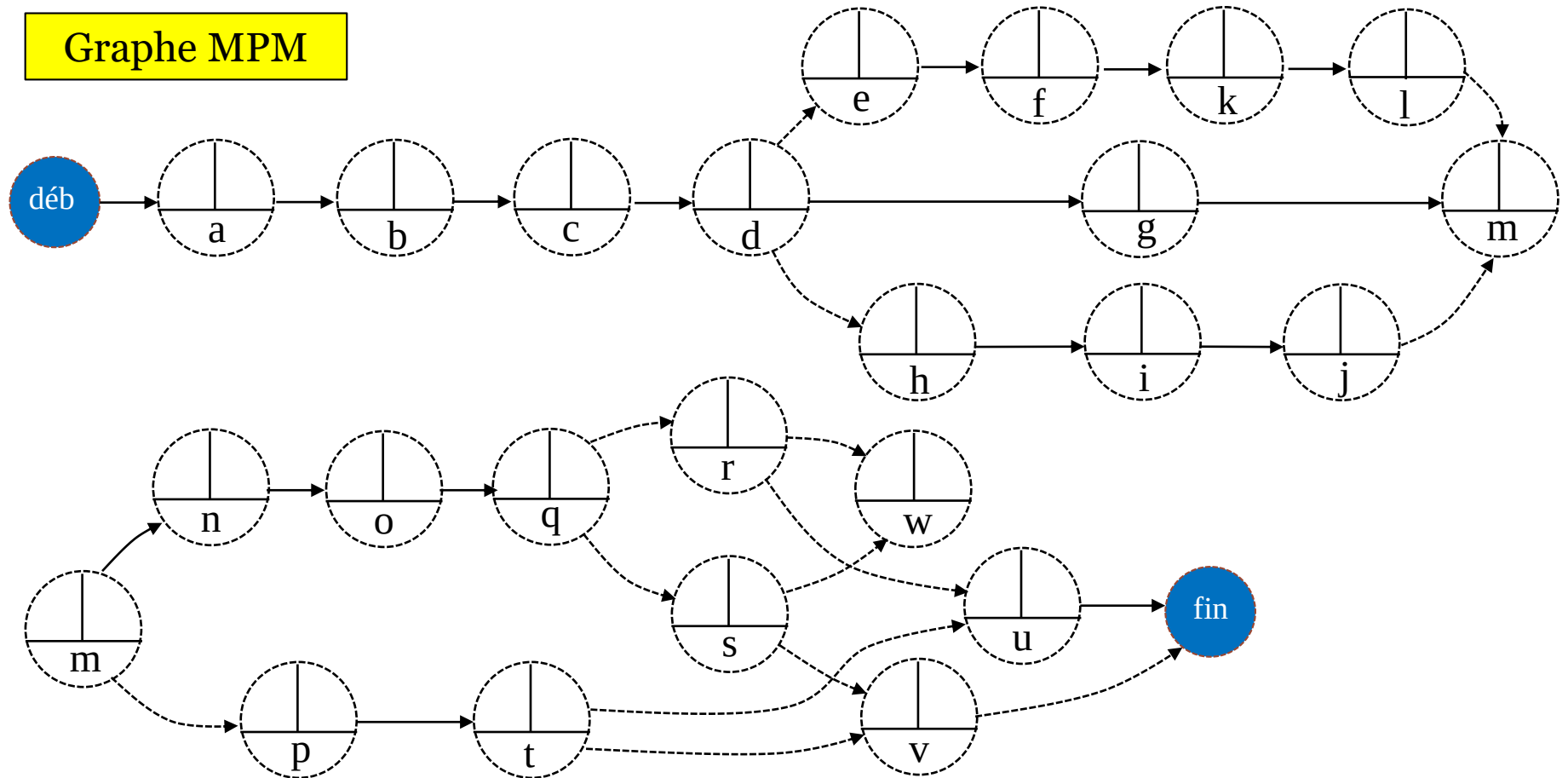
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Graphe MPM



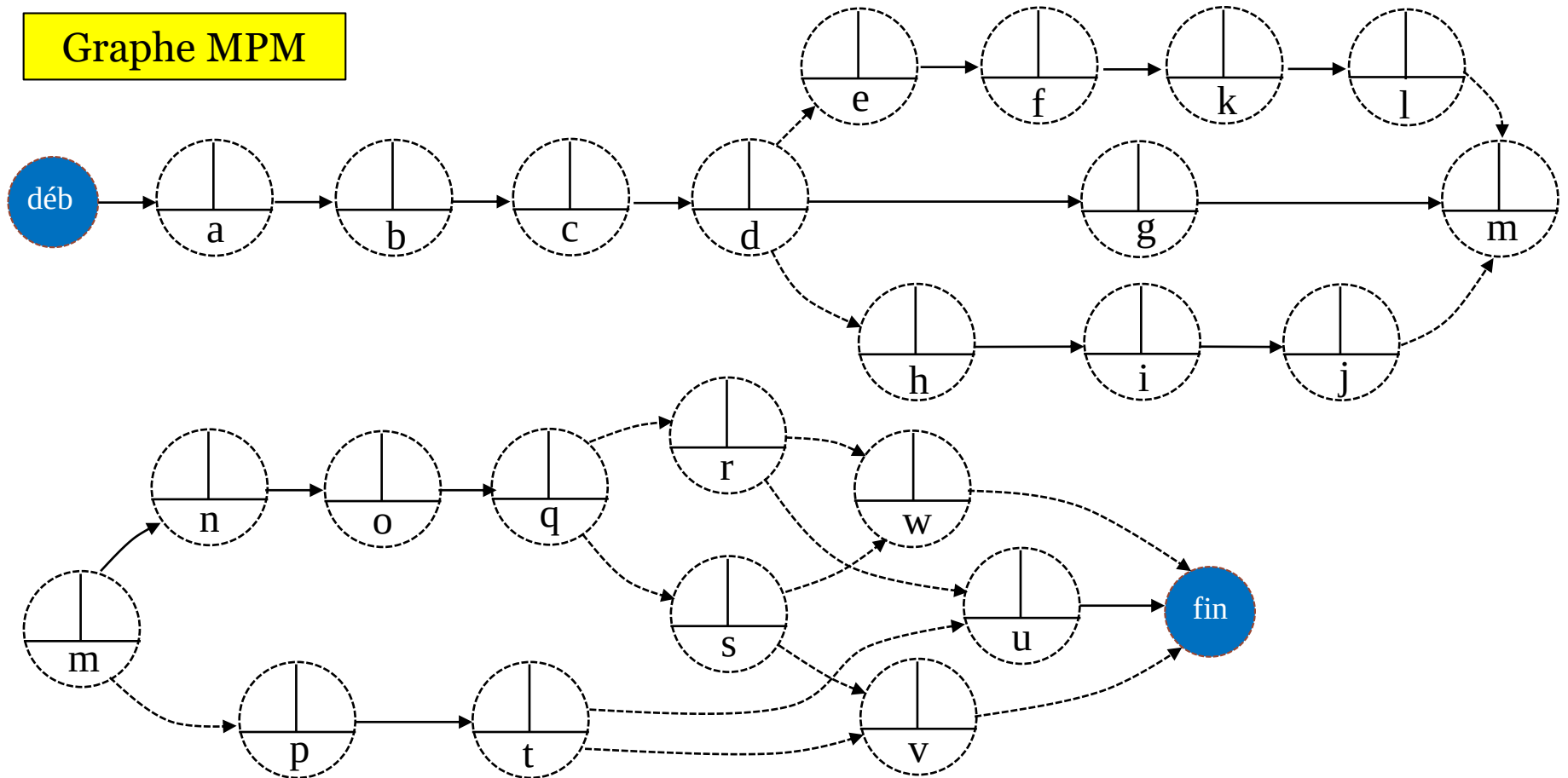
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Graphe MPM



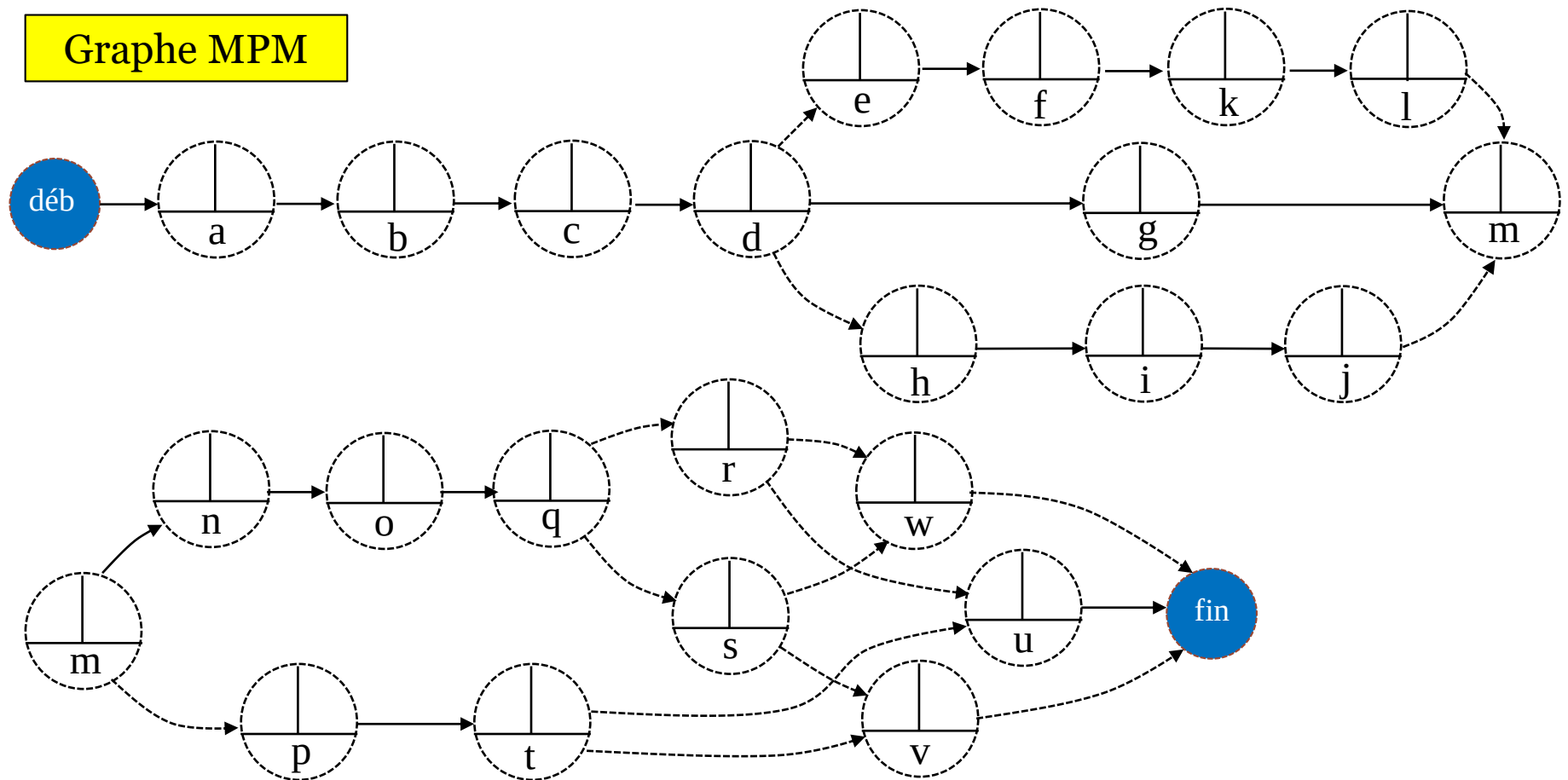
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Graphe MPM



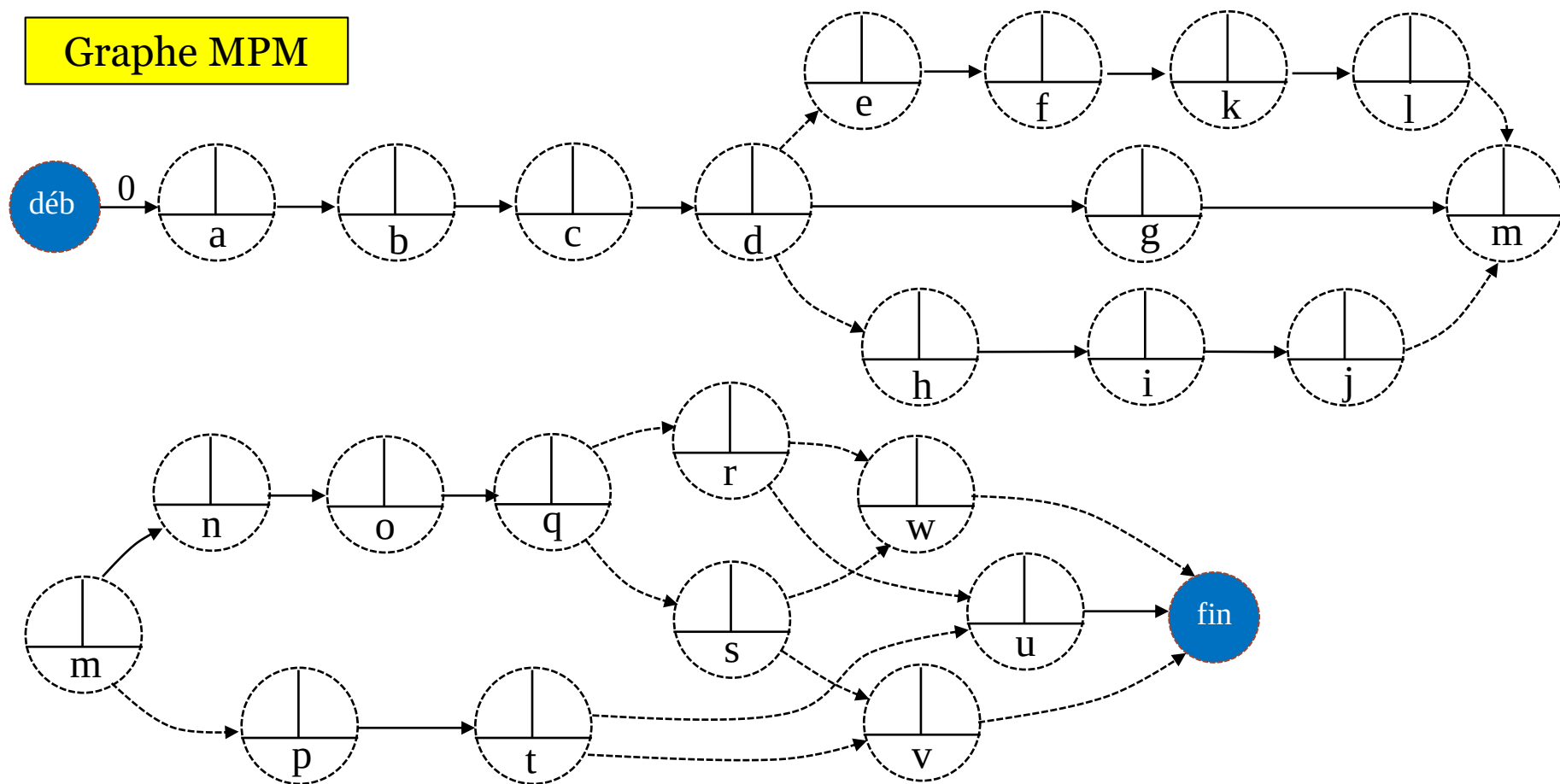
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Graphe MPM



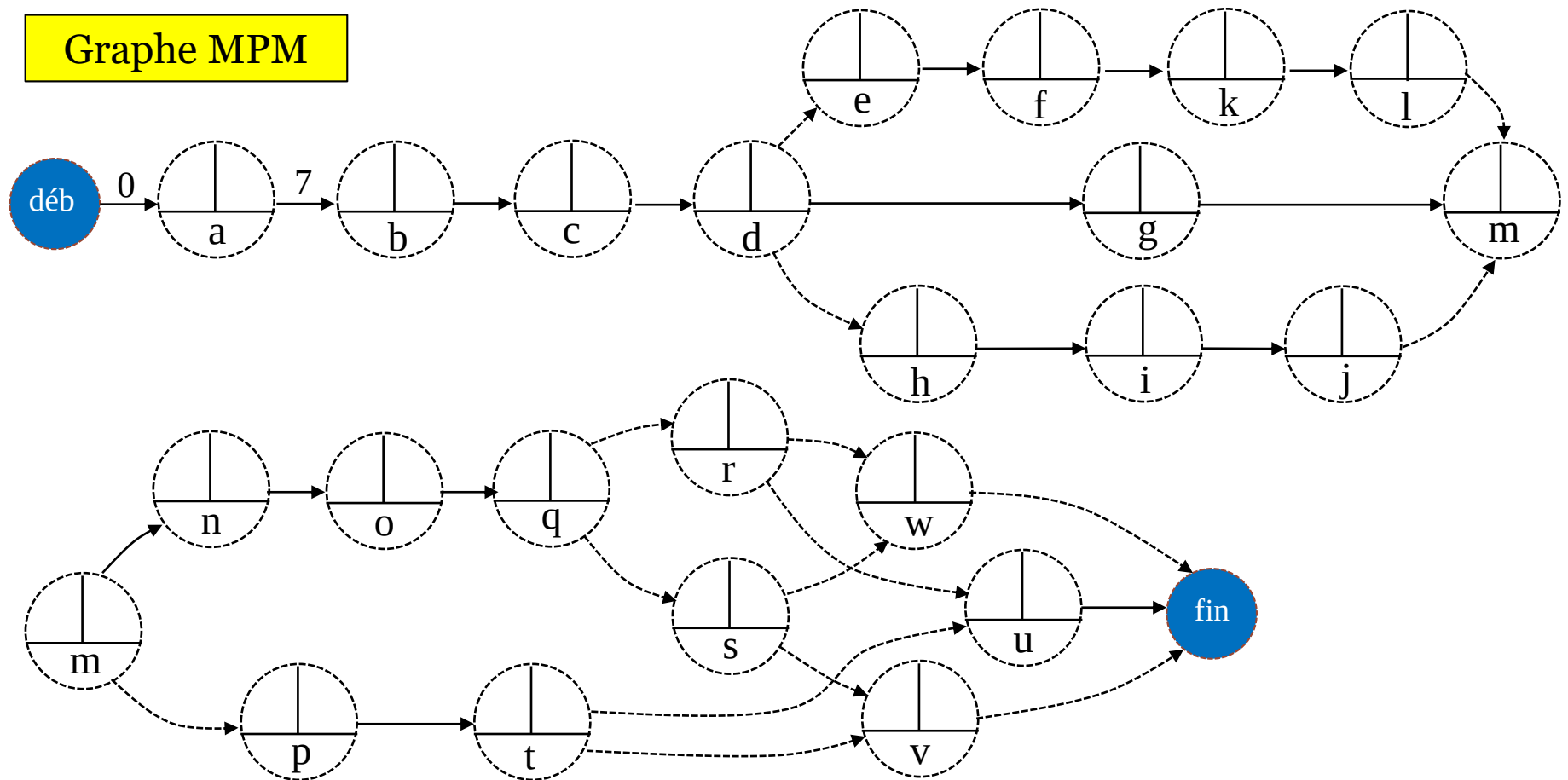
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Graphe MPM



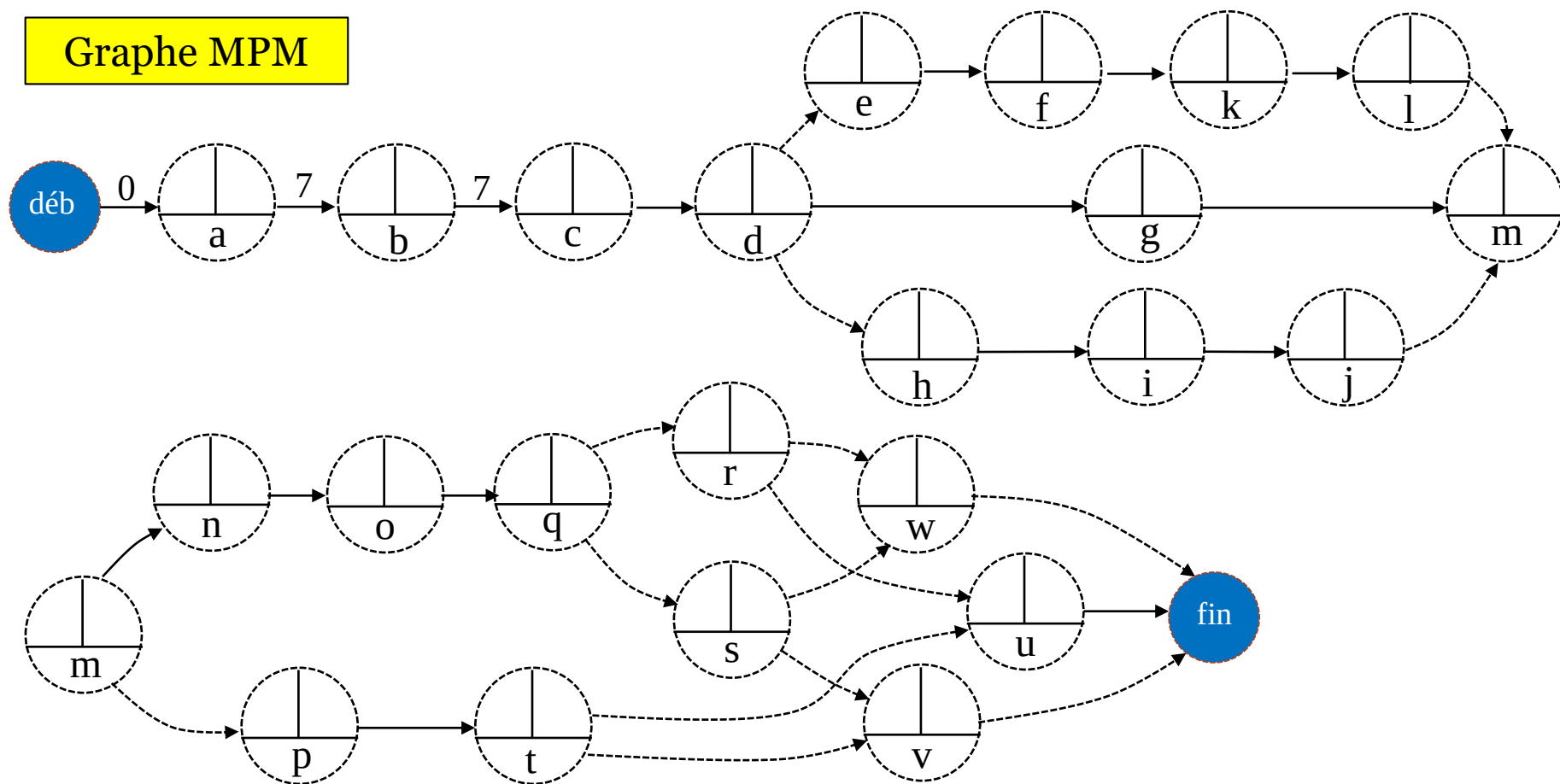
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Graphe MPM



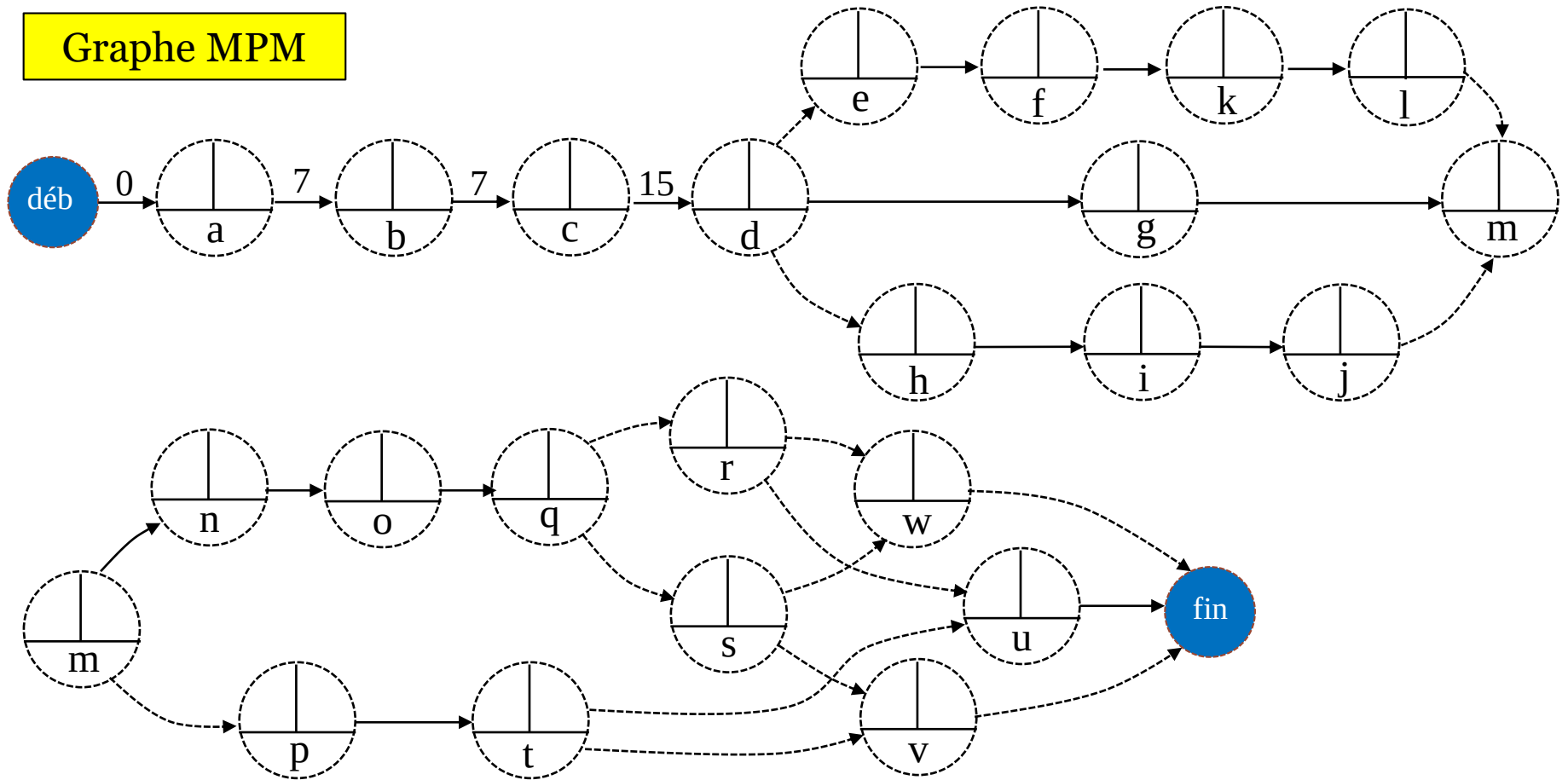
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Graphe MPM



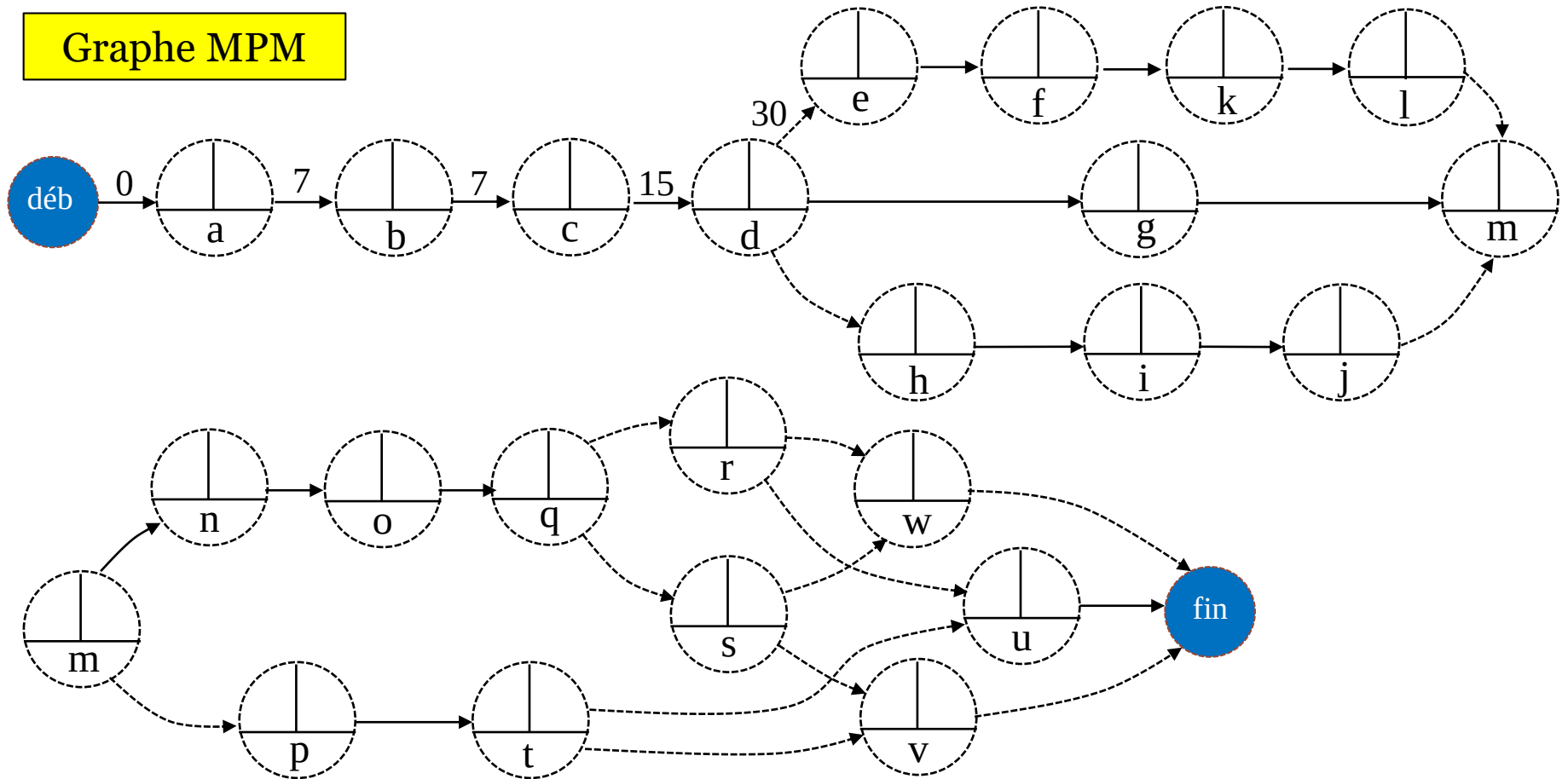
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Graphe MPM



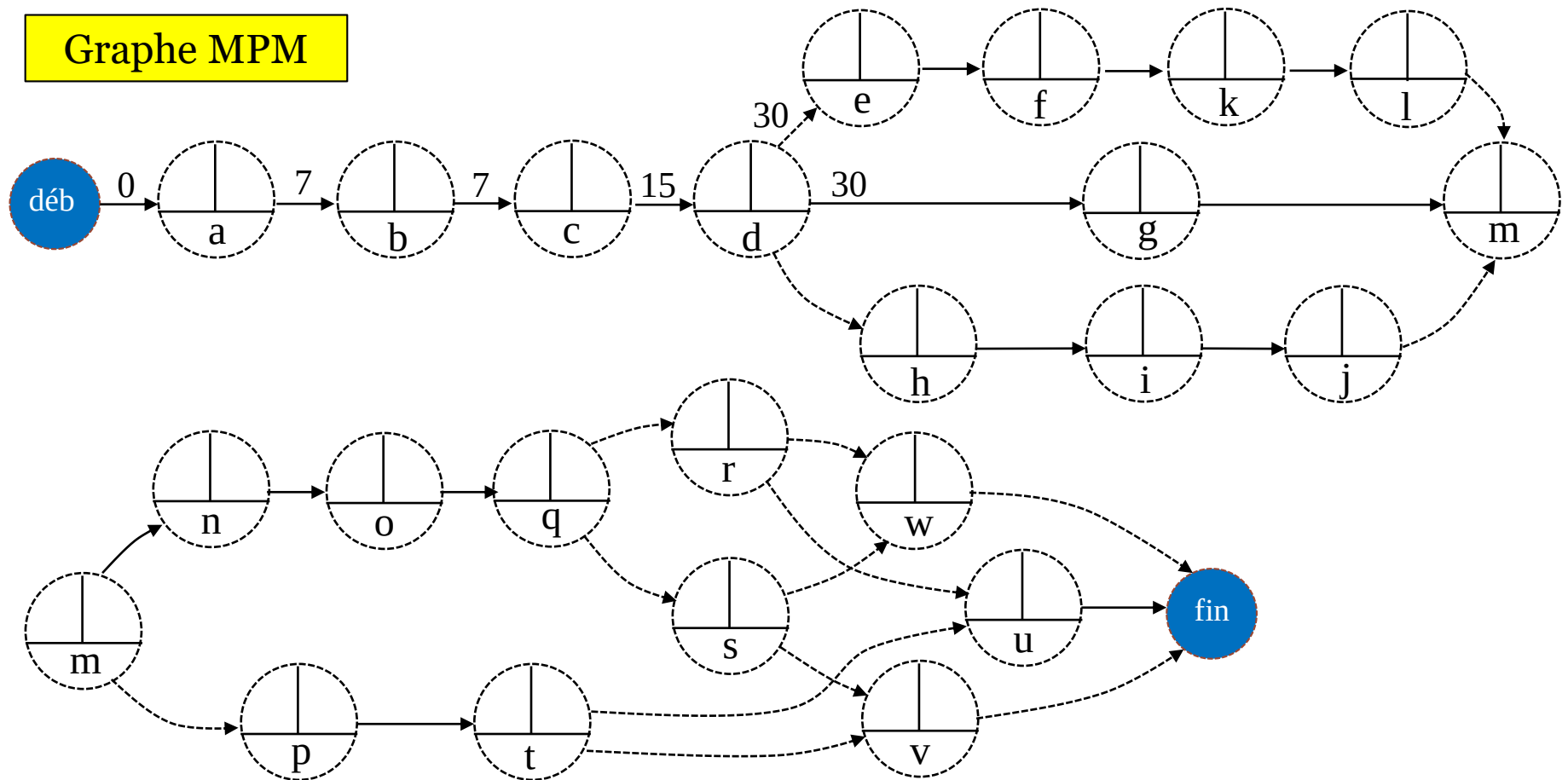
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Graphe MPM



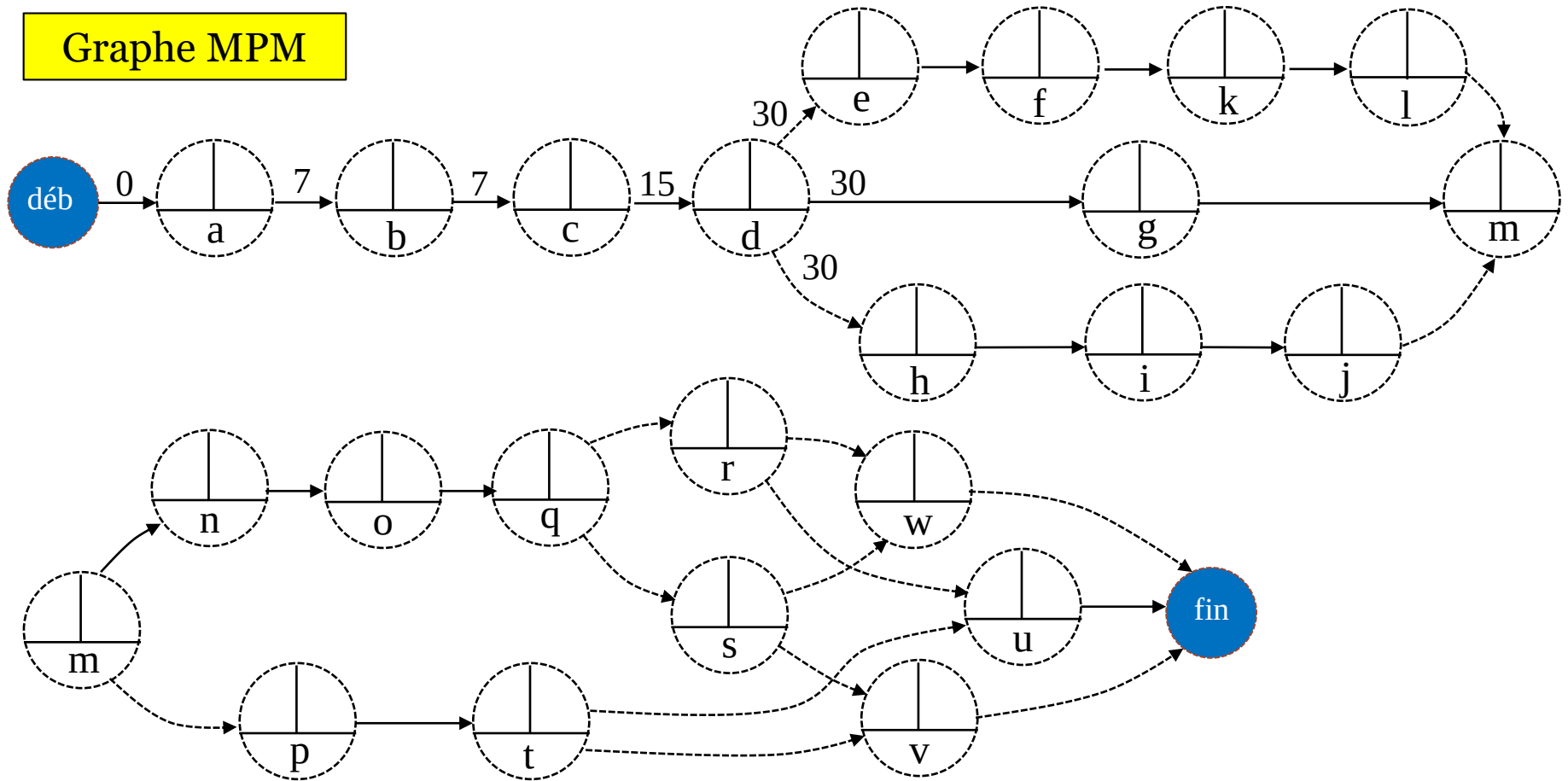
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Graphe MPM



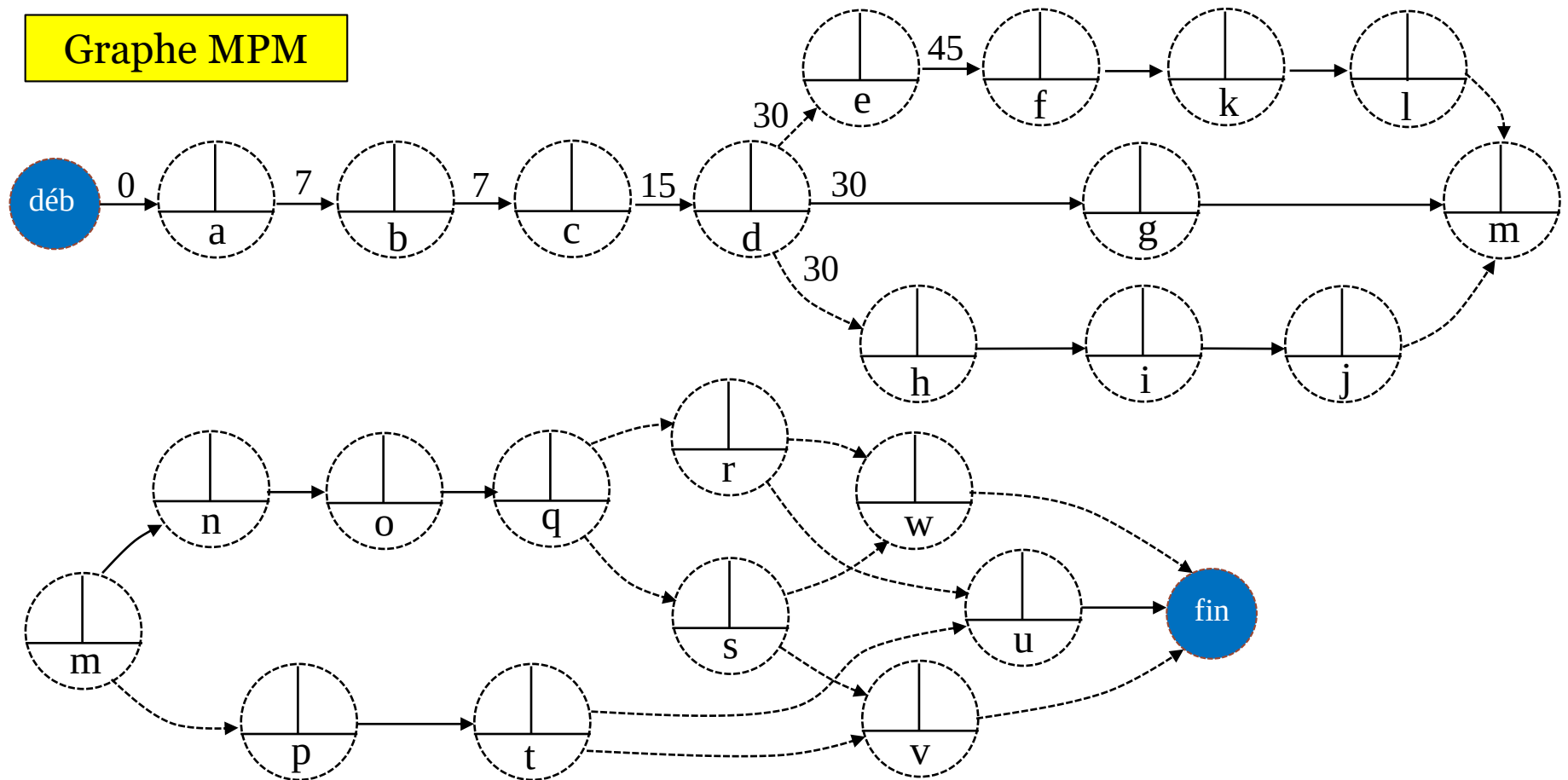
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Graphe MPM



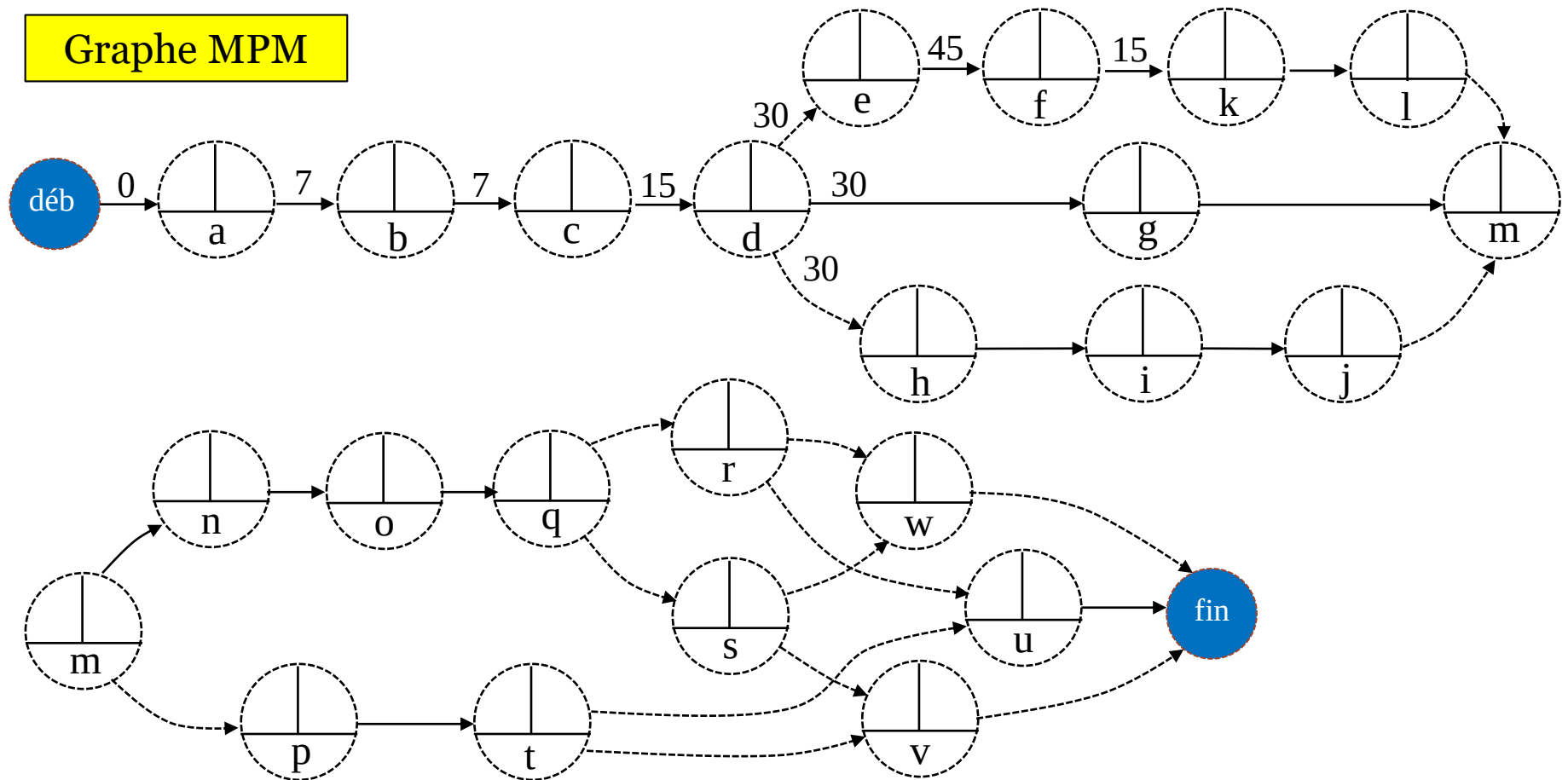
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Graphe MPM



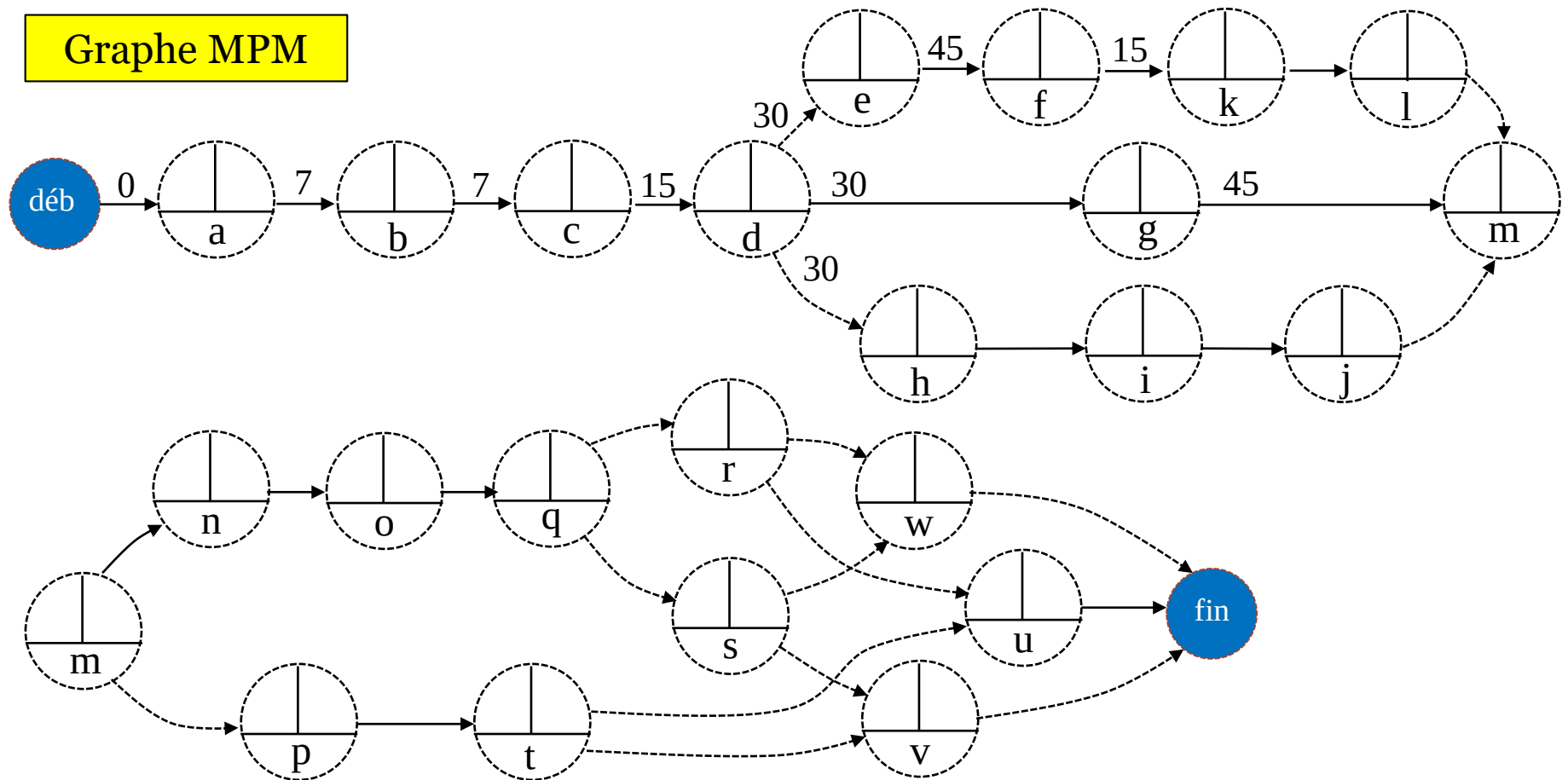
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Graphe MPM



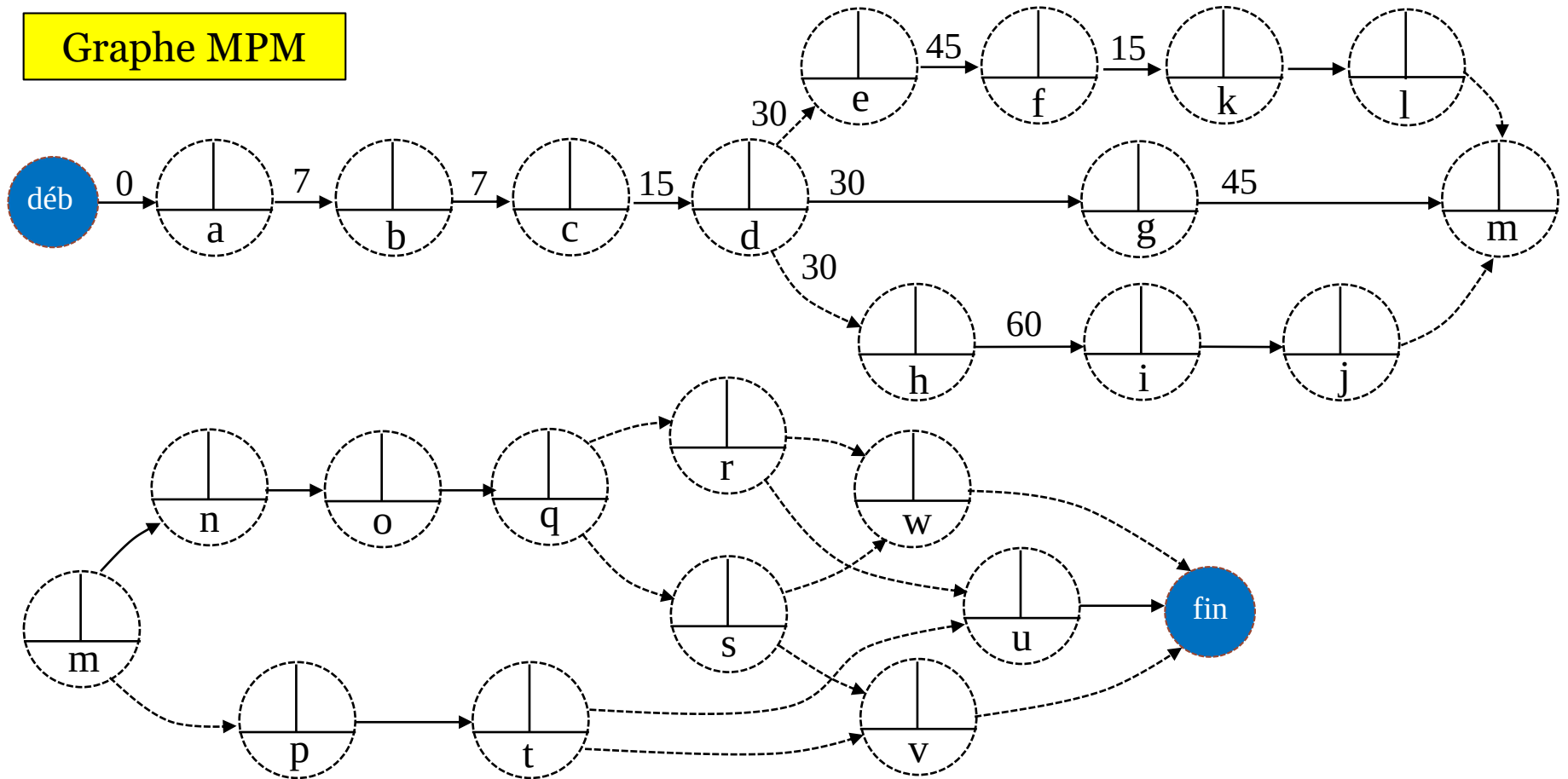
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Graphe MPM



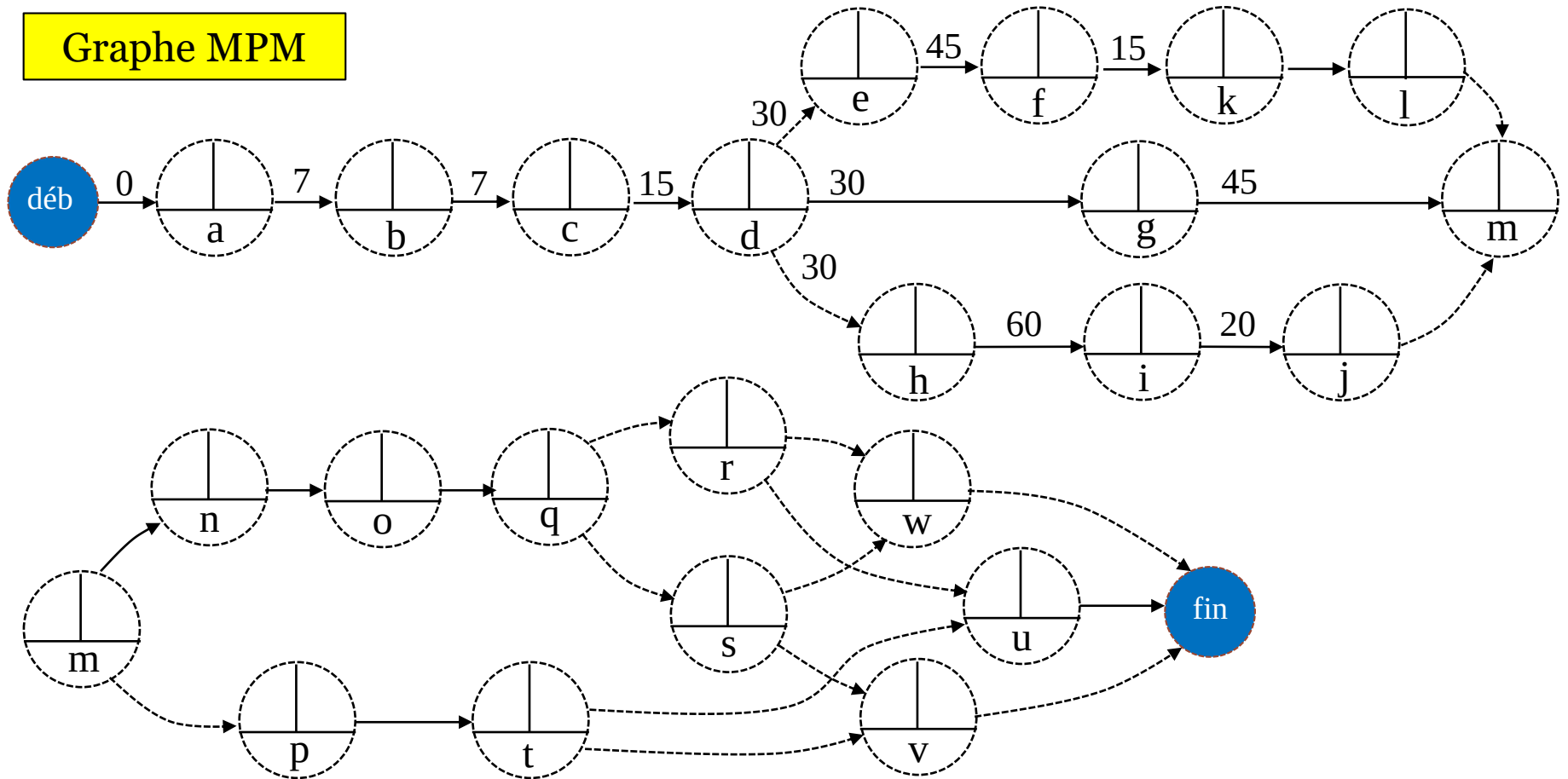
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Graphe MPM



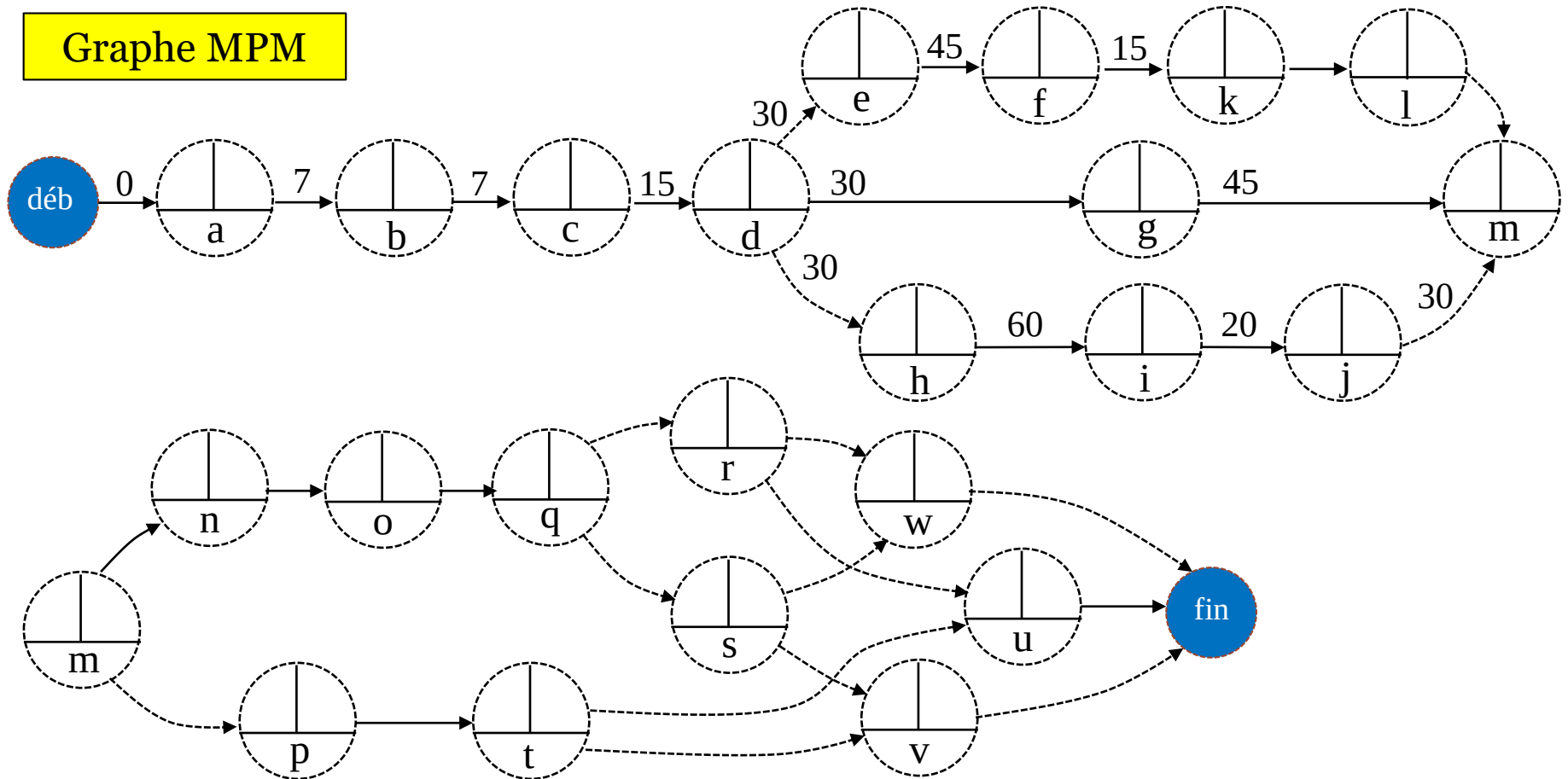
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Durée	7	7	15	30	45	15	45	60	20	30	30	15	30	15	30	15	15	15	30	7	4	2	7

Graphe MPM



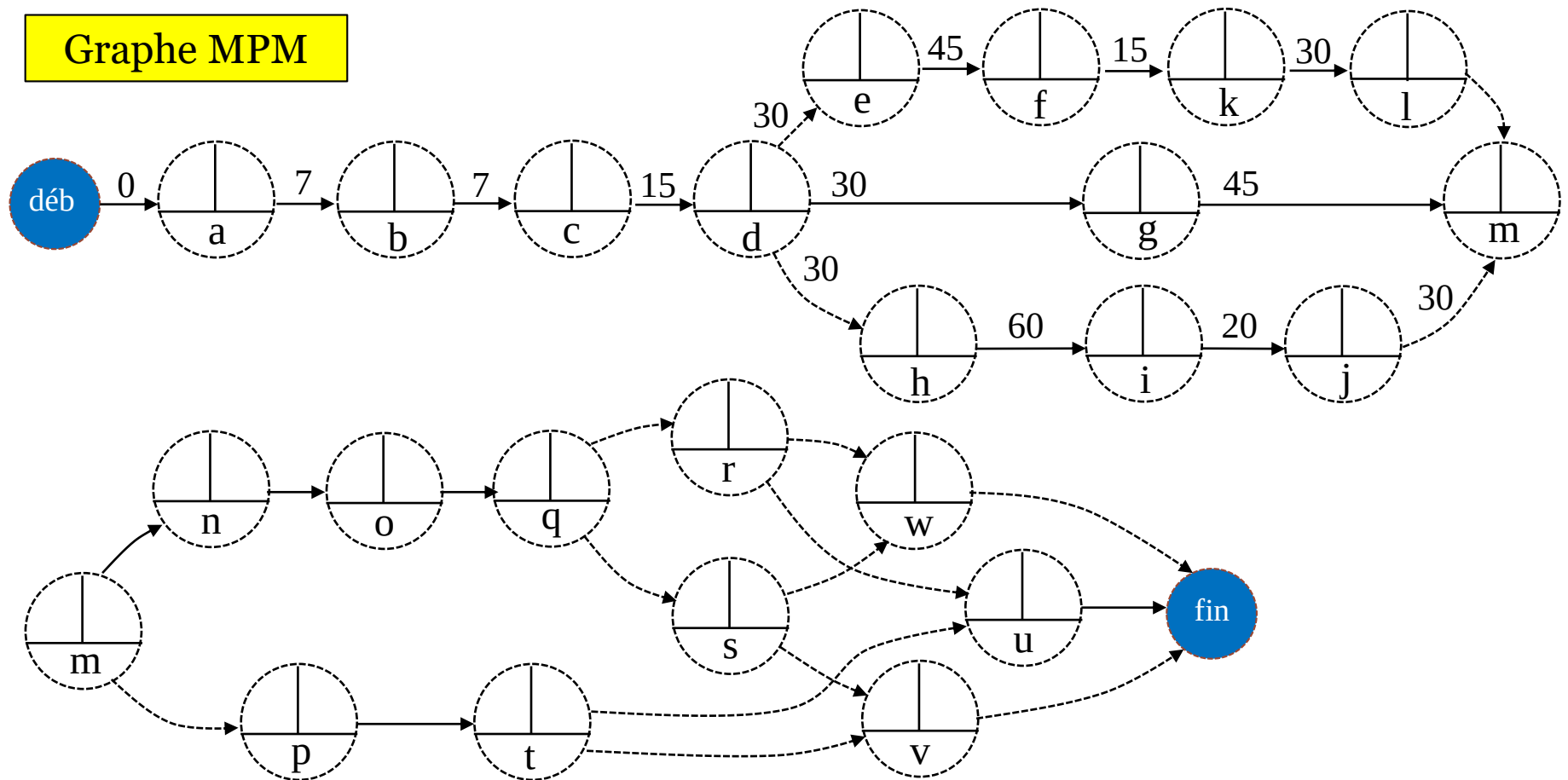
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Graphe MPM



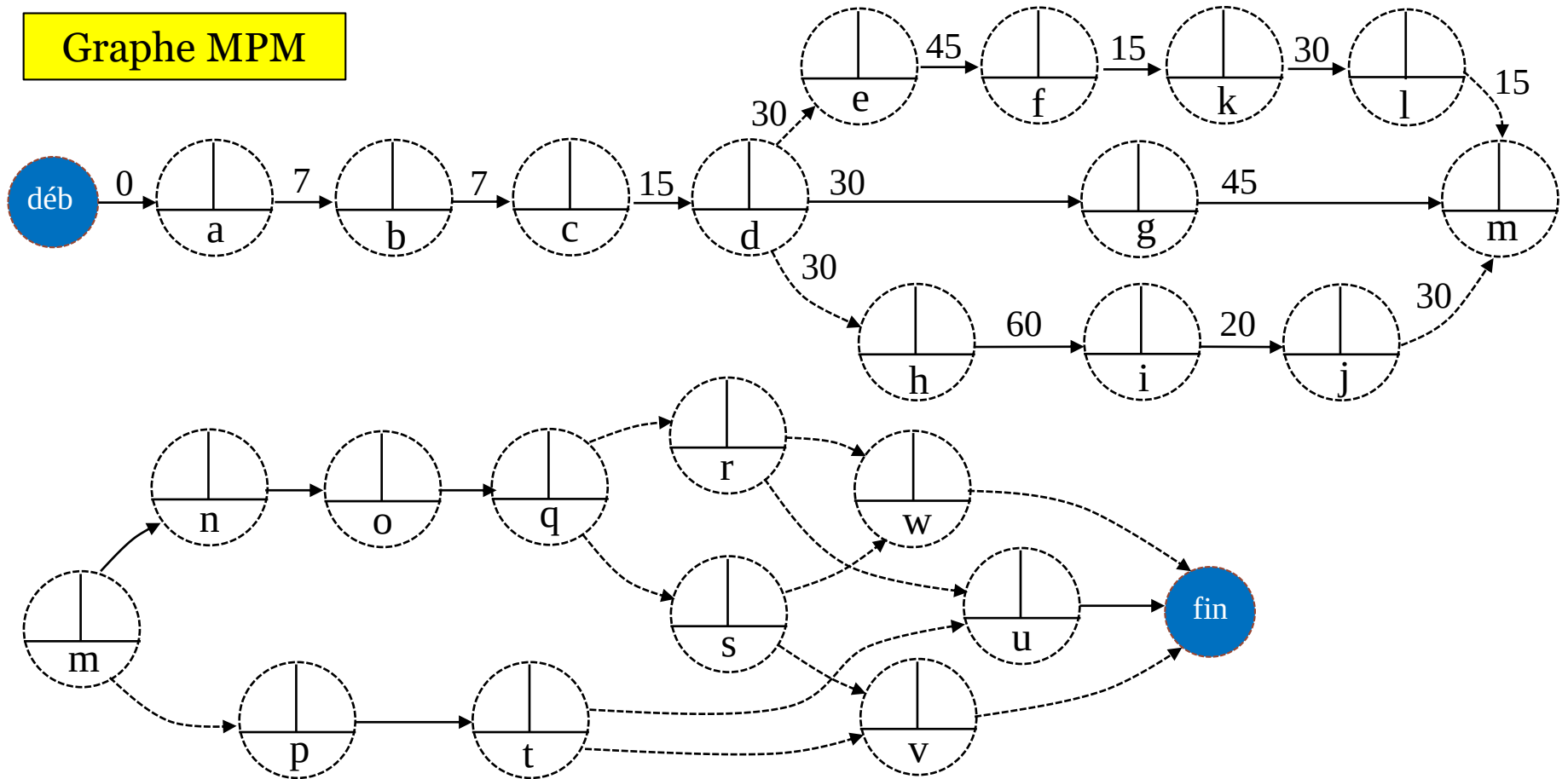
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Graphe MPM



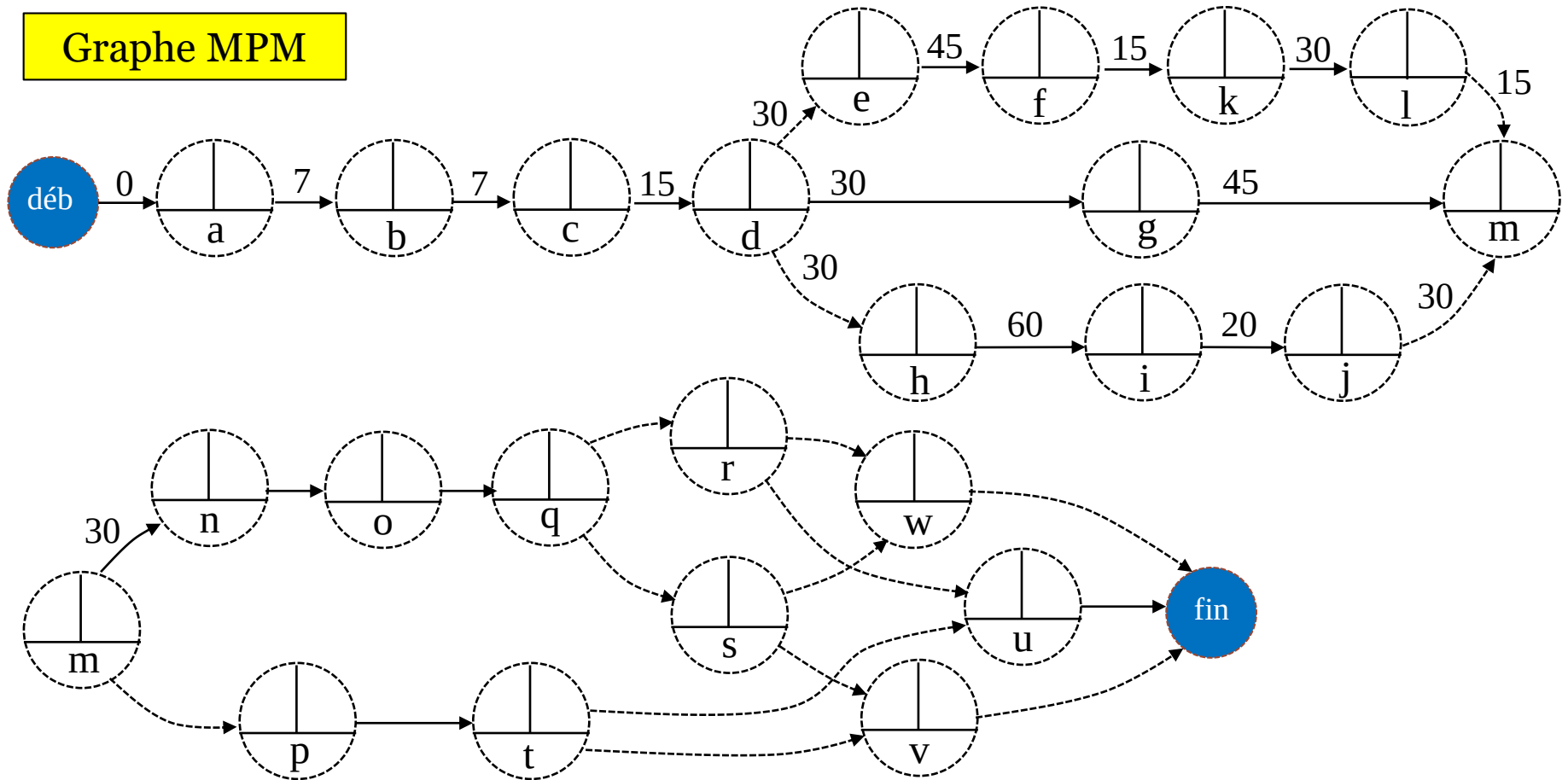
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Graphe MPM



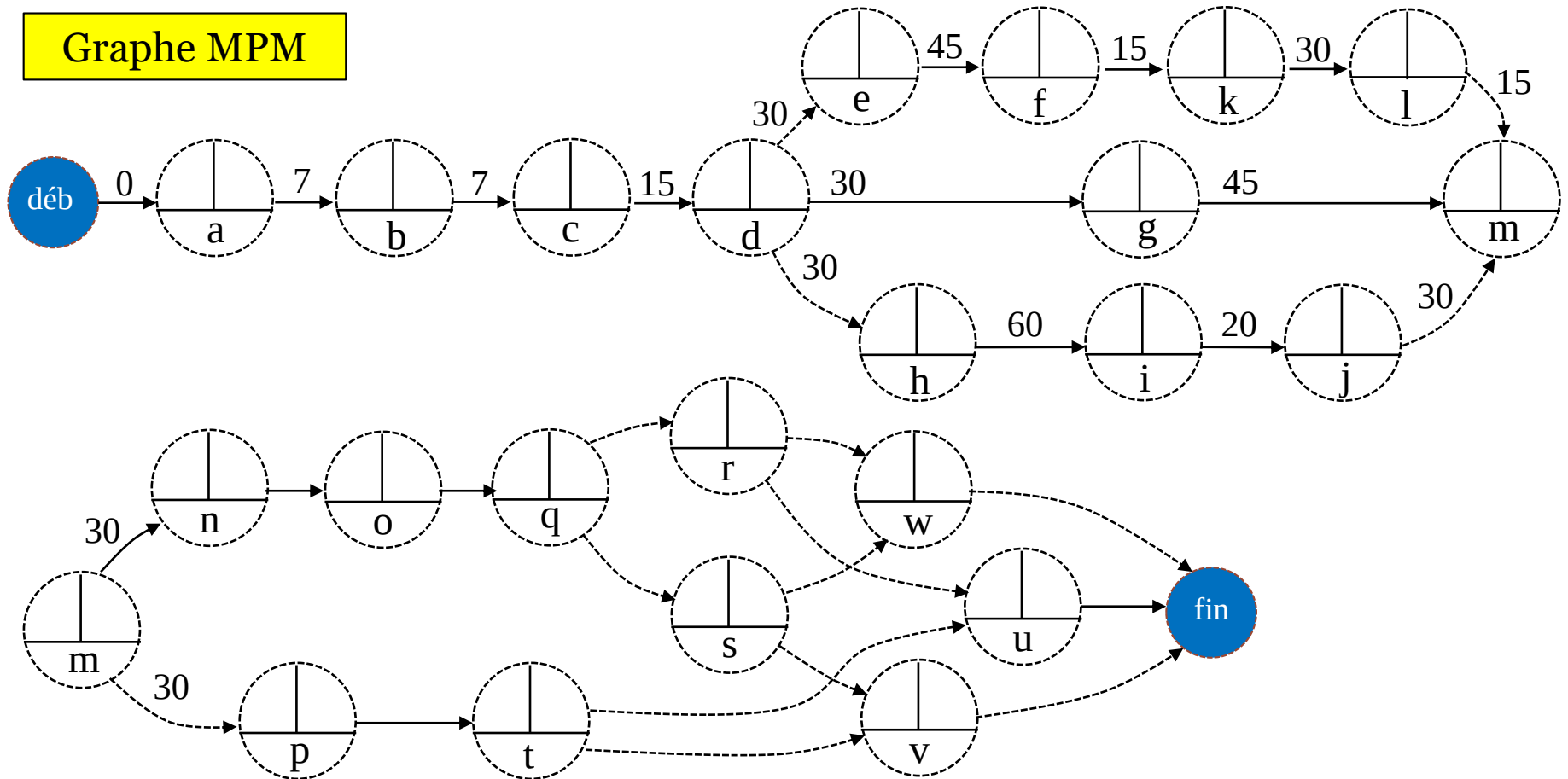
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Graphe MPM



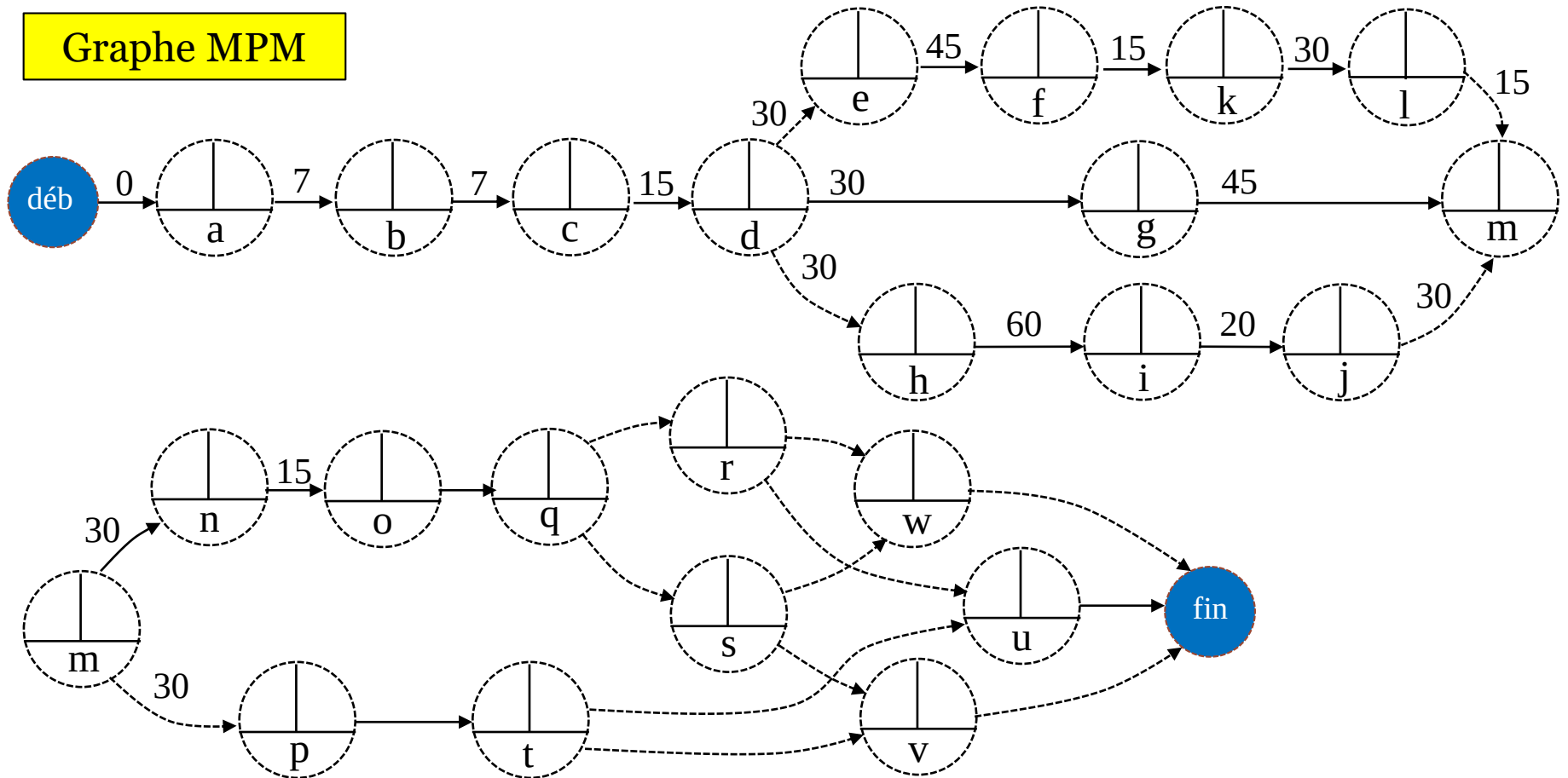
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Graphe MPM



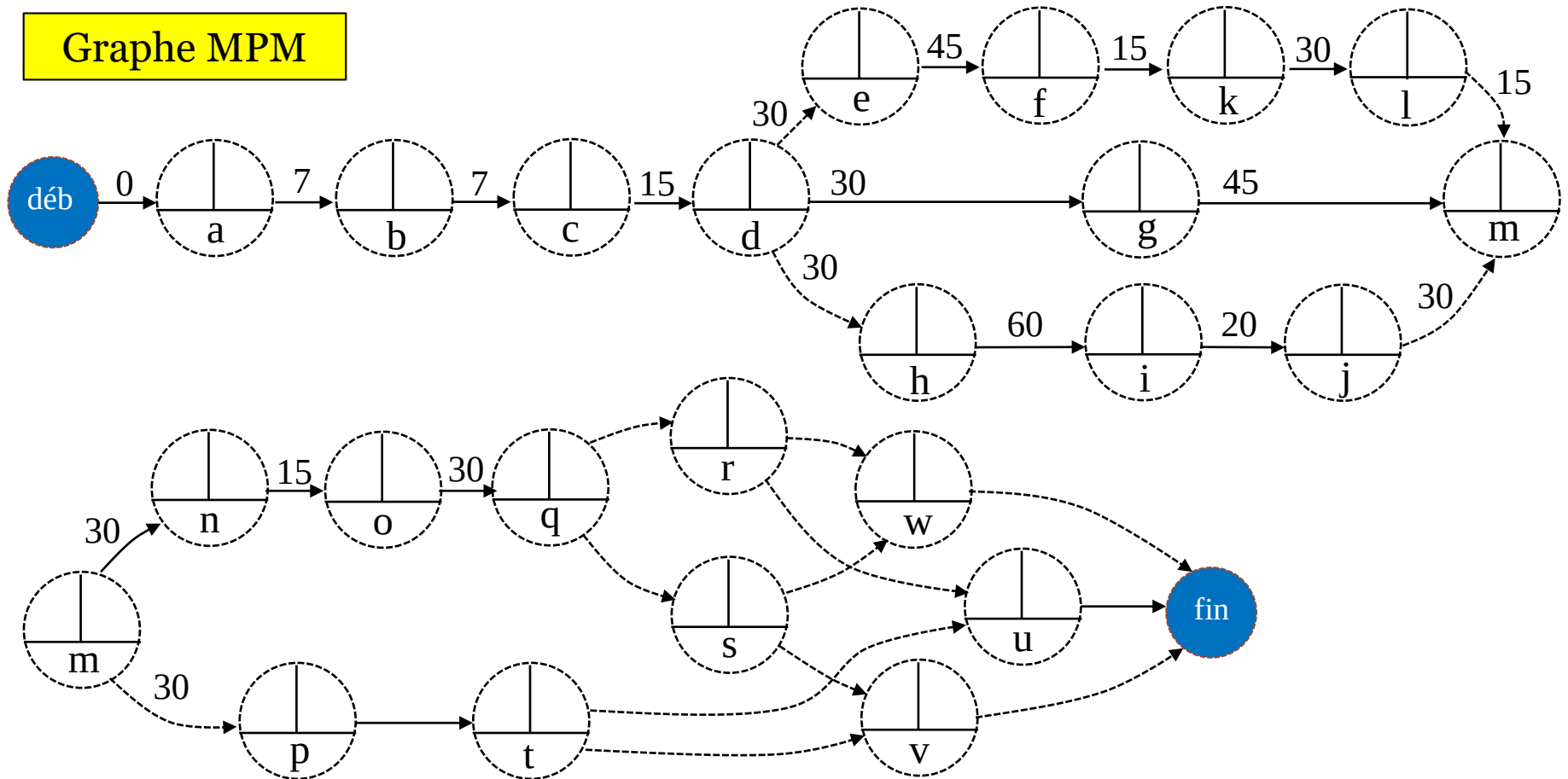
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Graphe MPM



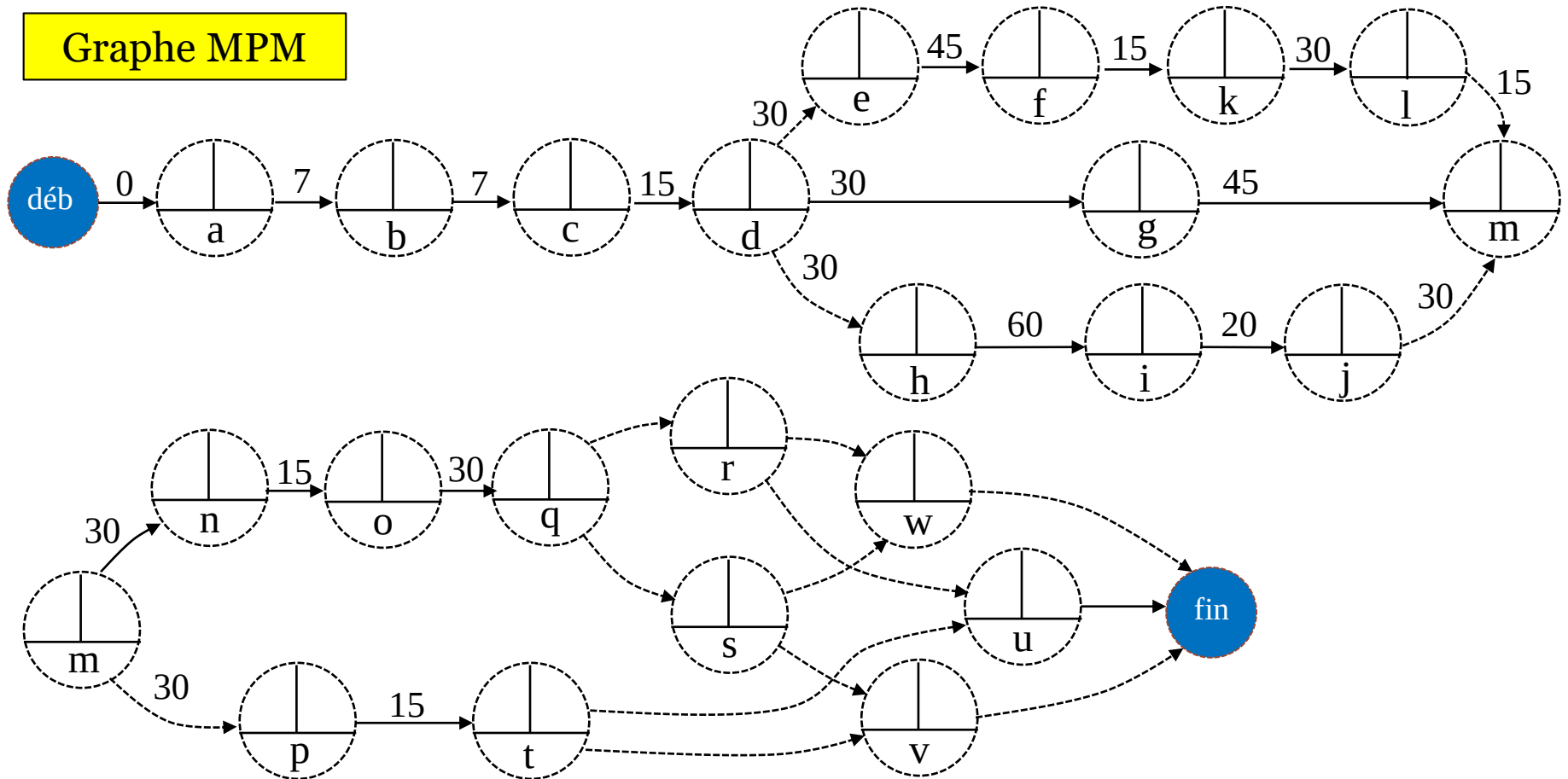
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Graphe MPM



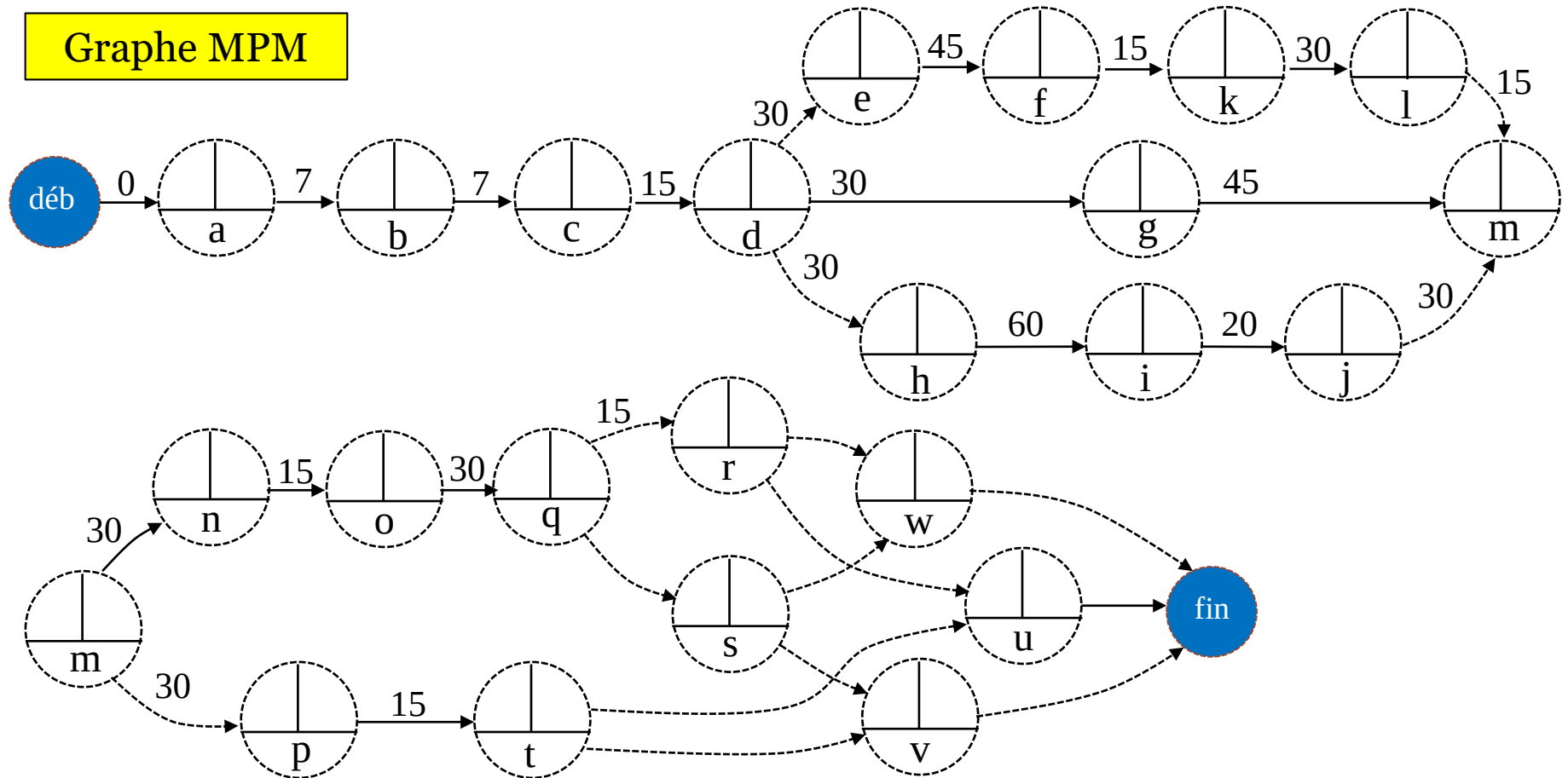
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Graphe MPM



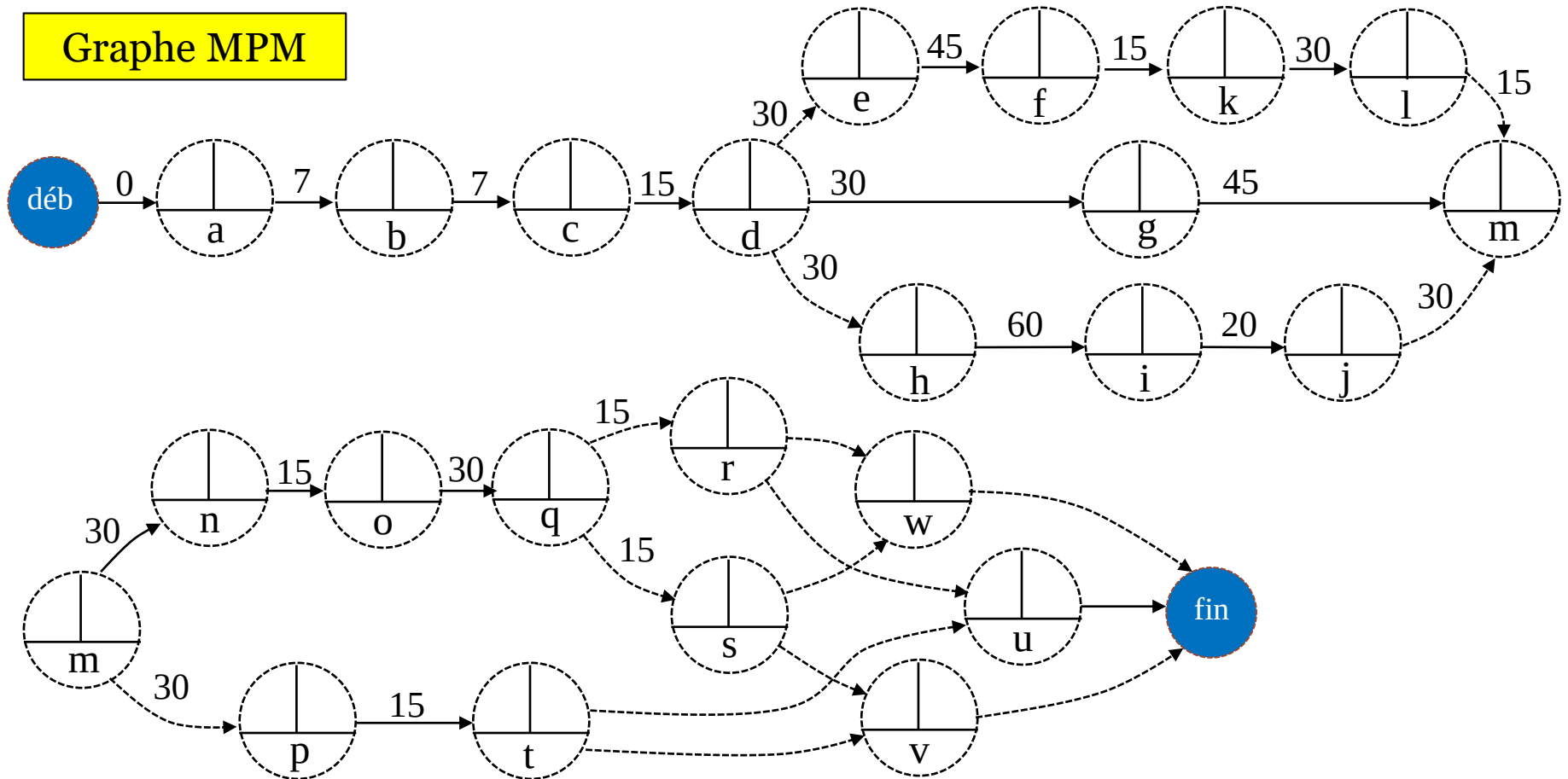
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Graphe MPM



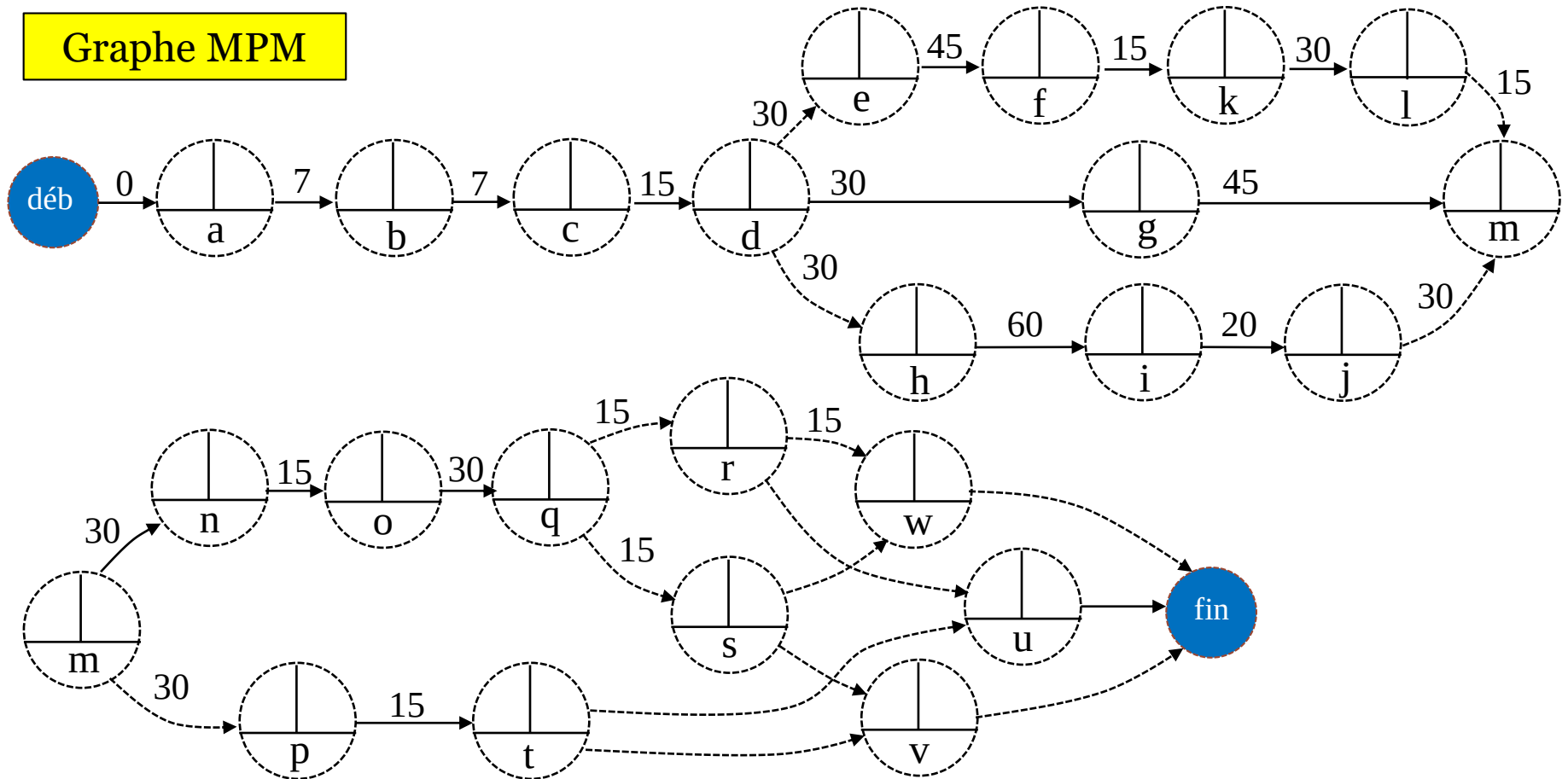
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Graphe MPM



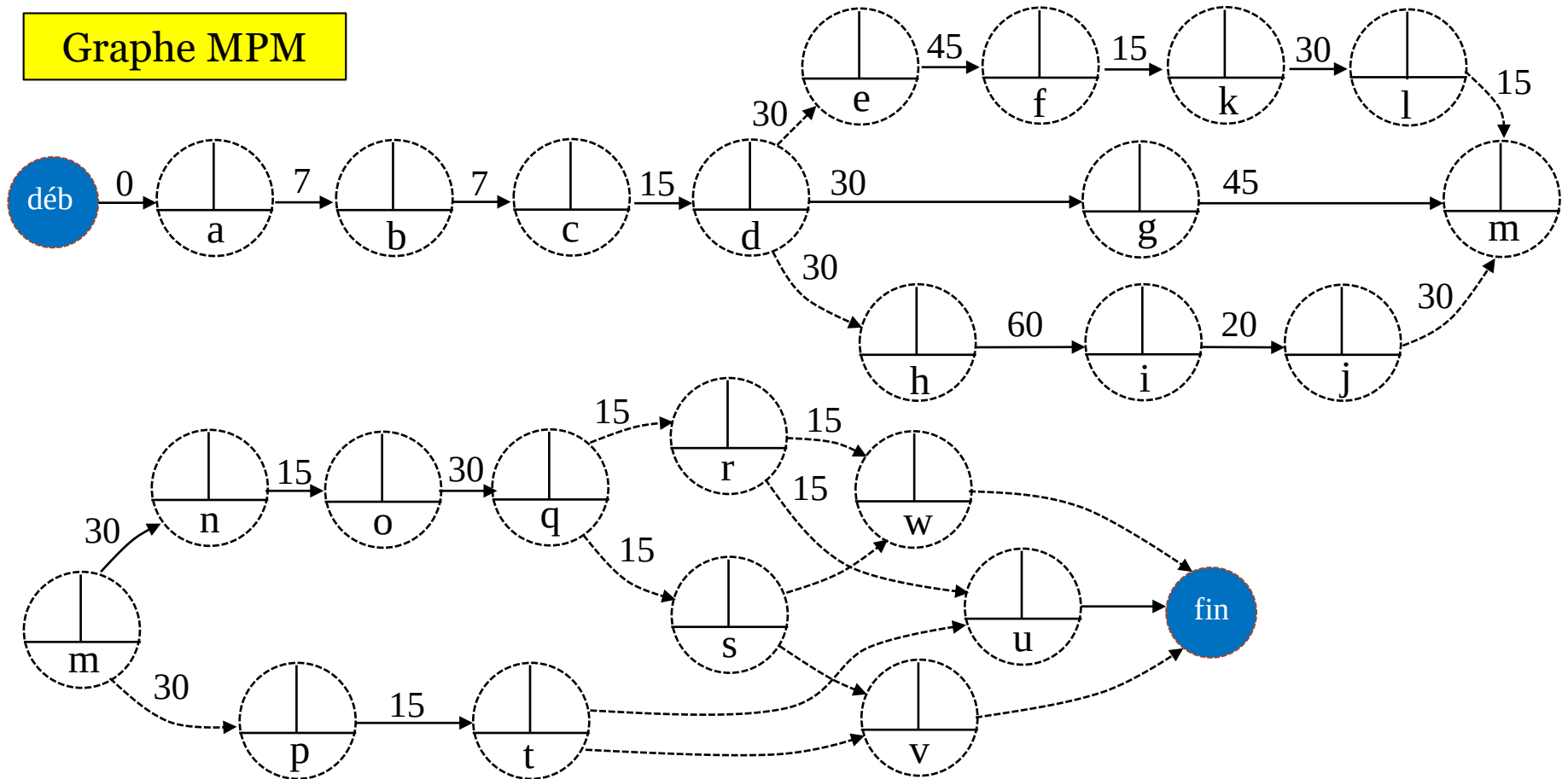
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Graphe MPM



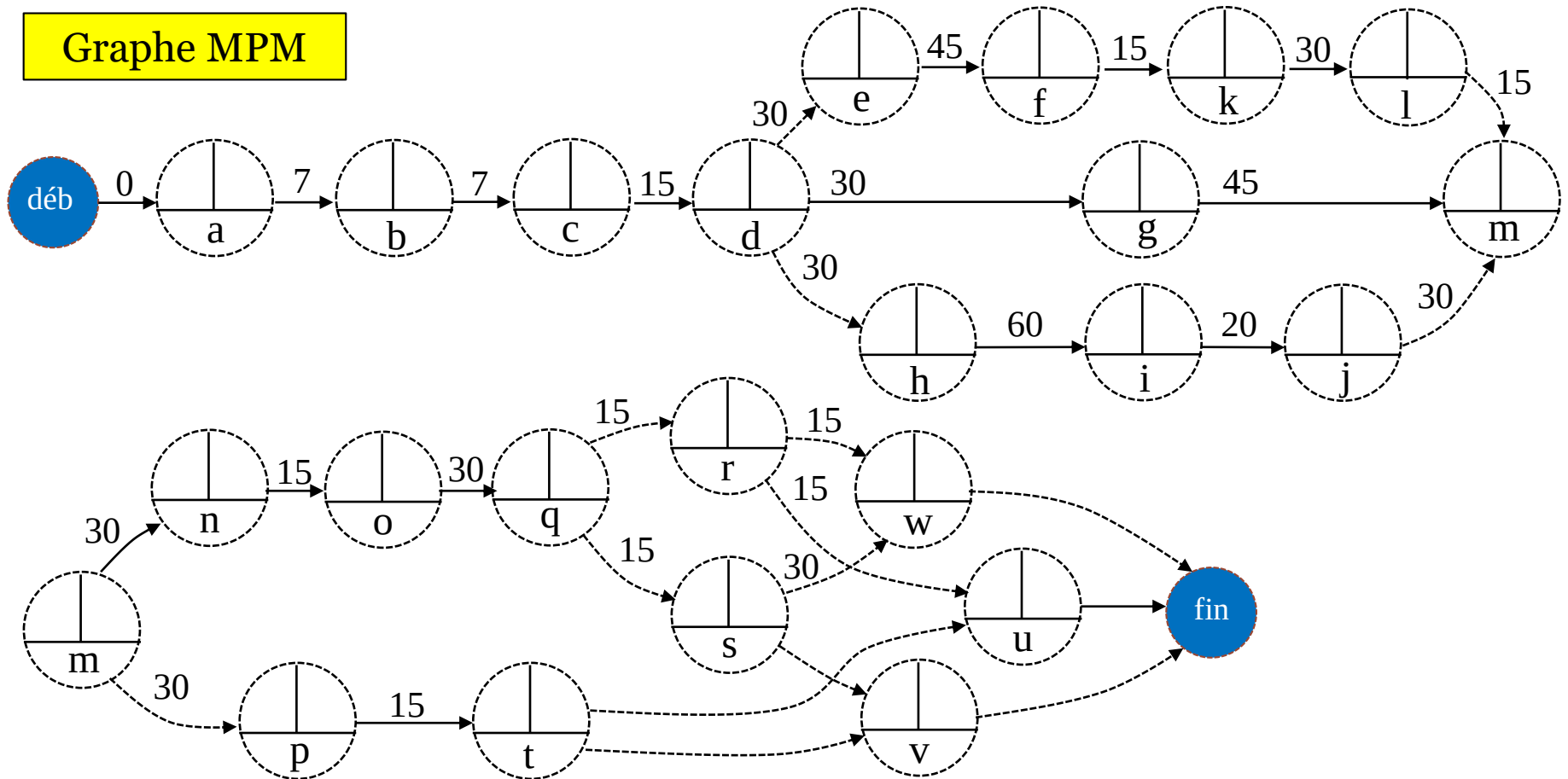
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Graphe MPM



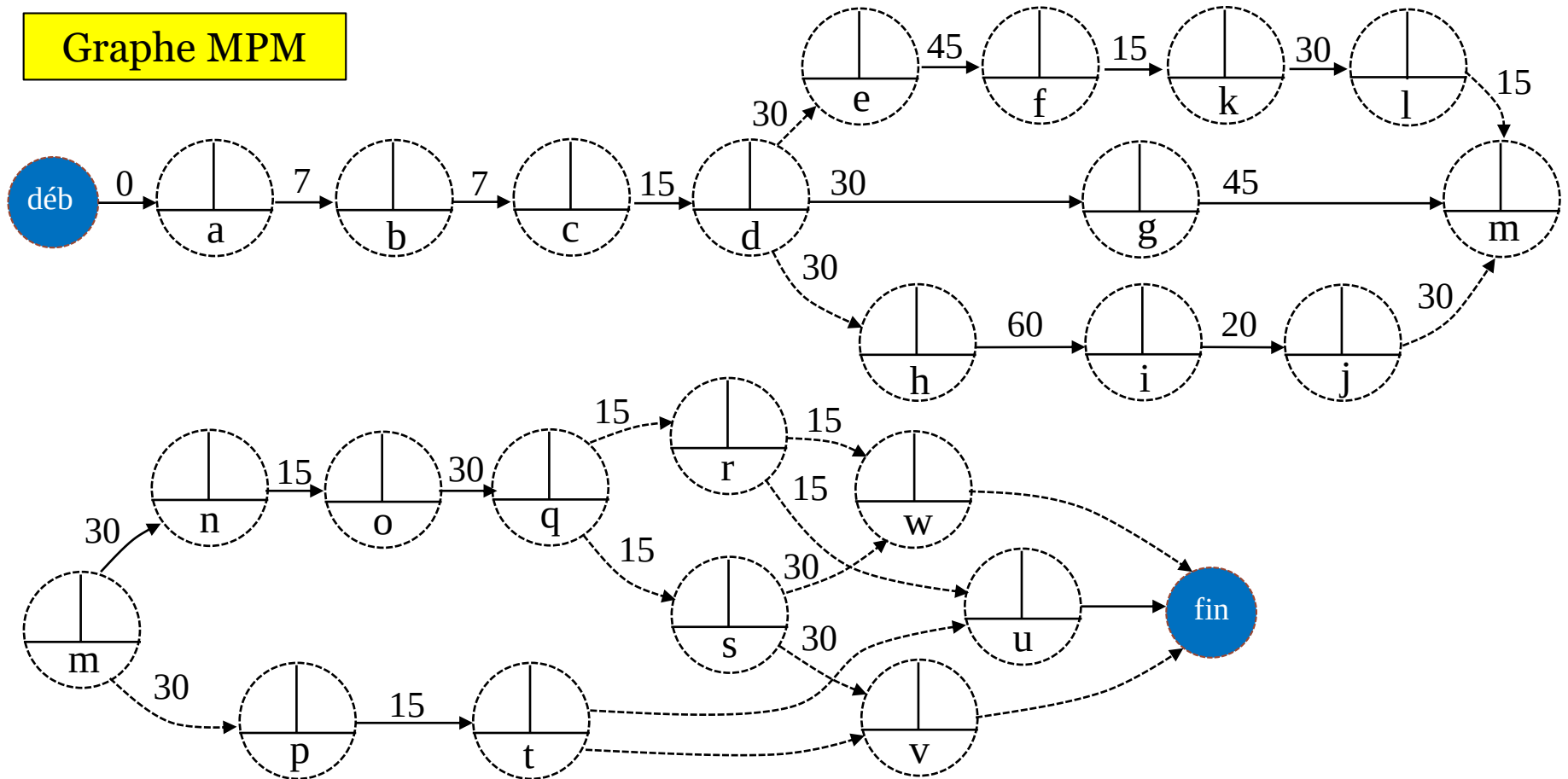
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Graphe MPM



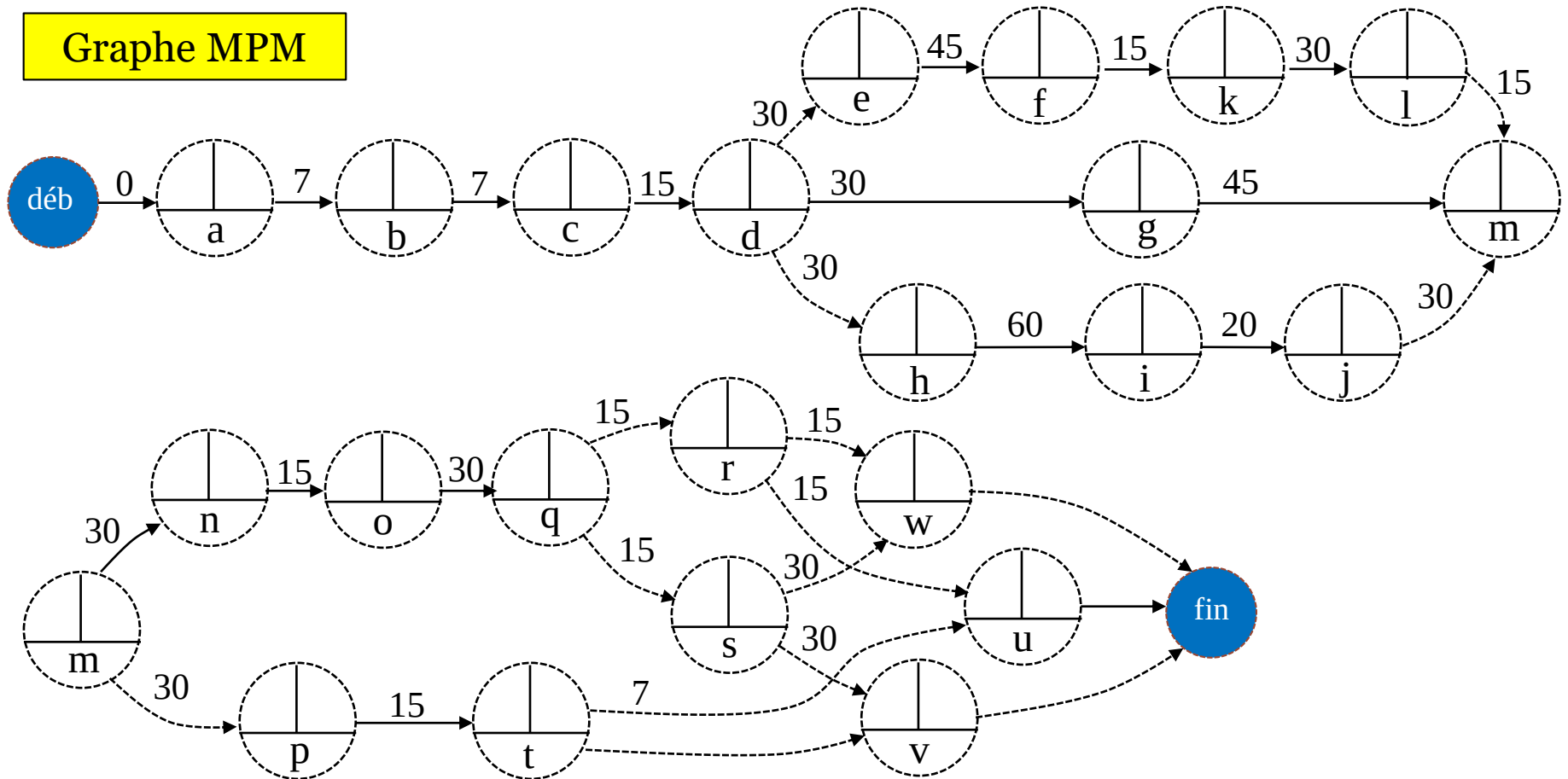
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Graphe MPM



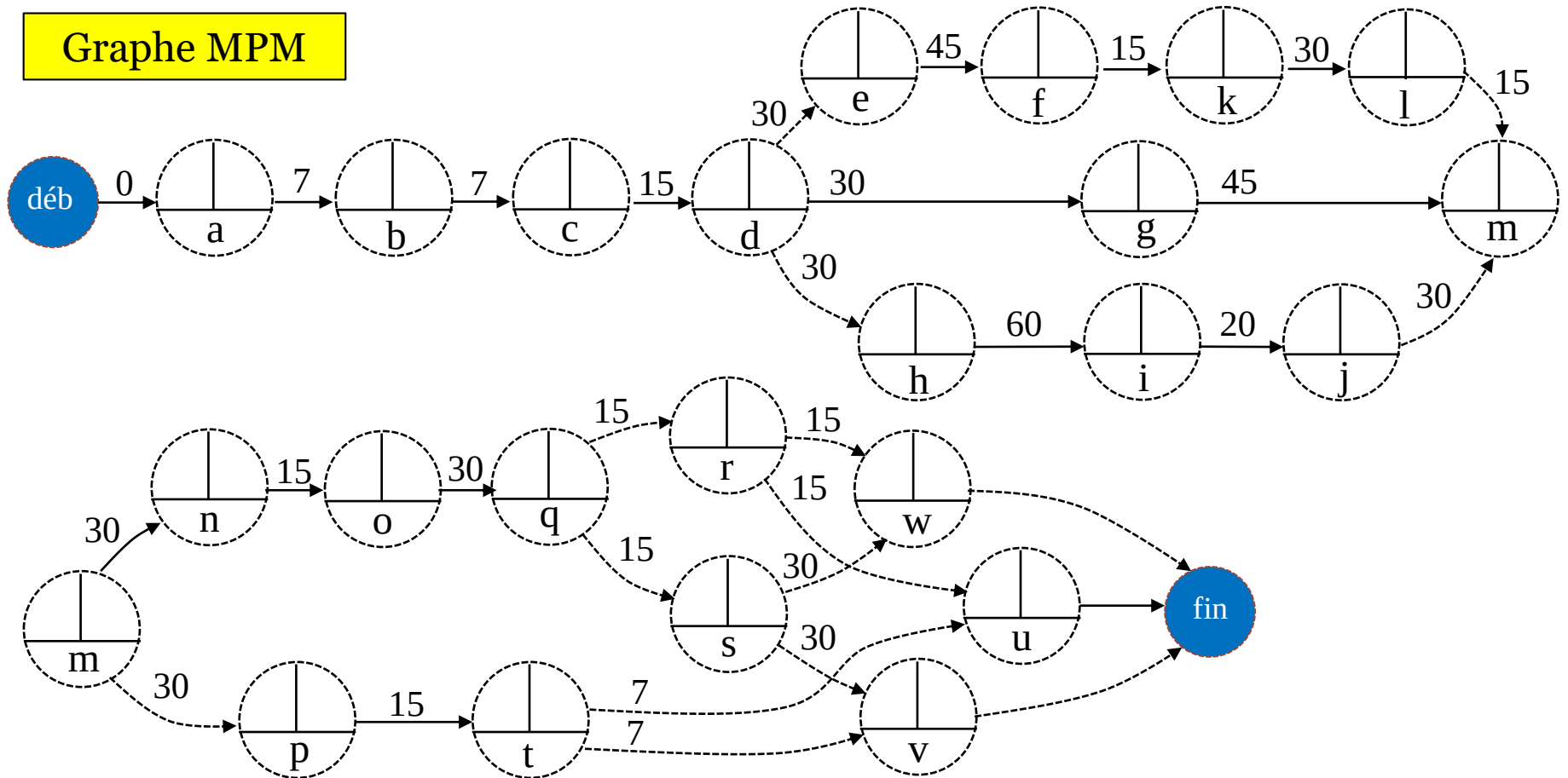
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Graphe MPM



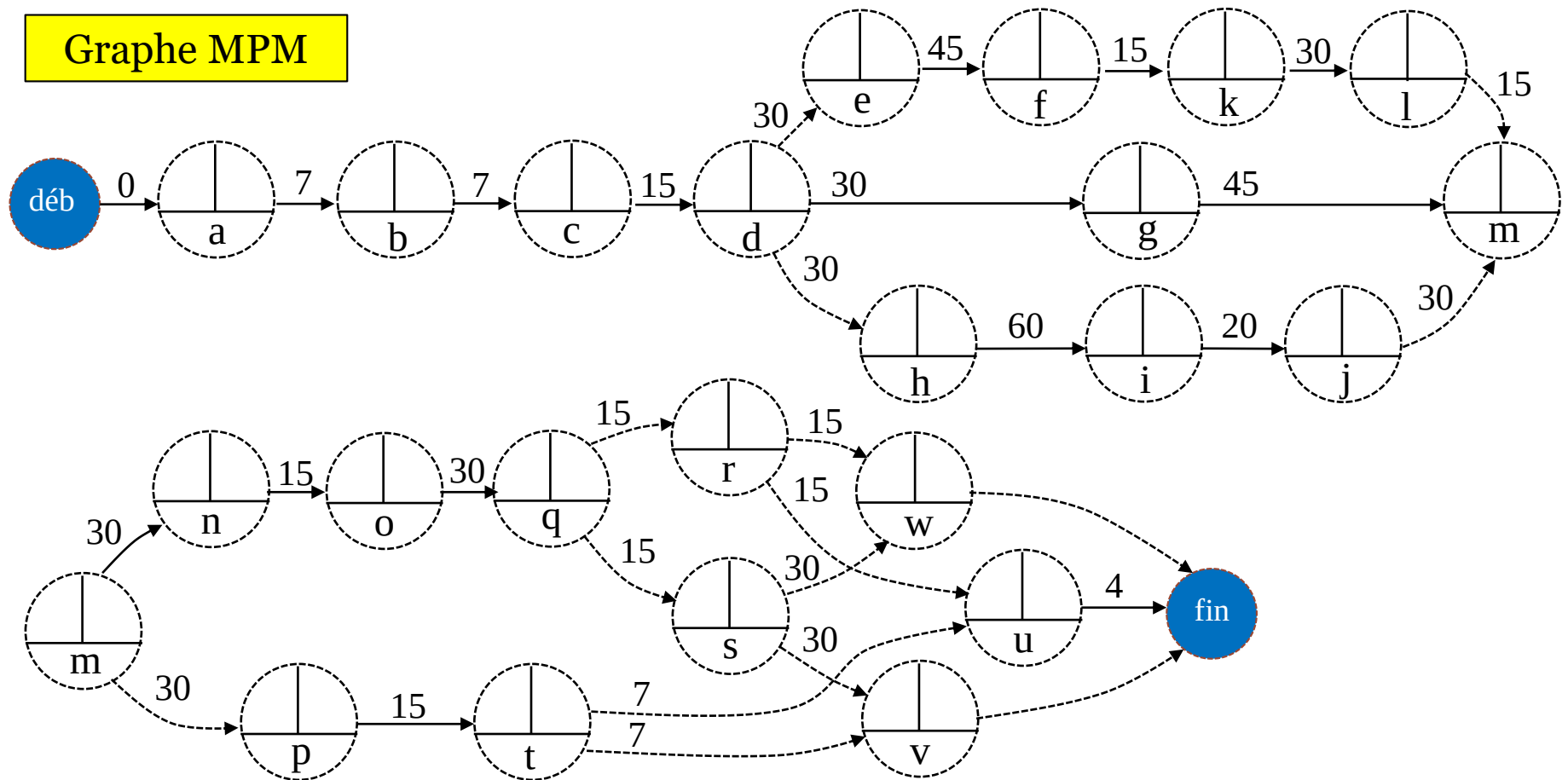
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Graphe MPM



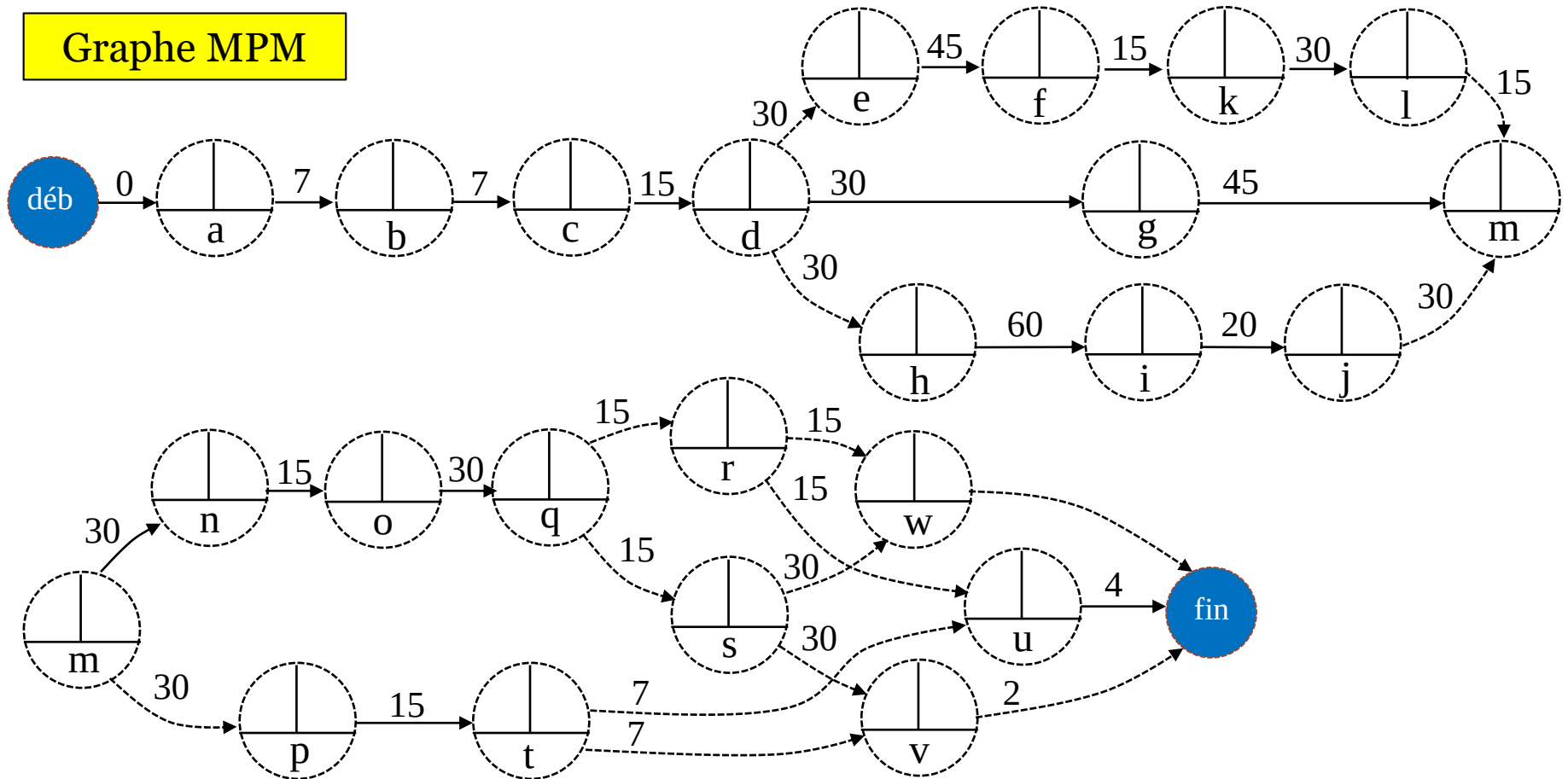
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Graphe MPM



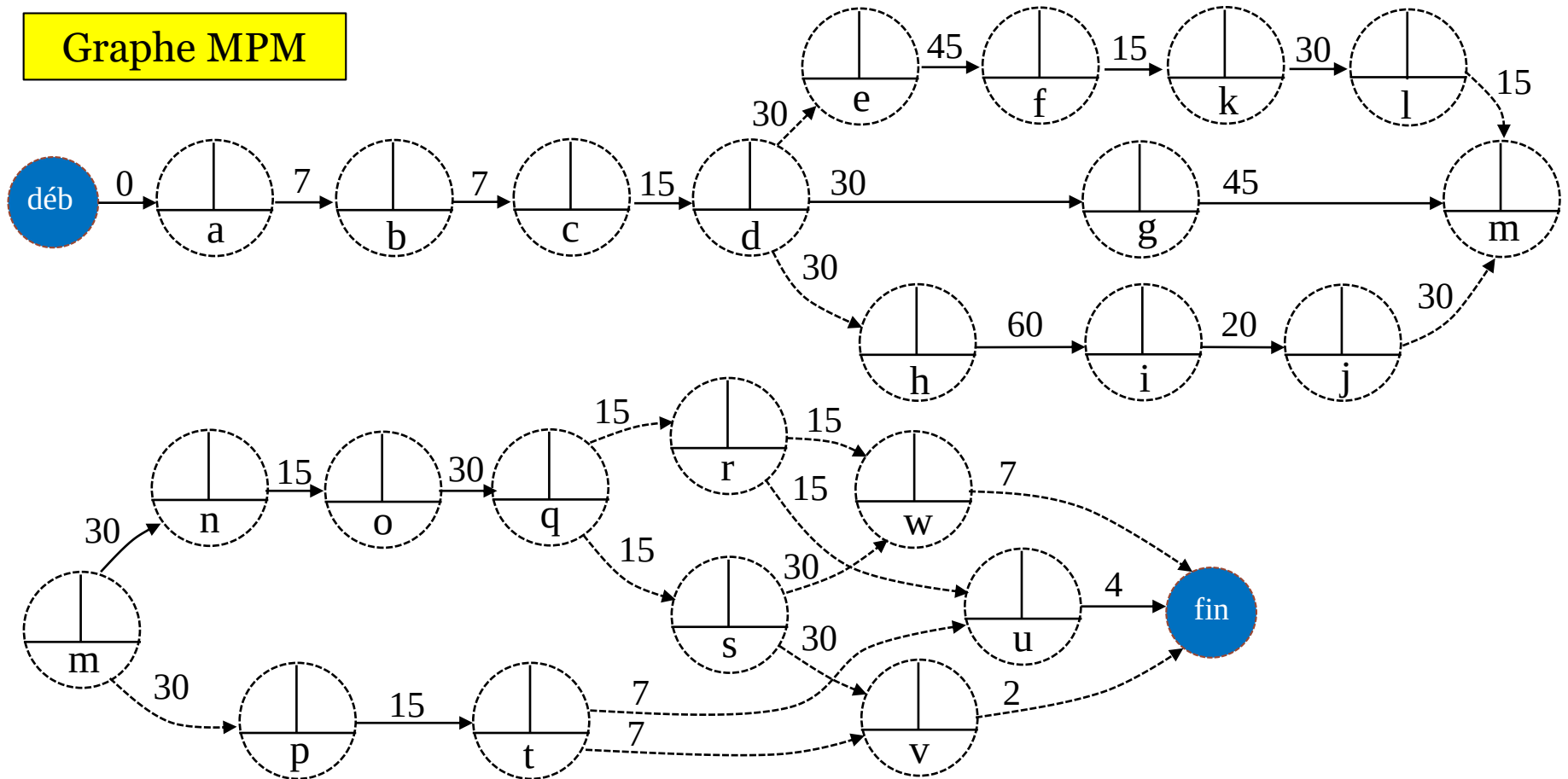
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Graphe MPM



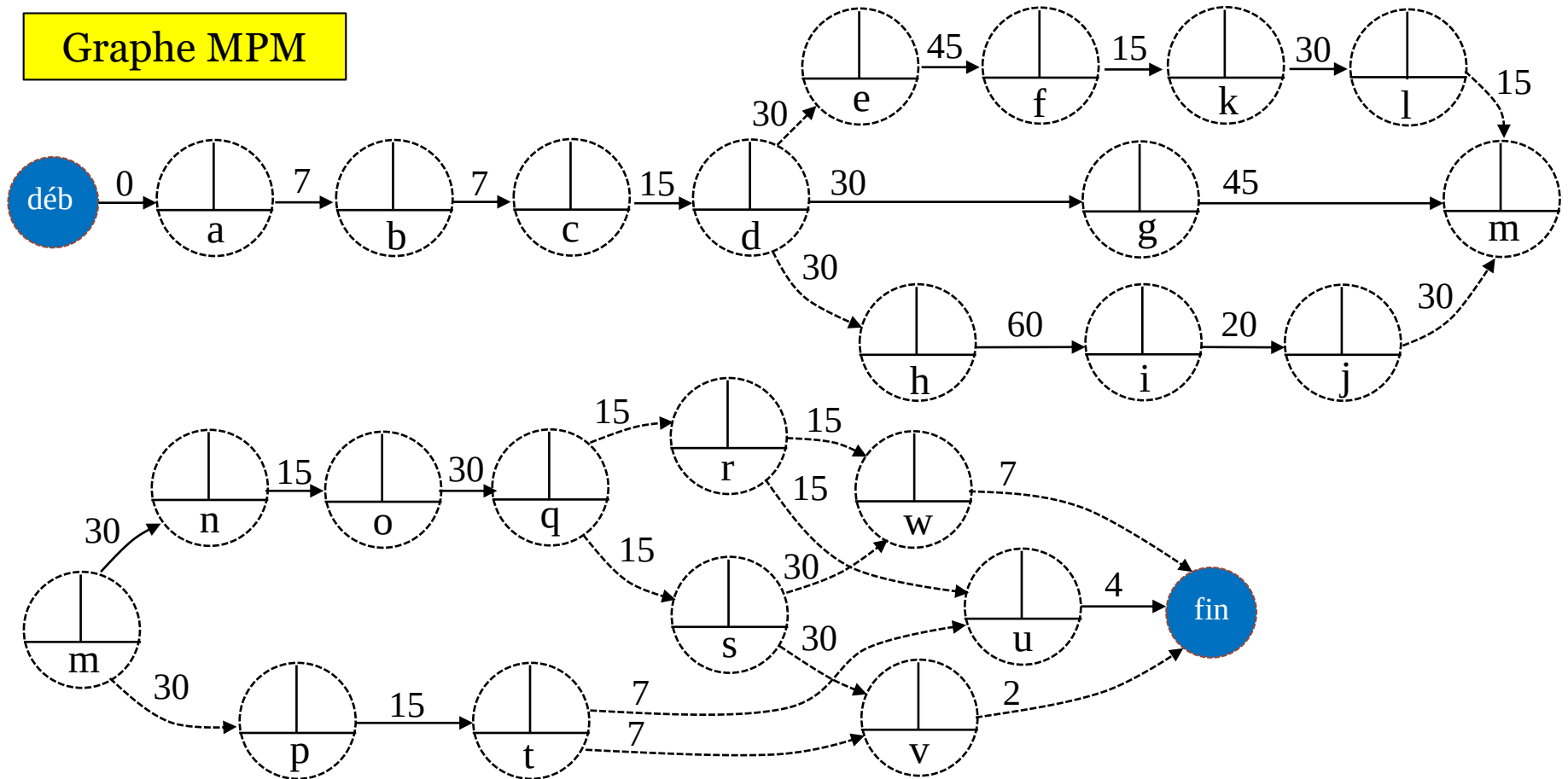
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Graphe MPM



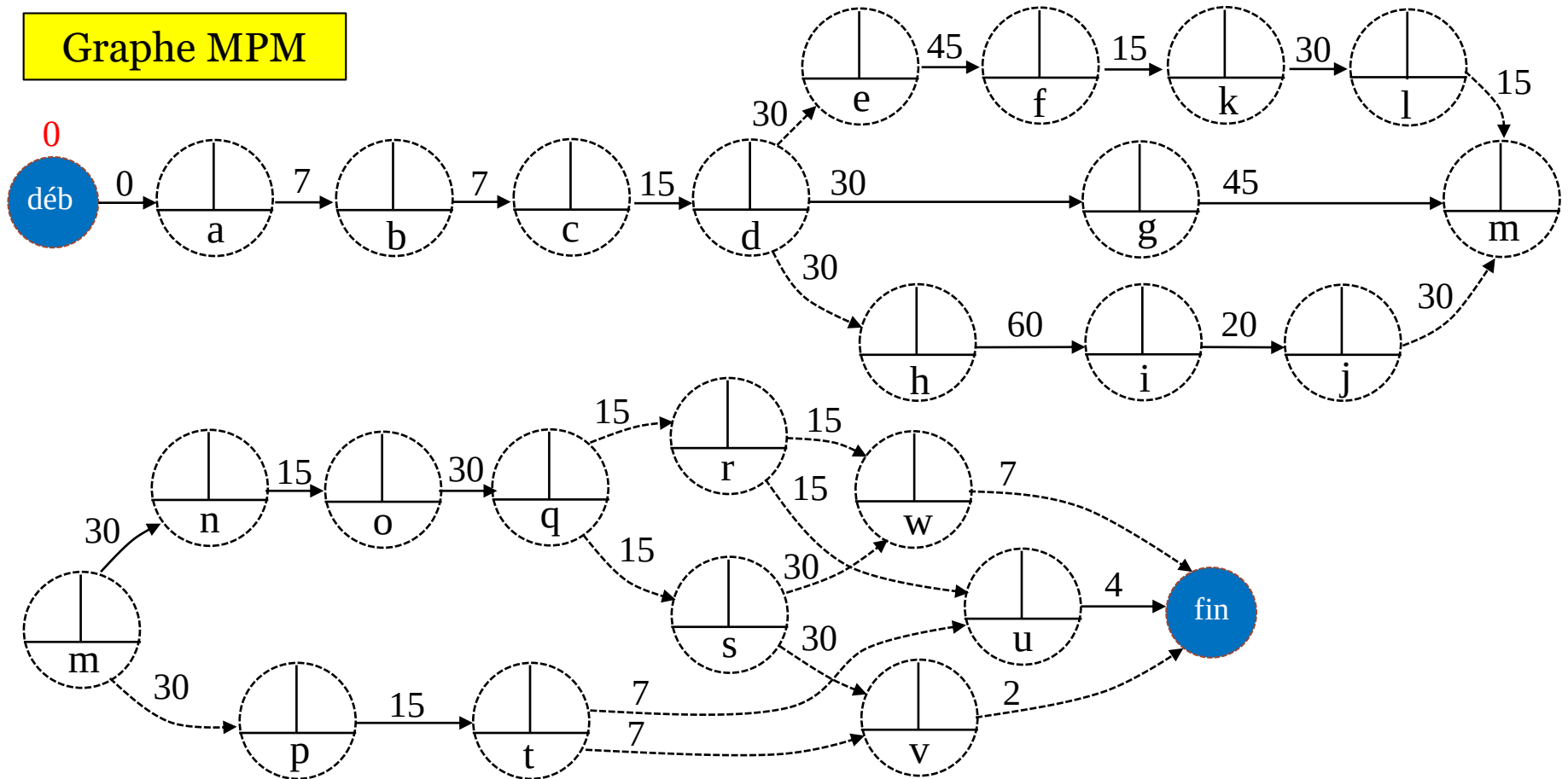
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Graphe MPM



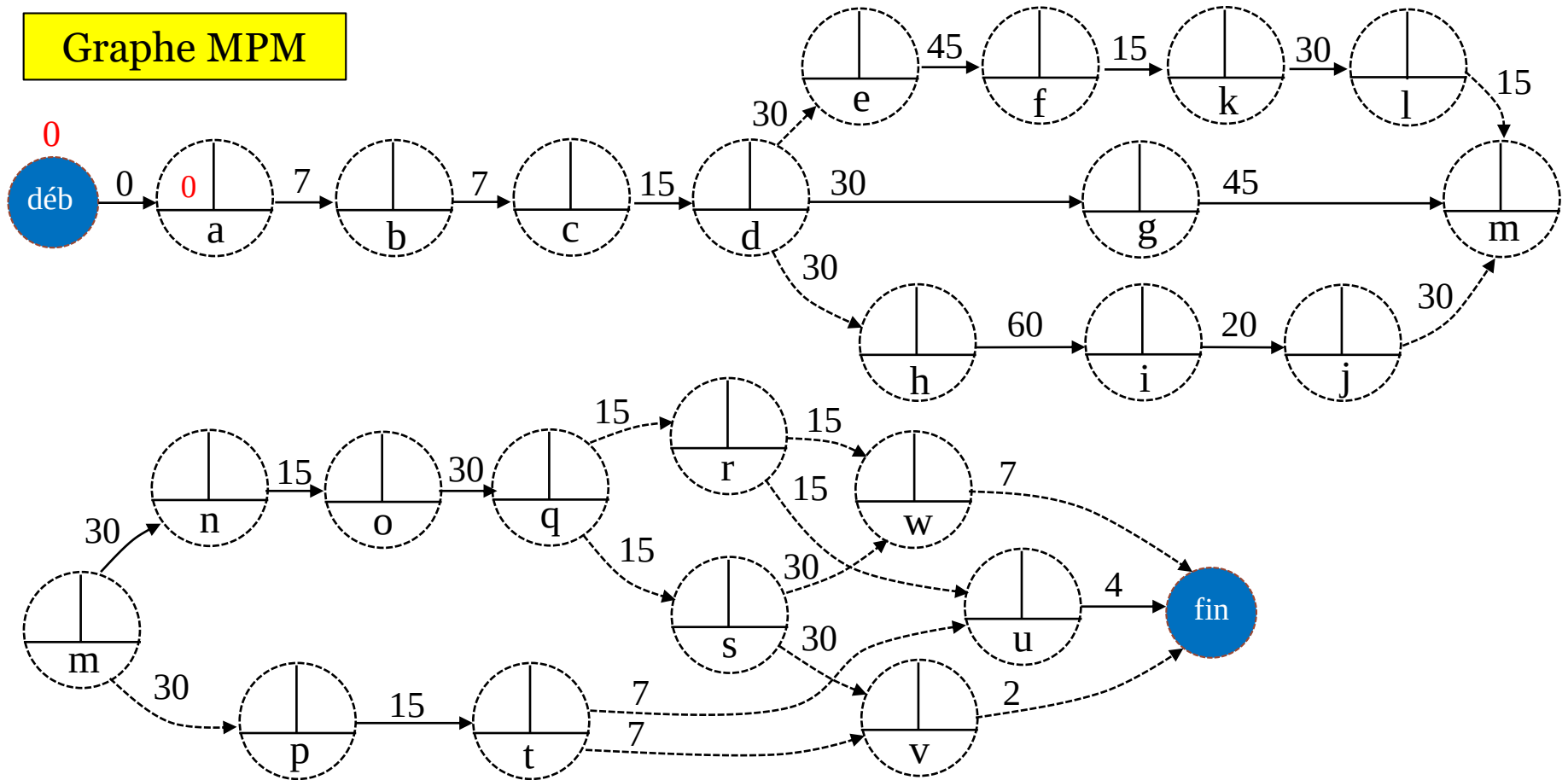
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Graphe MPM



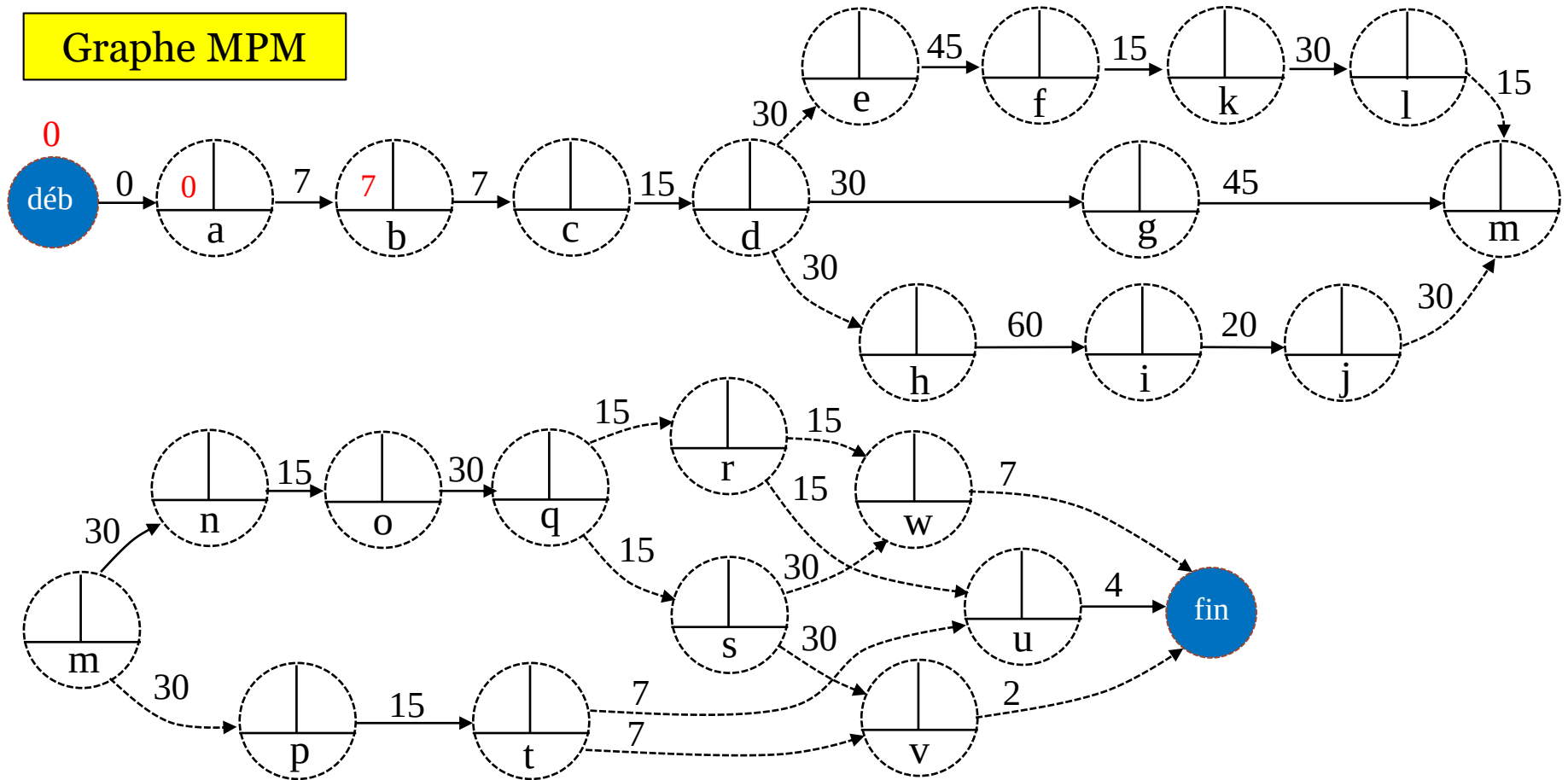
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Graphe MPM



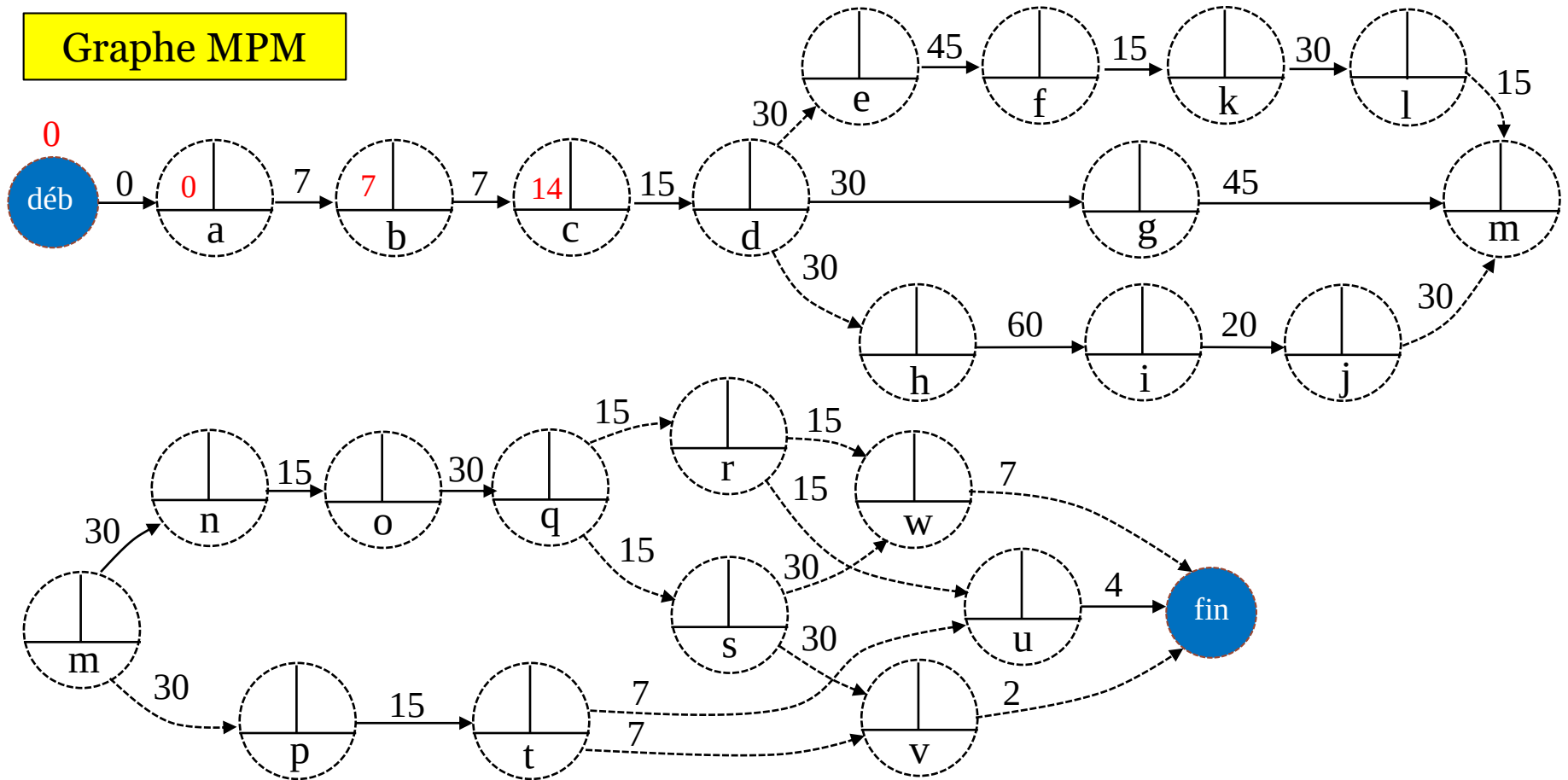
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Graphe MPM



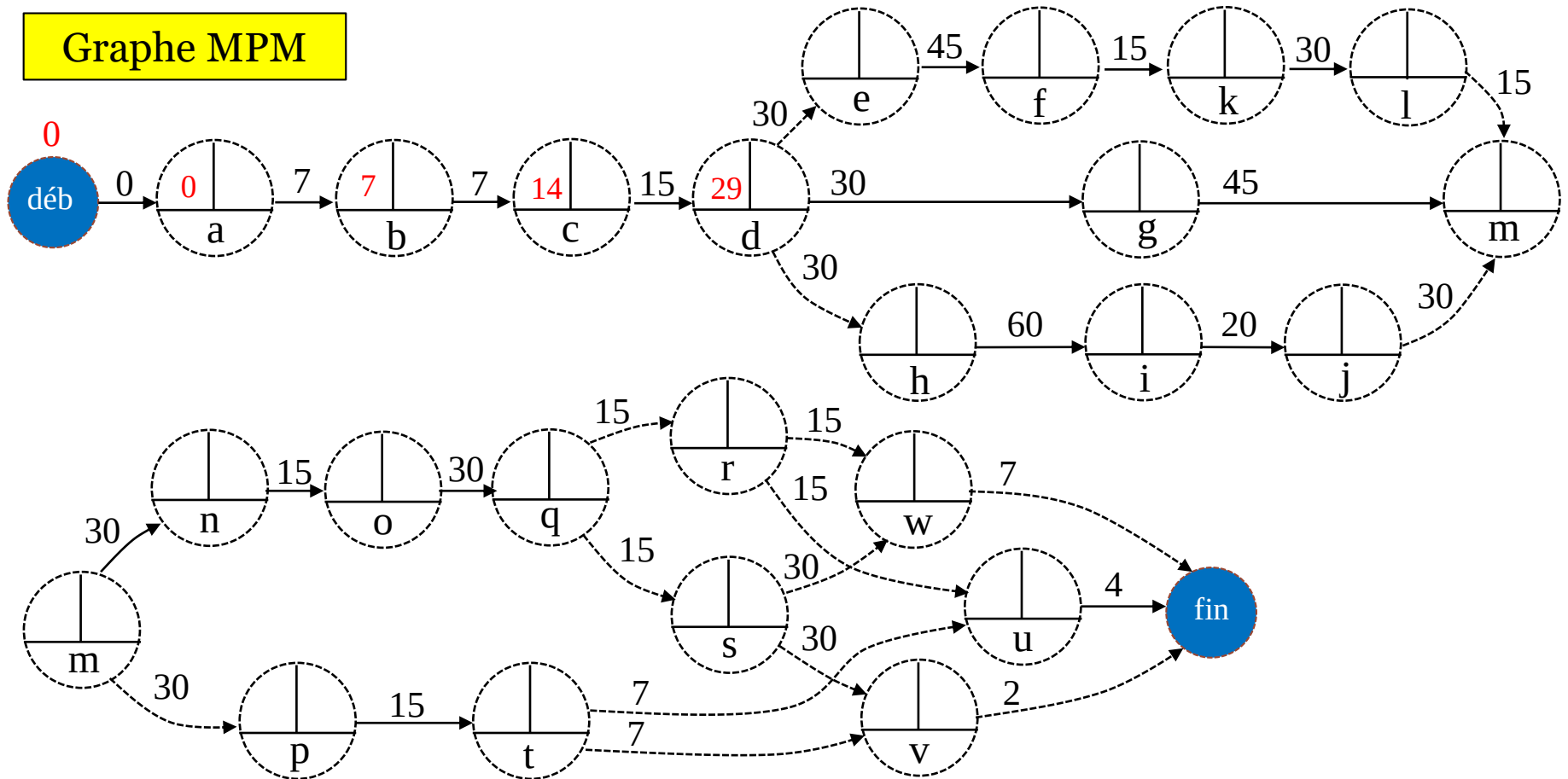
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Graphe MPM



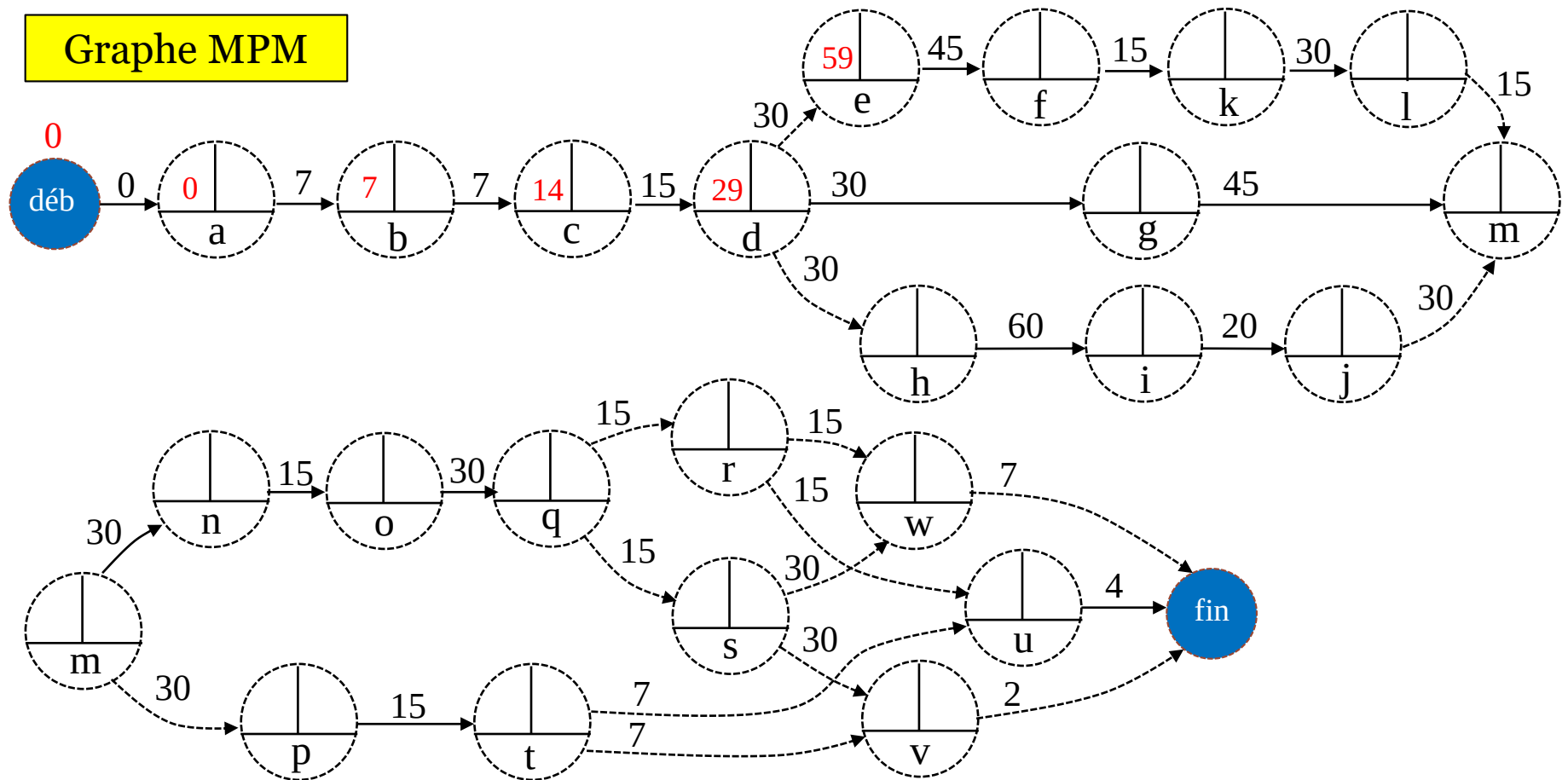
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Graphe MPM



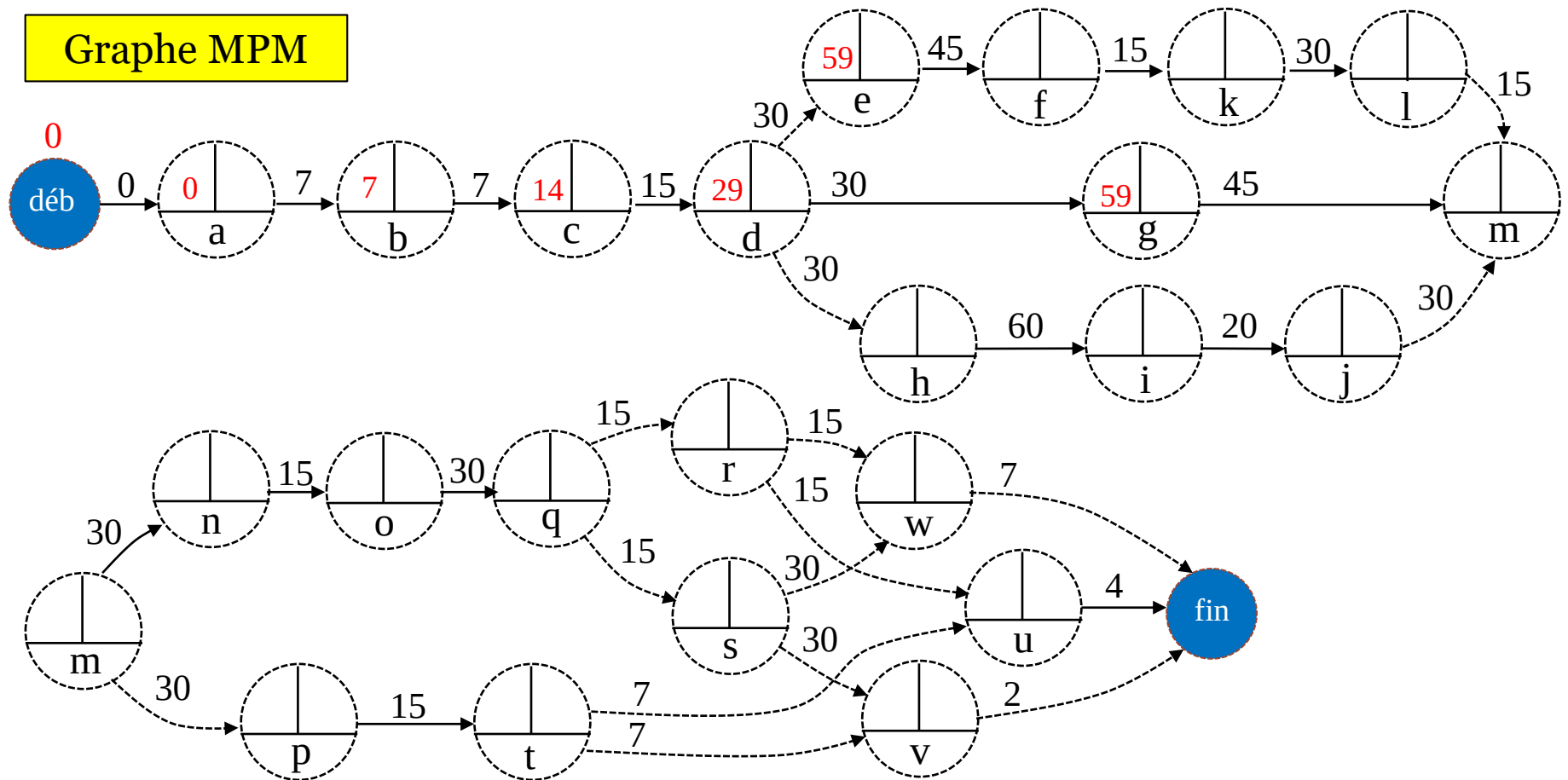
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Graphe MPM



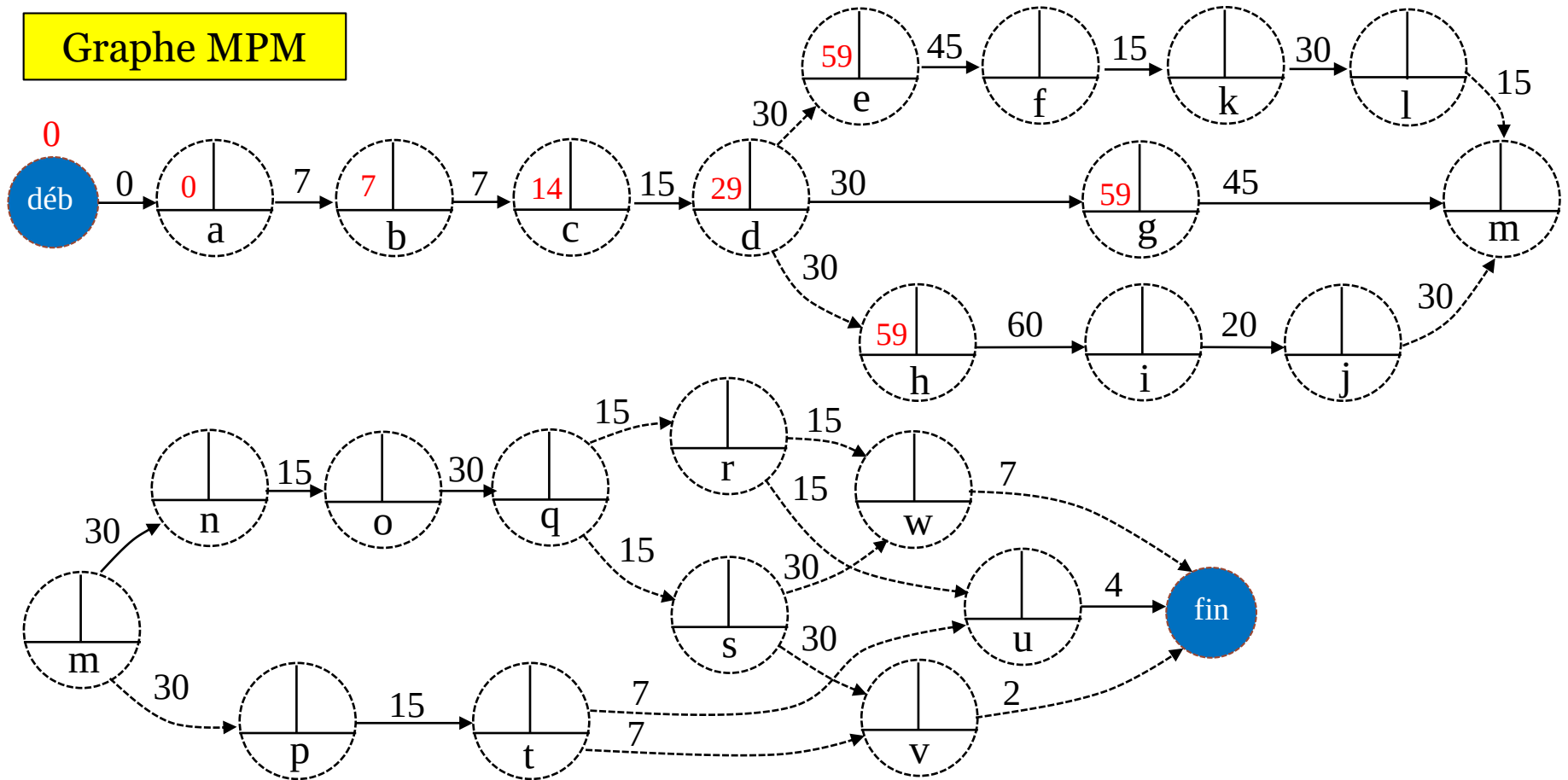
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Graphe MPM



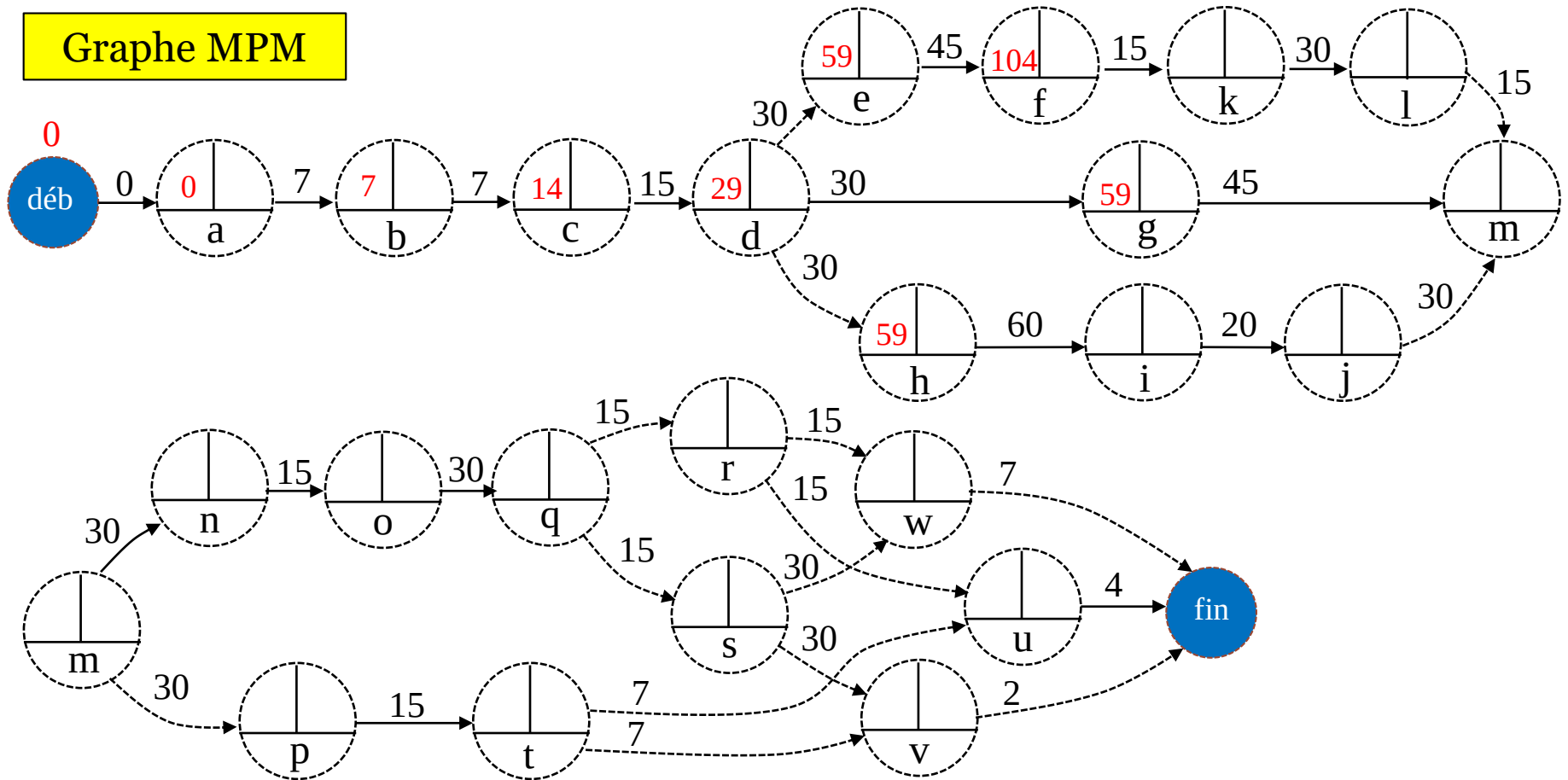
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Graphe MPM



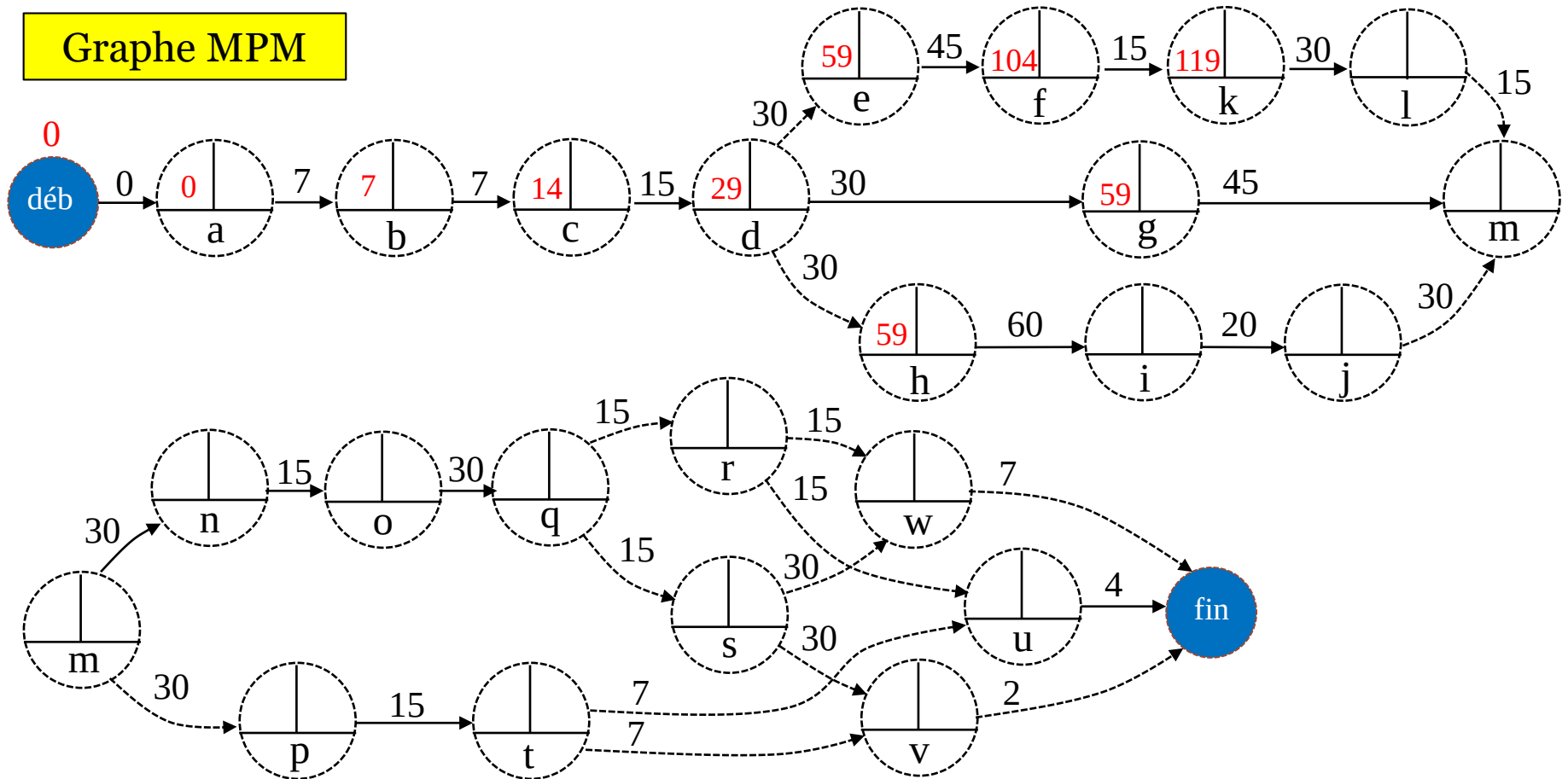
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Graphe MPM



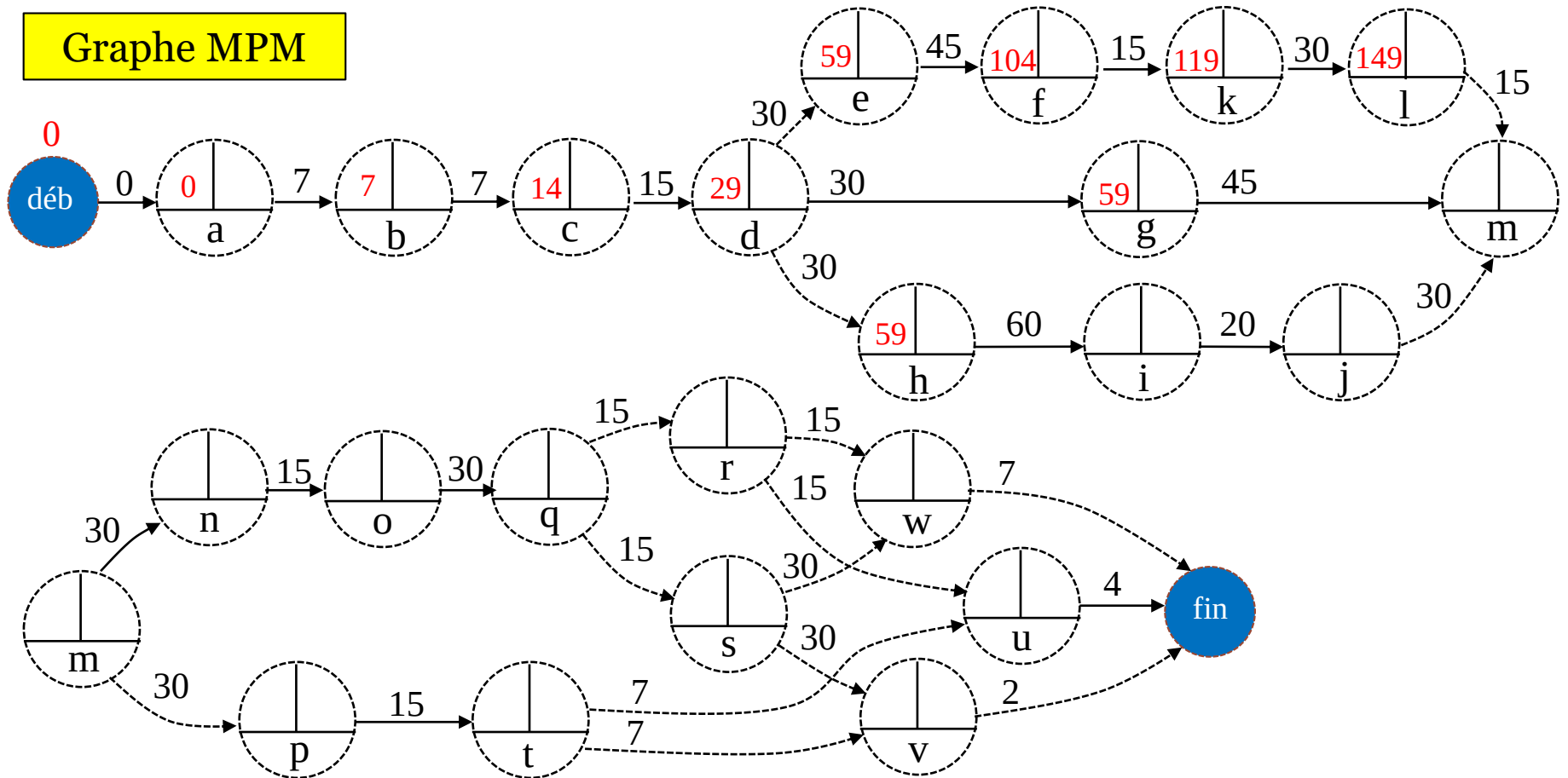
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Graphe MPM



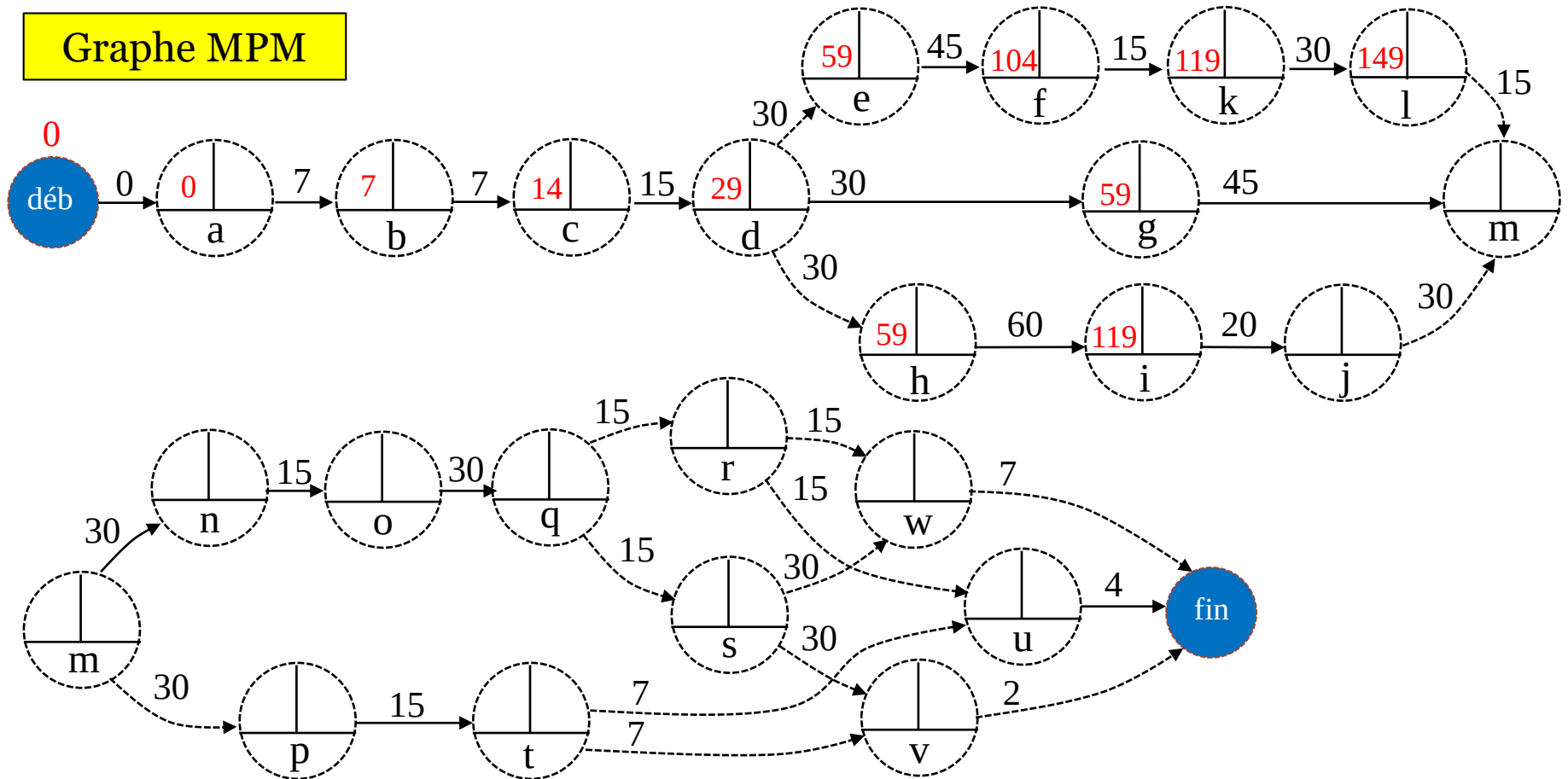
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Graphe MPM



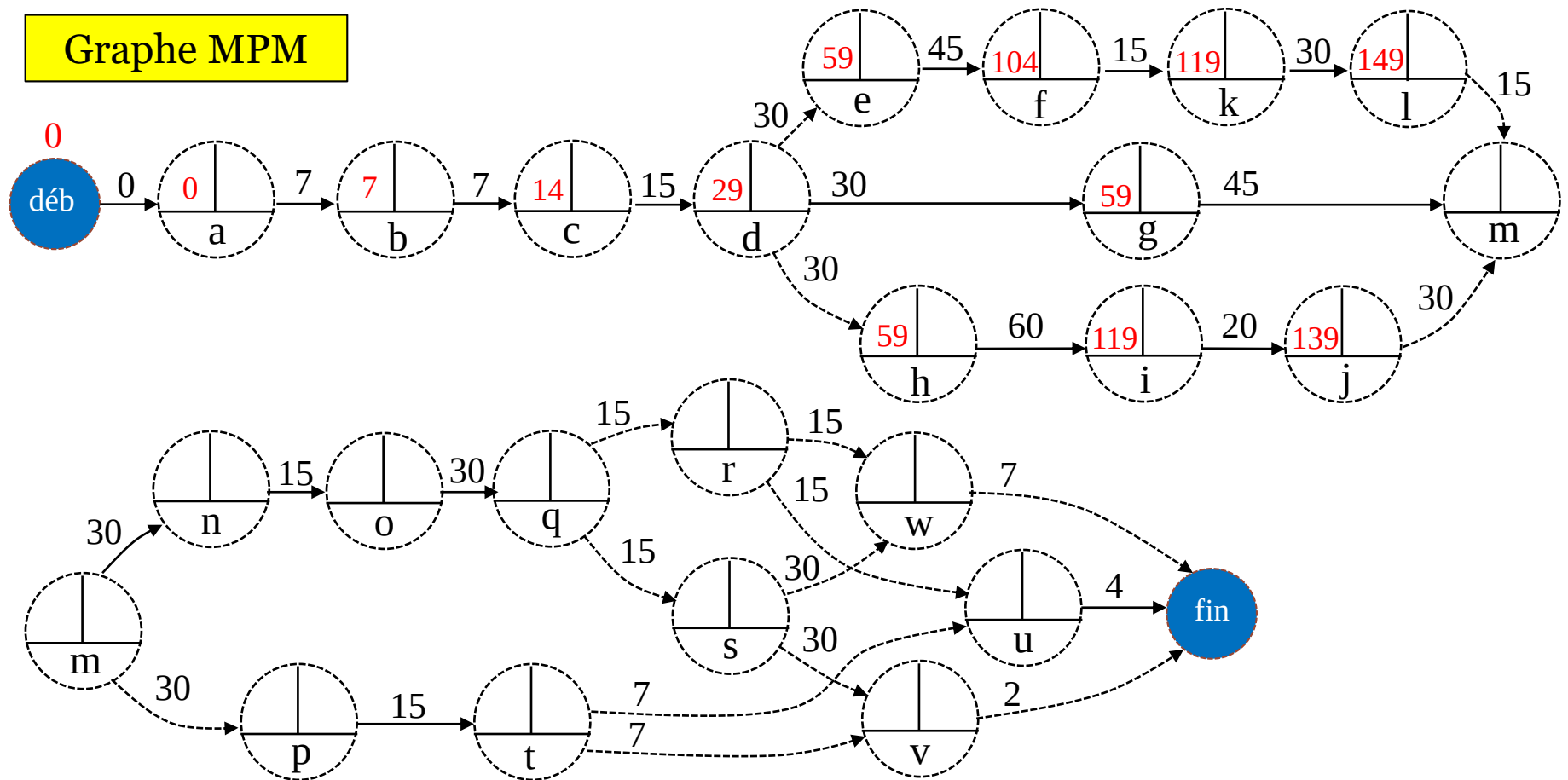
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Graphe MPM



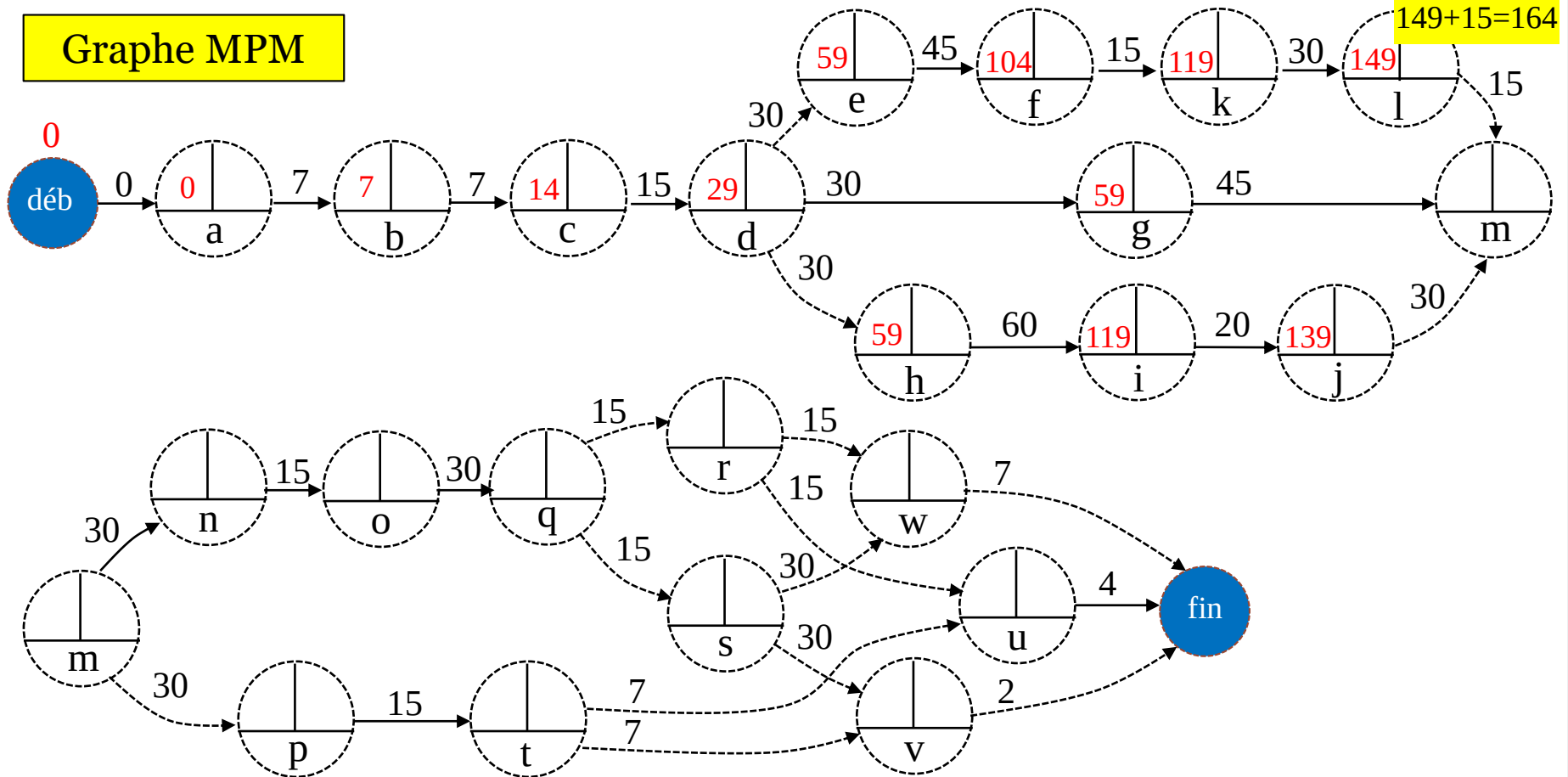
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Graphe MPM



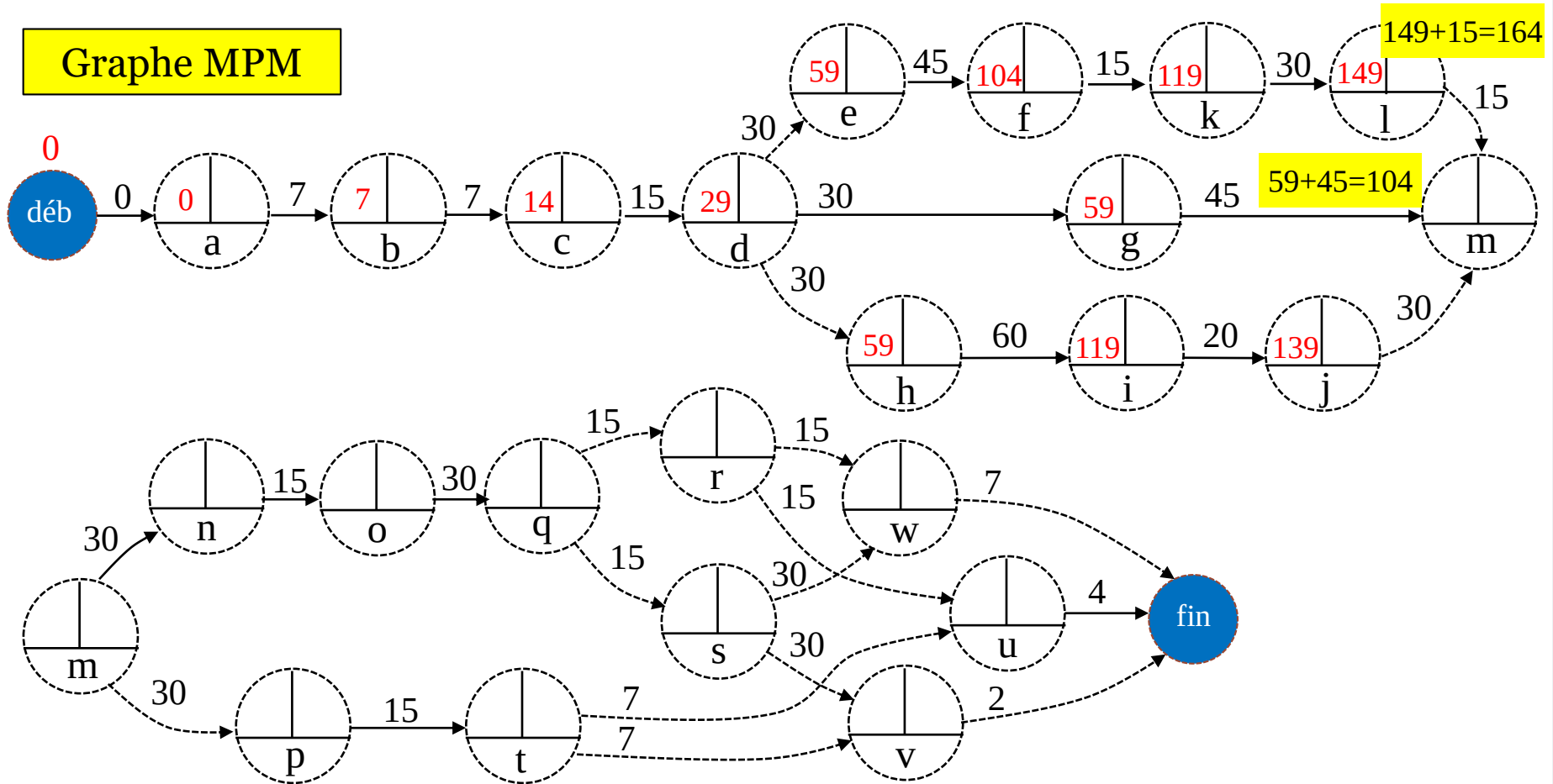
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Graphe MPM



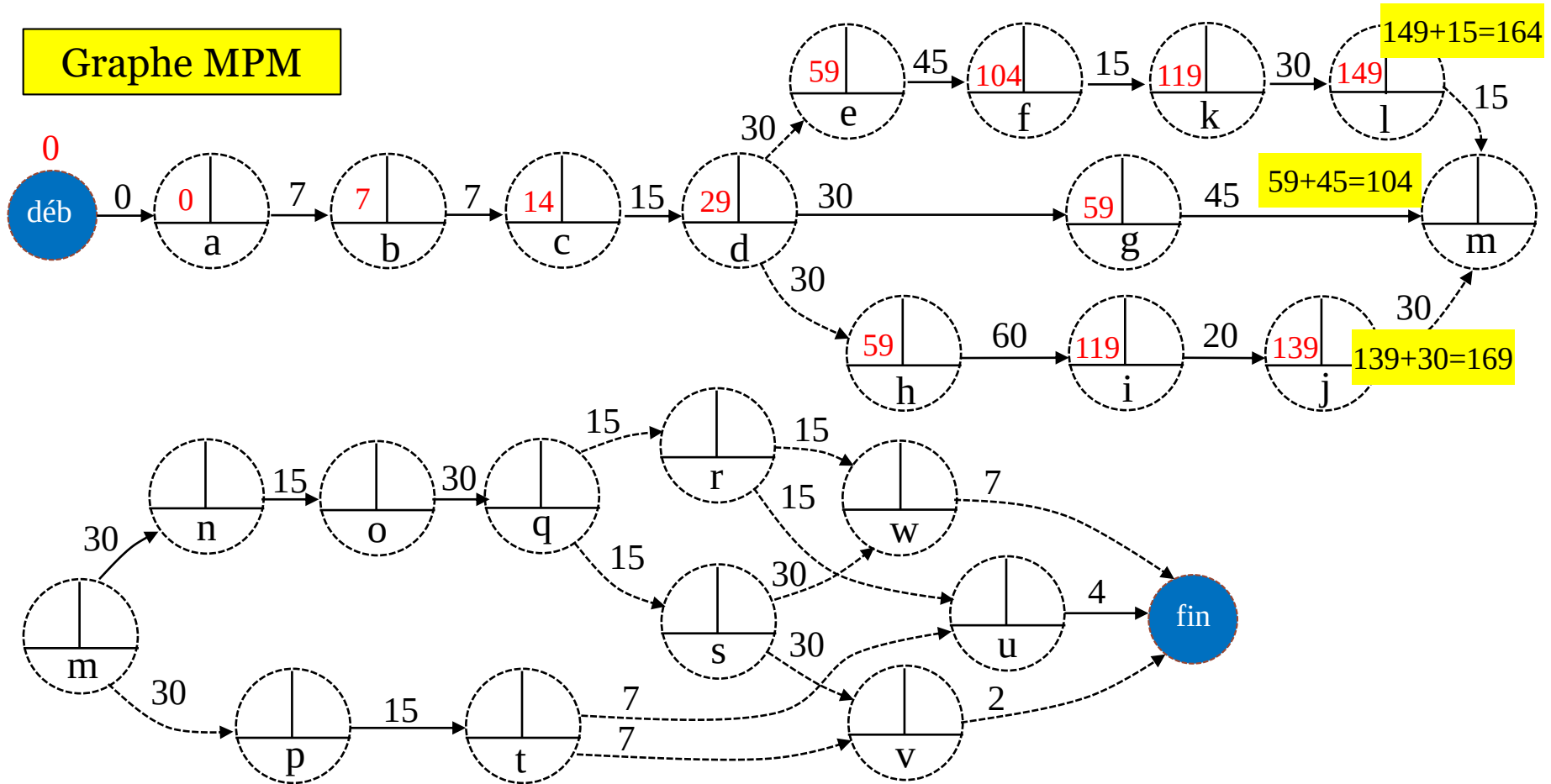
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Graphe MPM



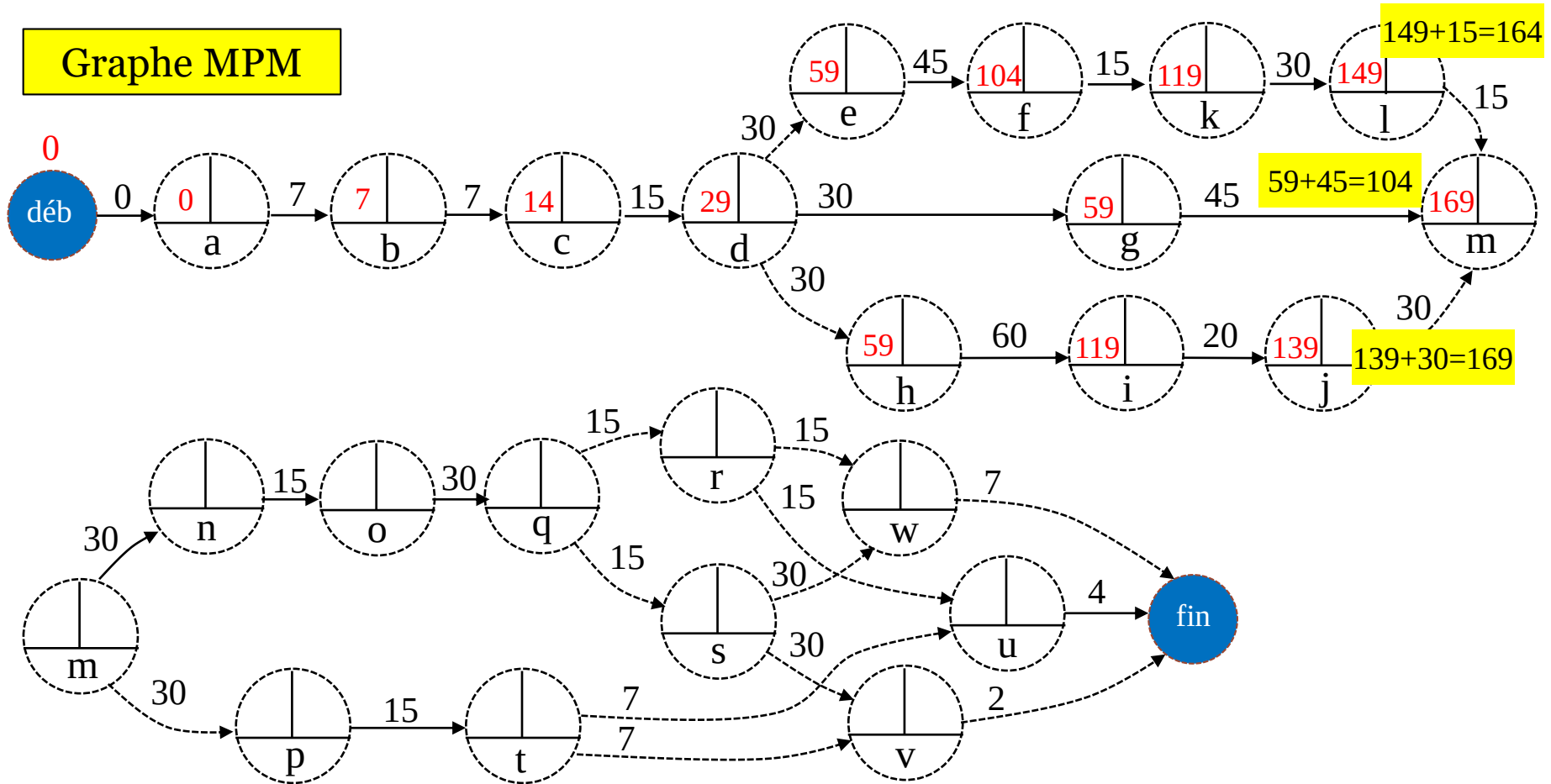
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Graphe MPM



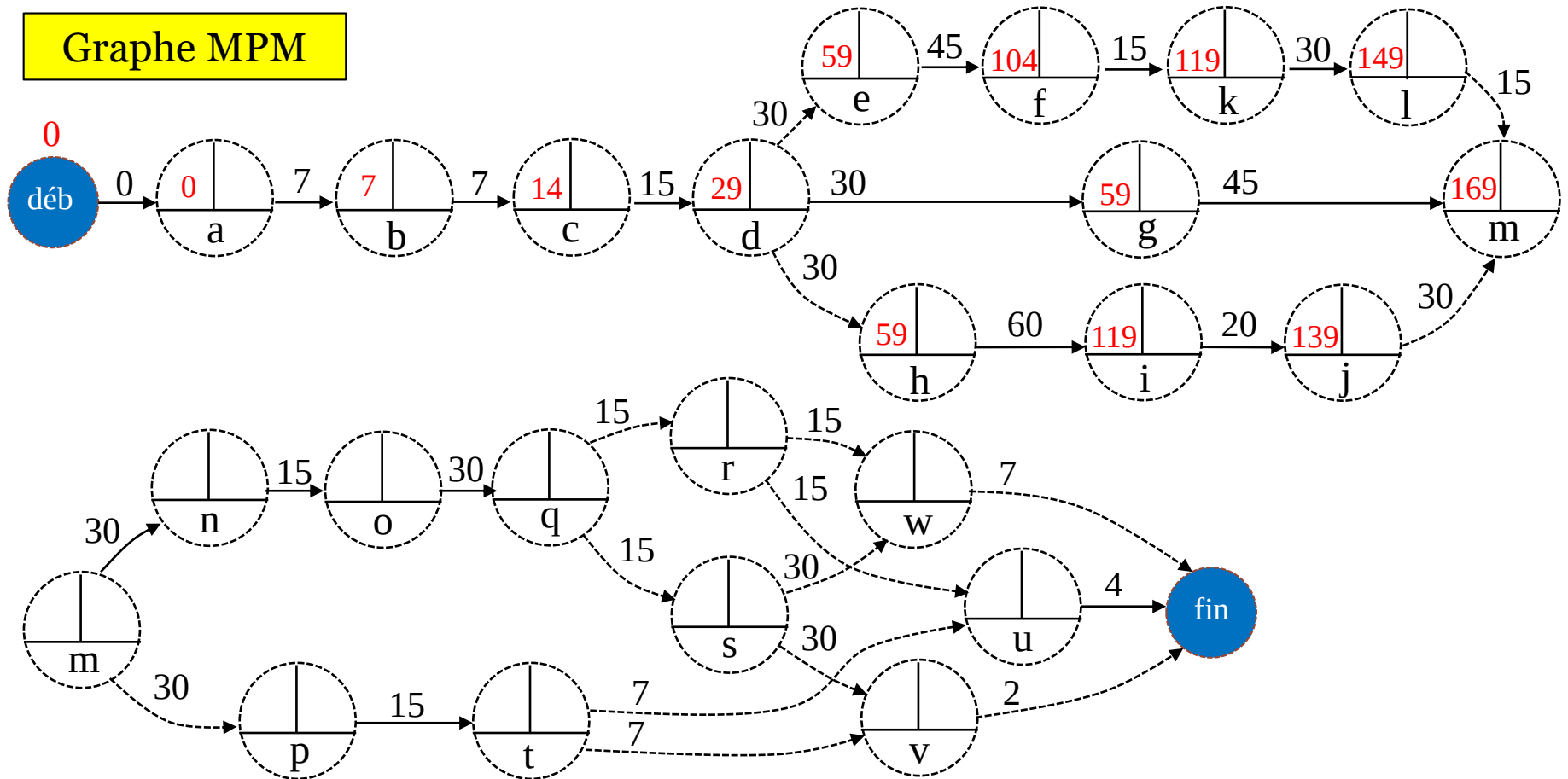
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Graphe MPM



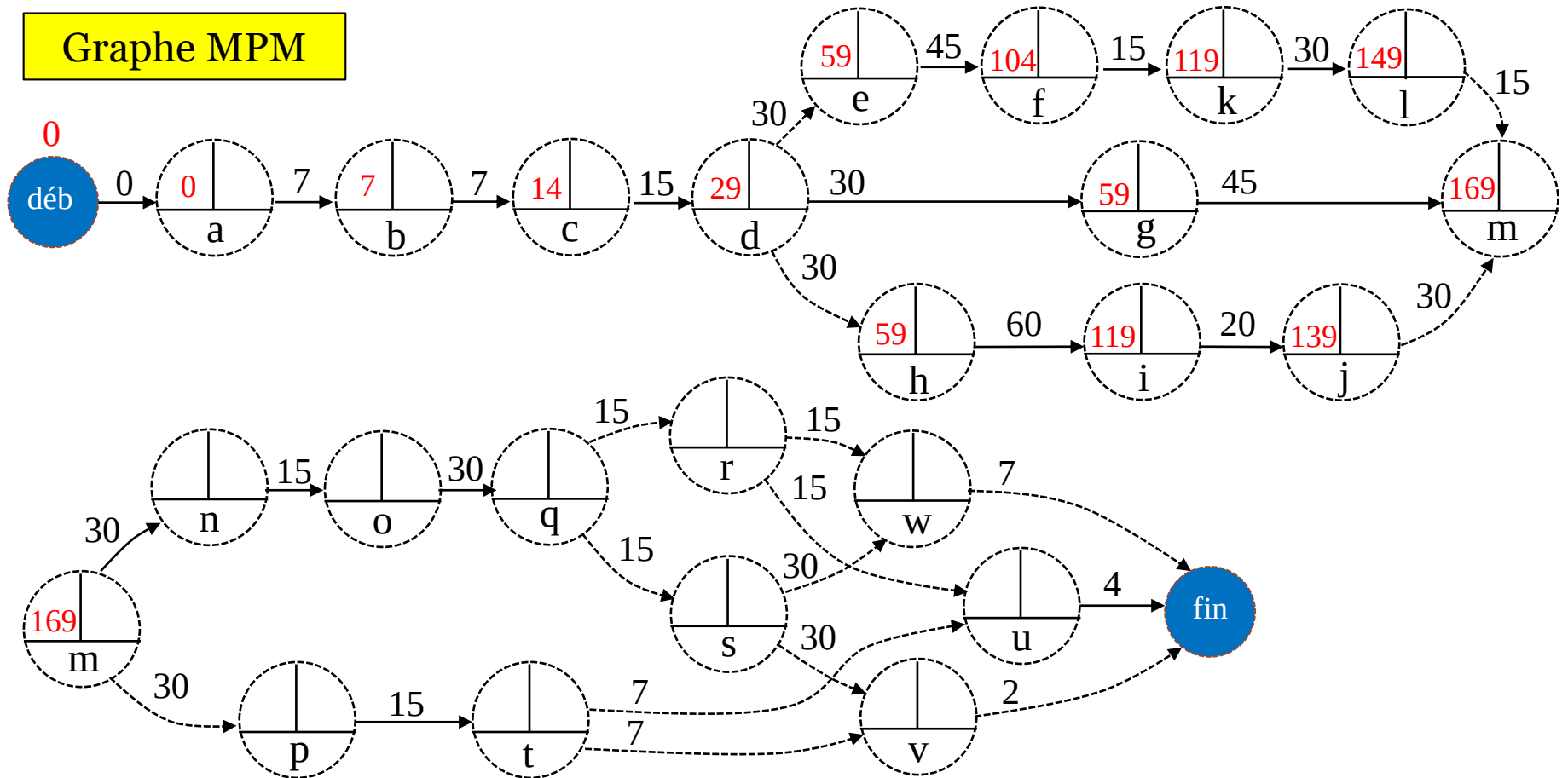
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Graphe MPM



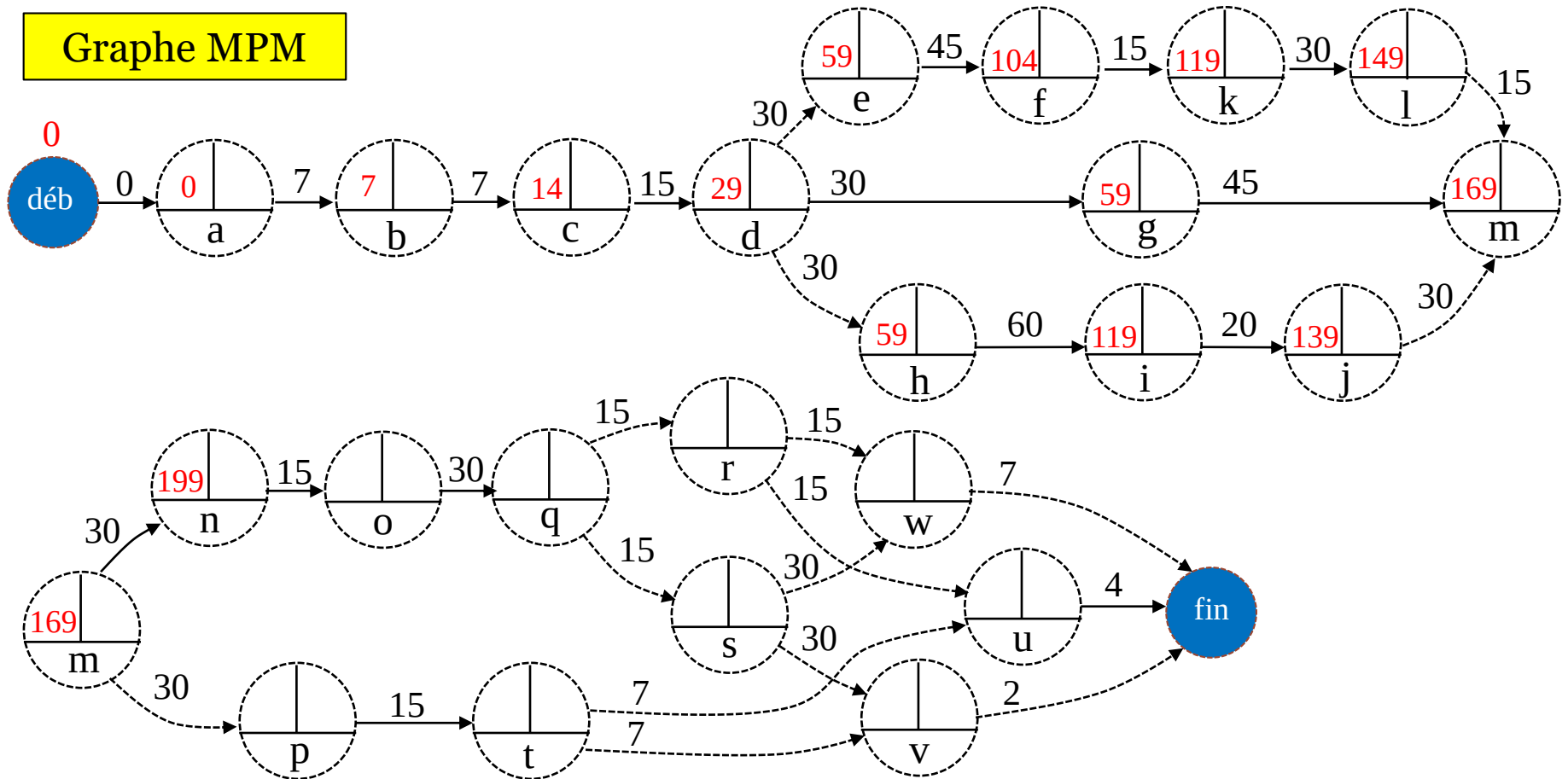
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Graphe MPM



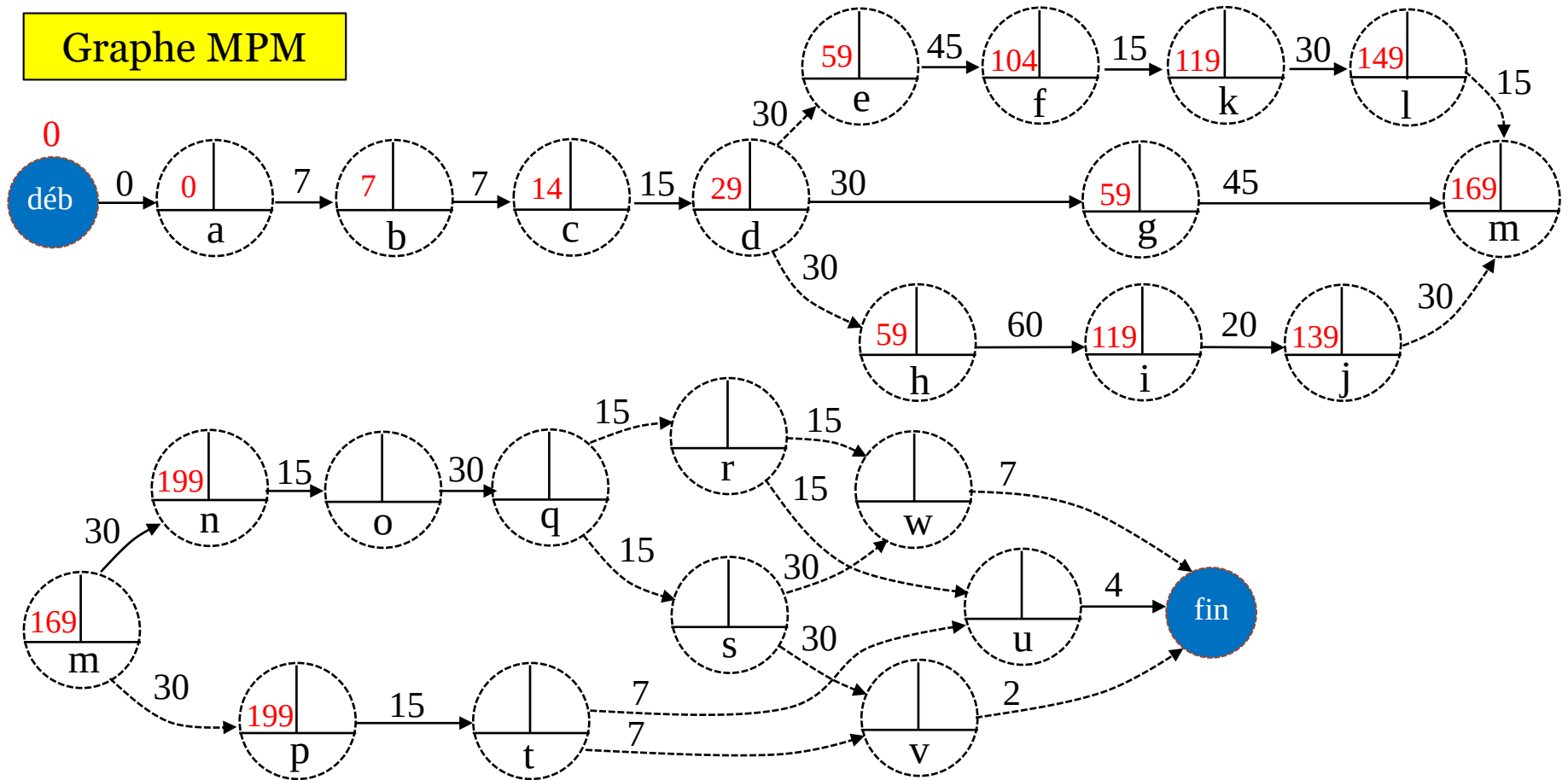
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Graphe MPM



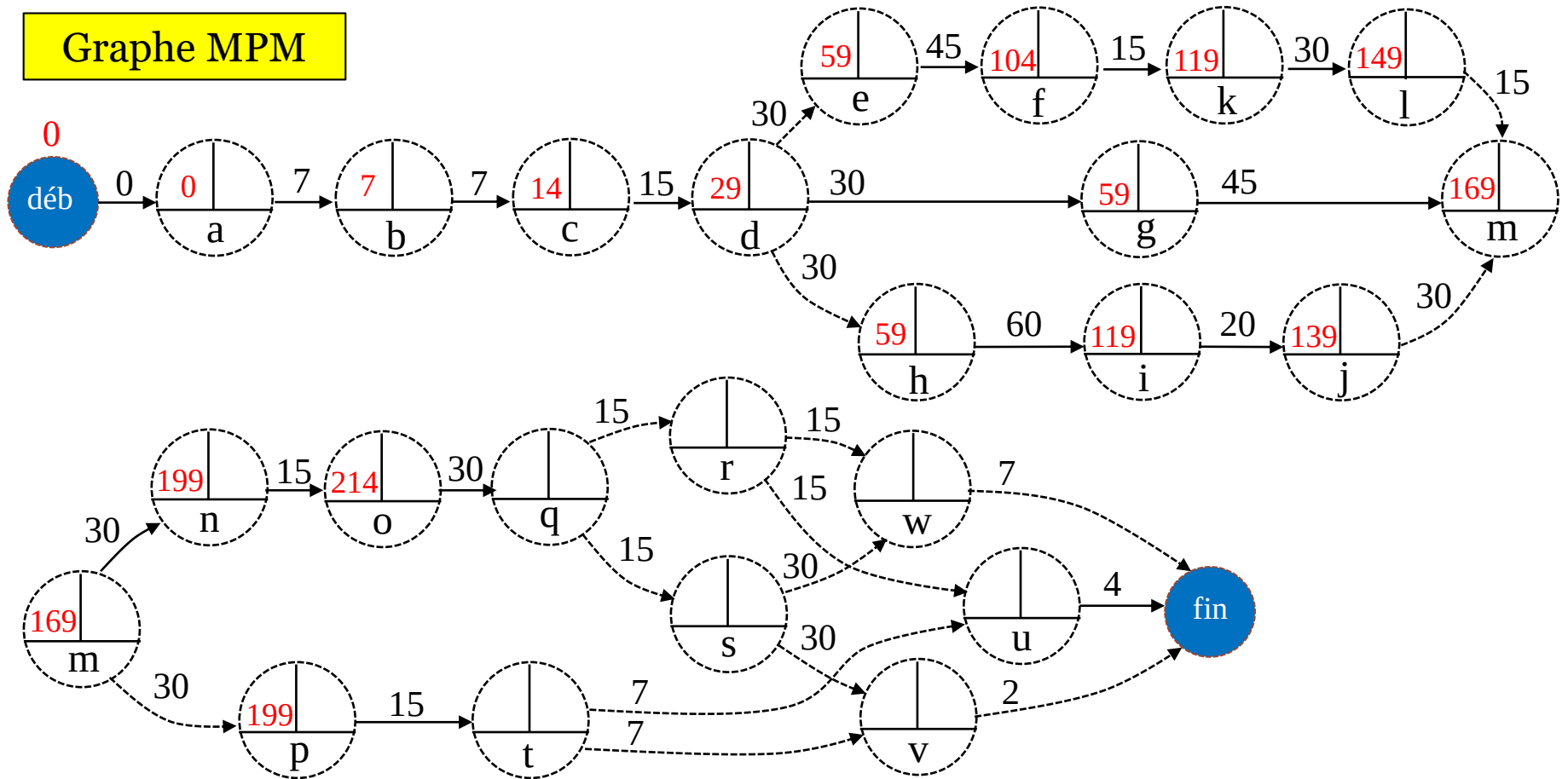
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Graphe MPM



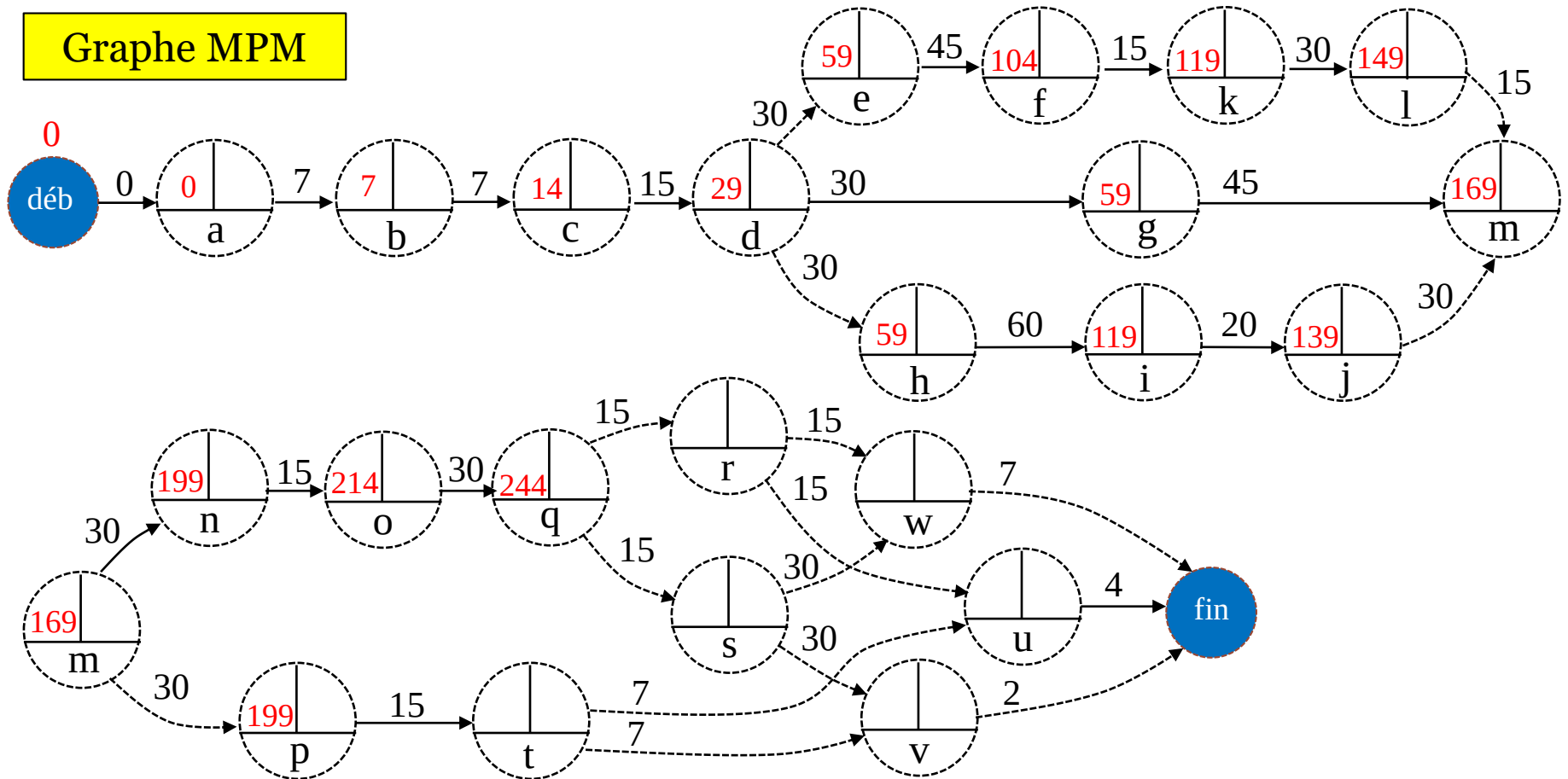
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Graphe MPM



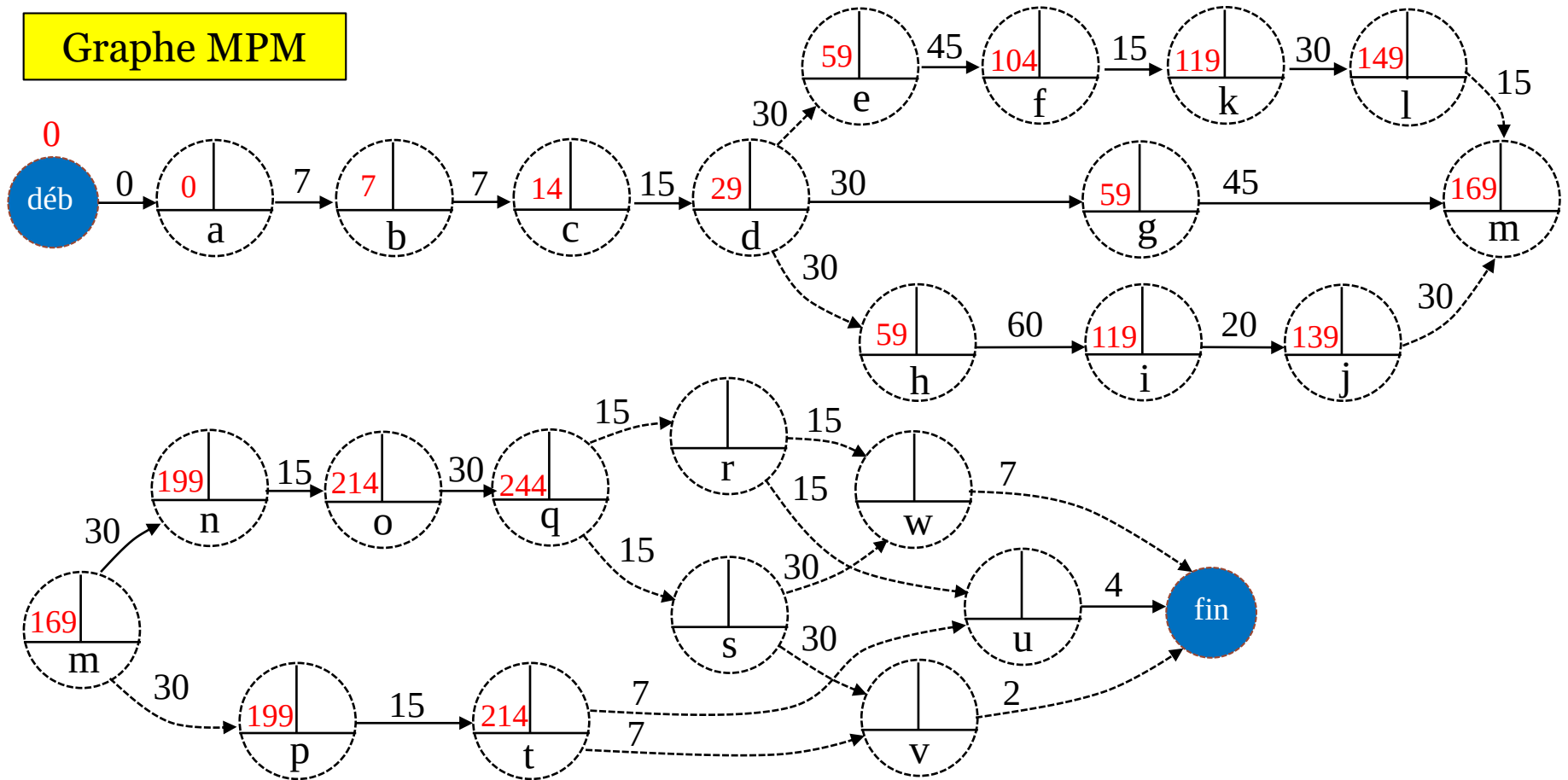
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Graphe MPM



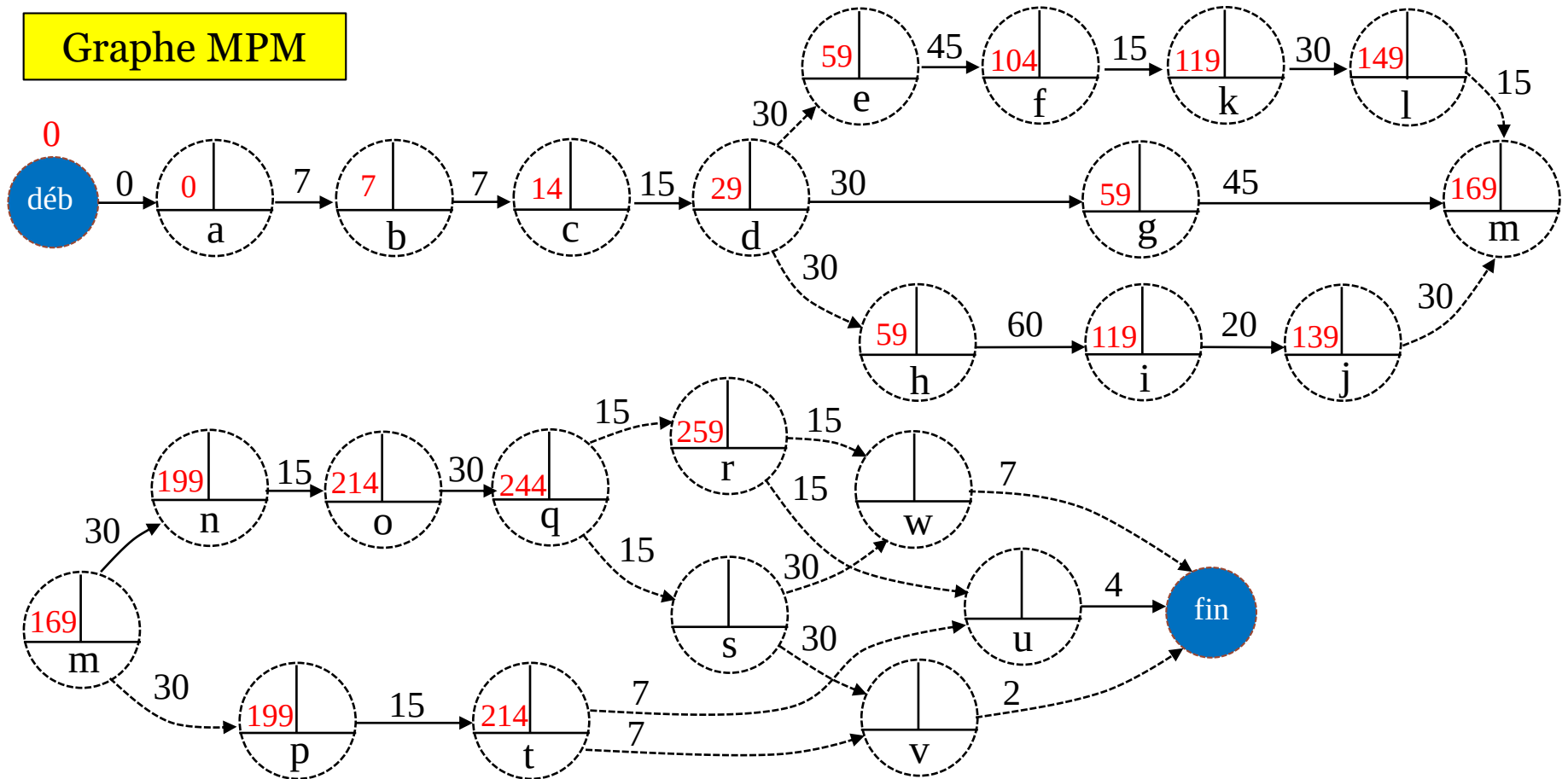
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Graphe MPM



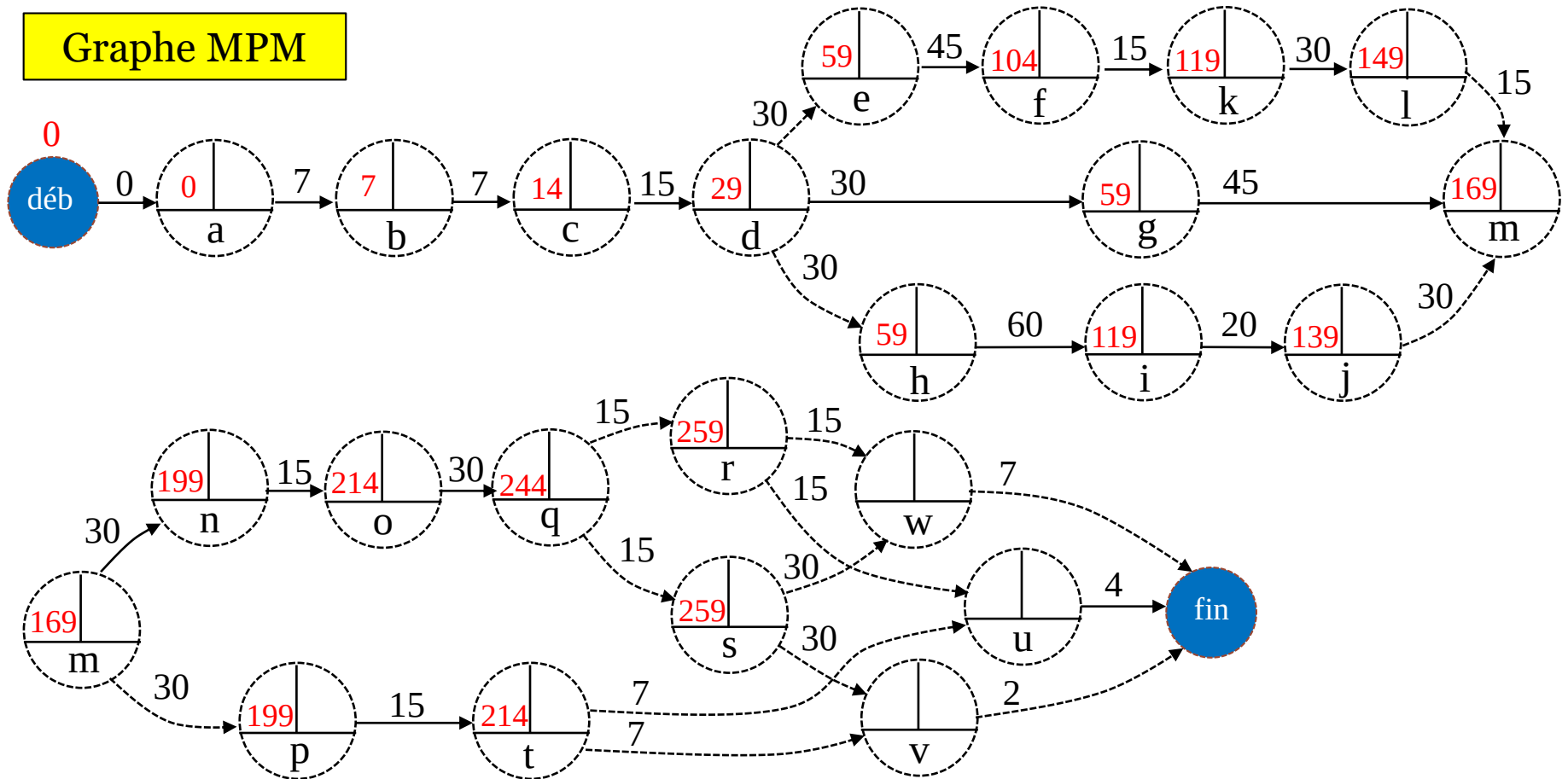
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Graphe MPM



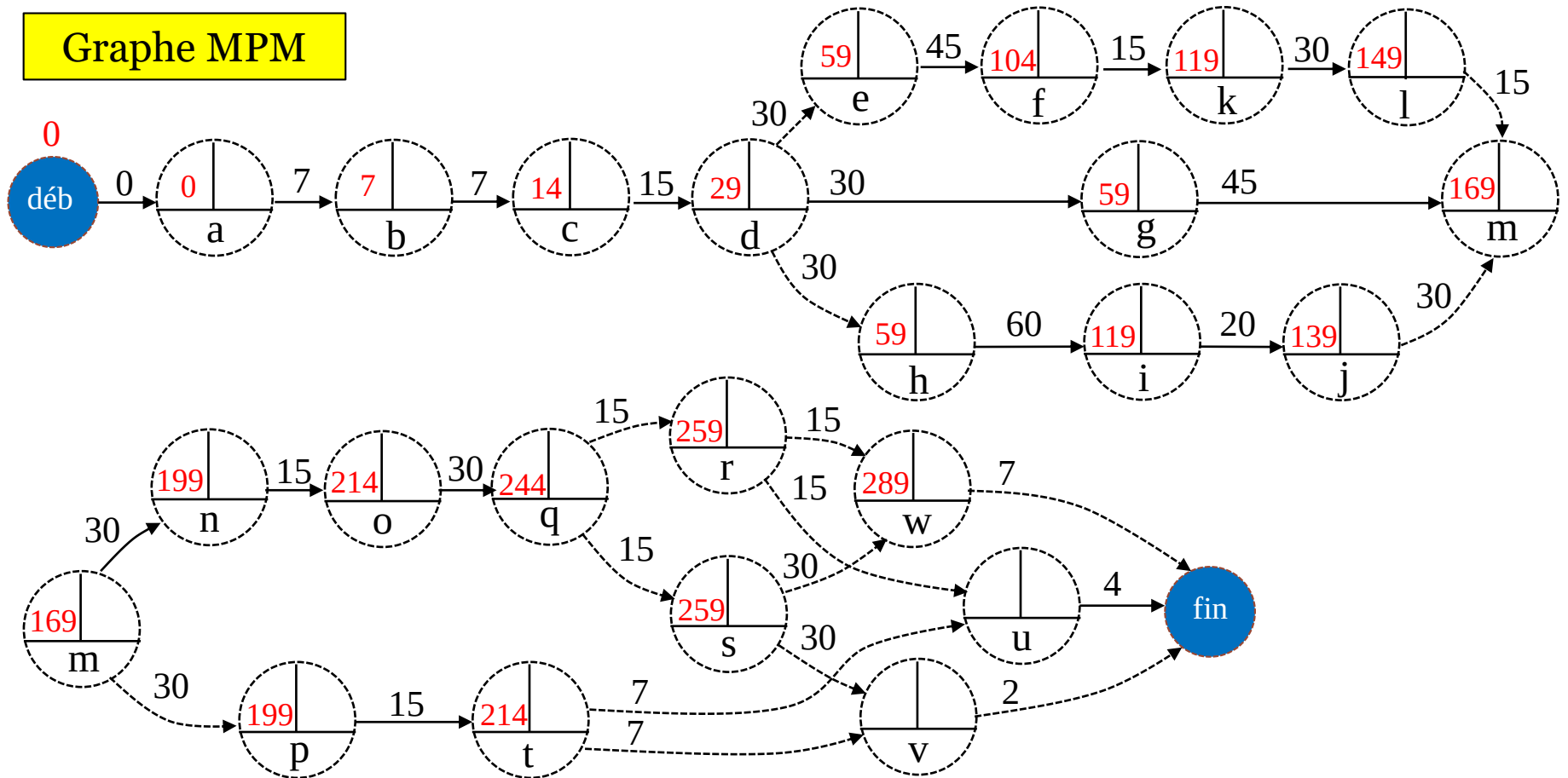
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Graphe MPM



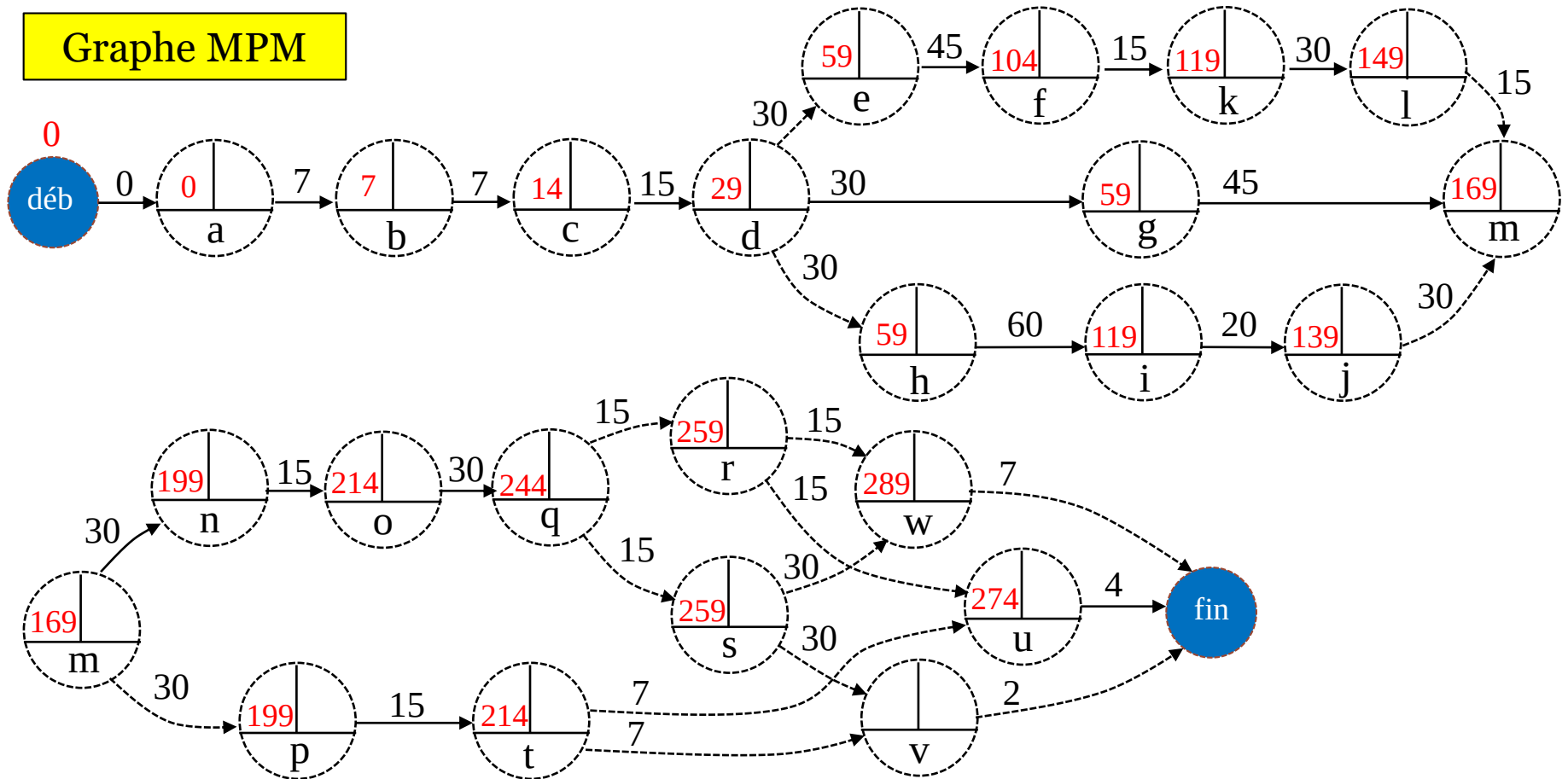
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Graphe MPM



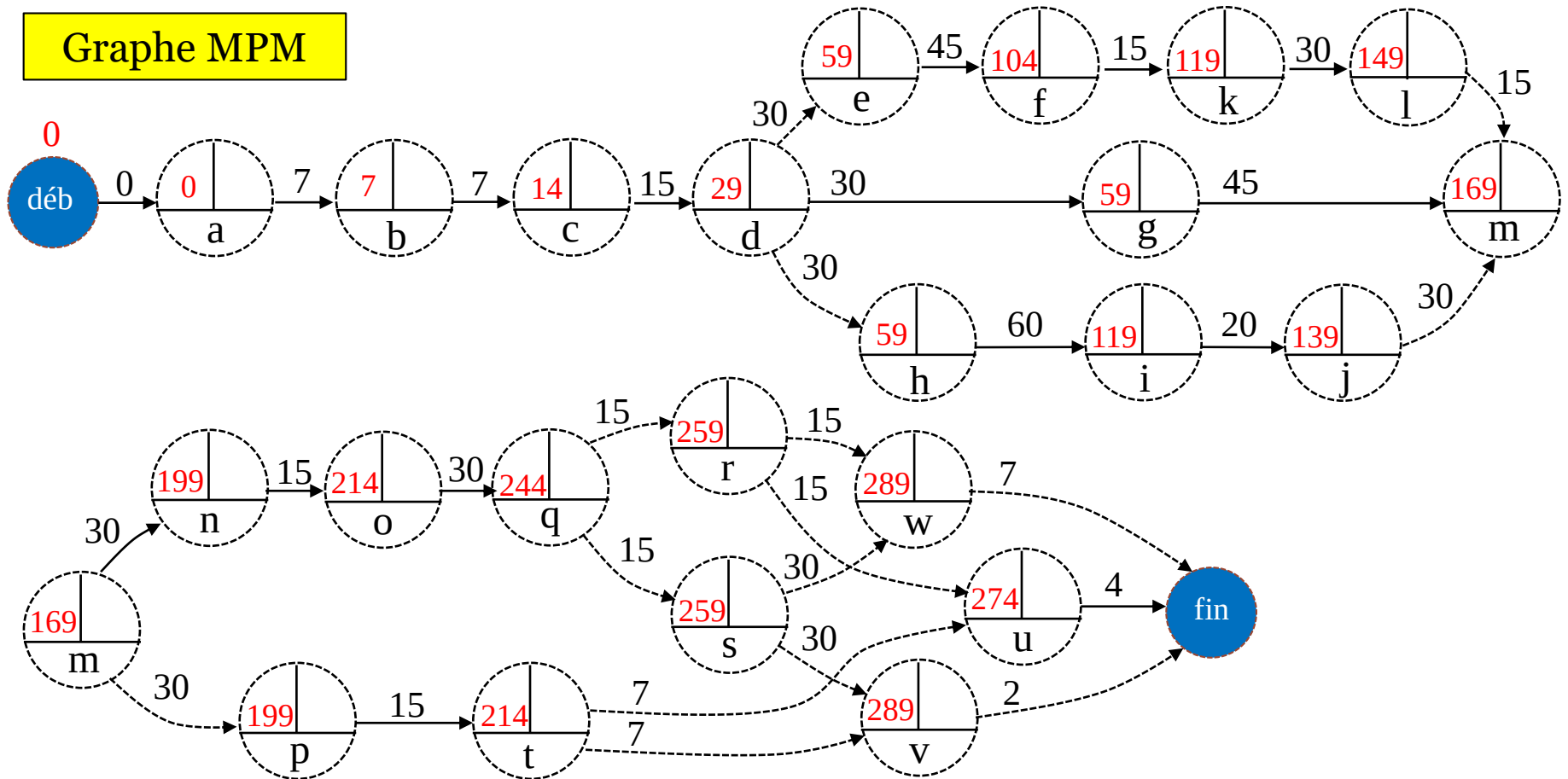
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Graphe MPM



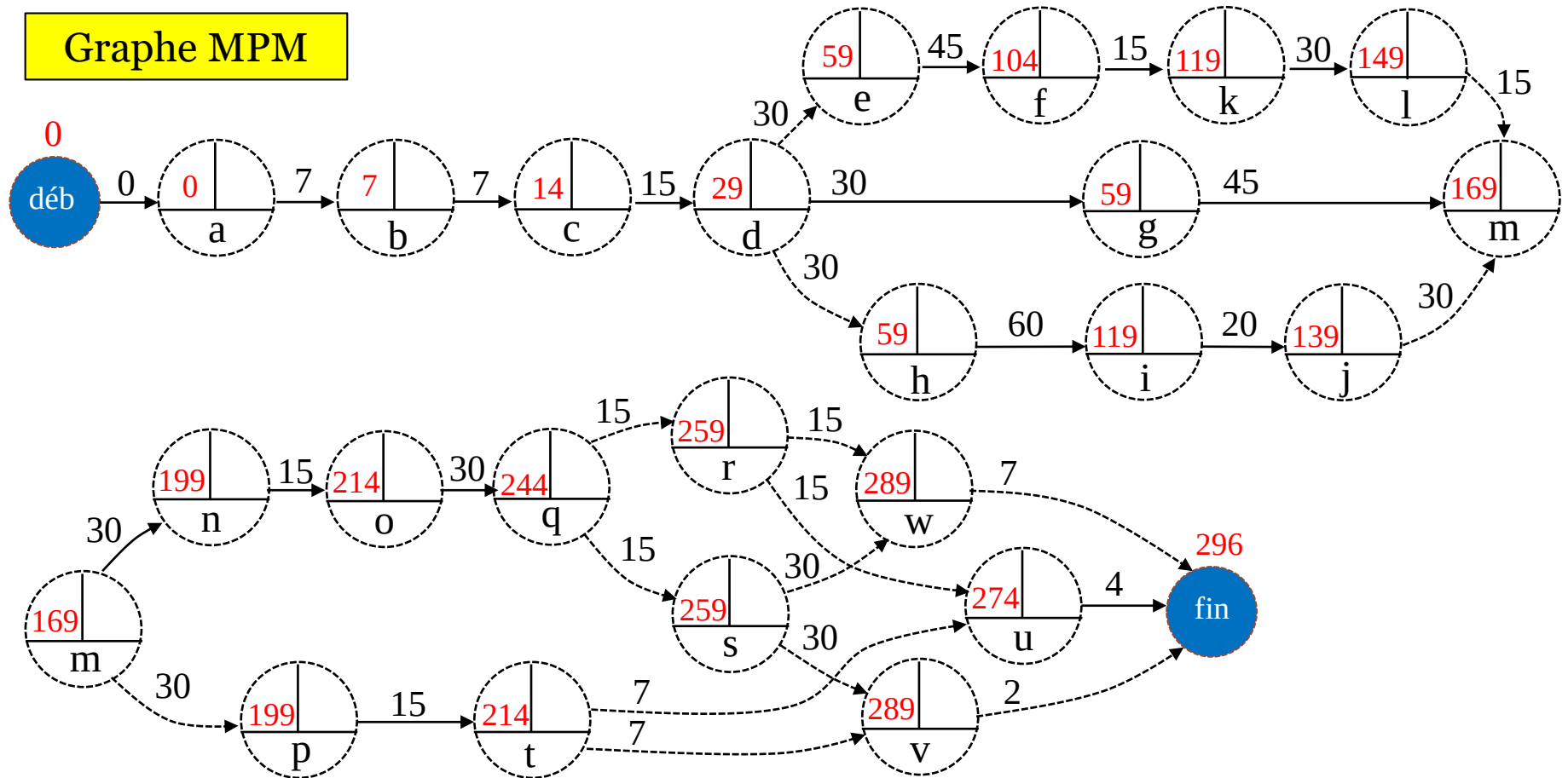
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Graphe MPM



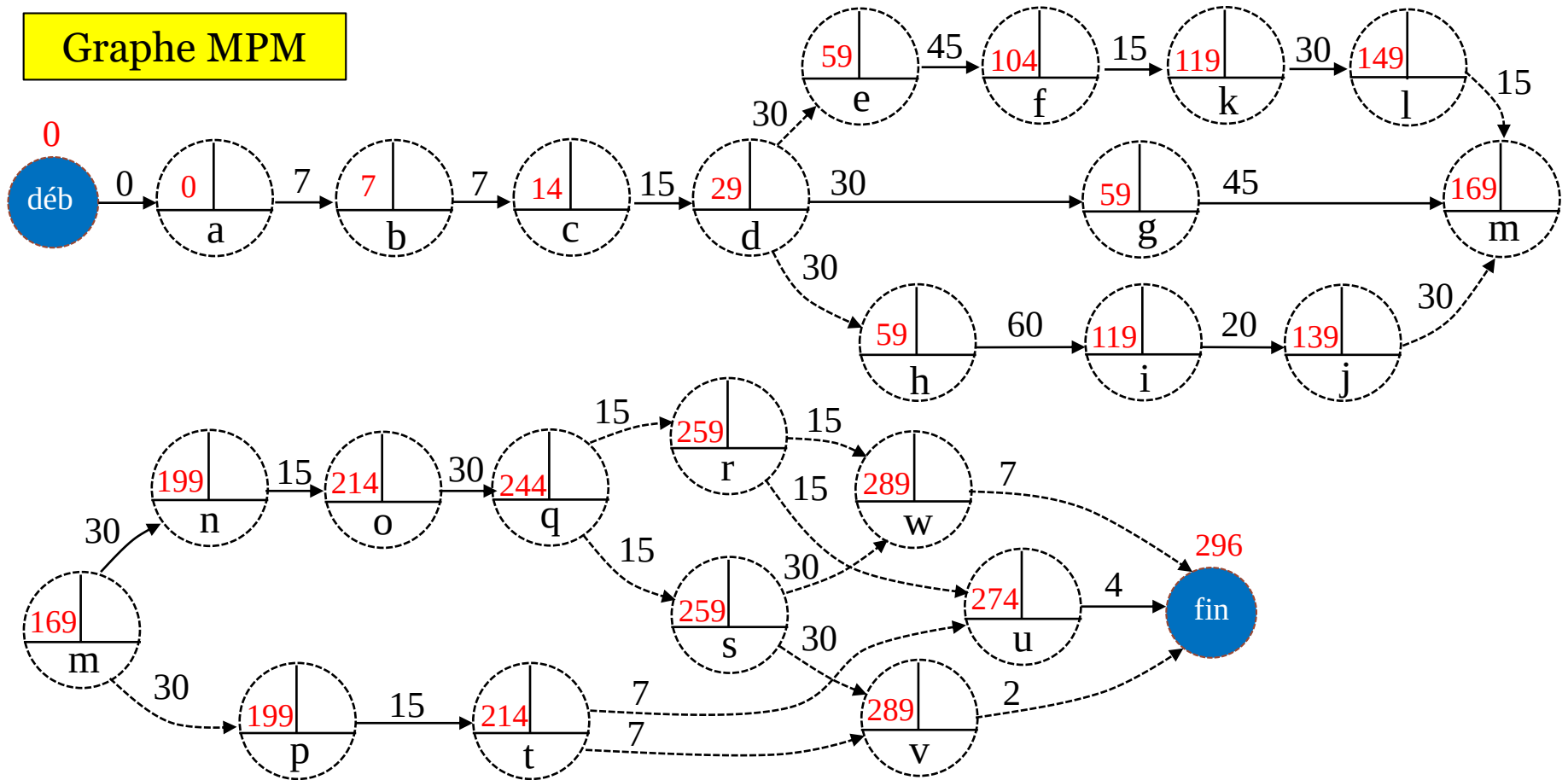
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Graphe MPM



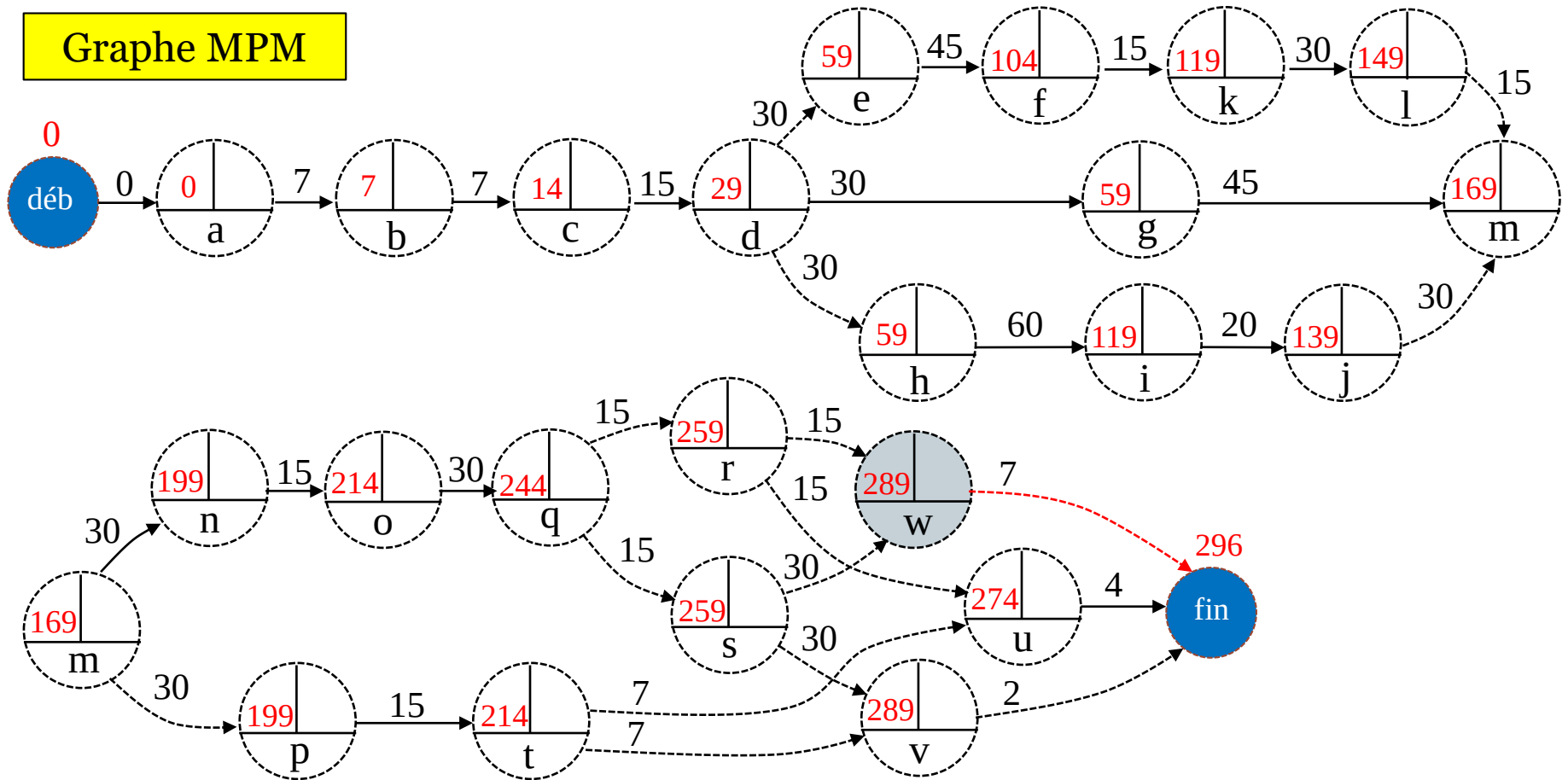
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Graphe MPM



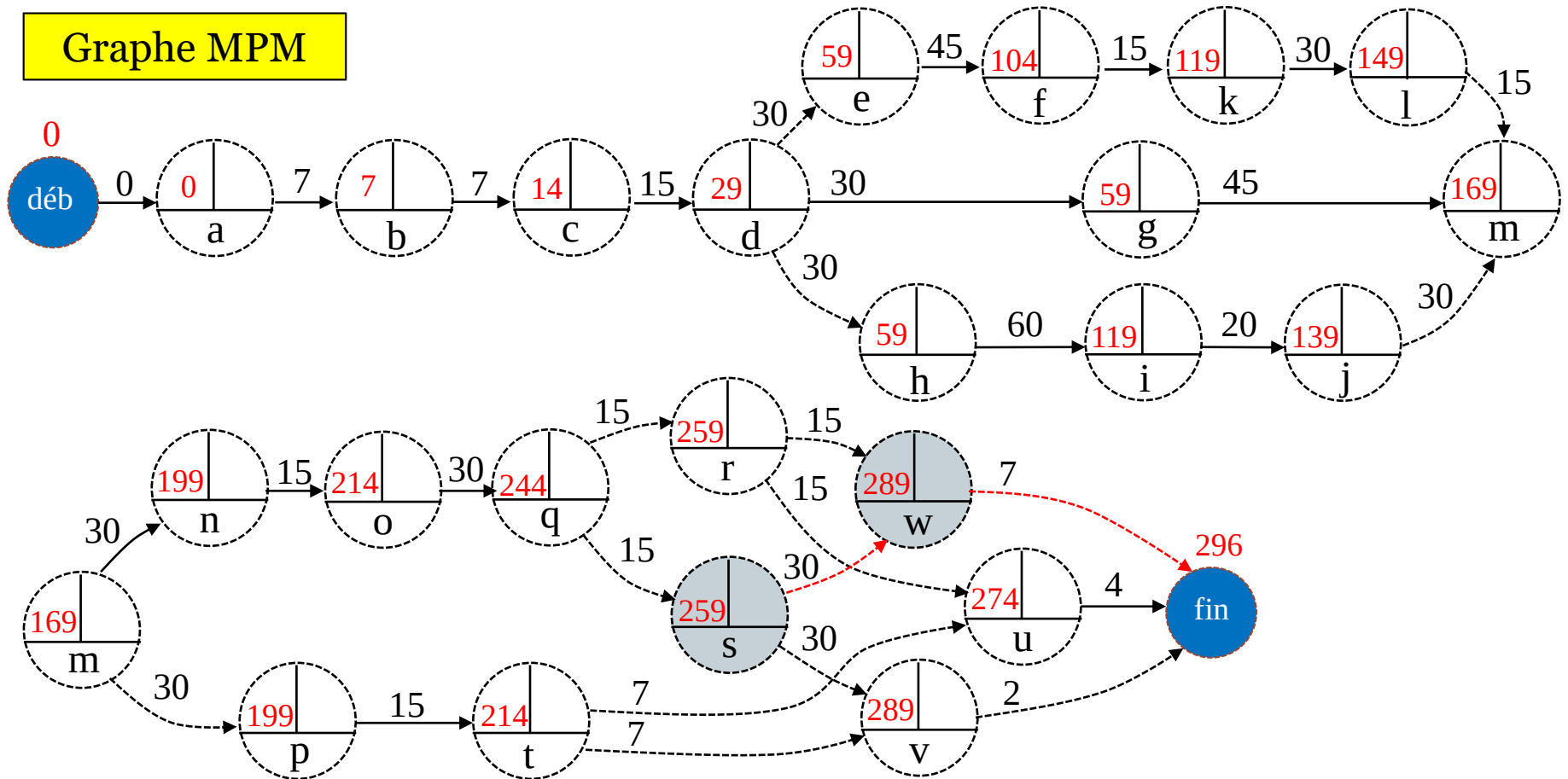
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Graphe MPM



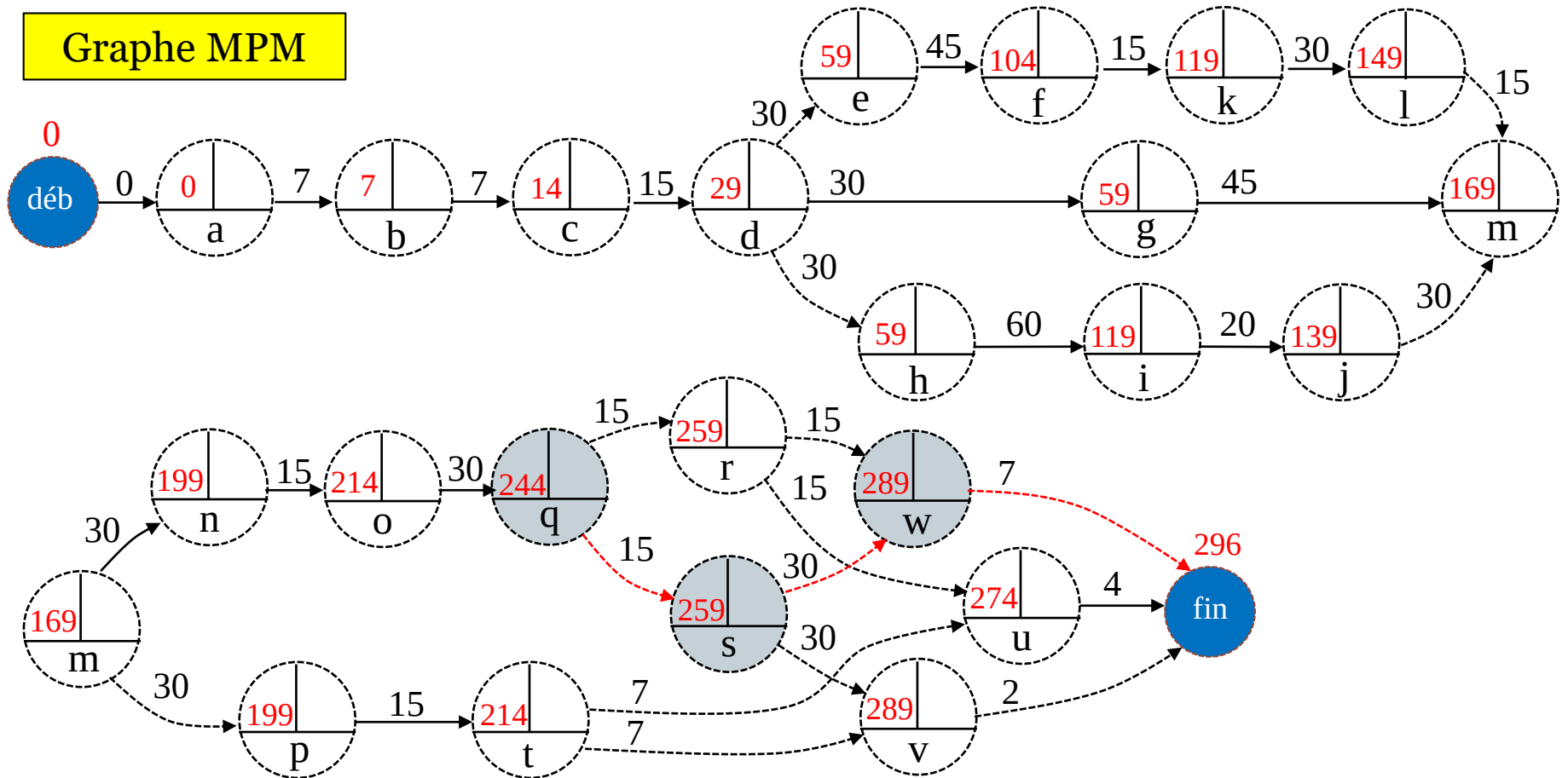
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Graphe MPM



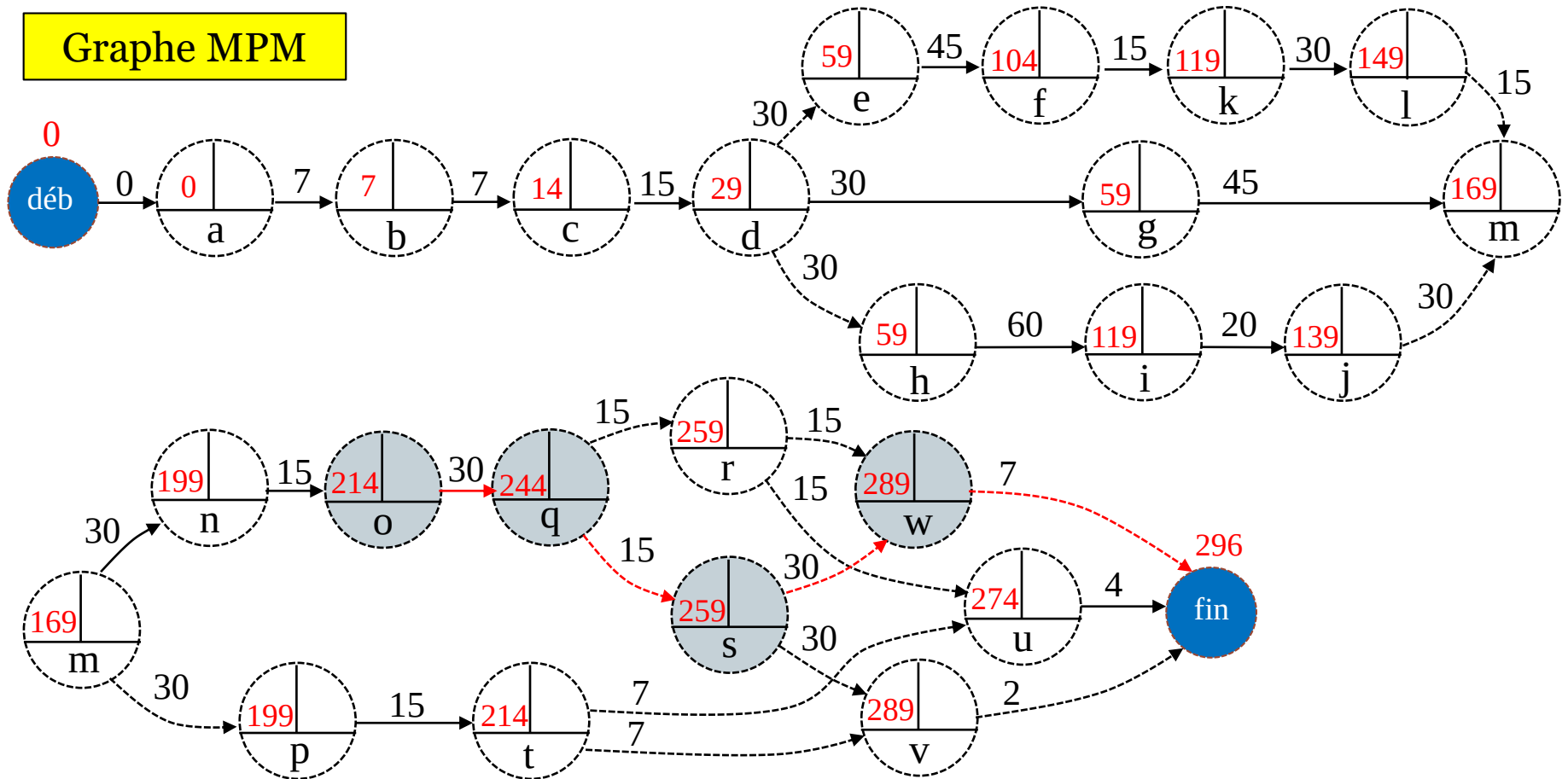
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Graphe MPM



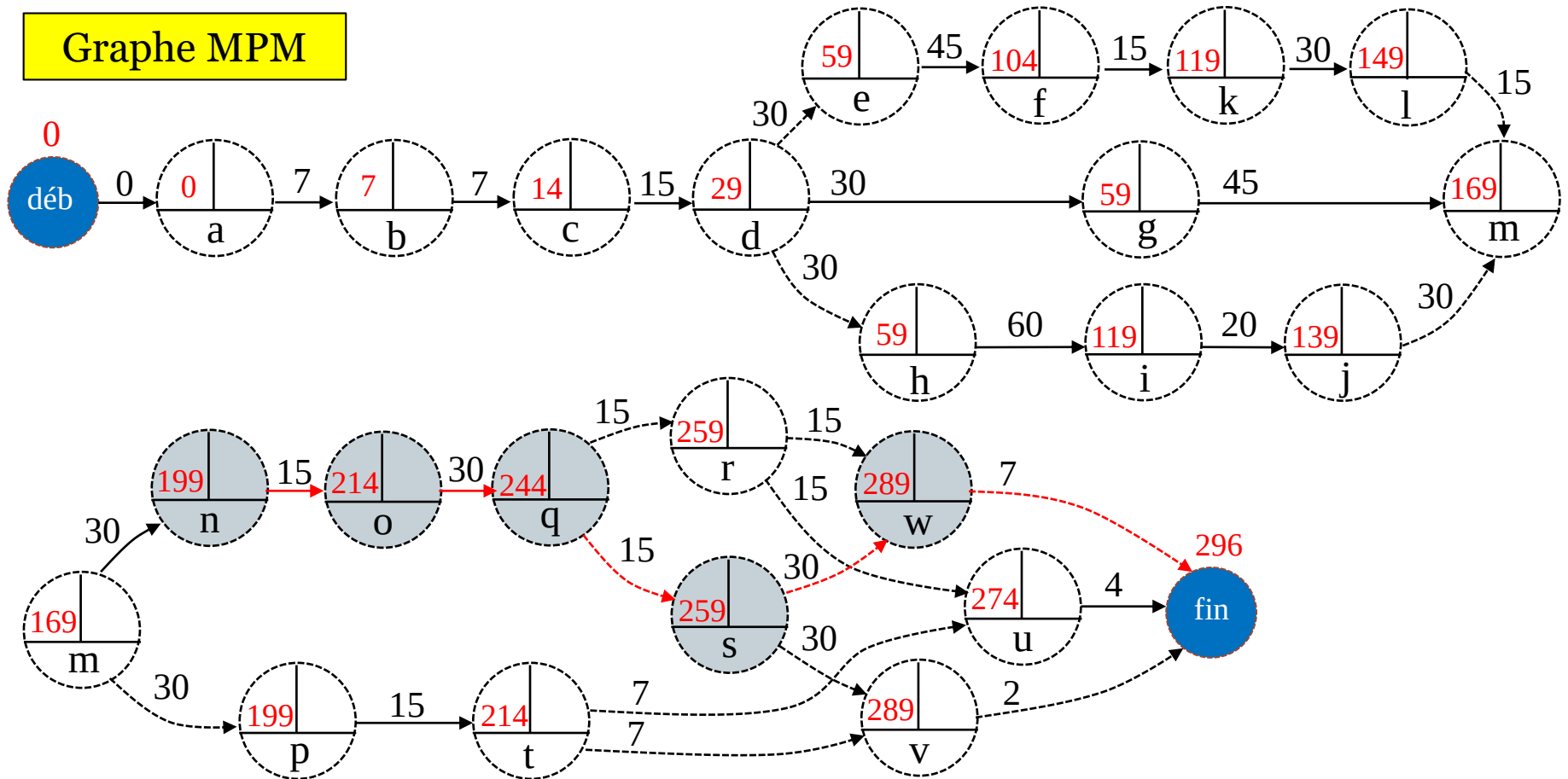
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Graphe MPM



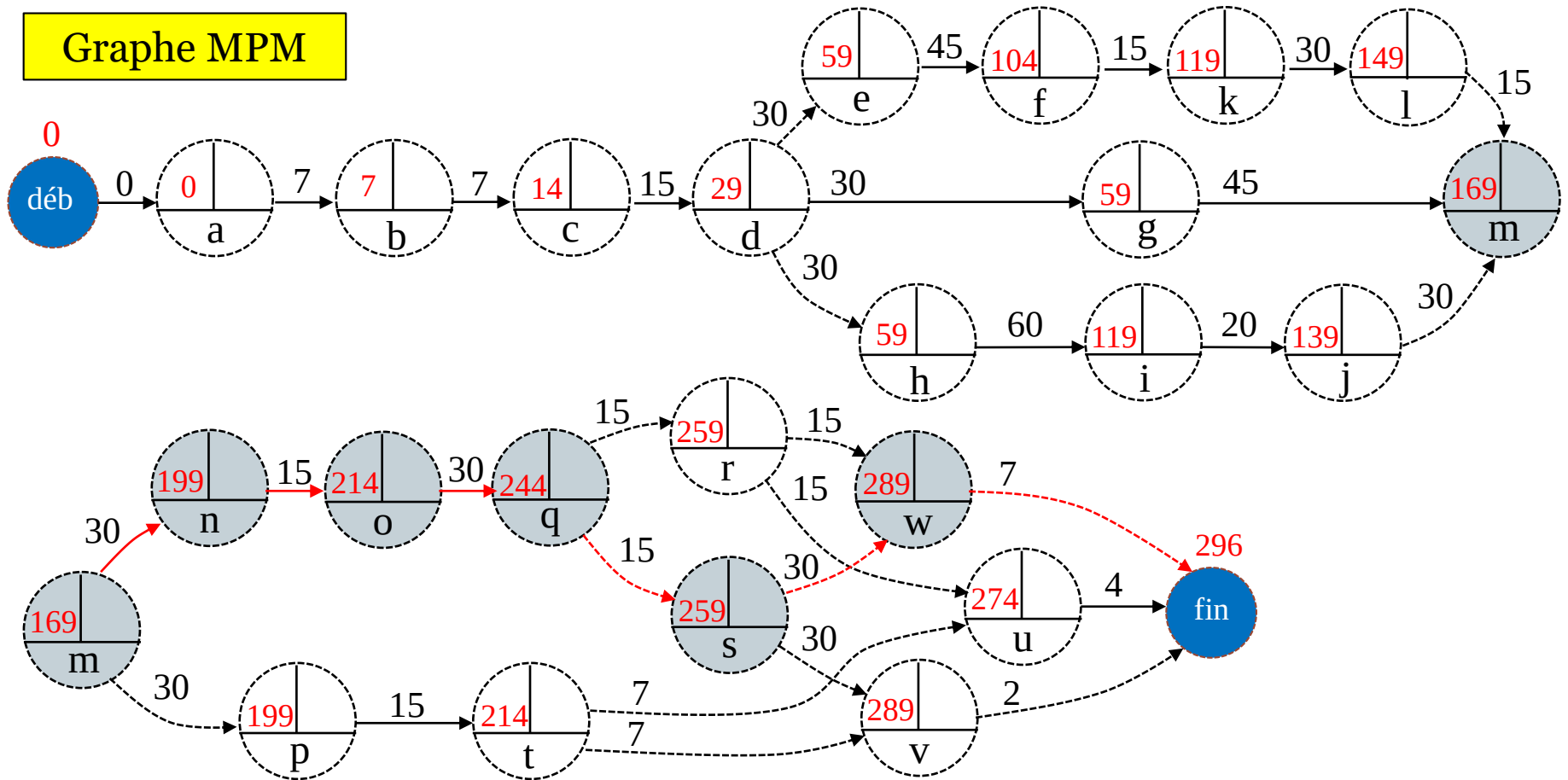
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Graphe MPM



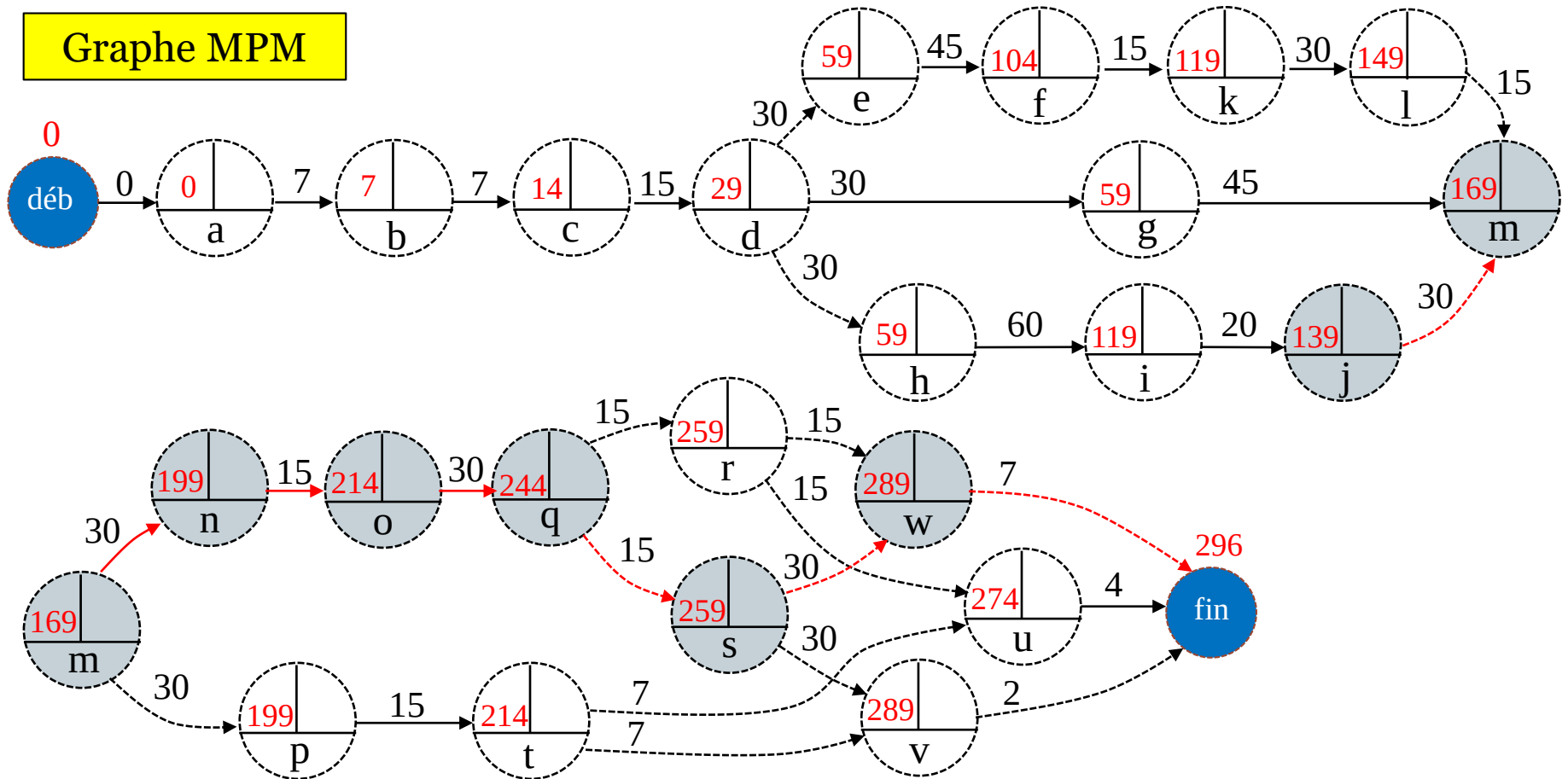
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Graphe MPM



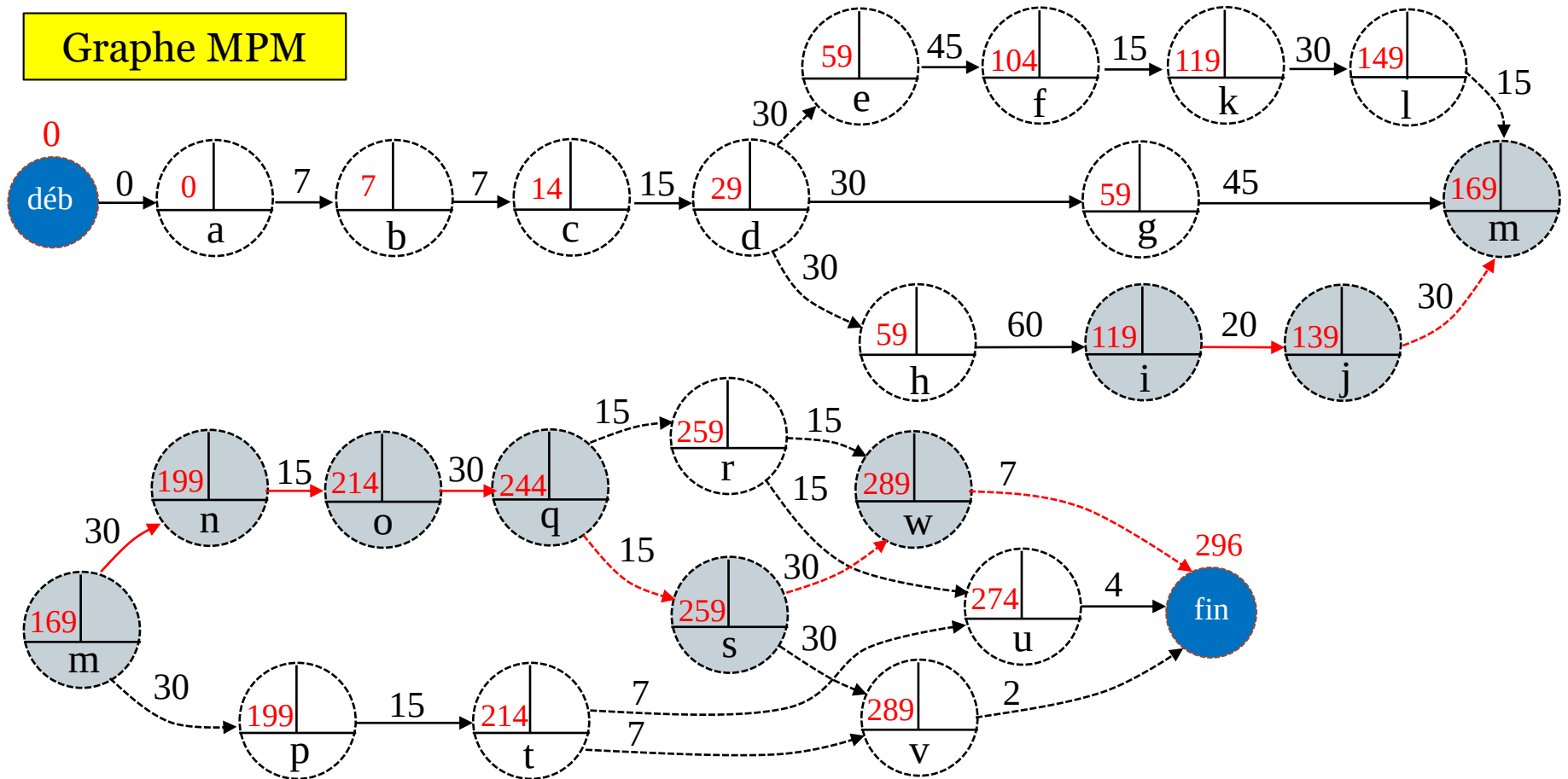
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Graphe MPM



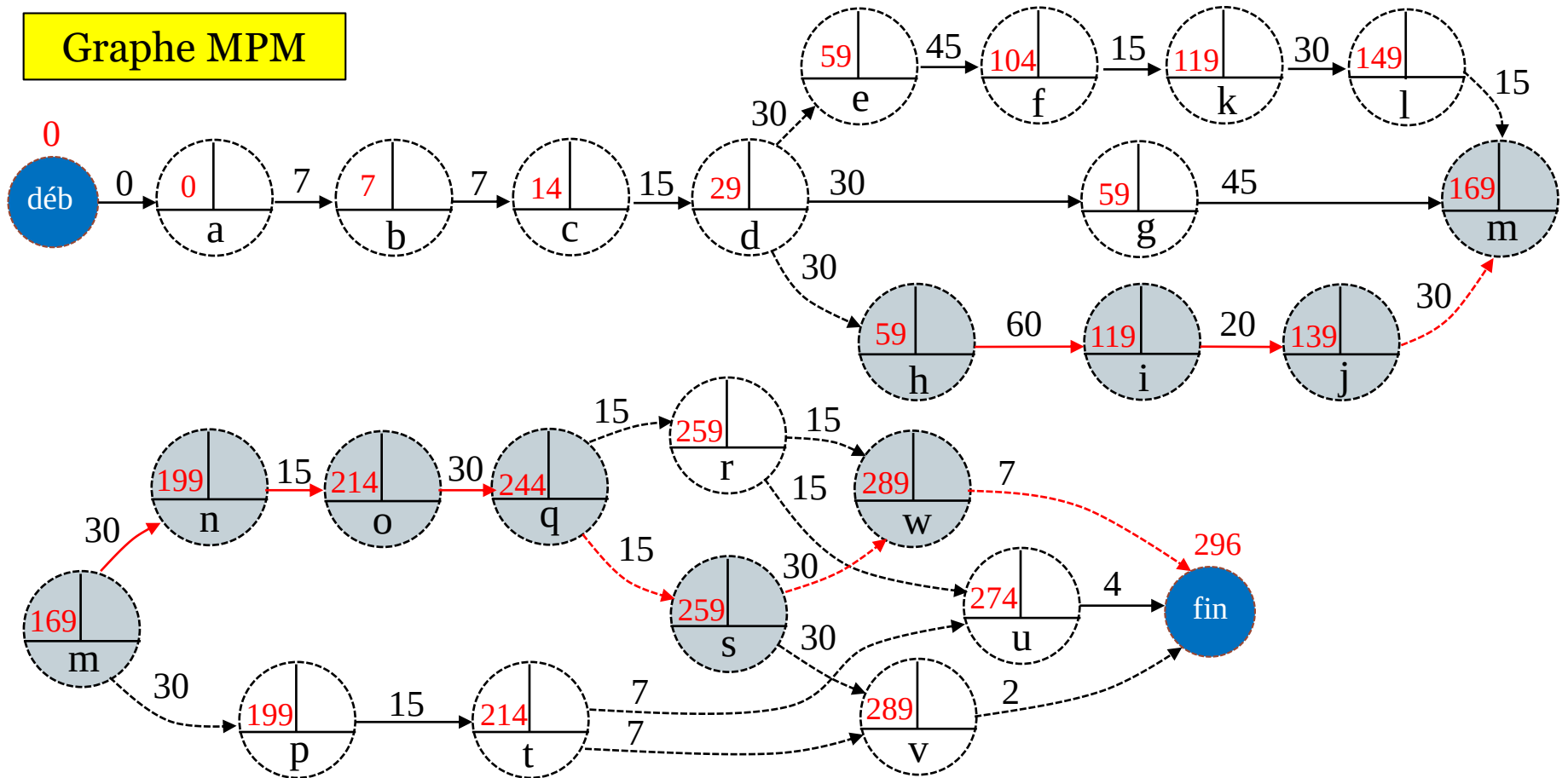
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Graphe MPM



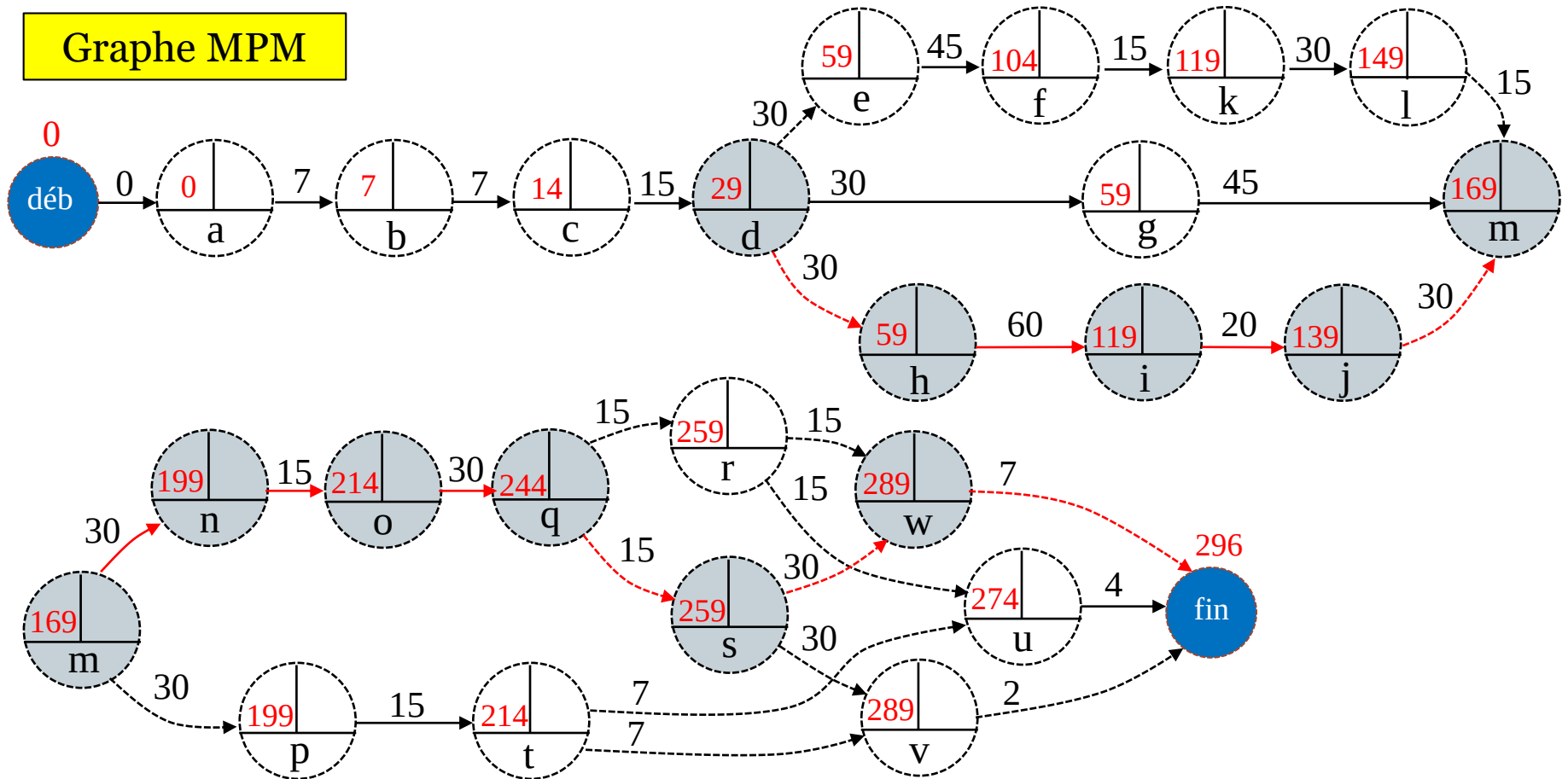
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Graphe MPM



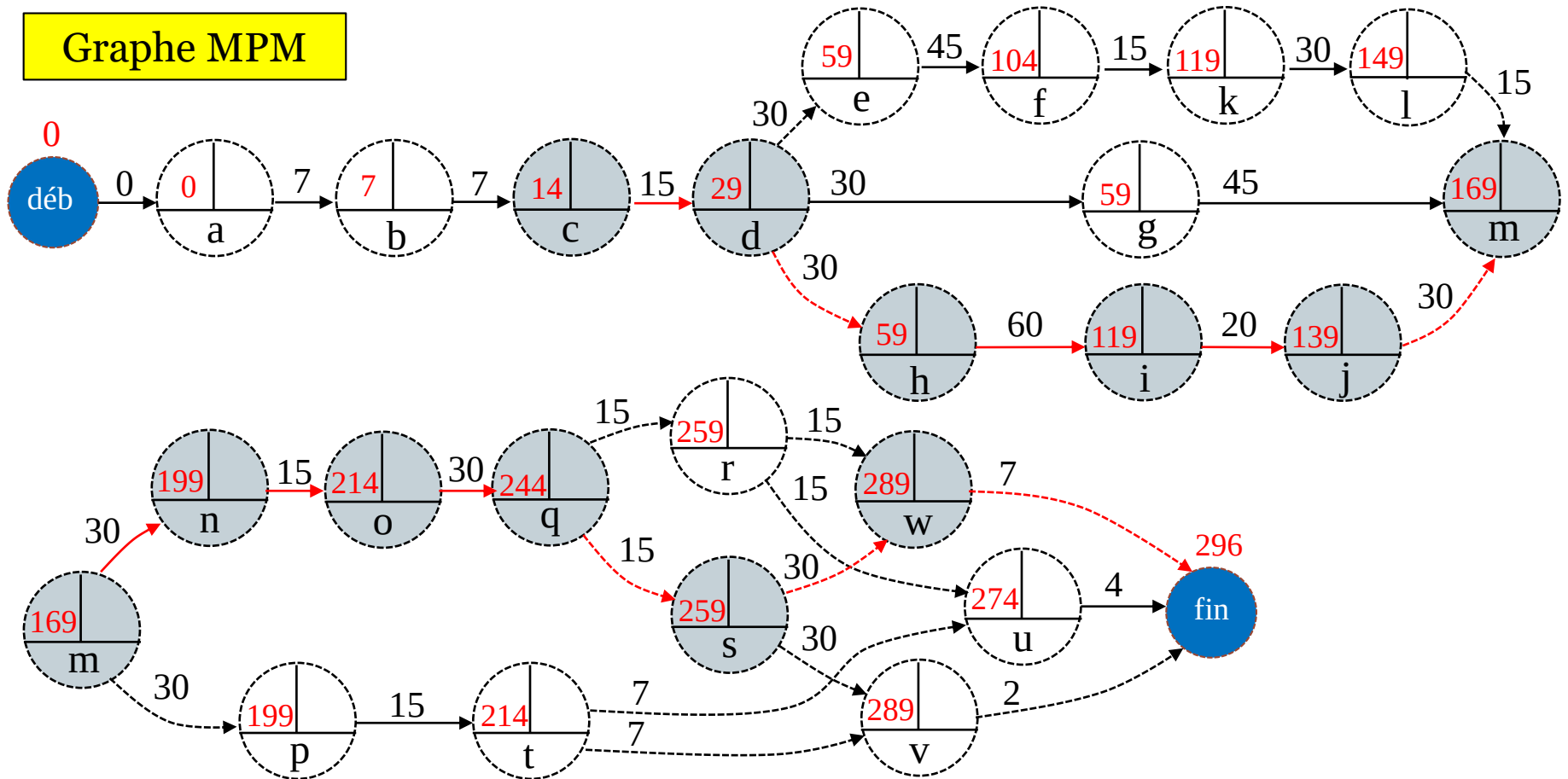
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Graphe MPM



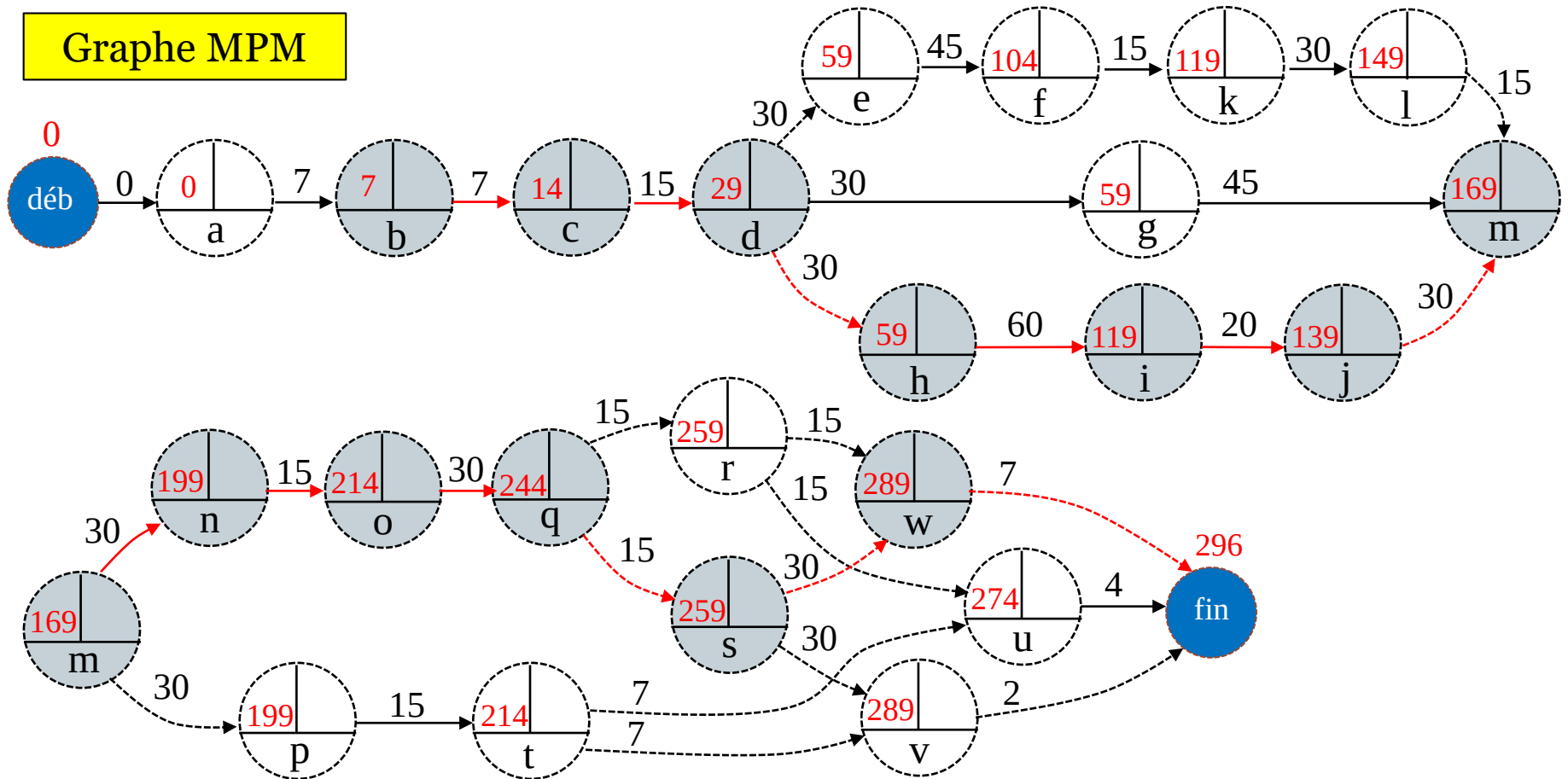
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Graphe MPM



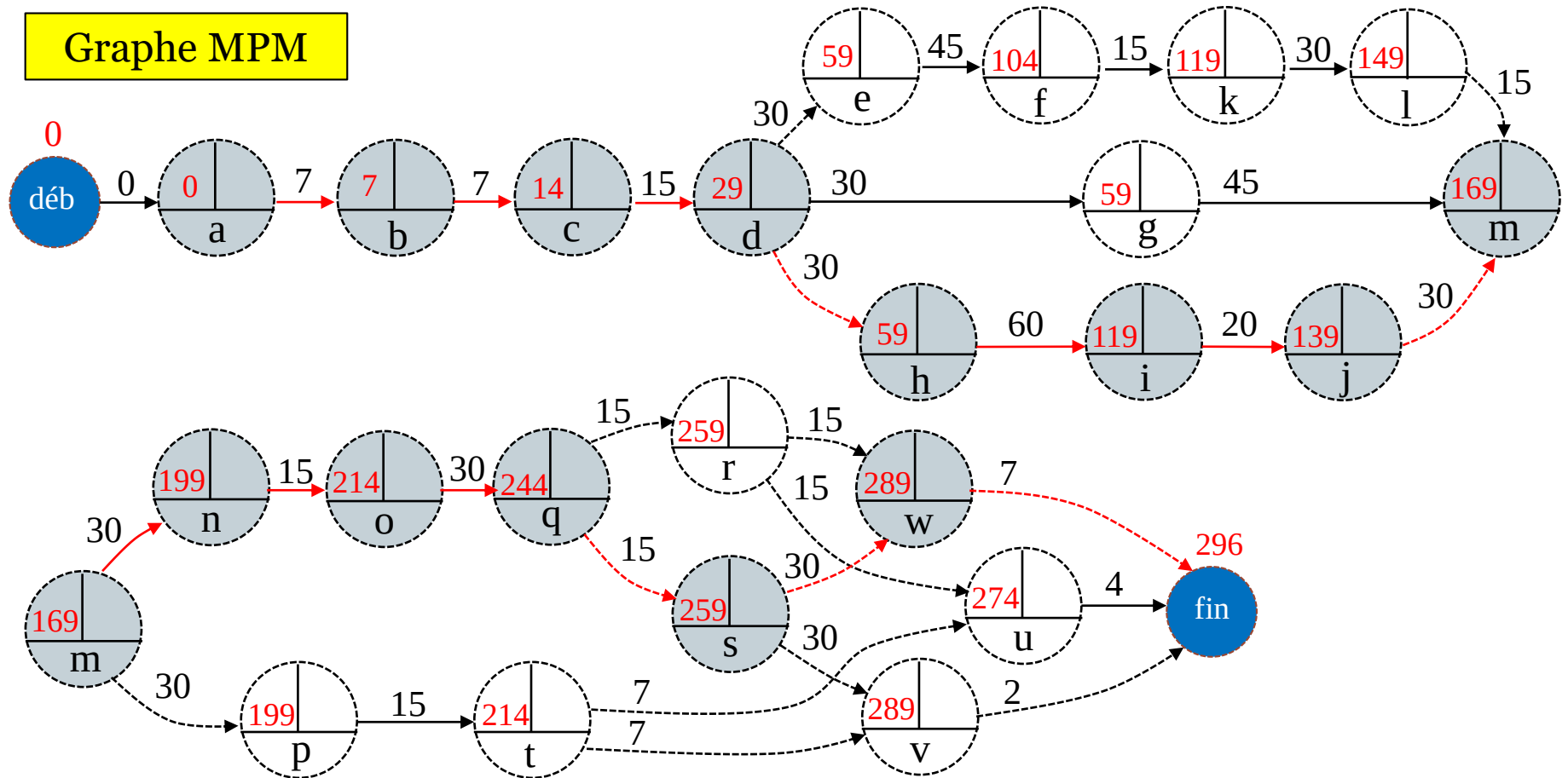
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Graphe MPM



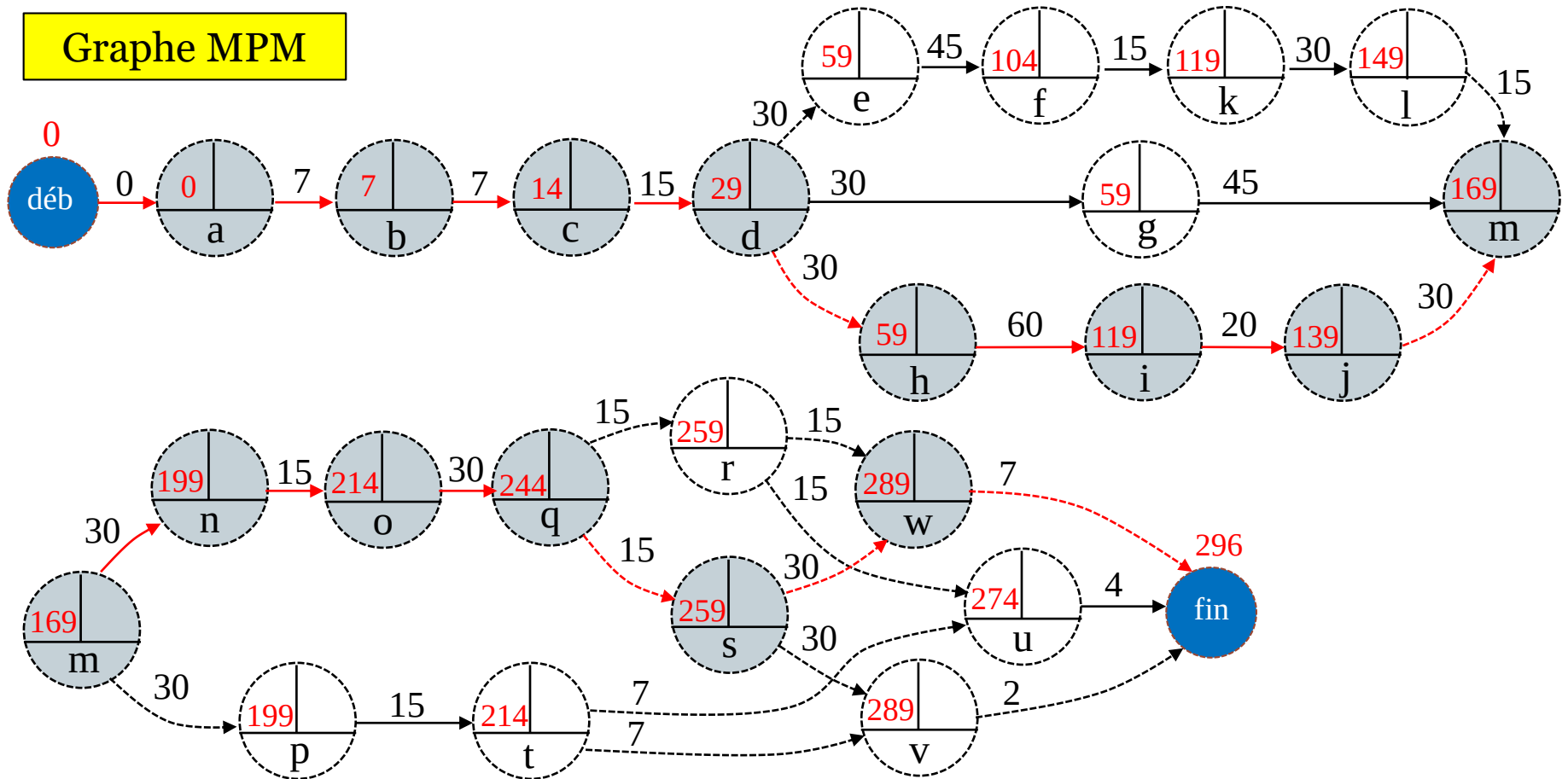
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Graphe MPM



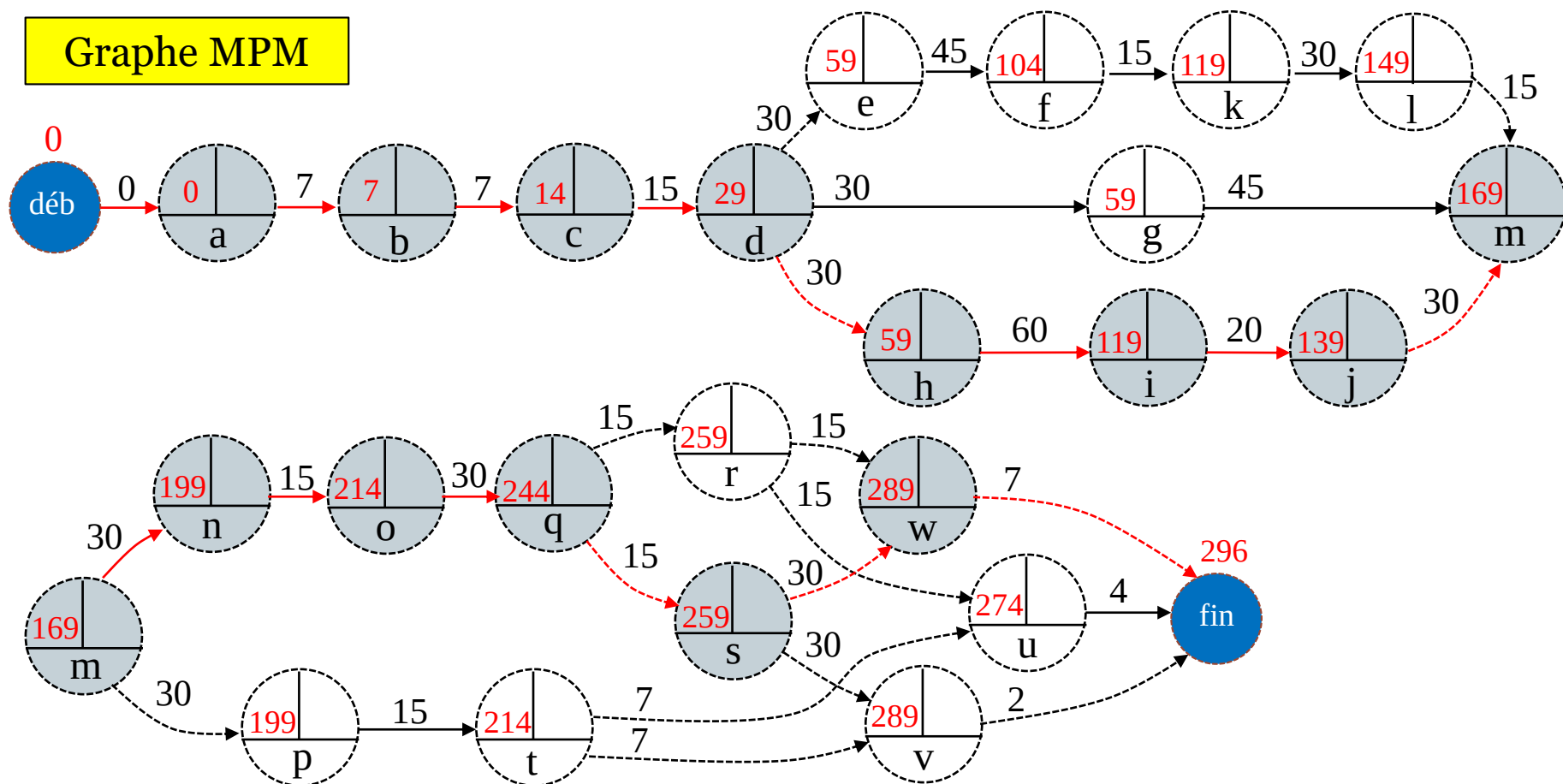
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Graphe MPM



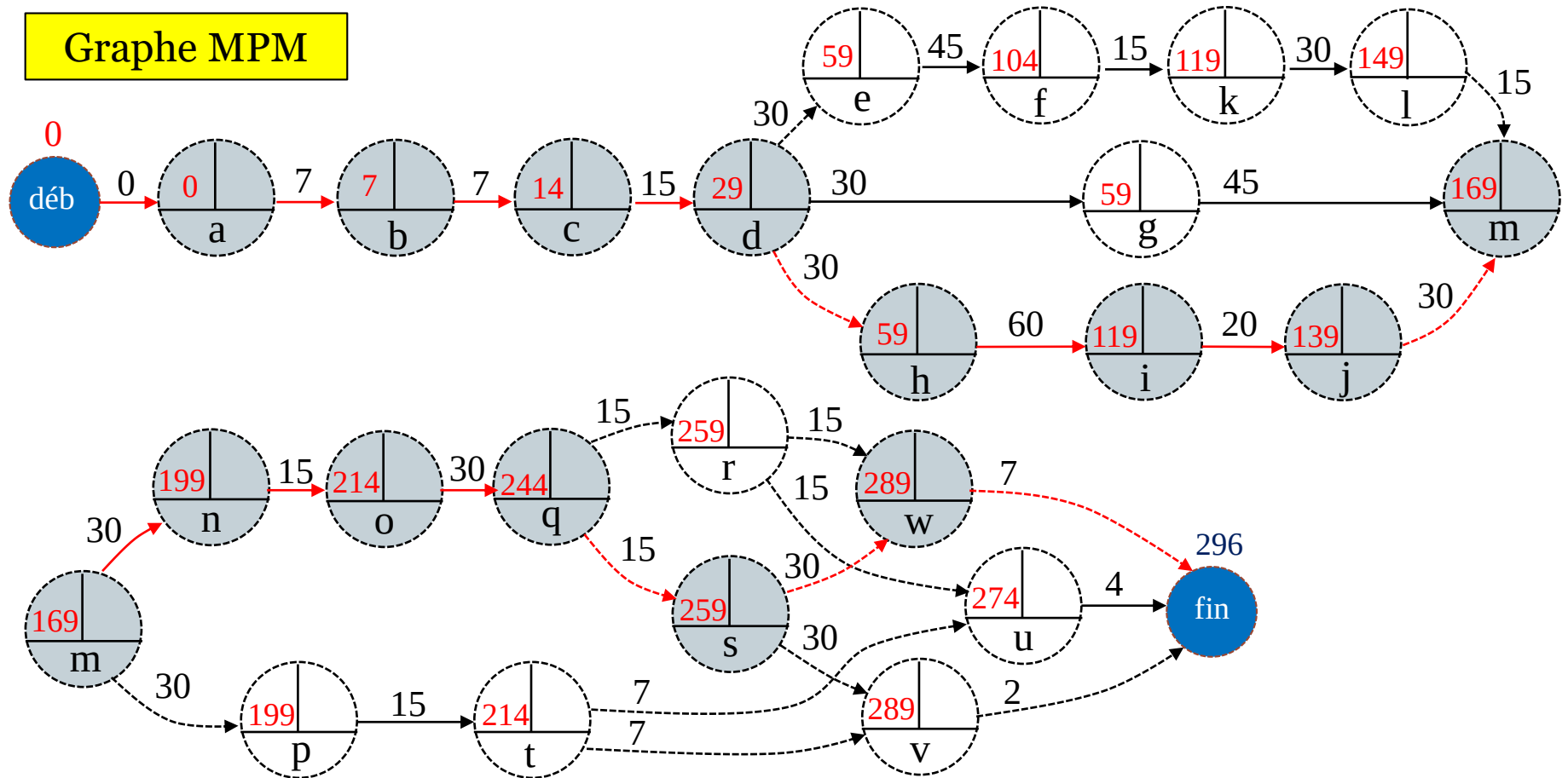
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Graphe MPM



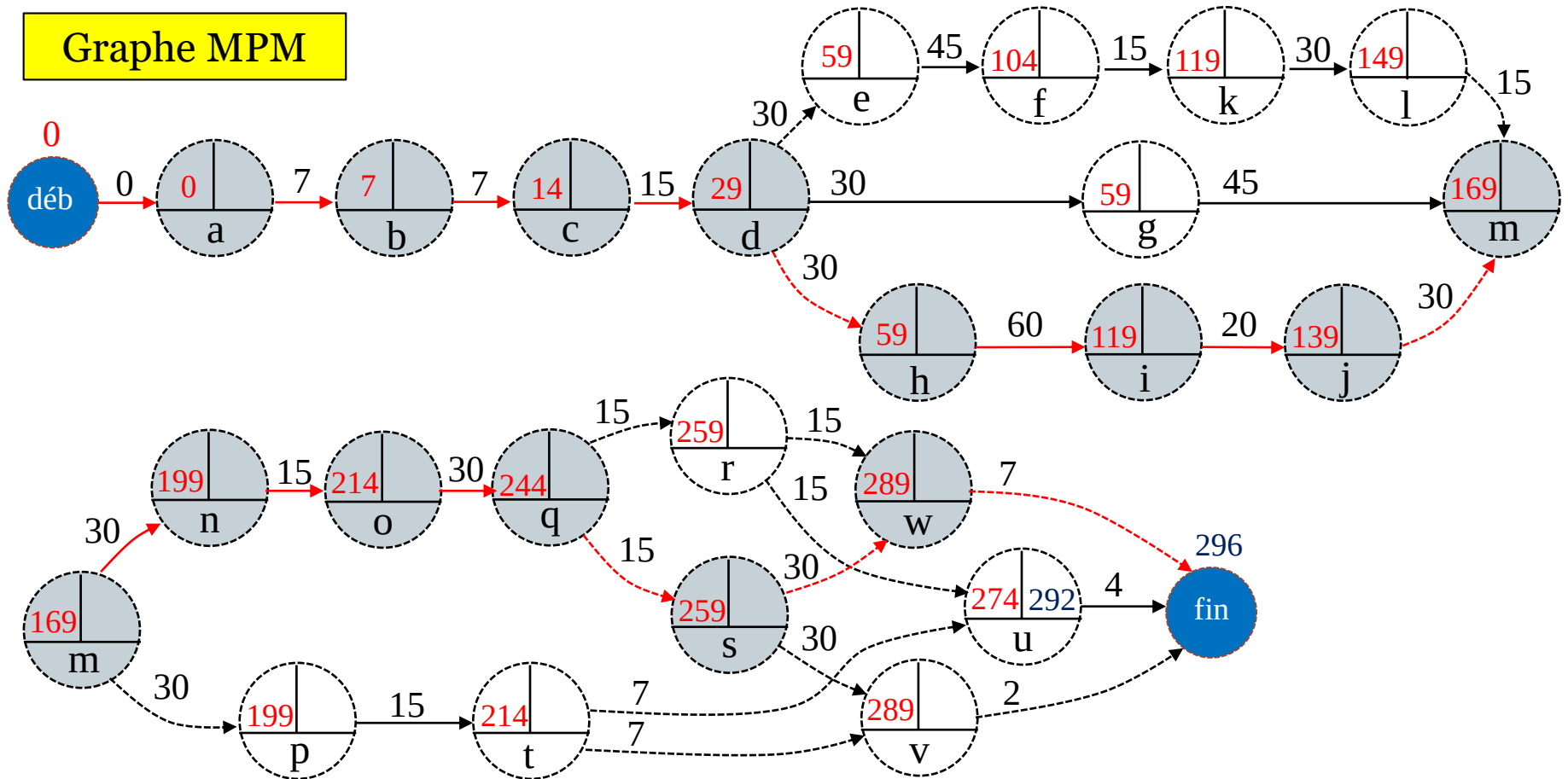
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Graphe MPM



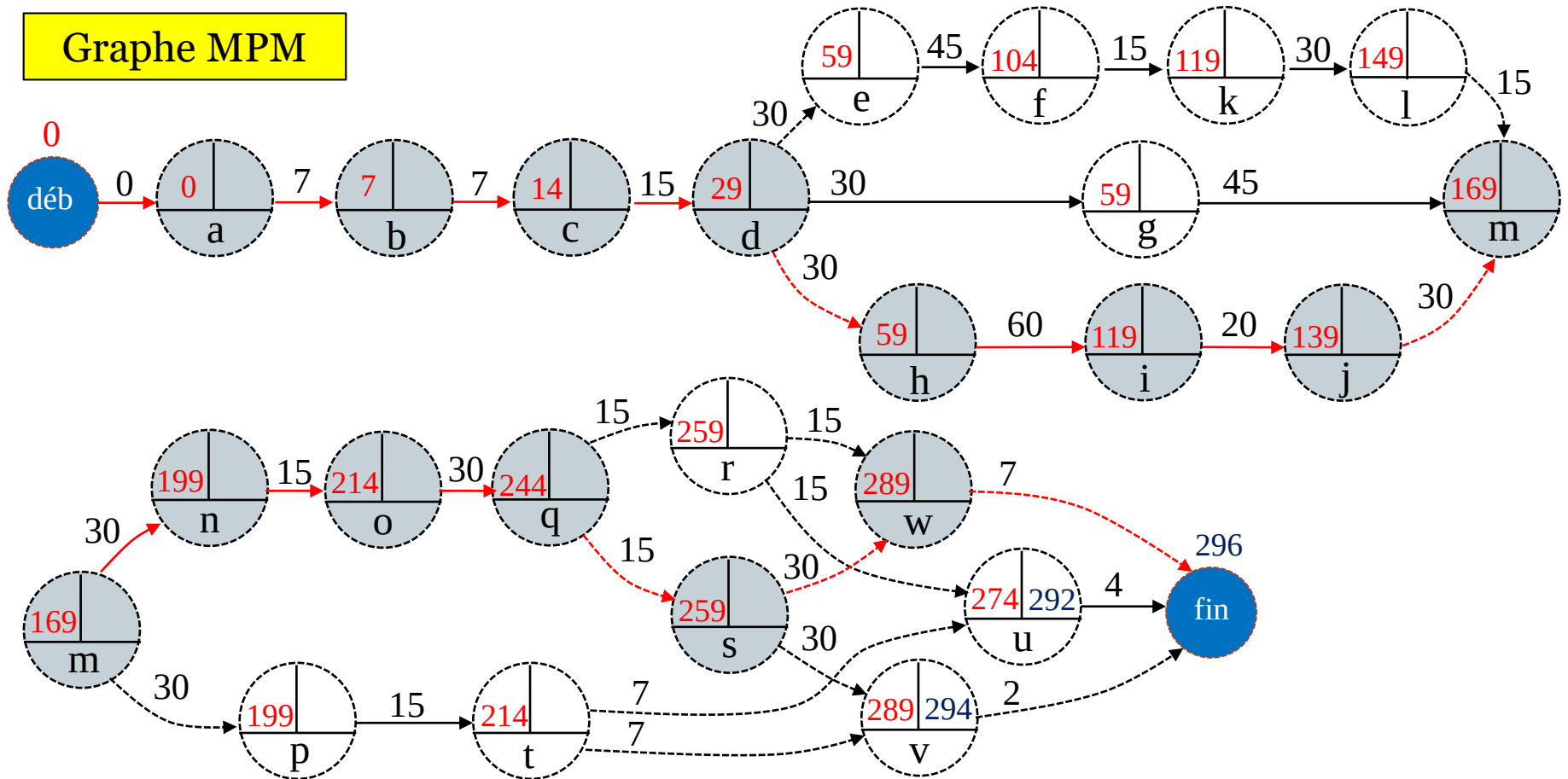
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Graphe MPM



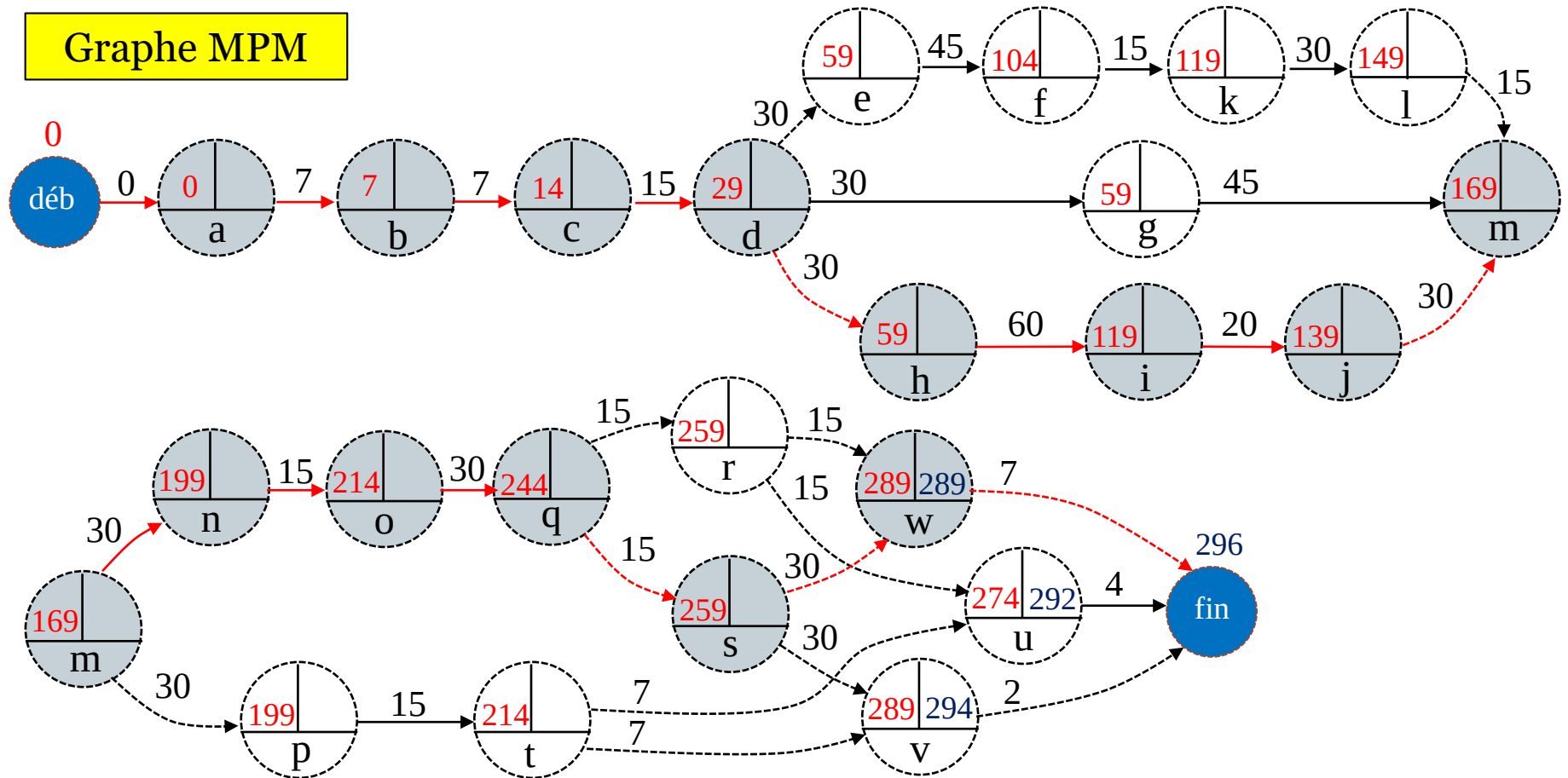
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Graphe MPM



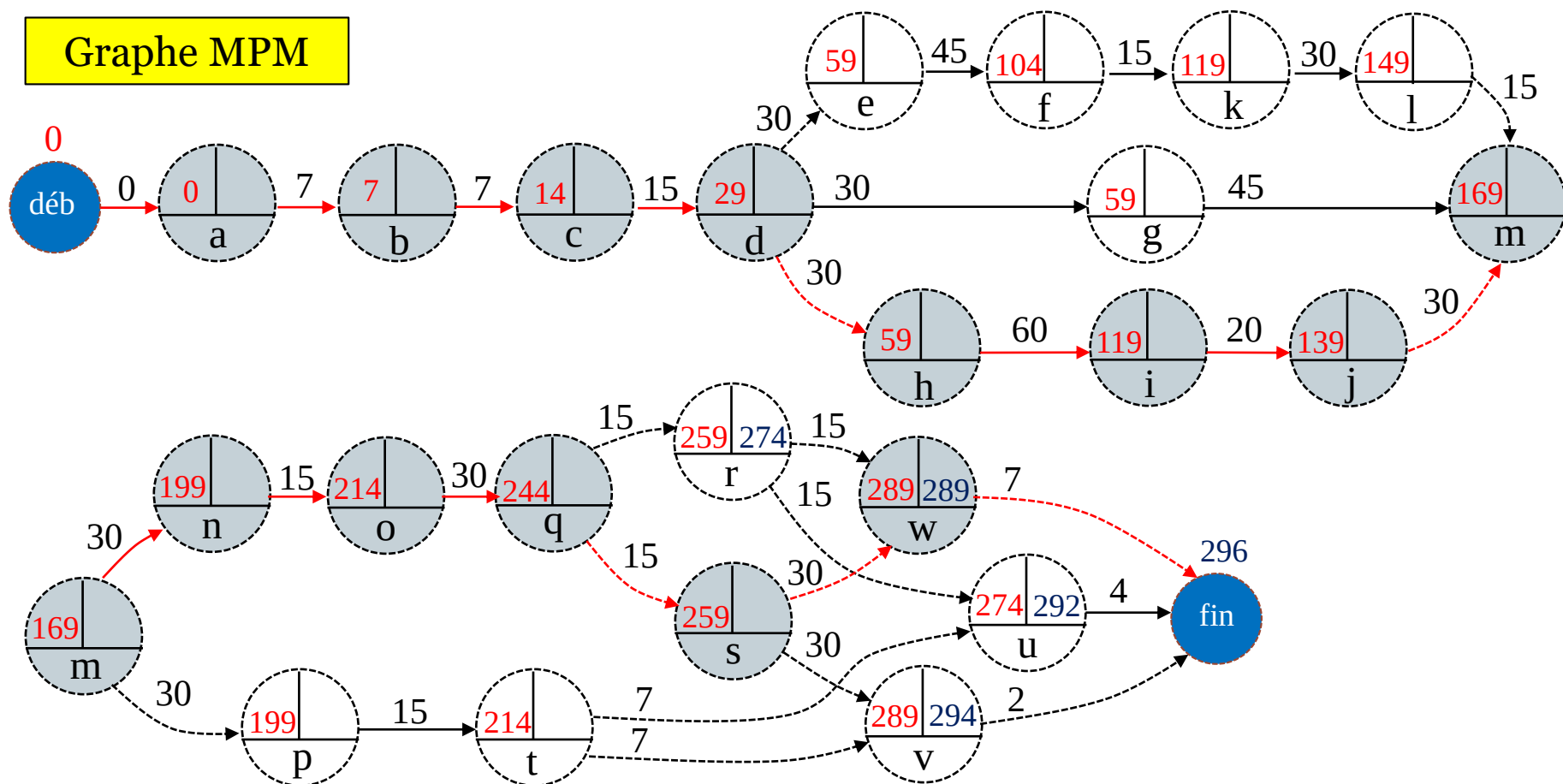
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Graphe MPM



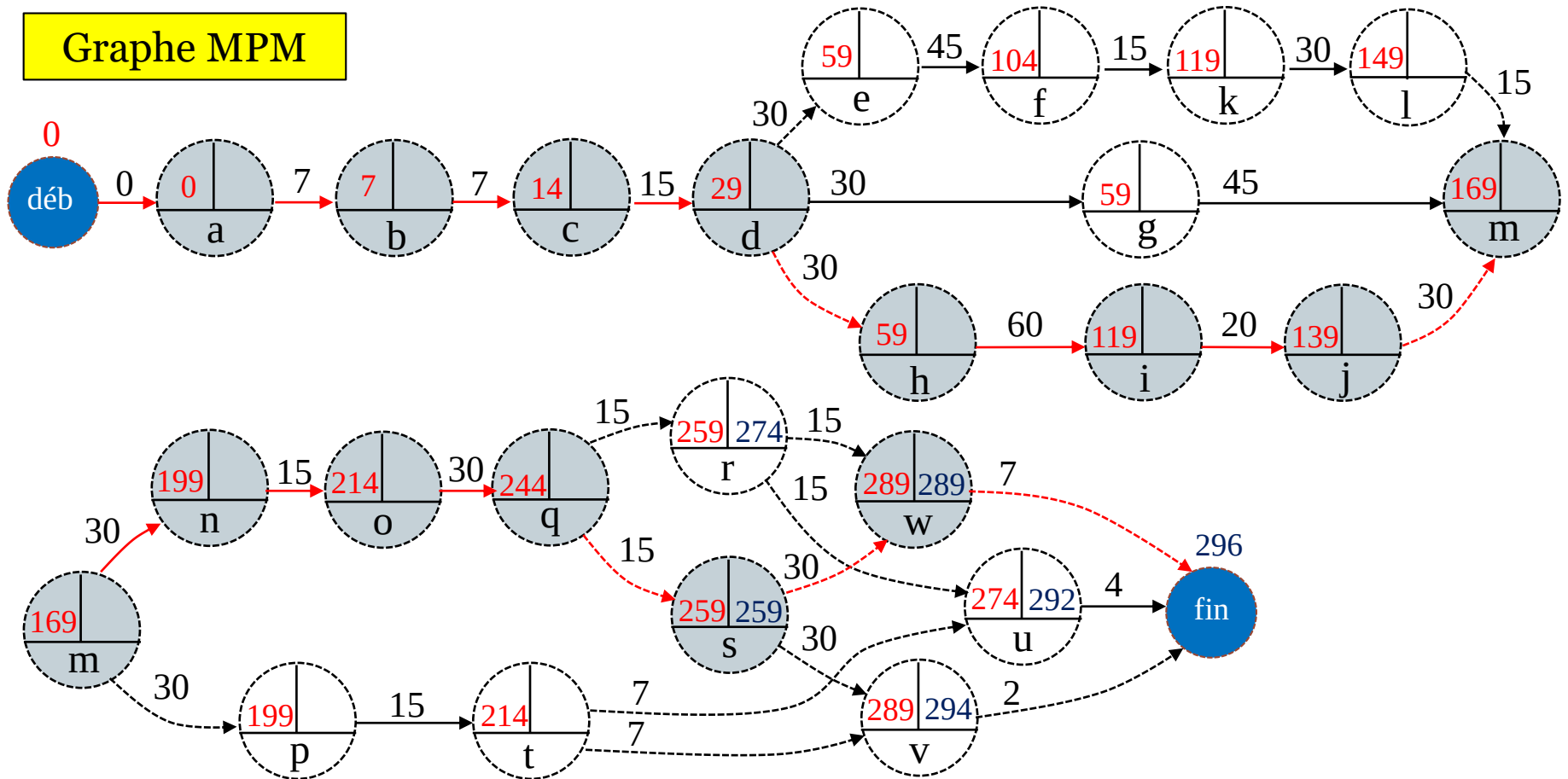
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Graphe MPM



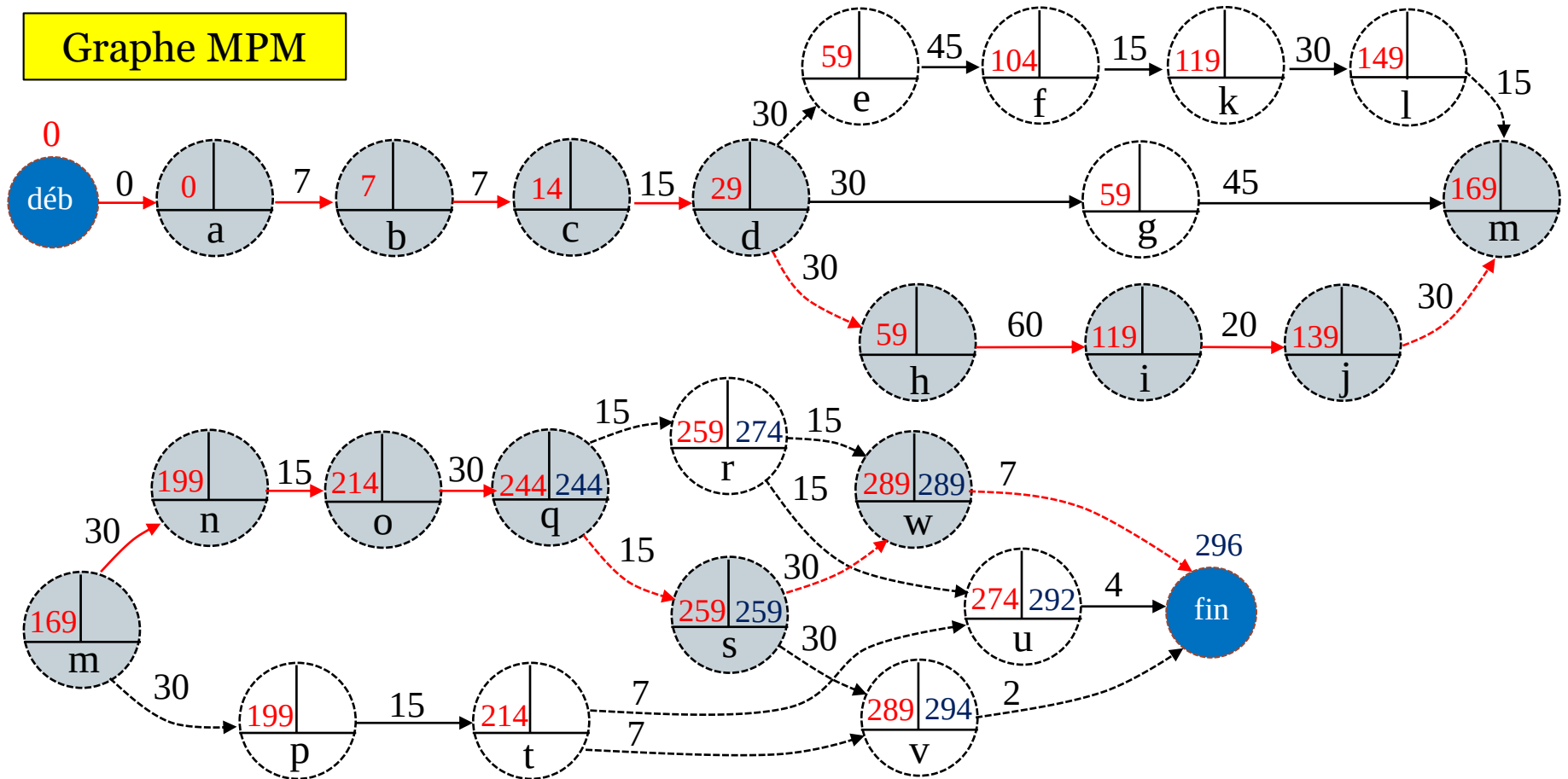
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Graphe MPM



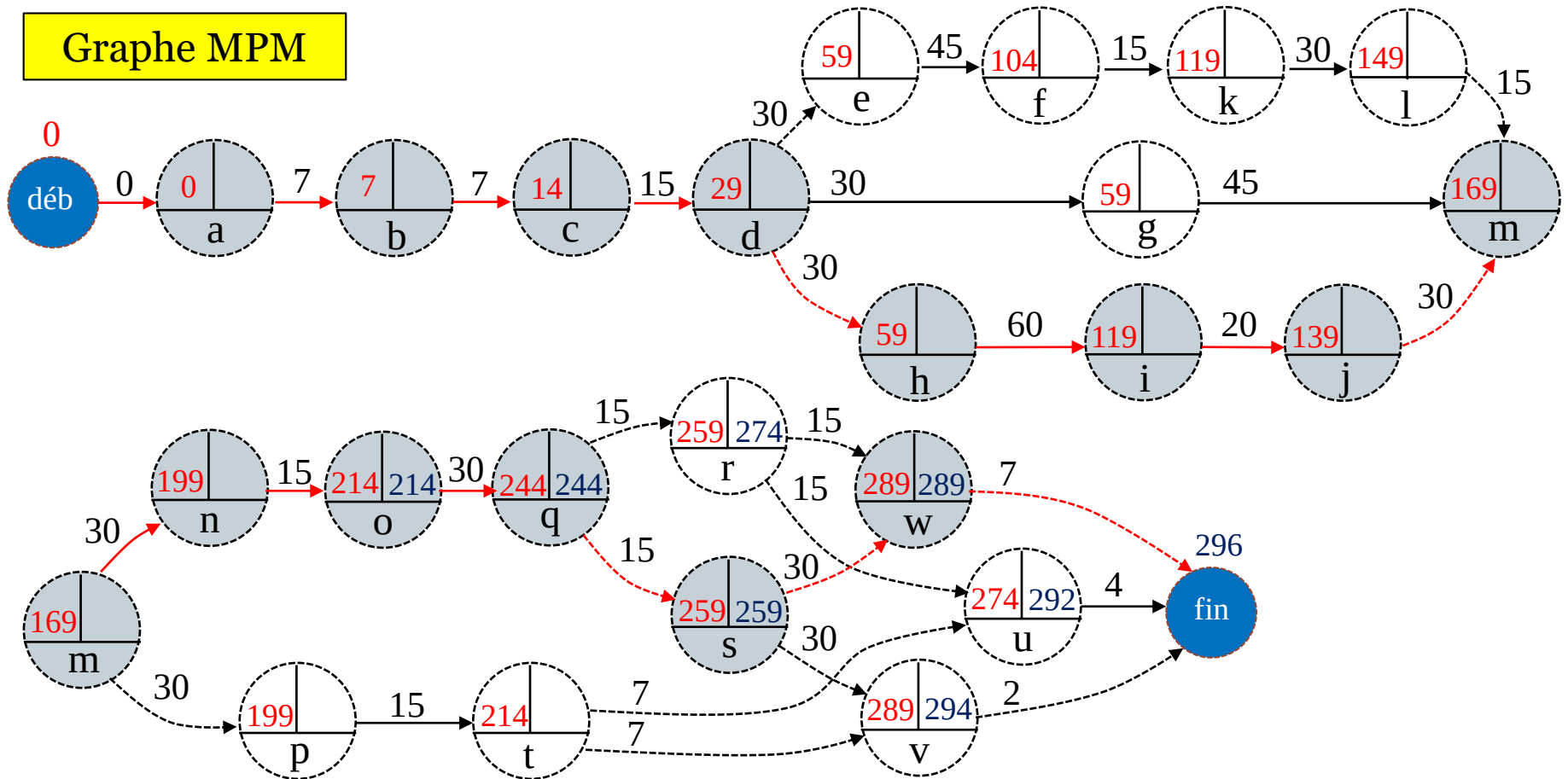
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Graphe MPM



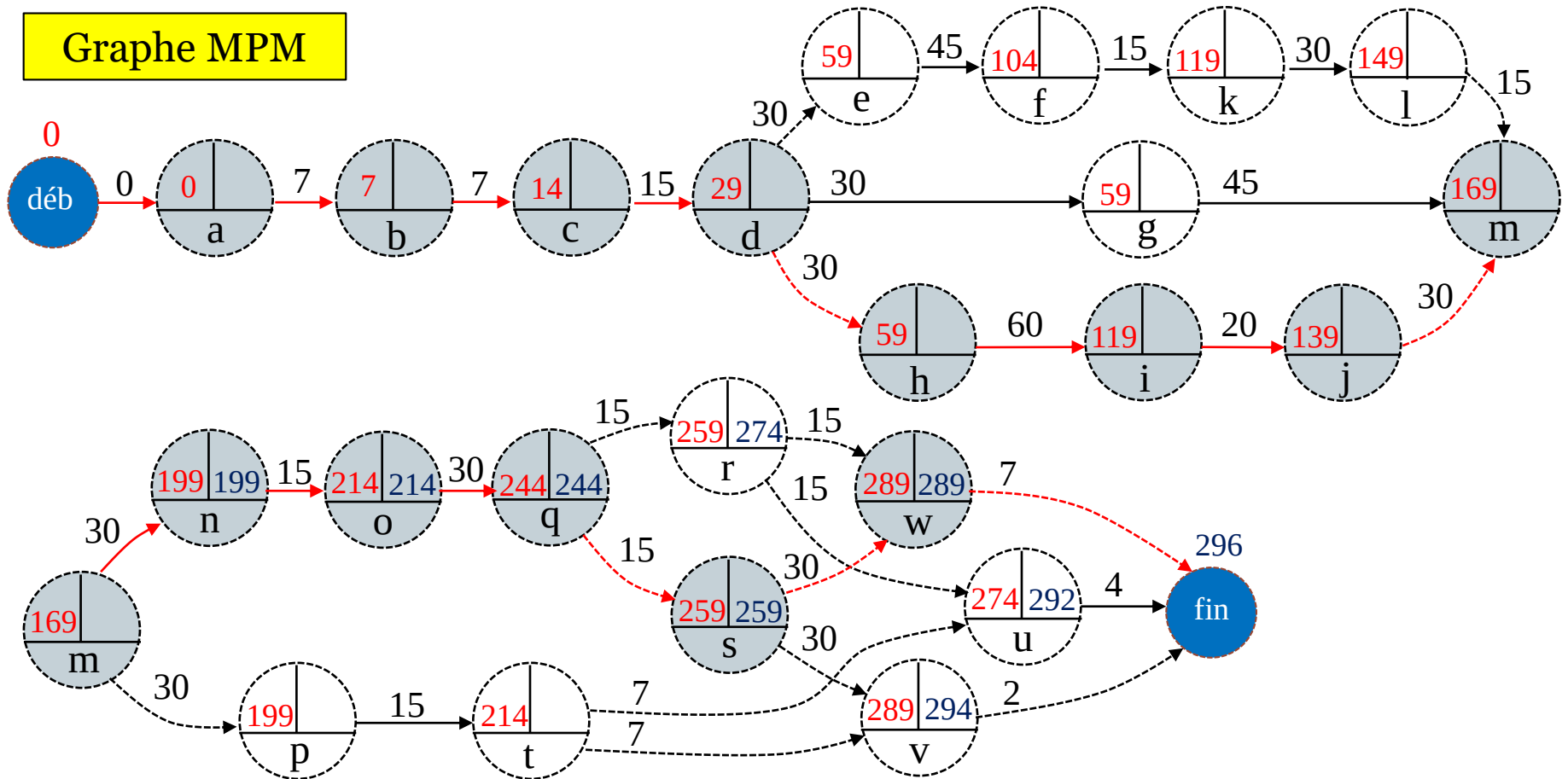
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Graphe MPM



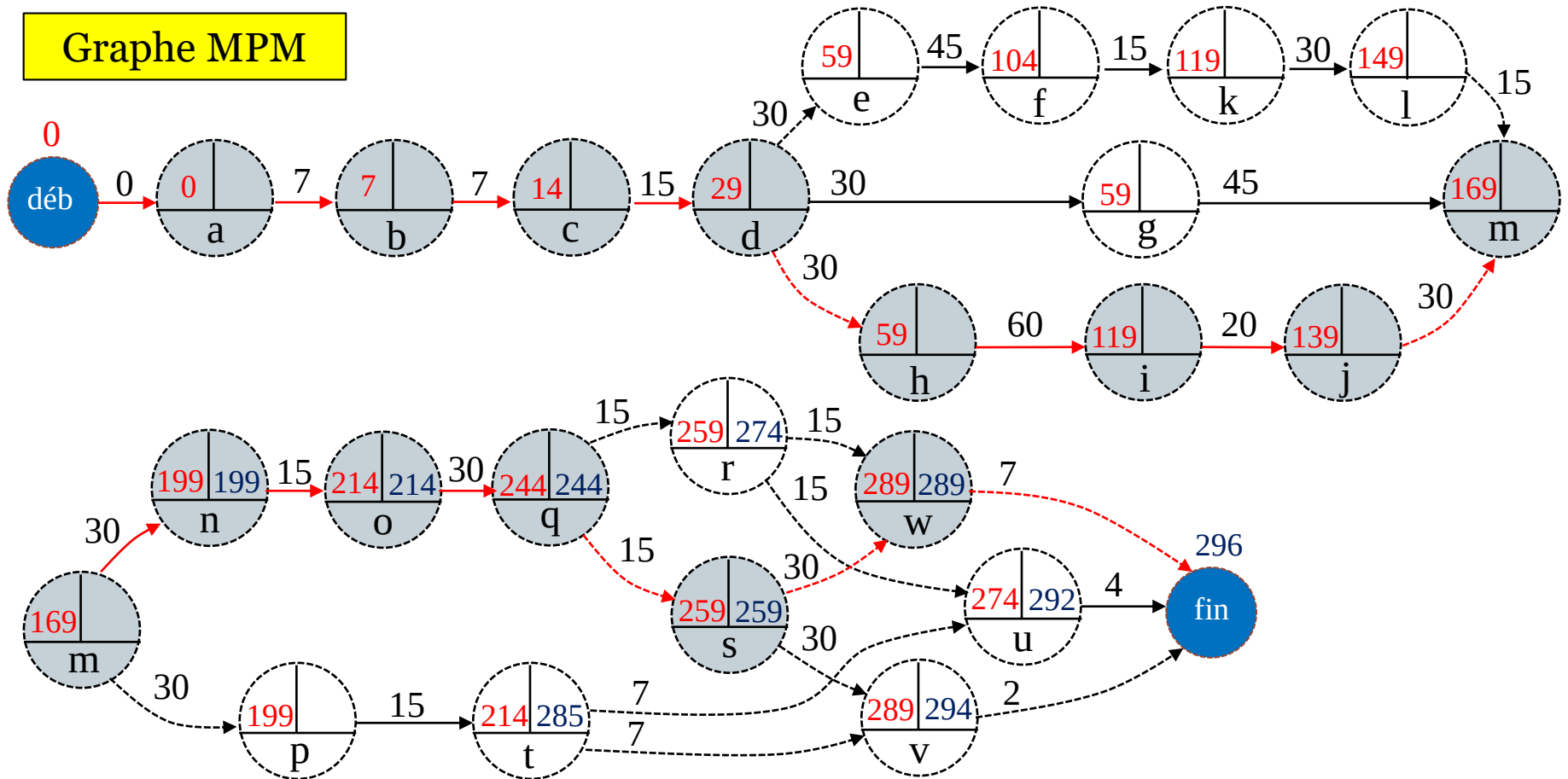
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Graphe MPM



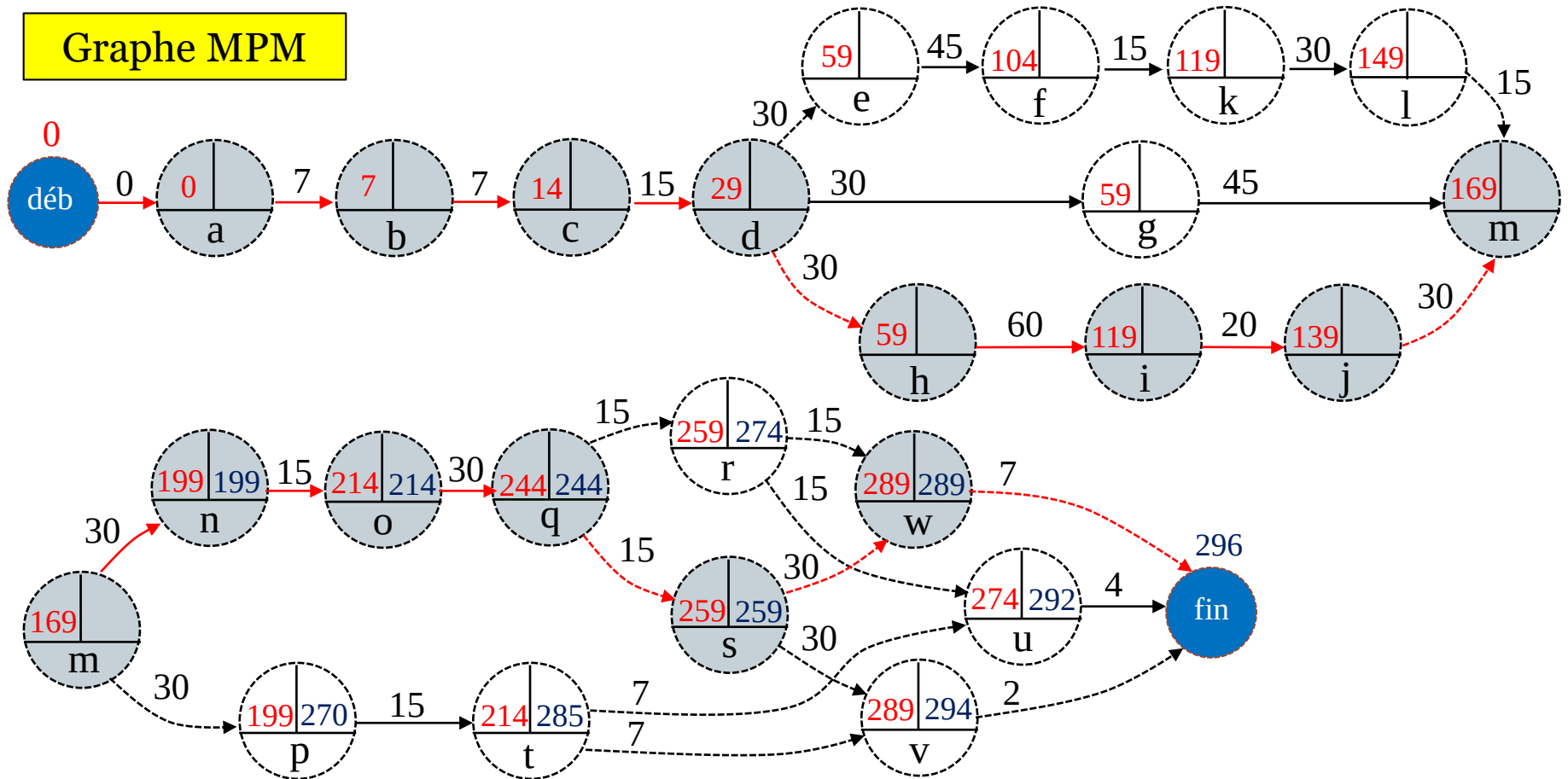
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Graphe MPM



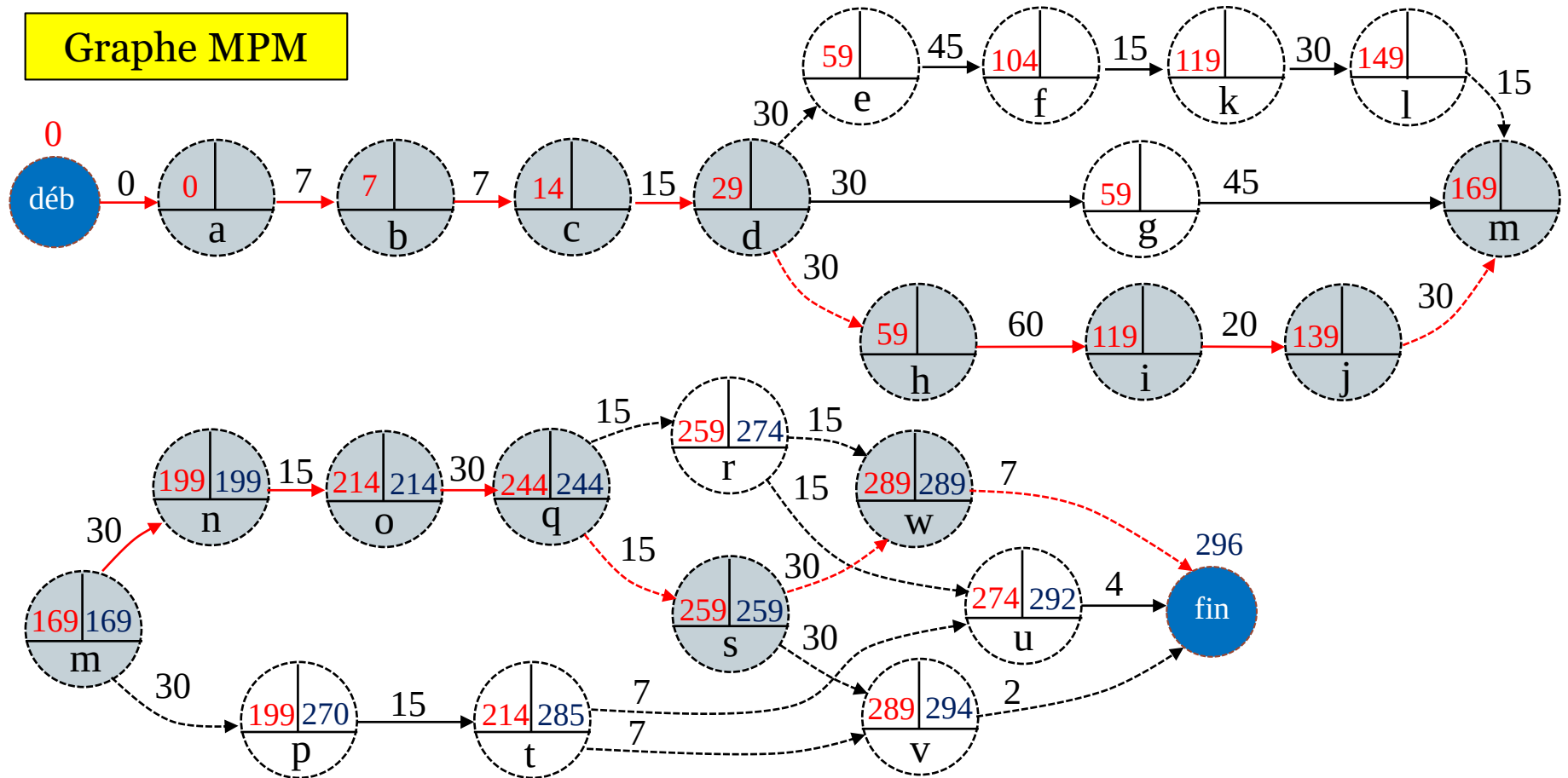
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Graphe MPM



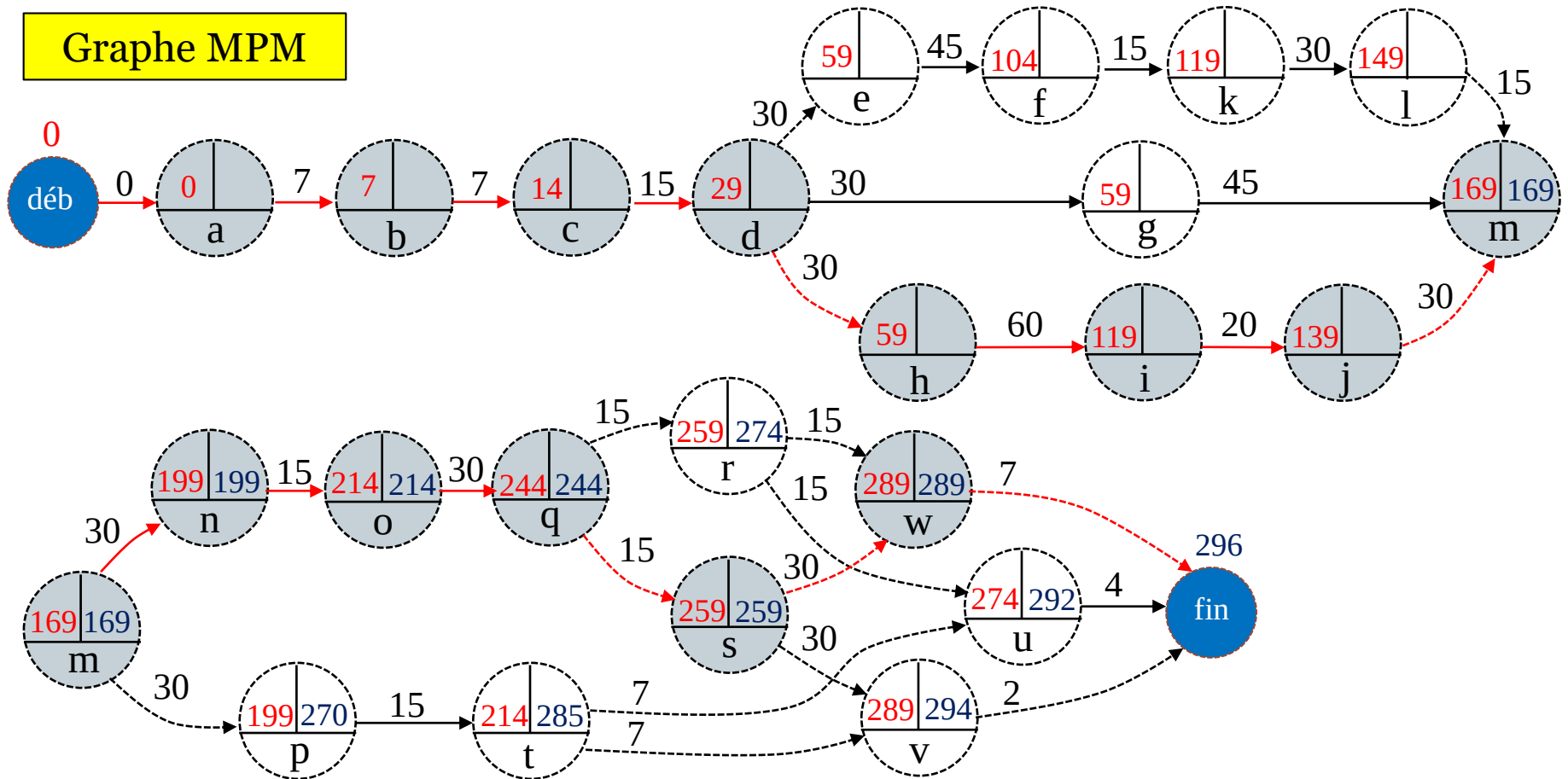
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Graphe MPM



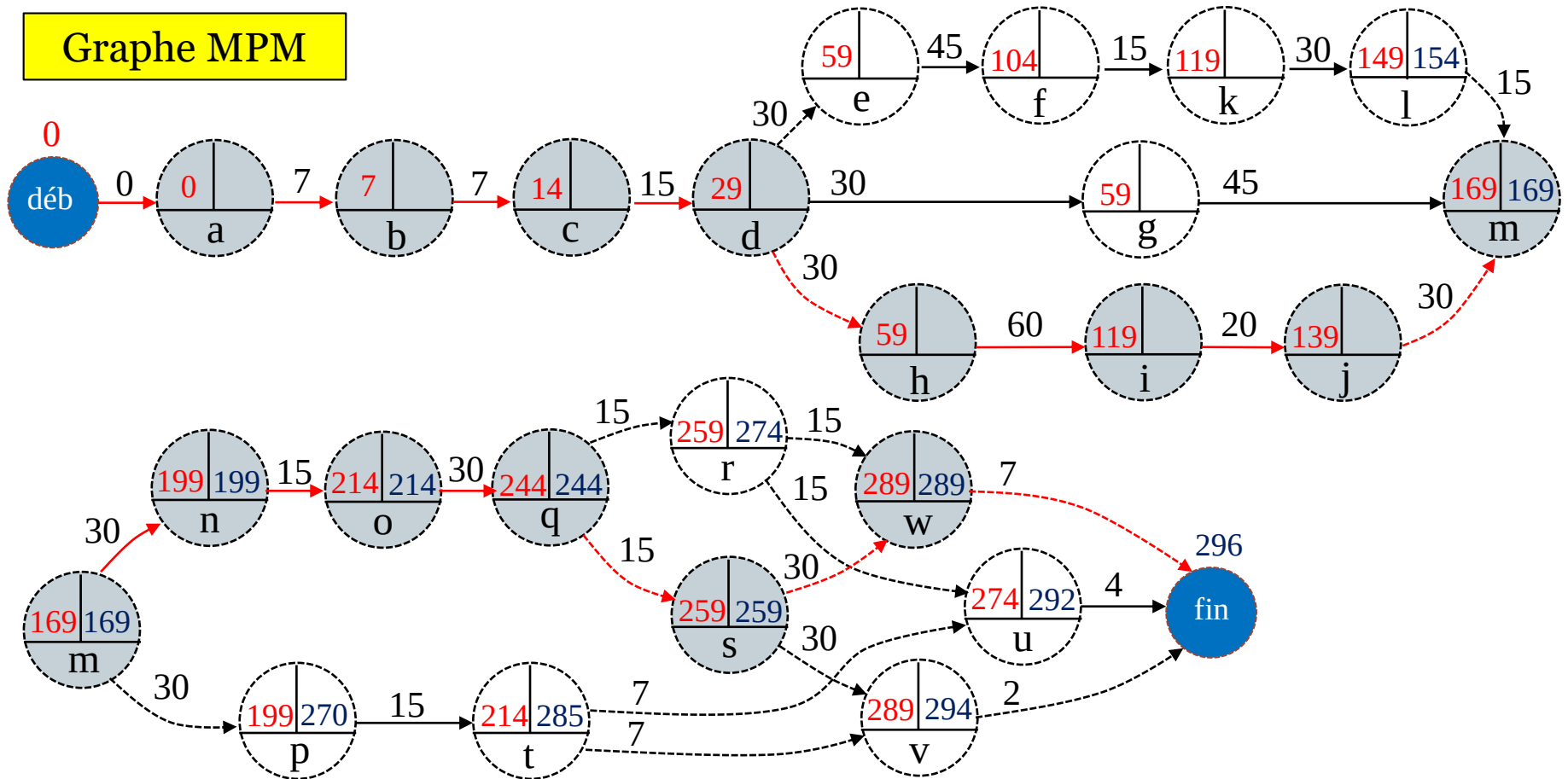
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Graphe MPM



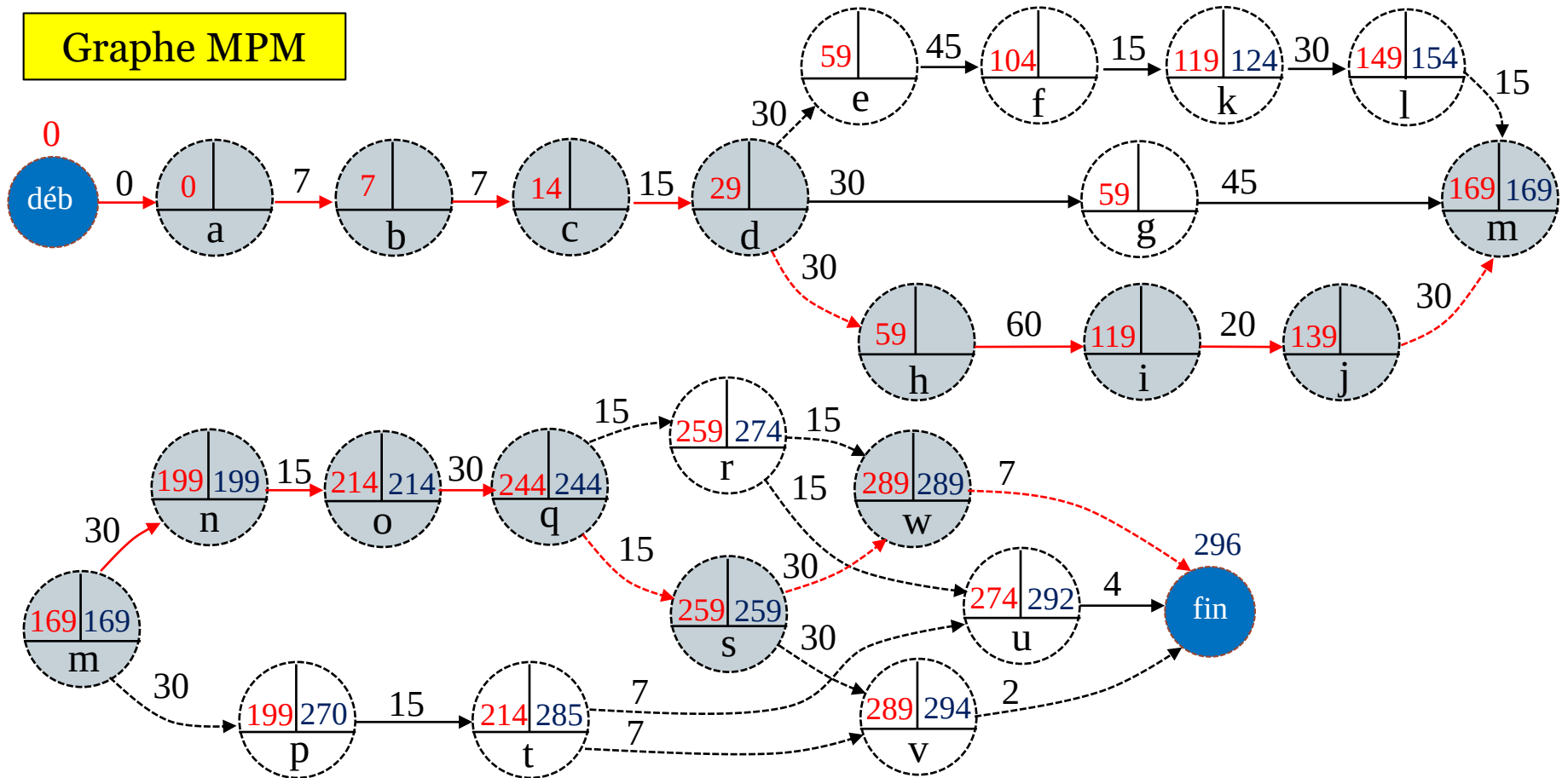
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Graphe MPM



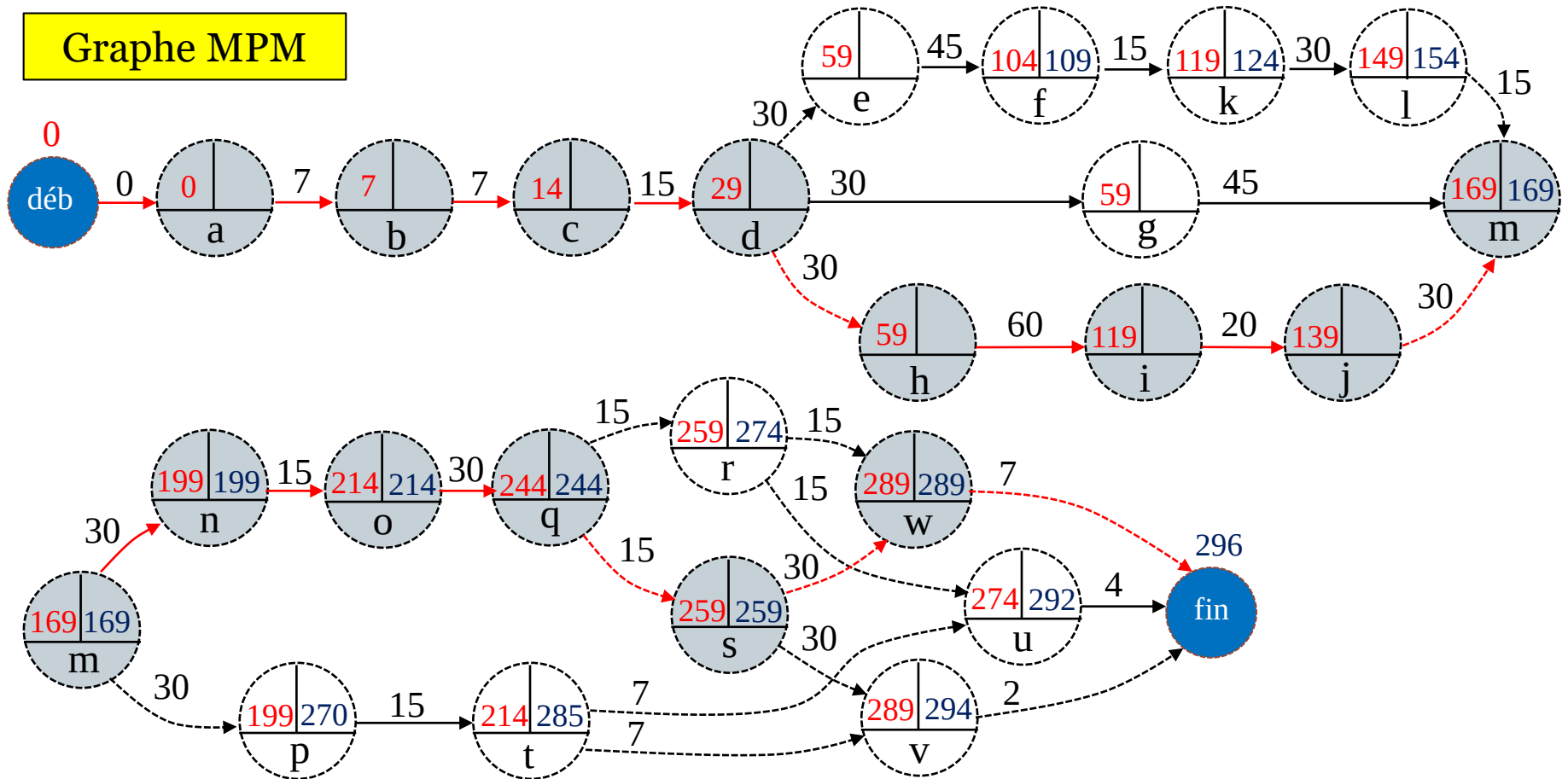
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Graphe MPM



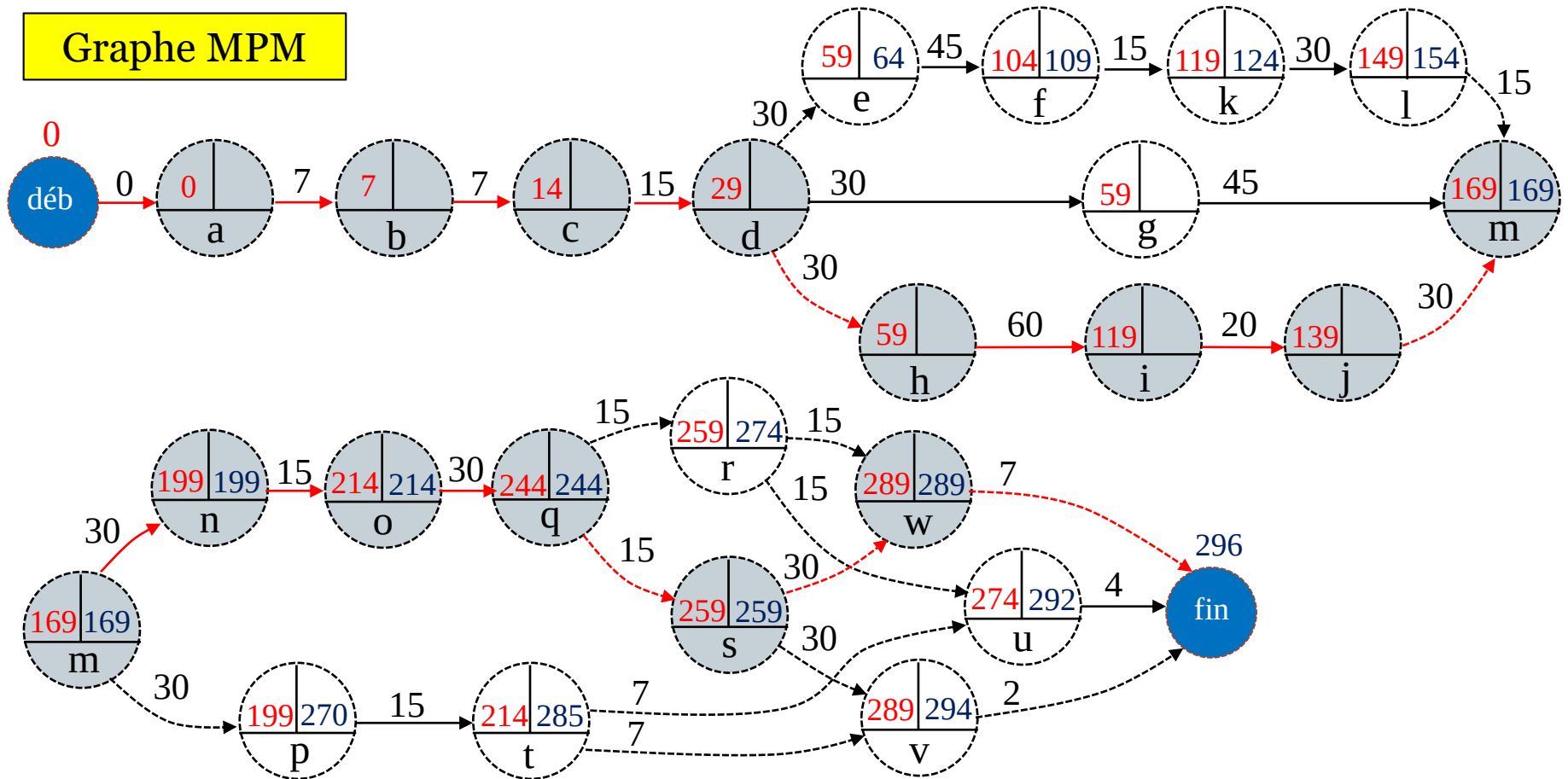
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Graphe MPM



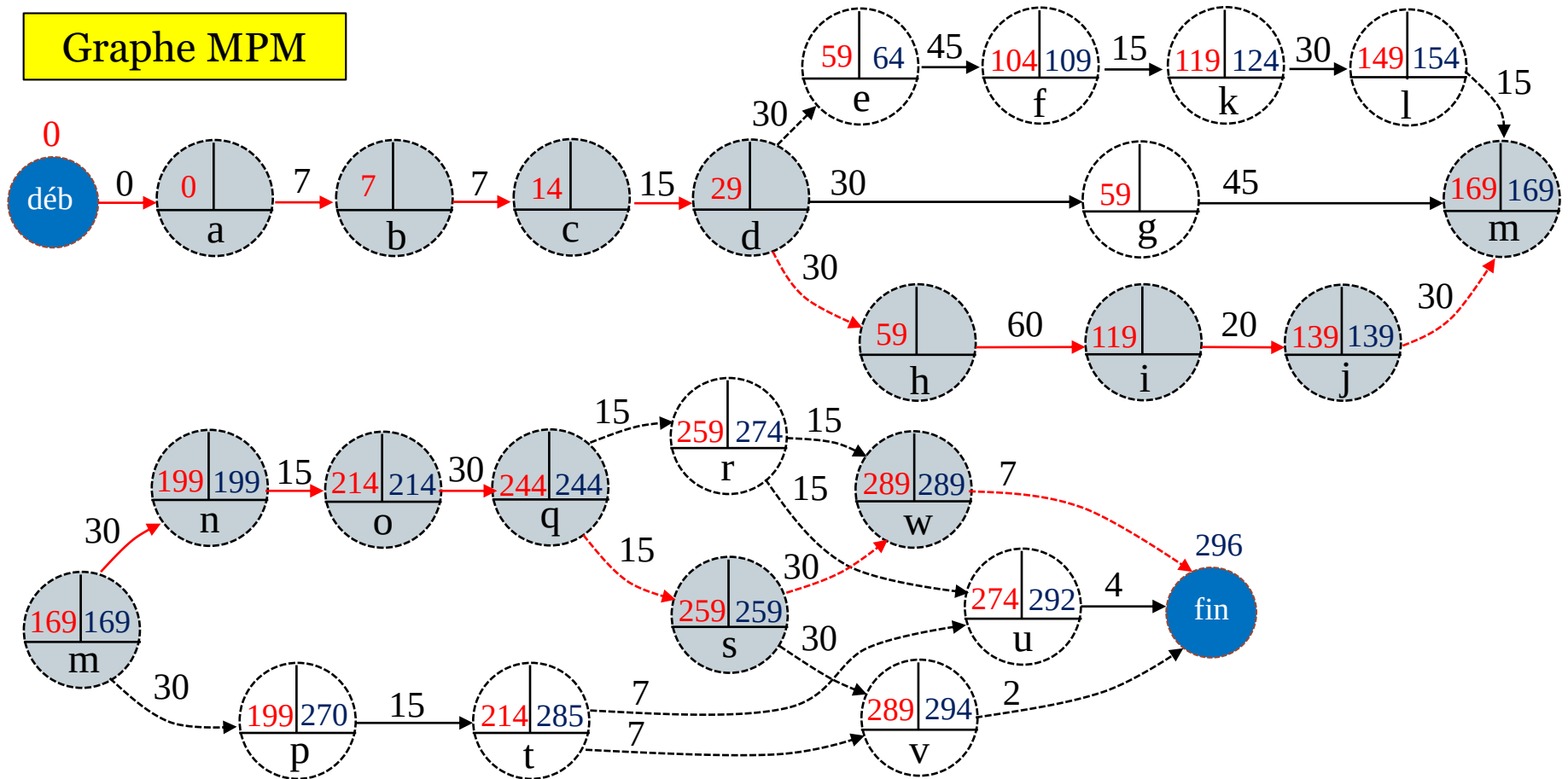
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Graphe MPM



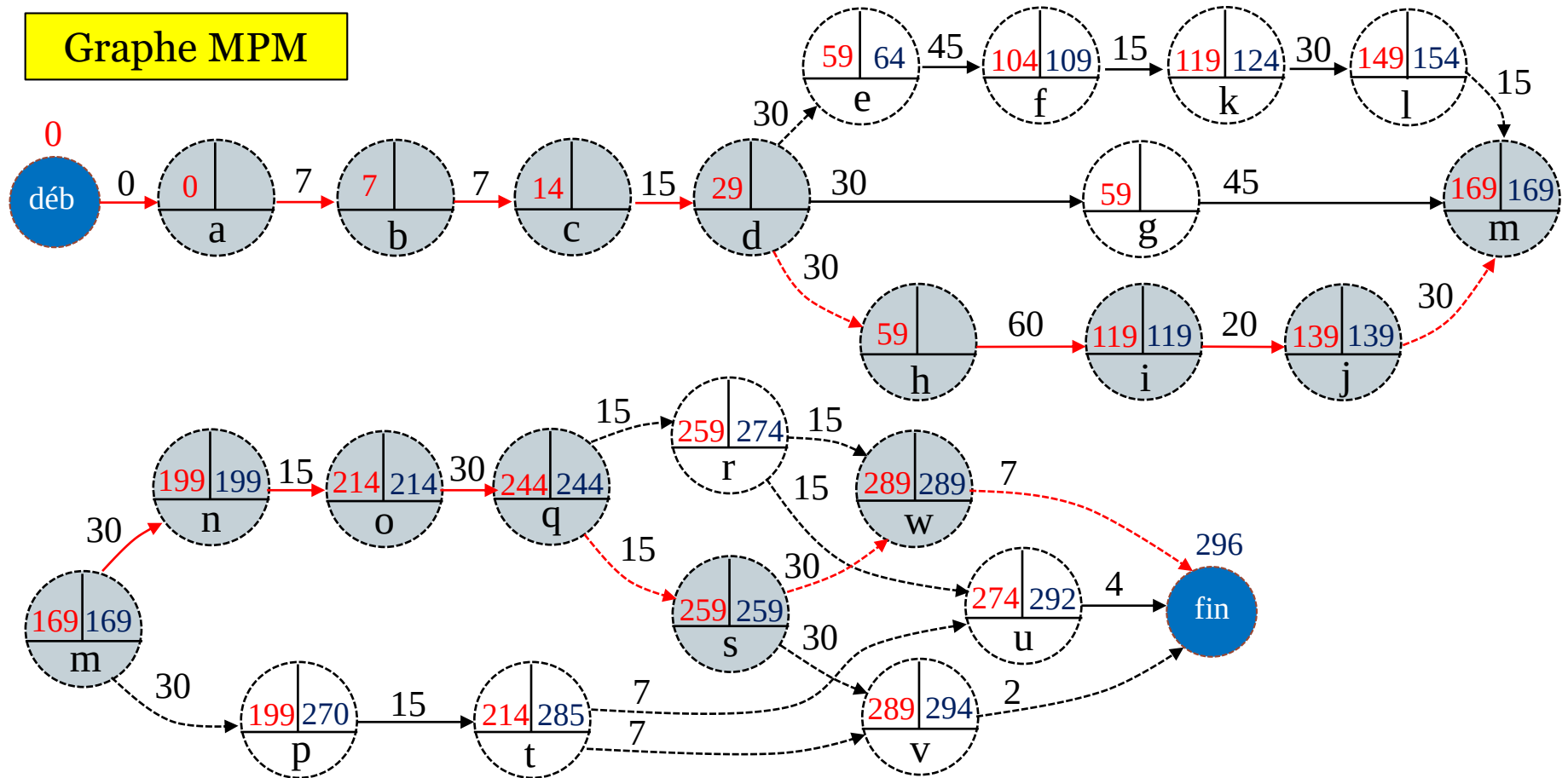
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Graphe MPM



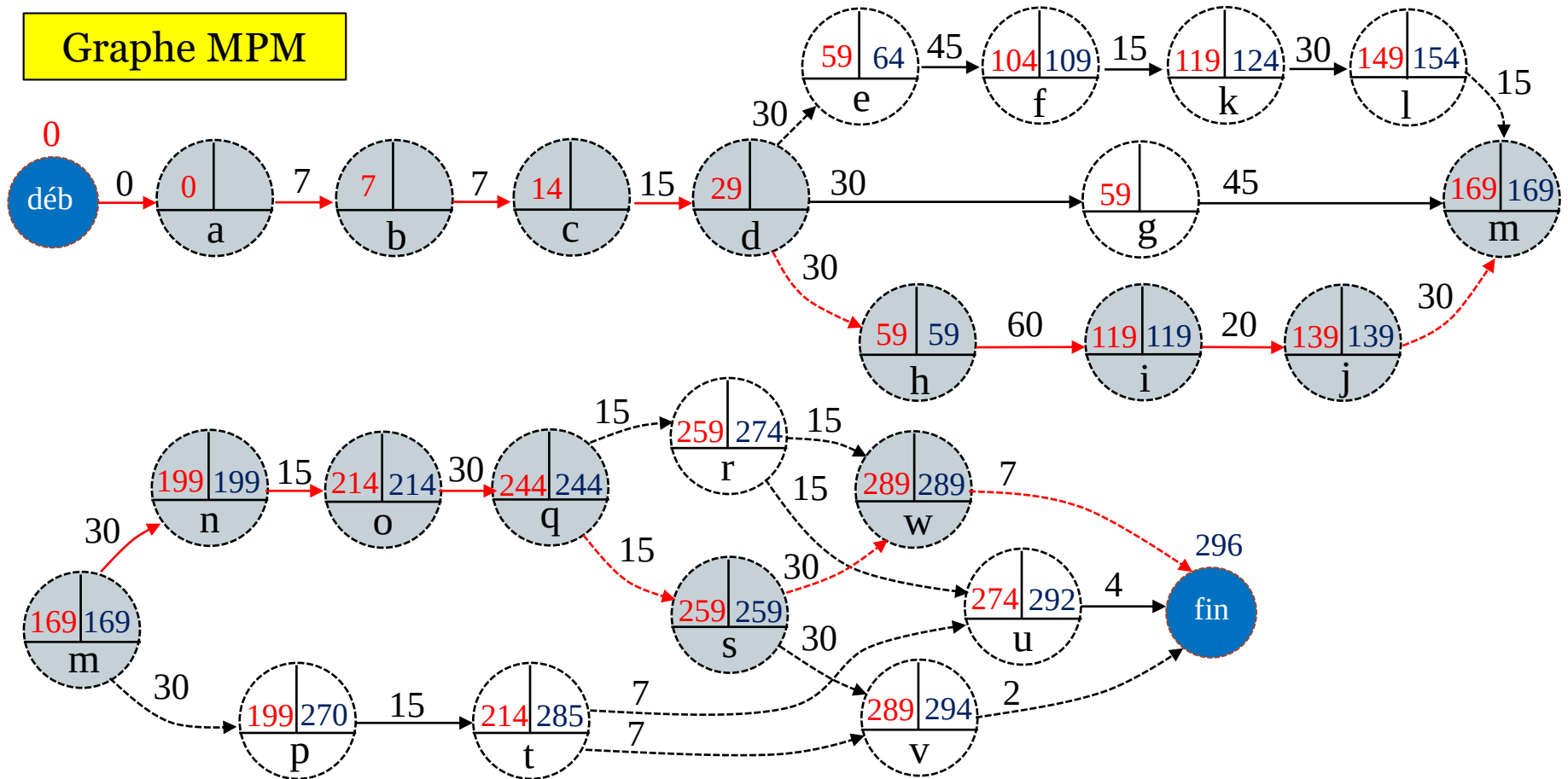
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Durée	7	7	15	30	45	15	45	60	20	30	30	15	30	15	30	15	15	15	30	7	4	2	7

Graphe MPM



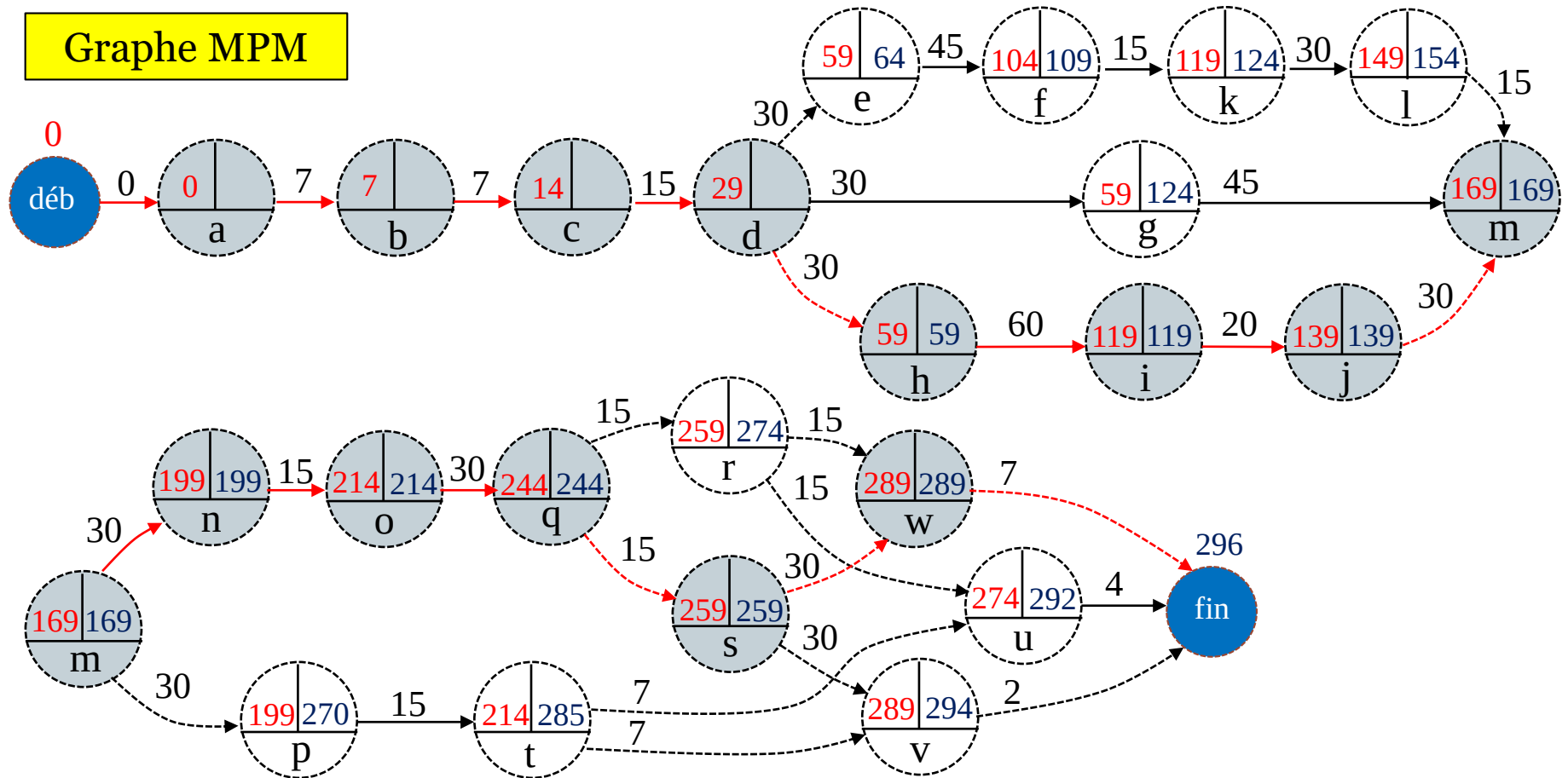
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Graphe MPM



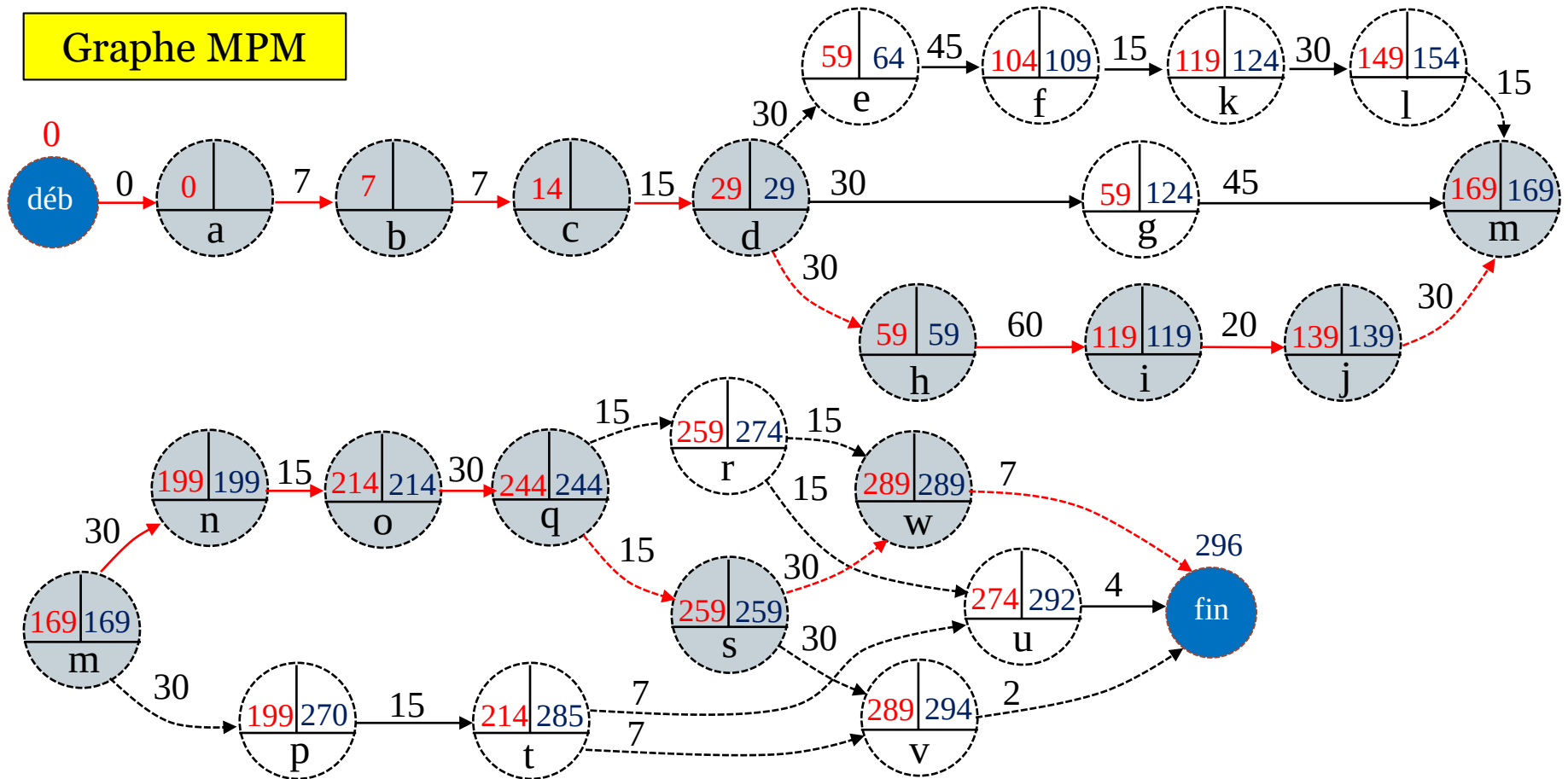
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Graphe MPM



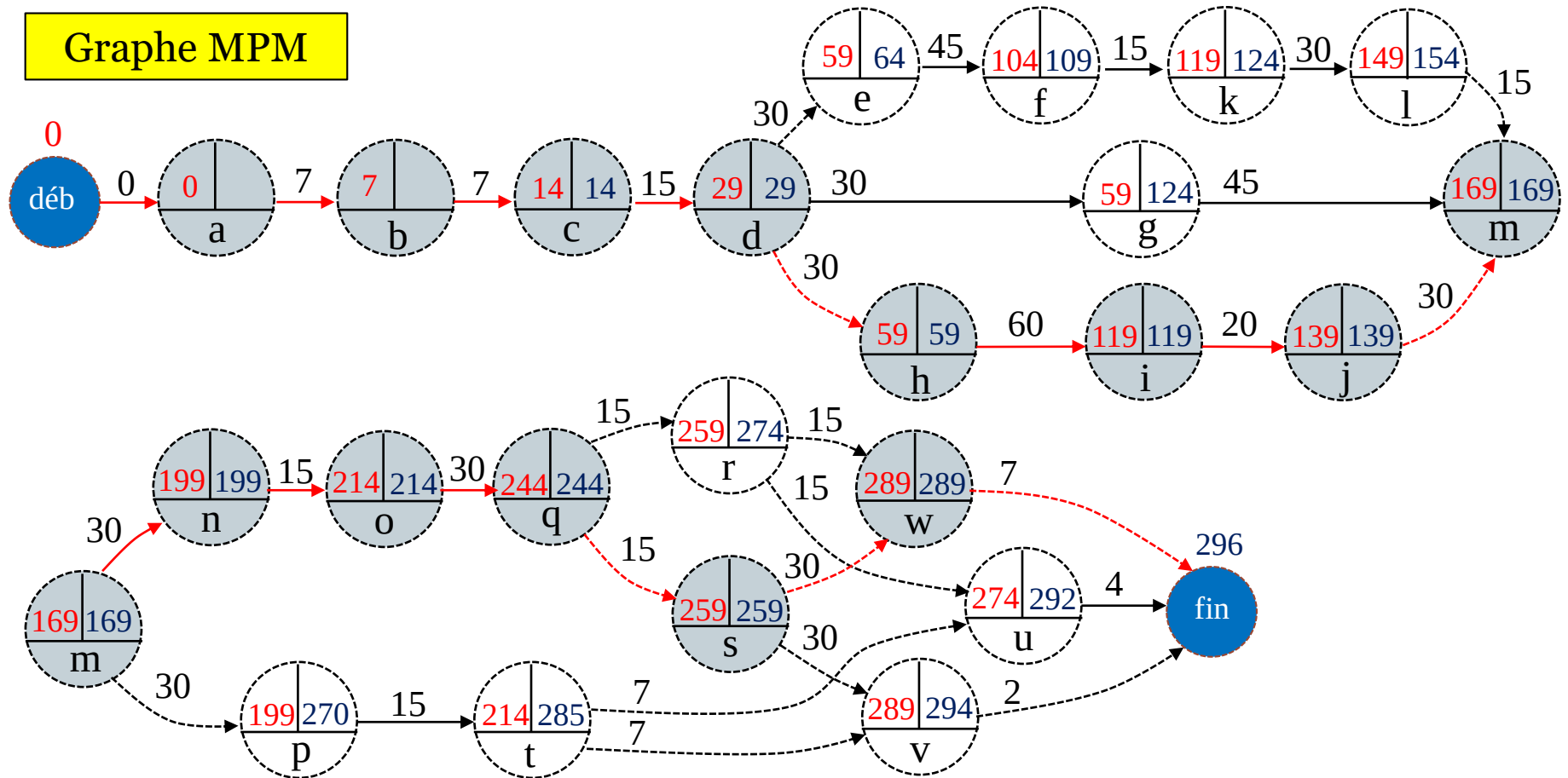
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Graphe MPM



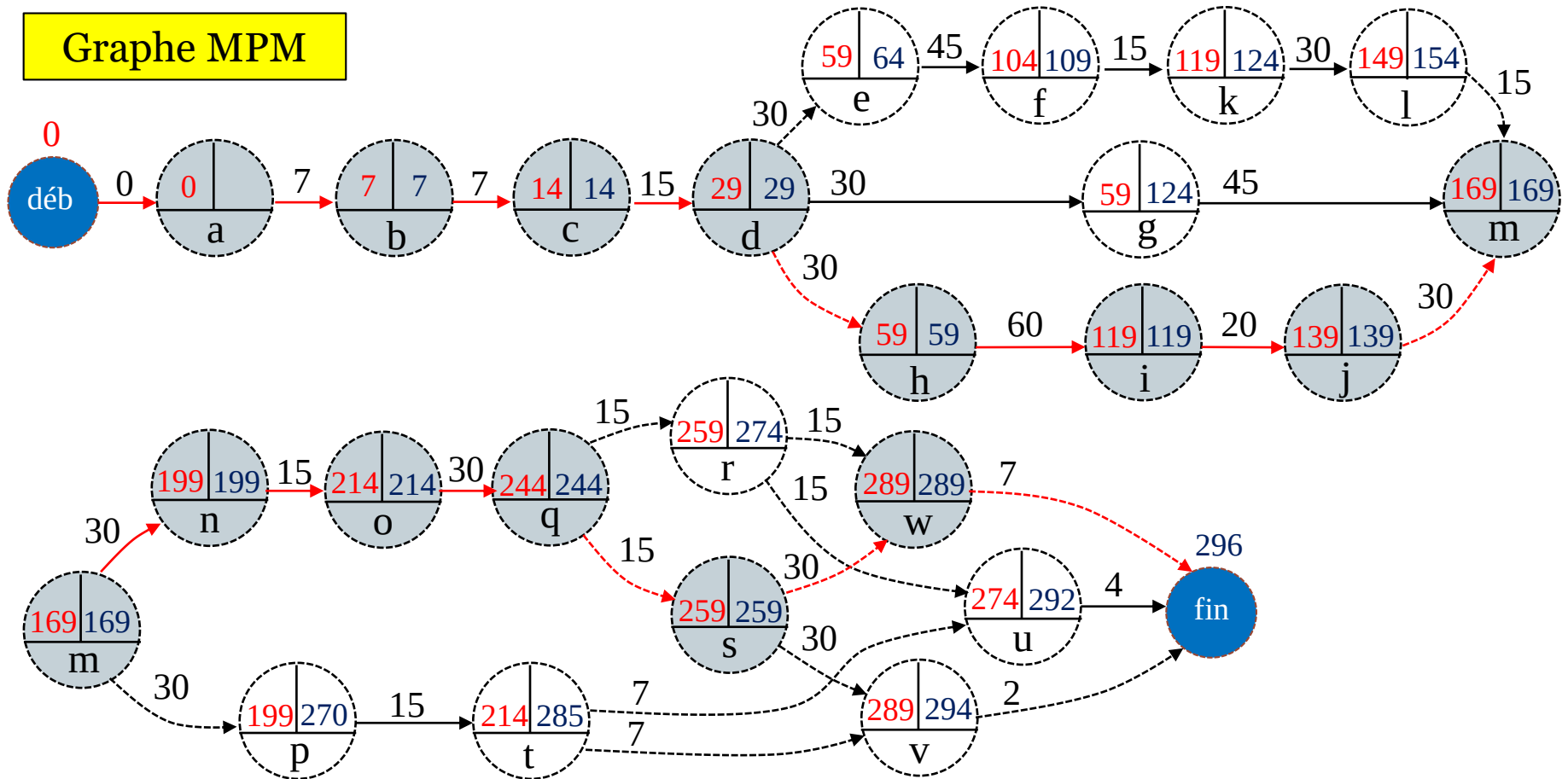
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Graphe MPM



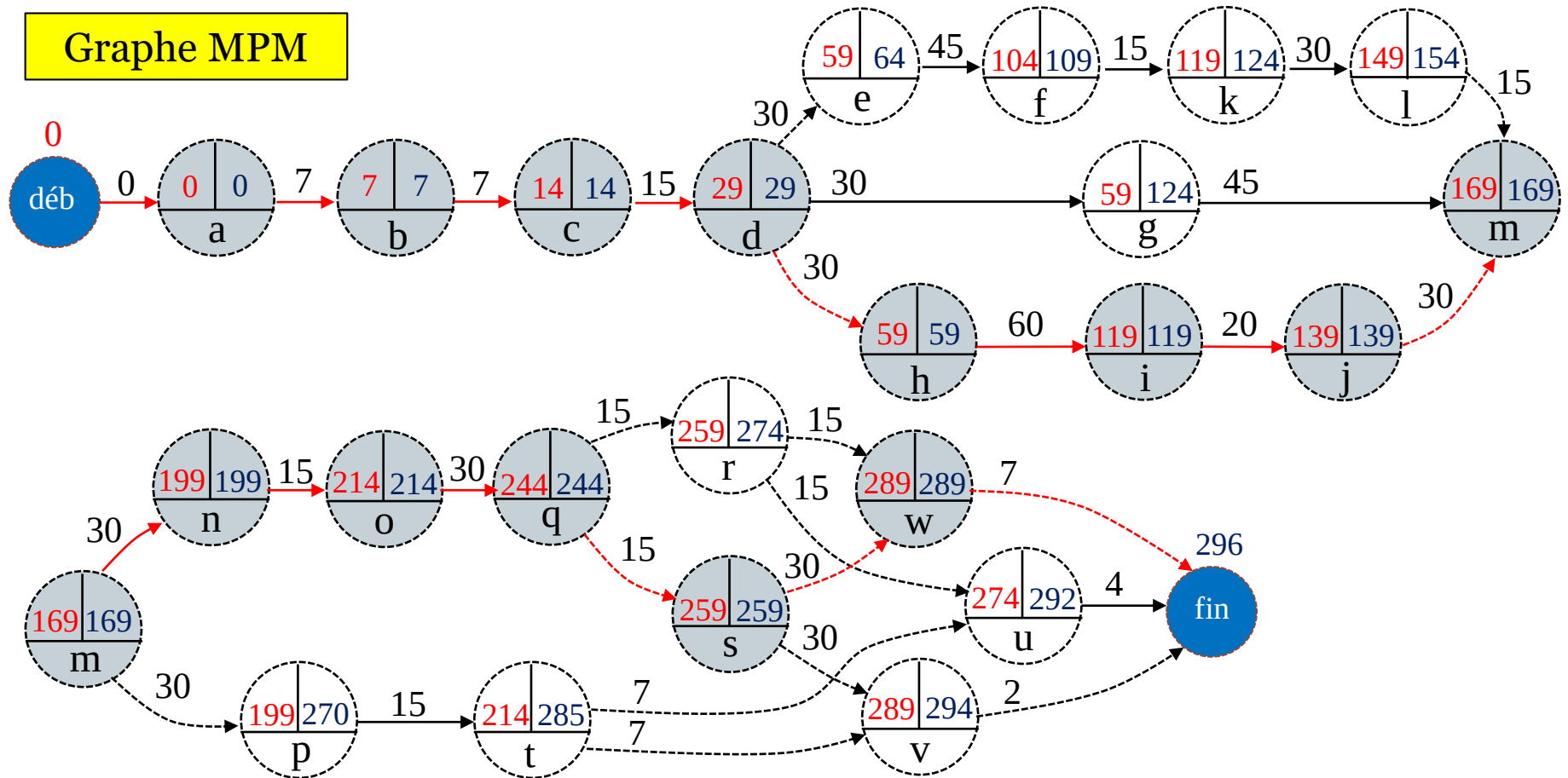
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Graphe MPM



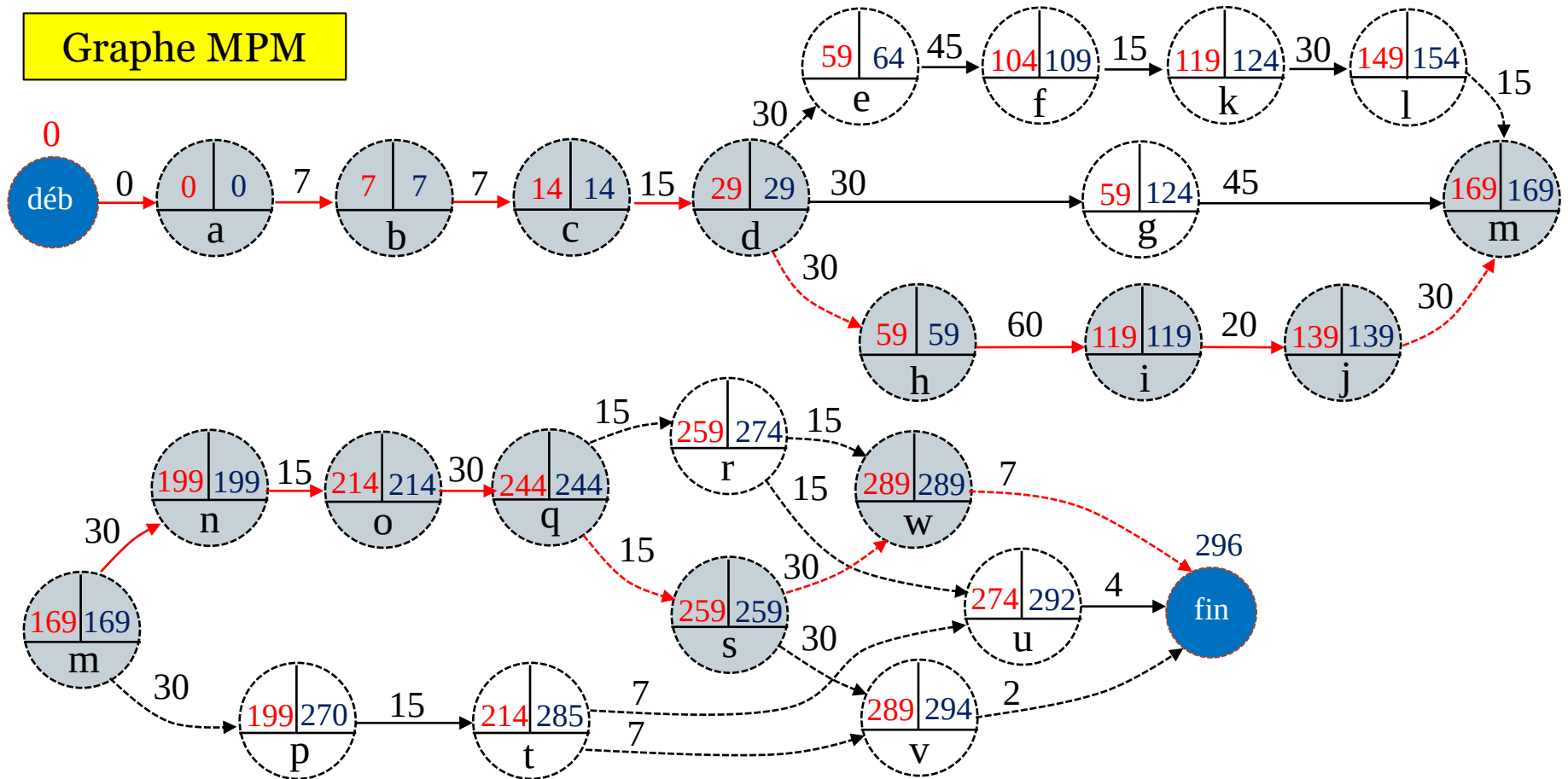
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Graphe MPM



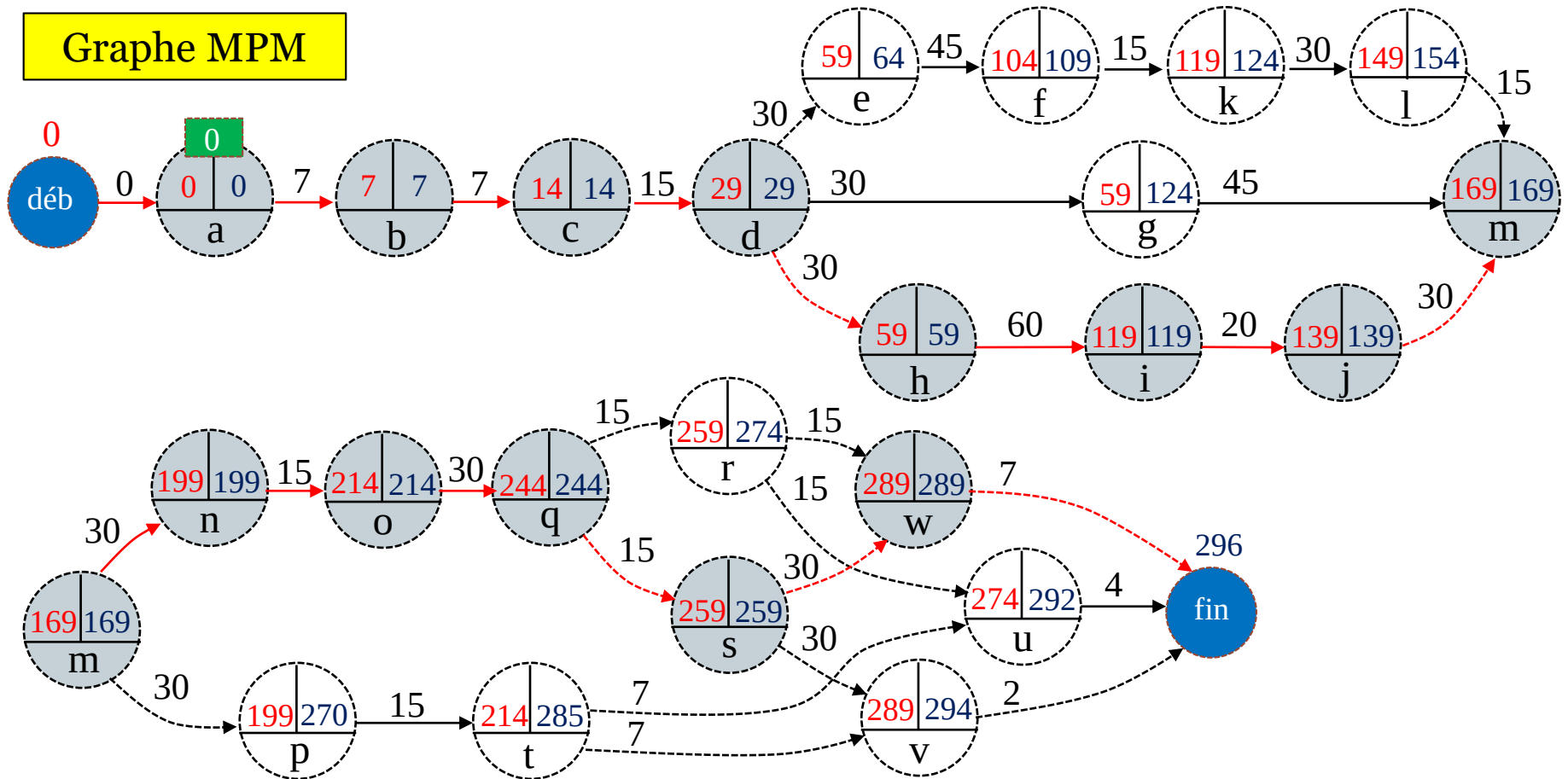
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Graphe MPM



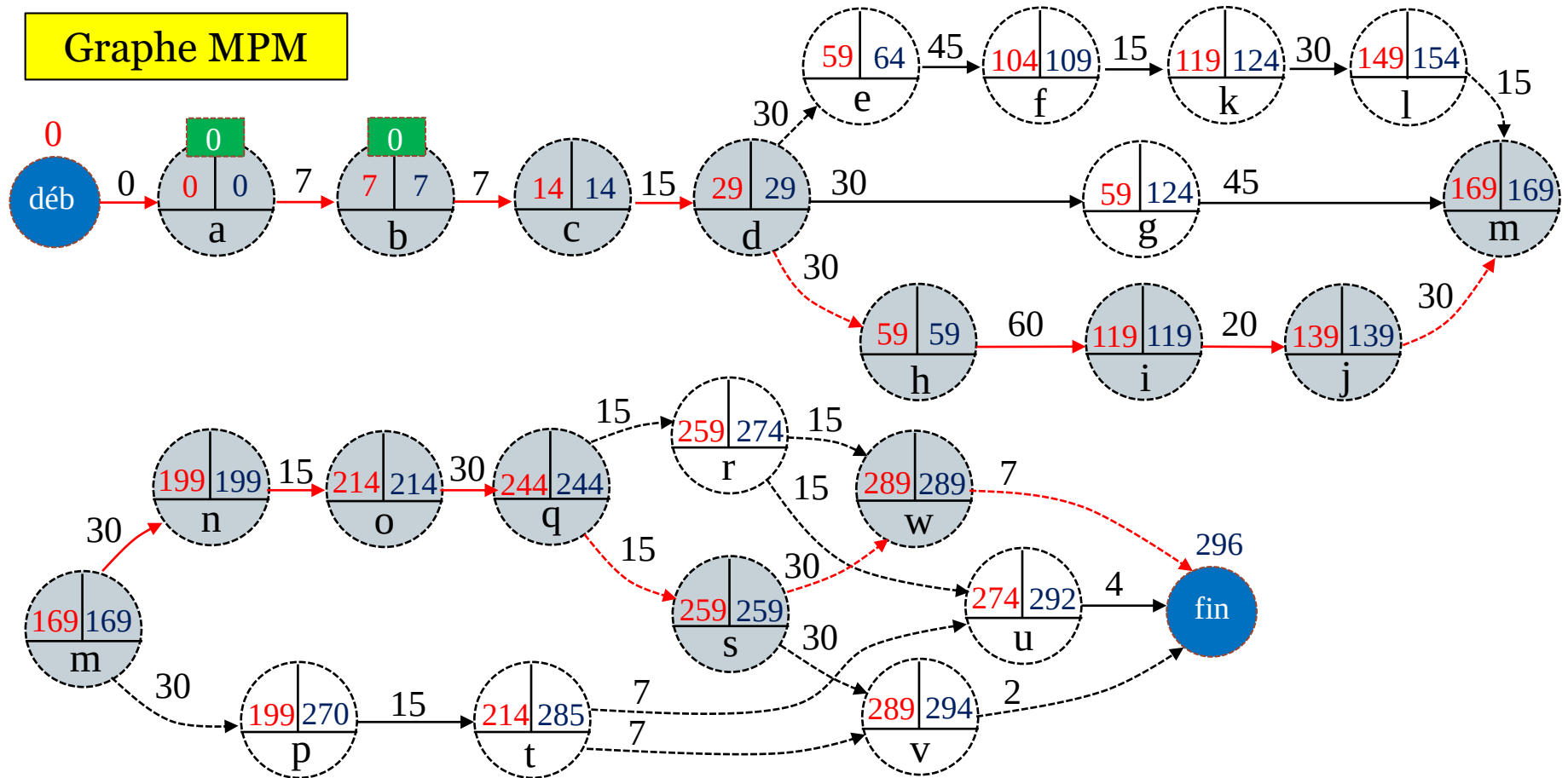
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Graphe MPM



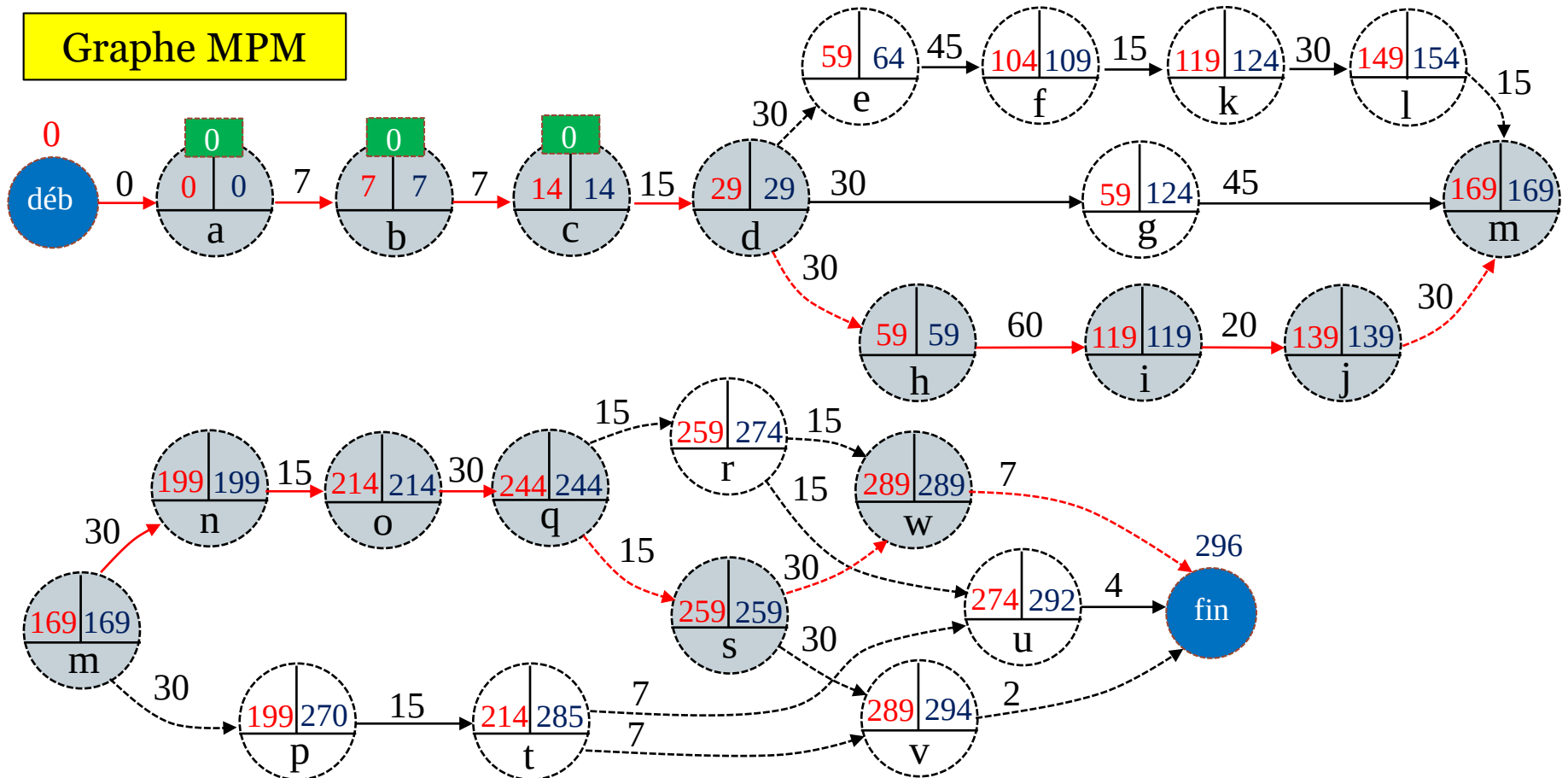
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Graphe MPM



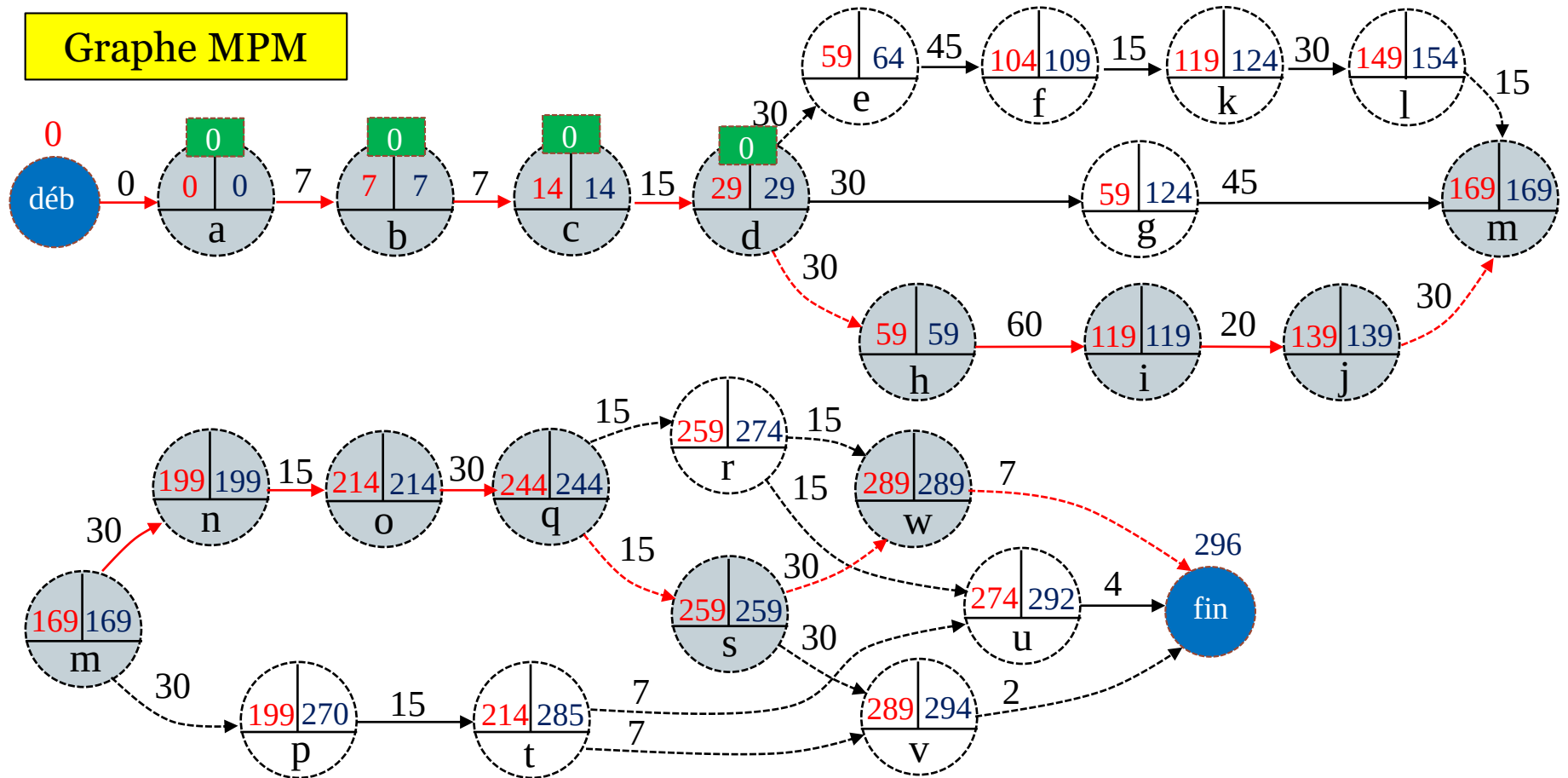
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Graphe MPM



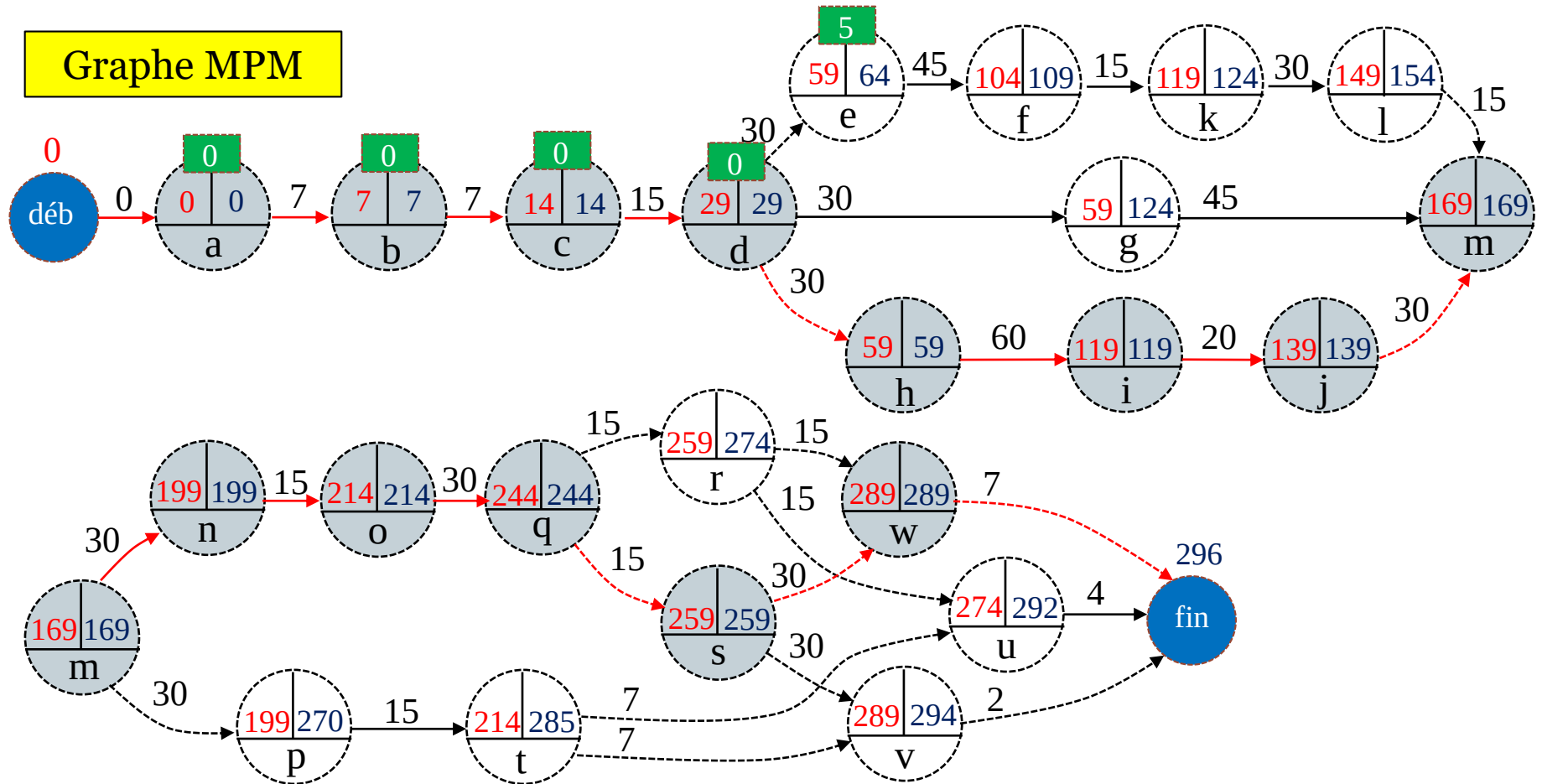
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Graphe MPM



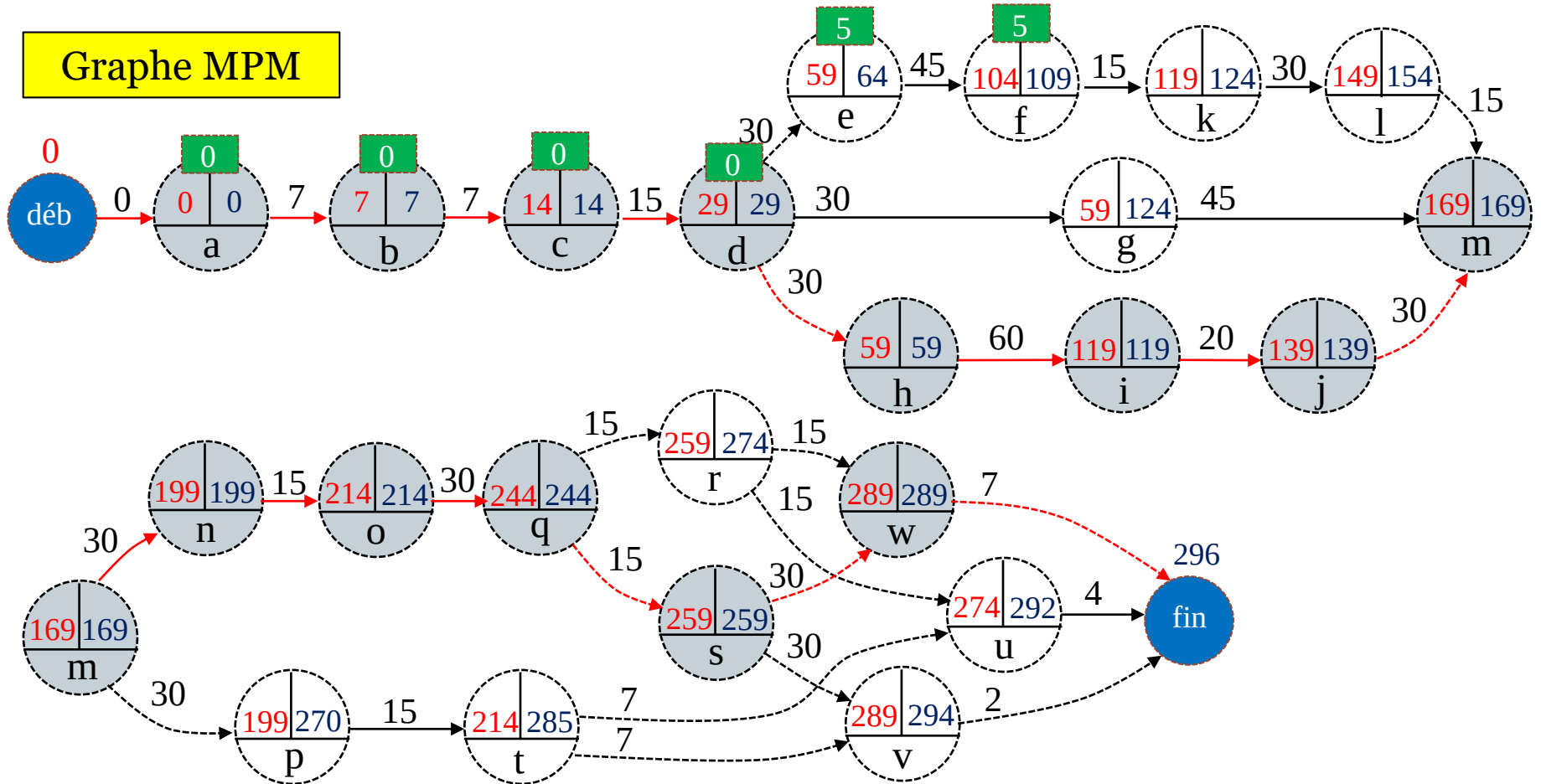
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Graphe MPM



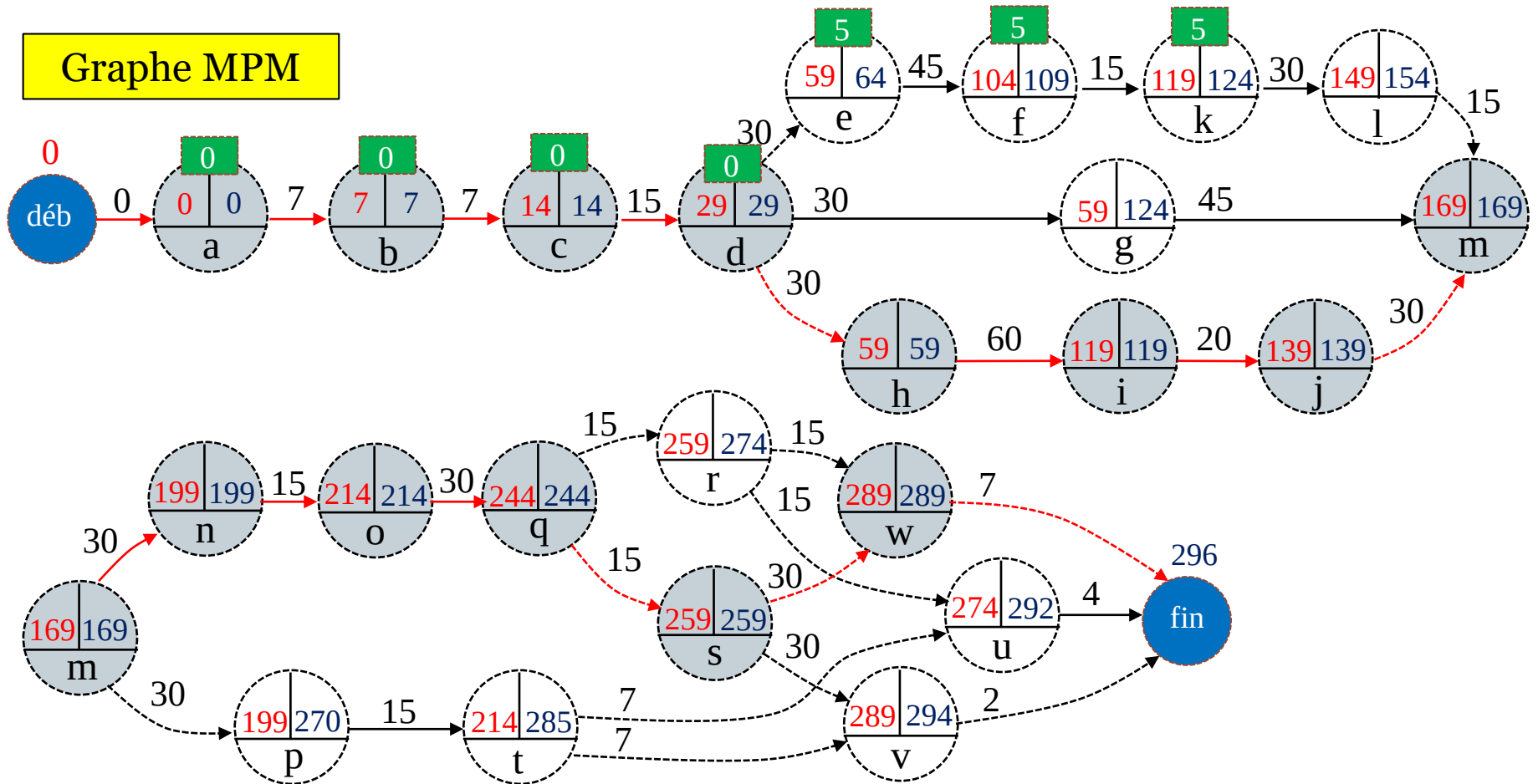
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Graphe MPM



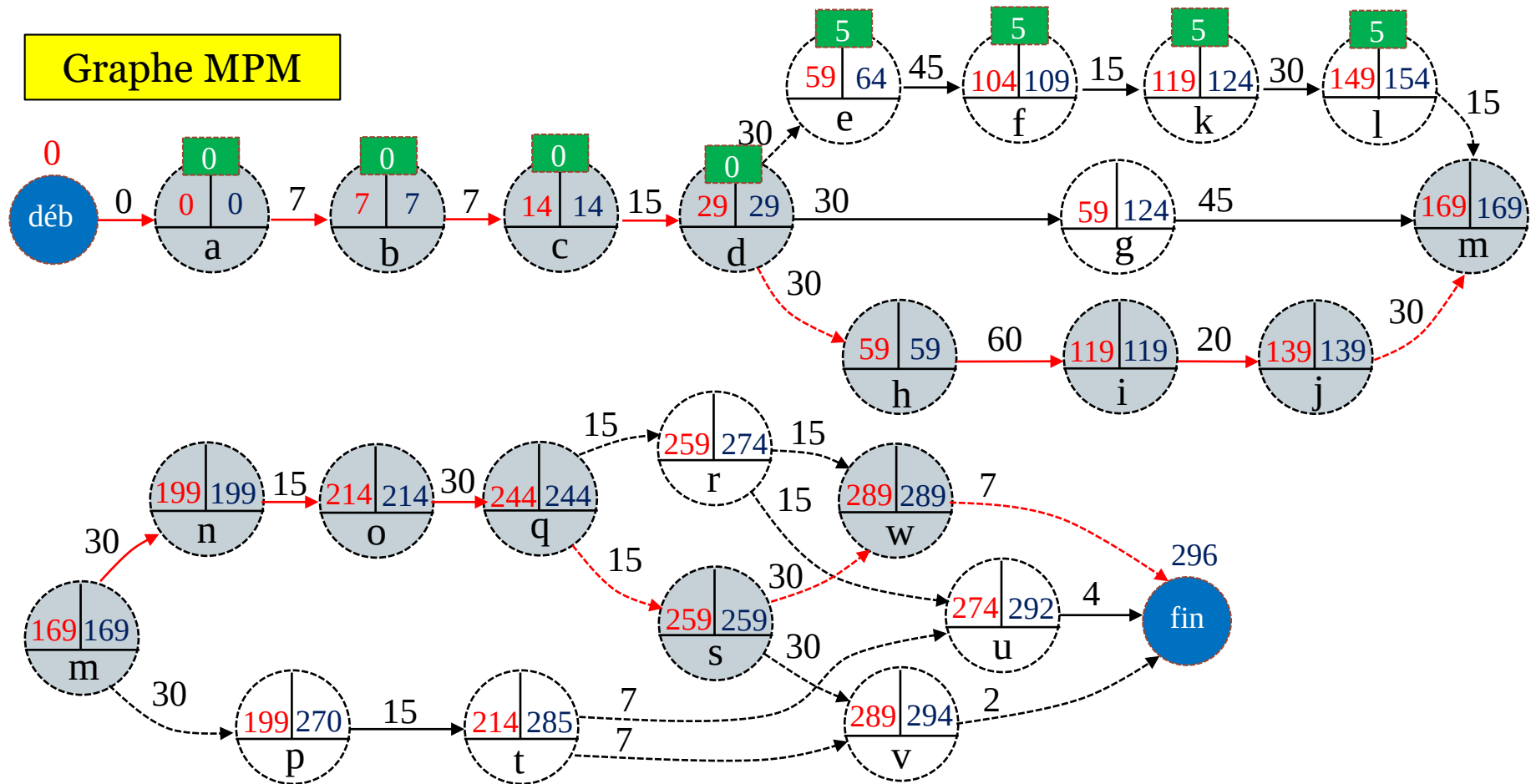
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Graphe MPM



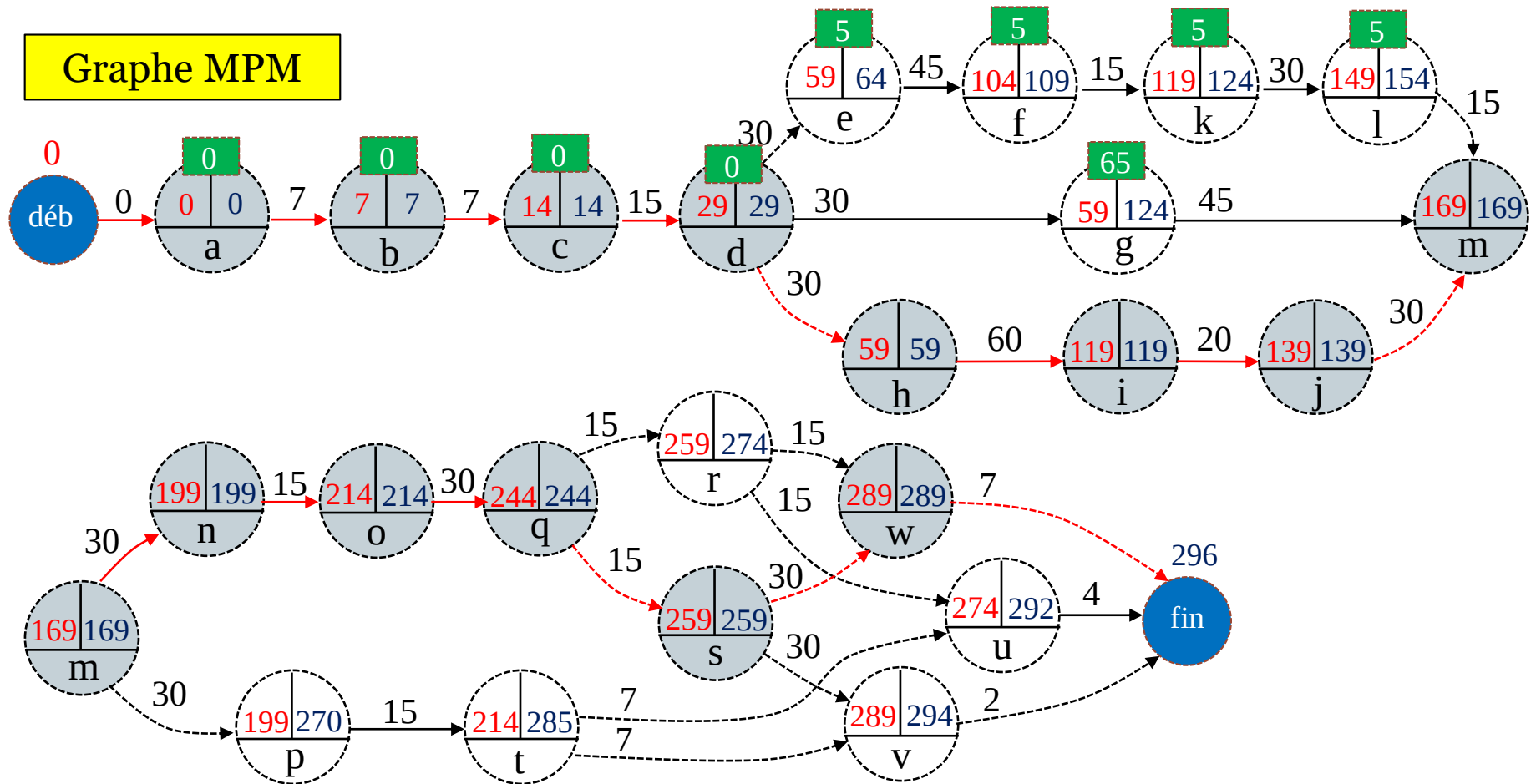
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Graphe MPM



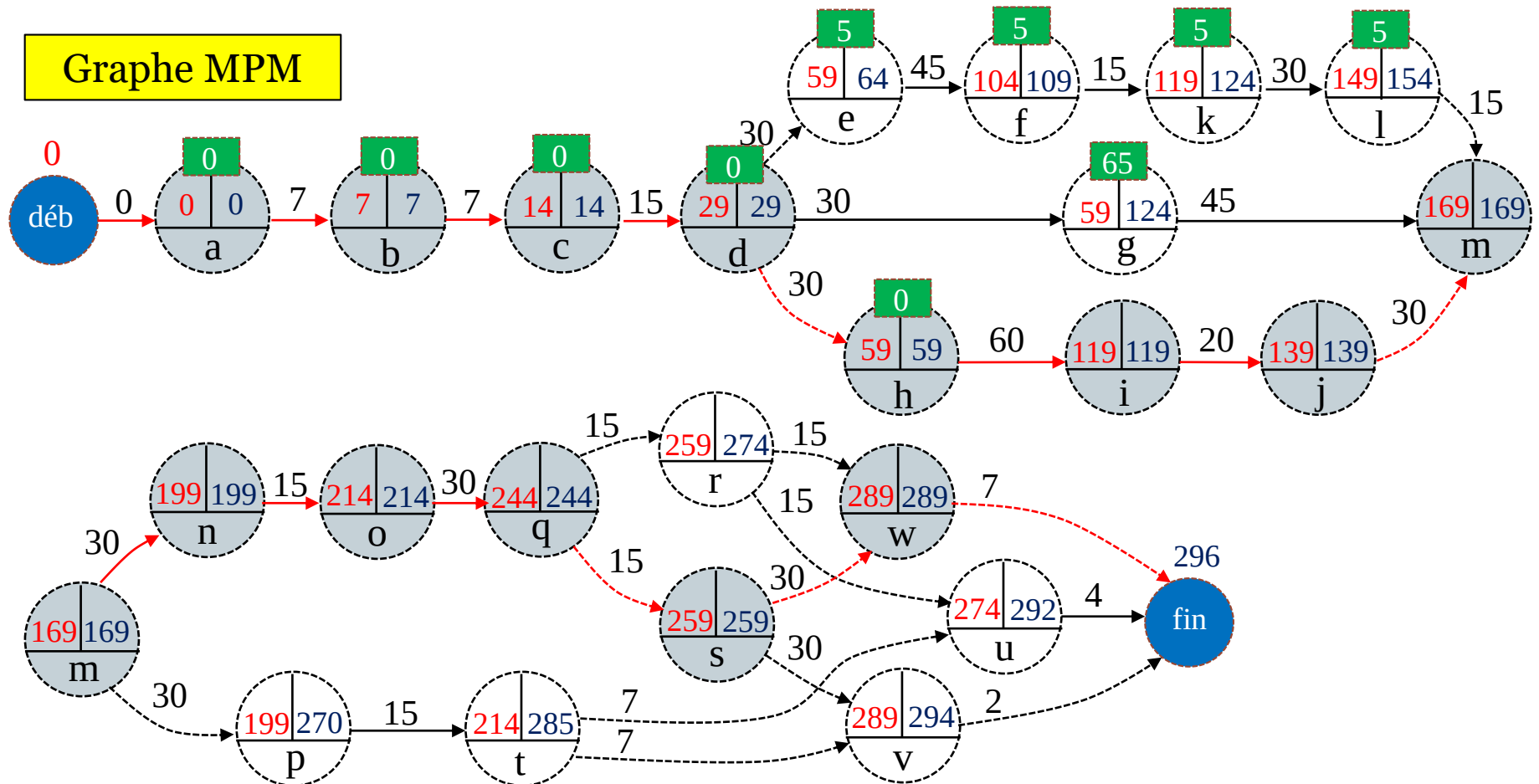
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Graphe MPM



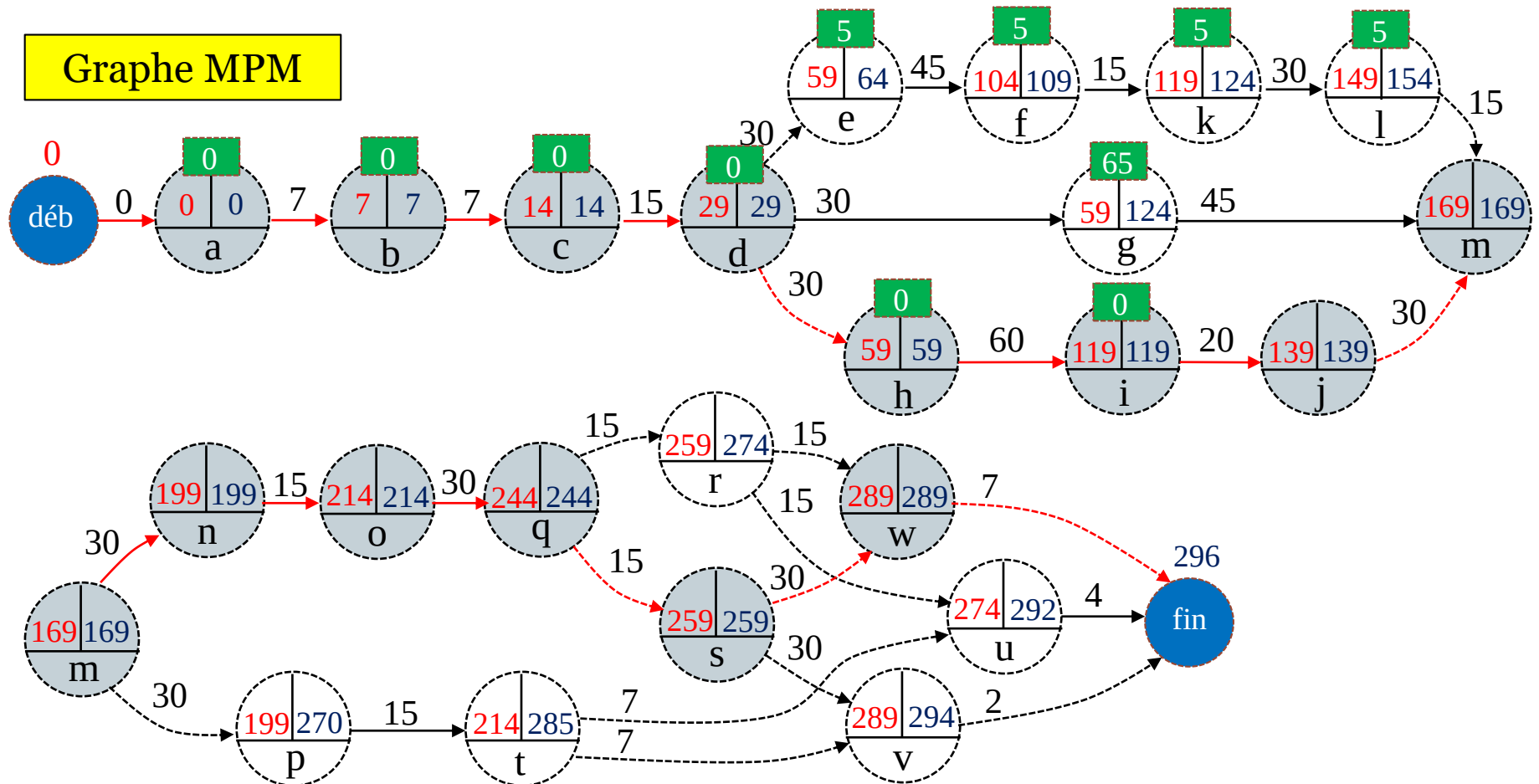
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Graphe MPM



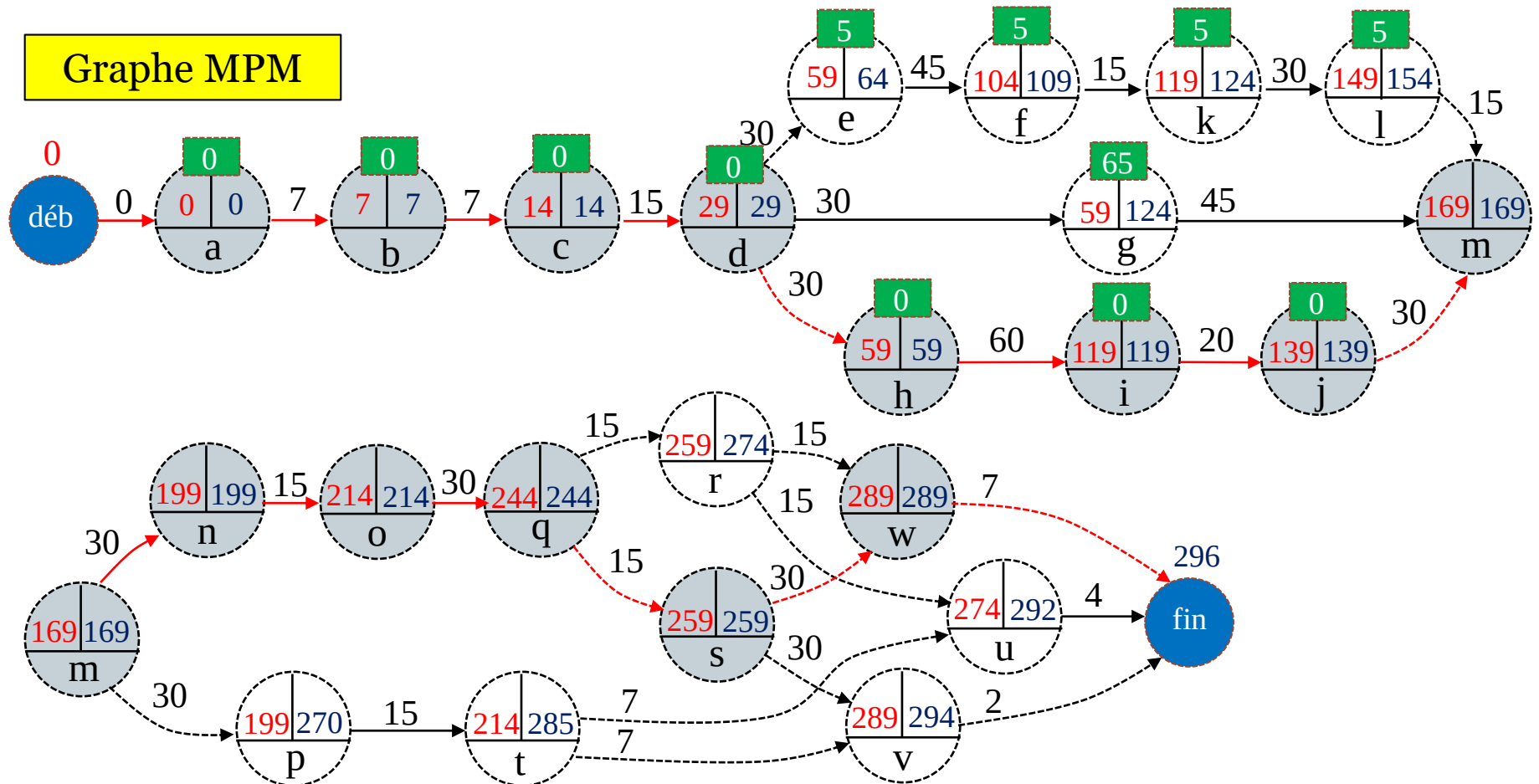
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Graphe MPM



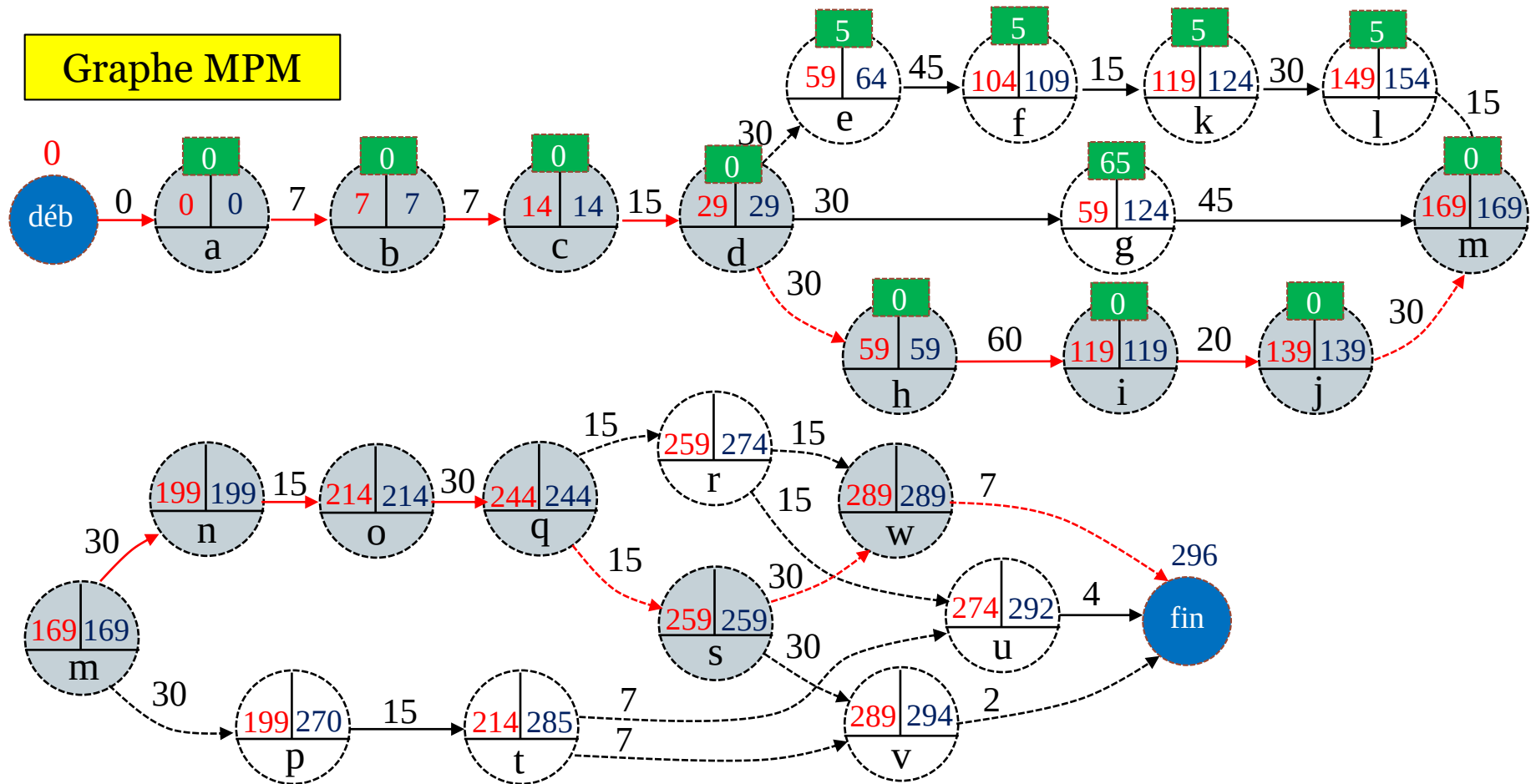
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Graphe MPM



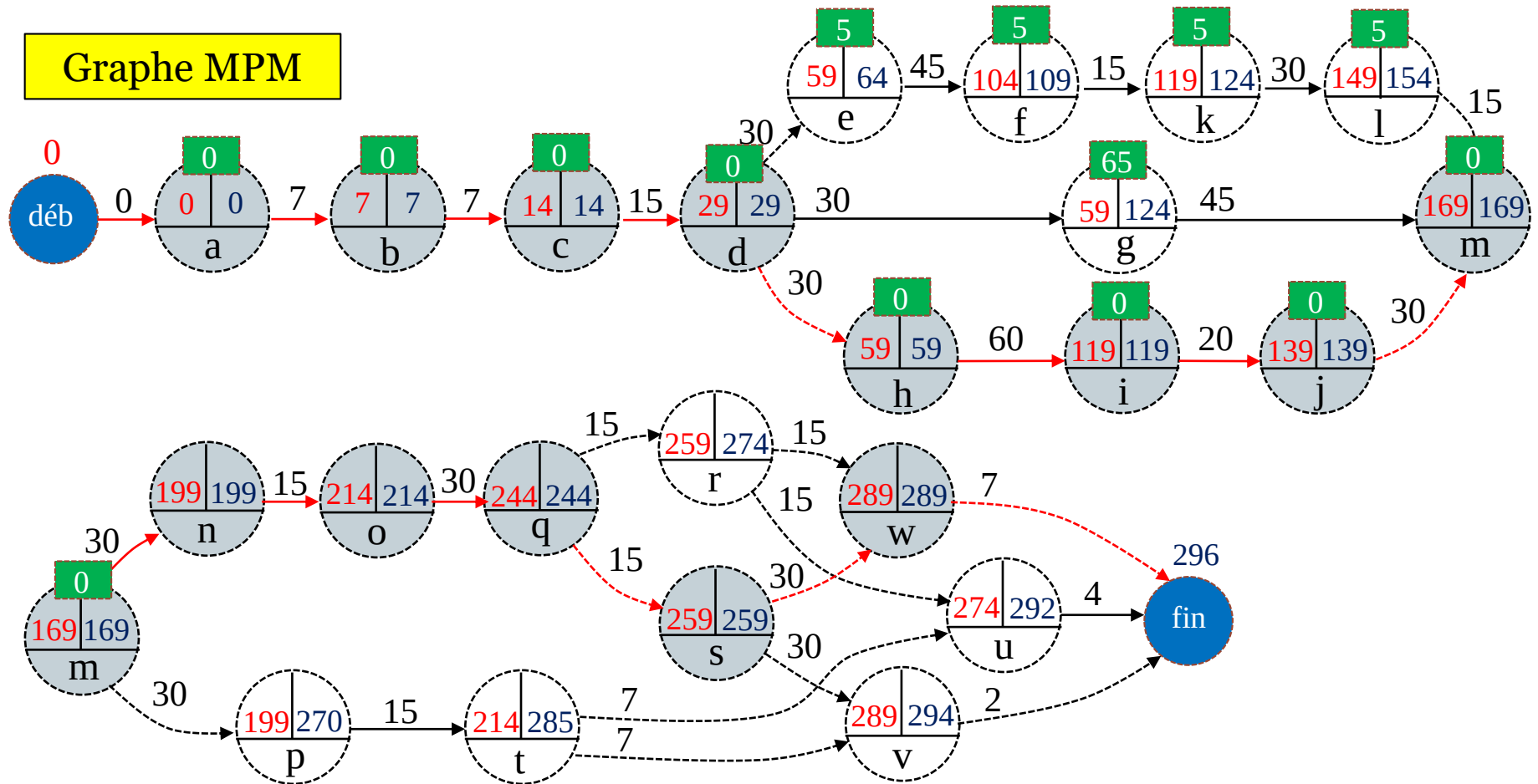
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Graphe MPM



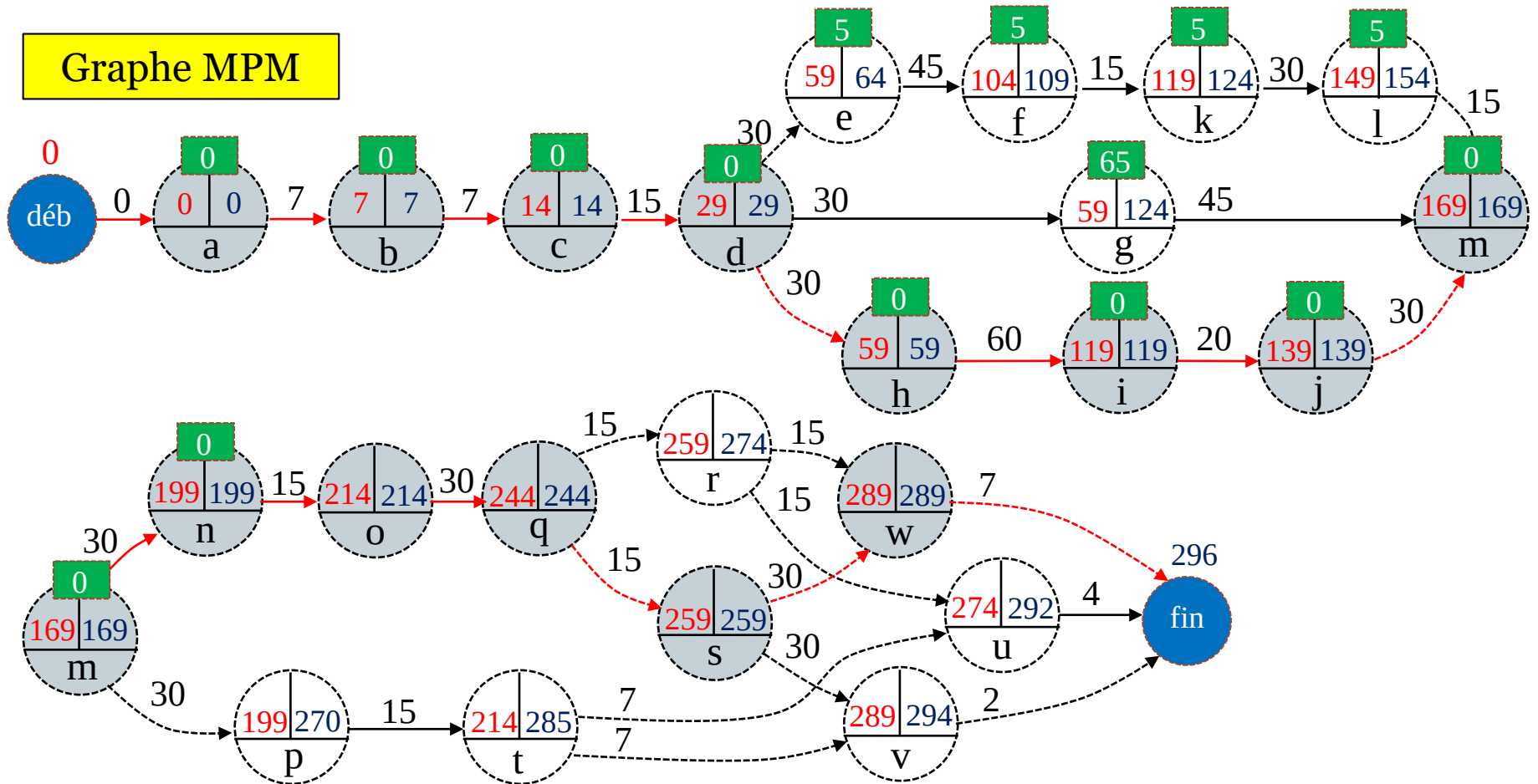
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Graphe MPM



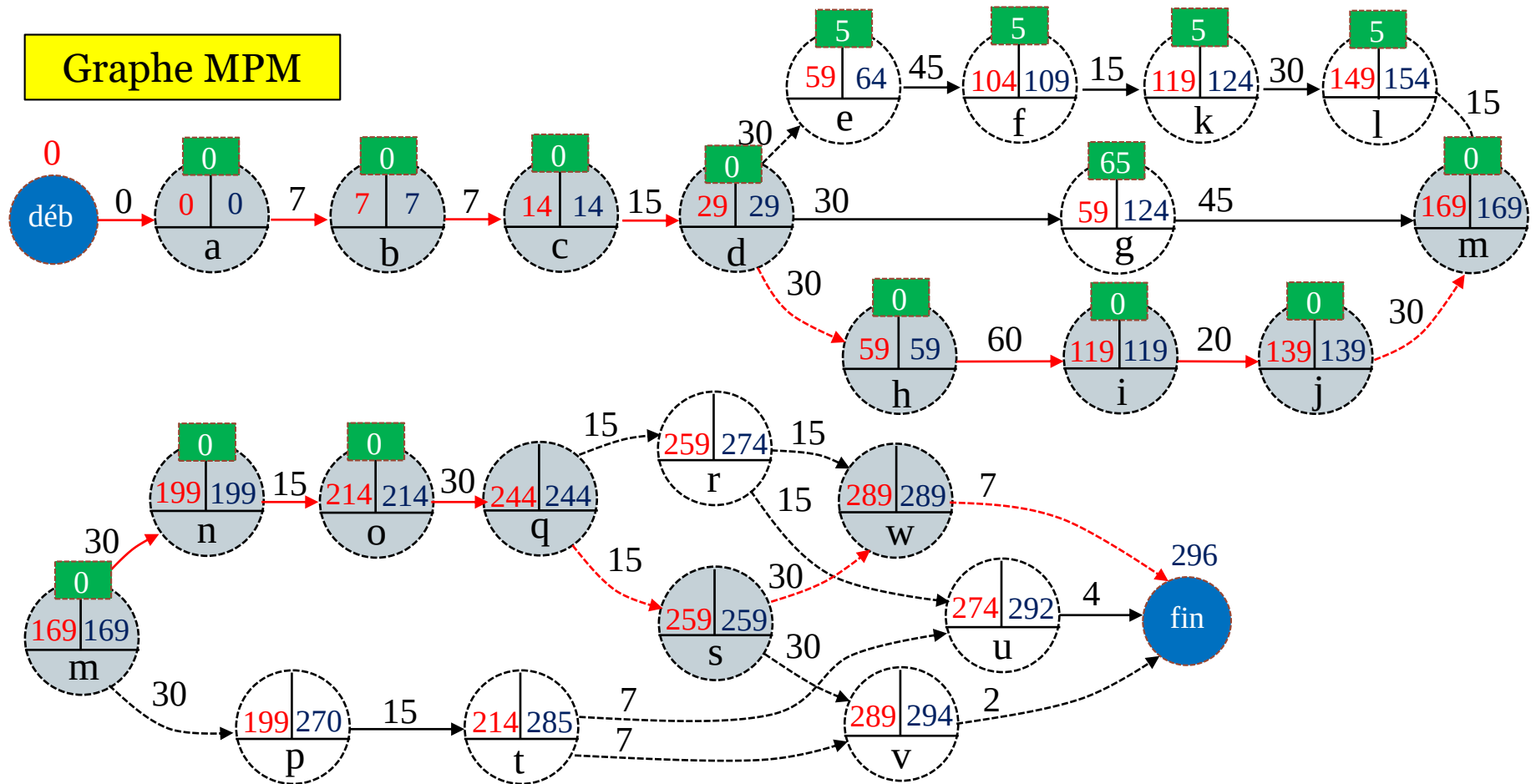
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Graphe MPM



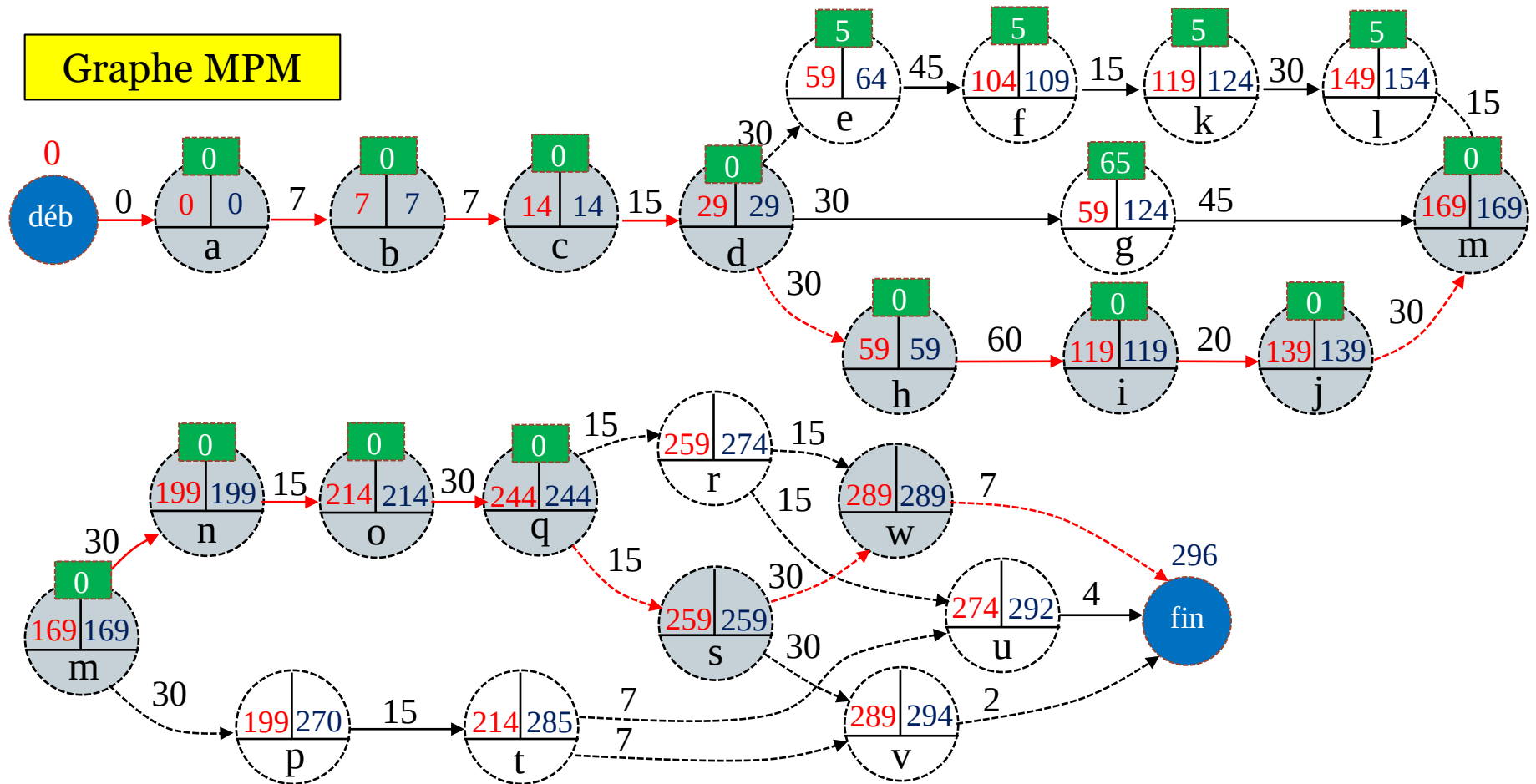
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Graphe MPM



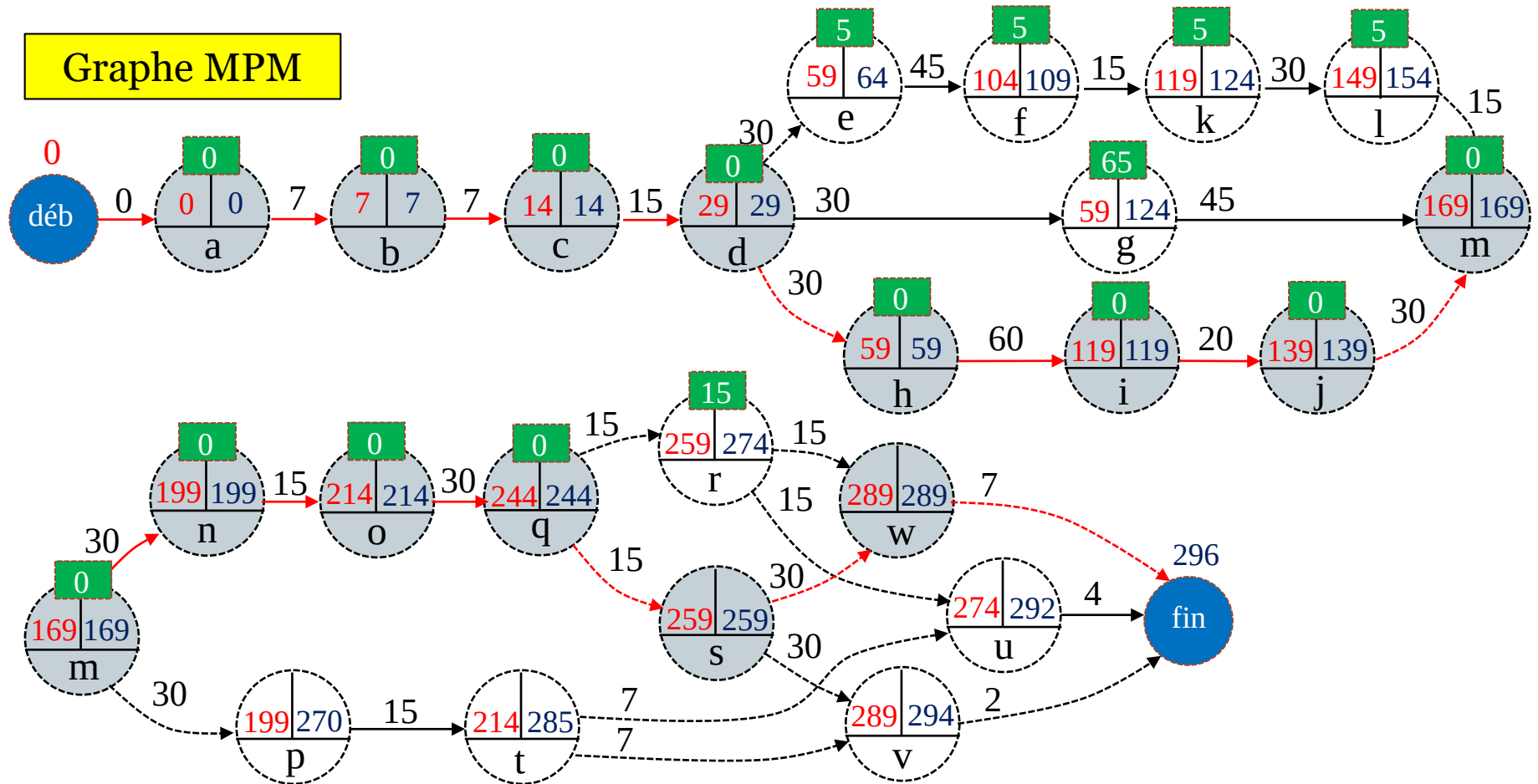
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Graphe MPM



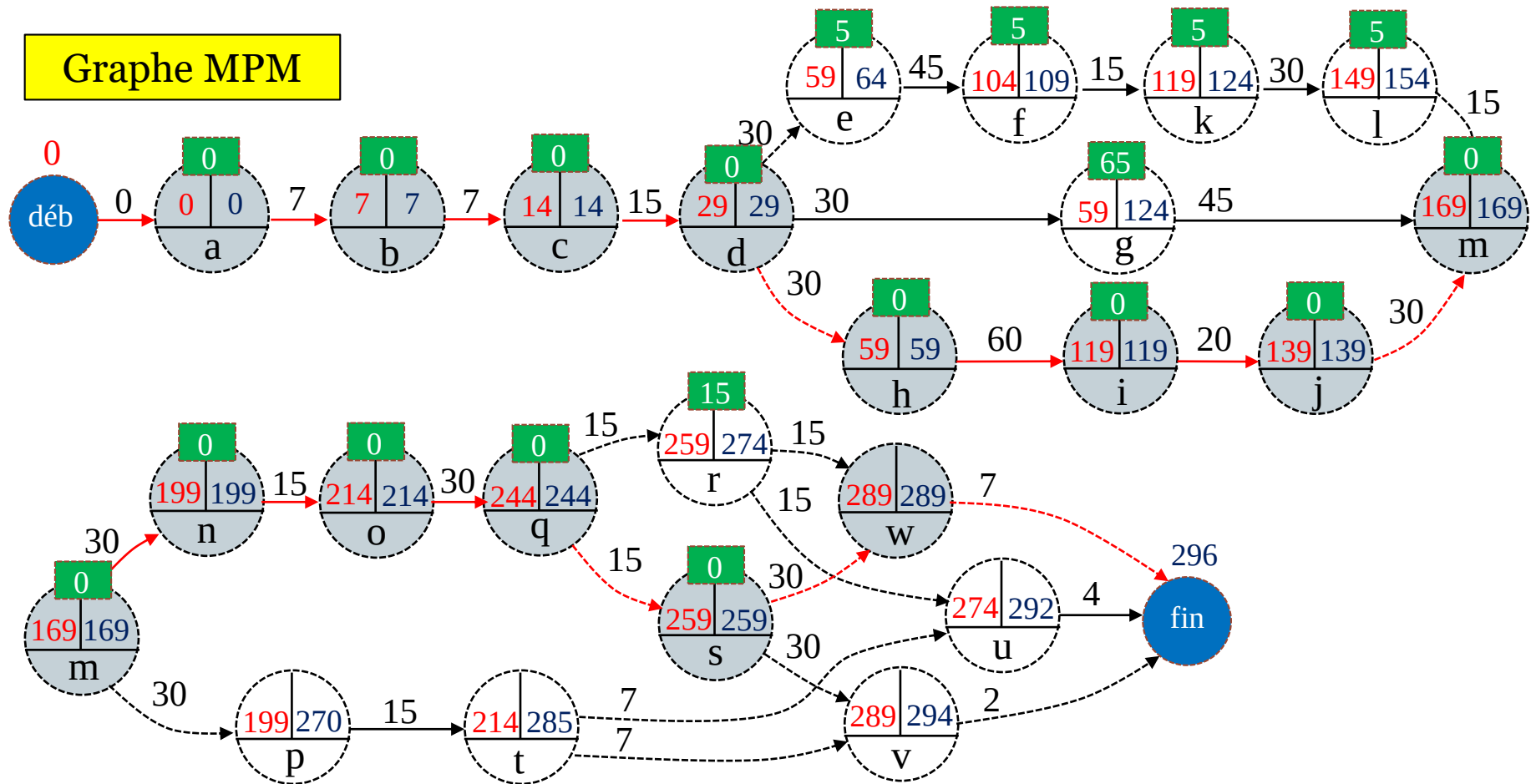
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Graphe MPM



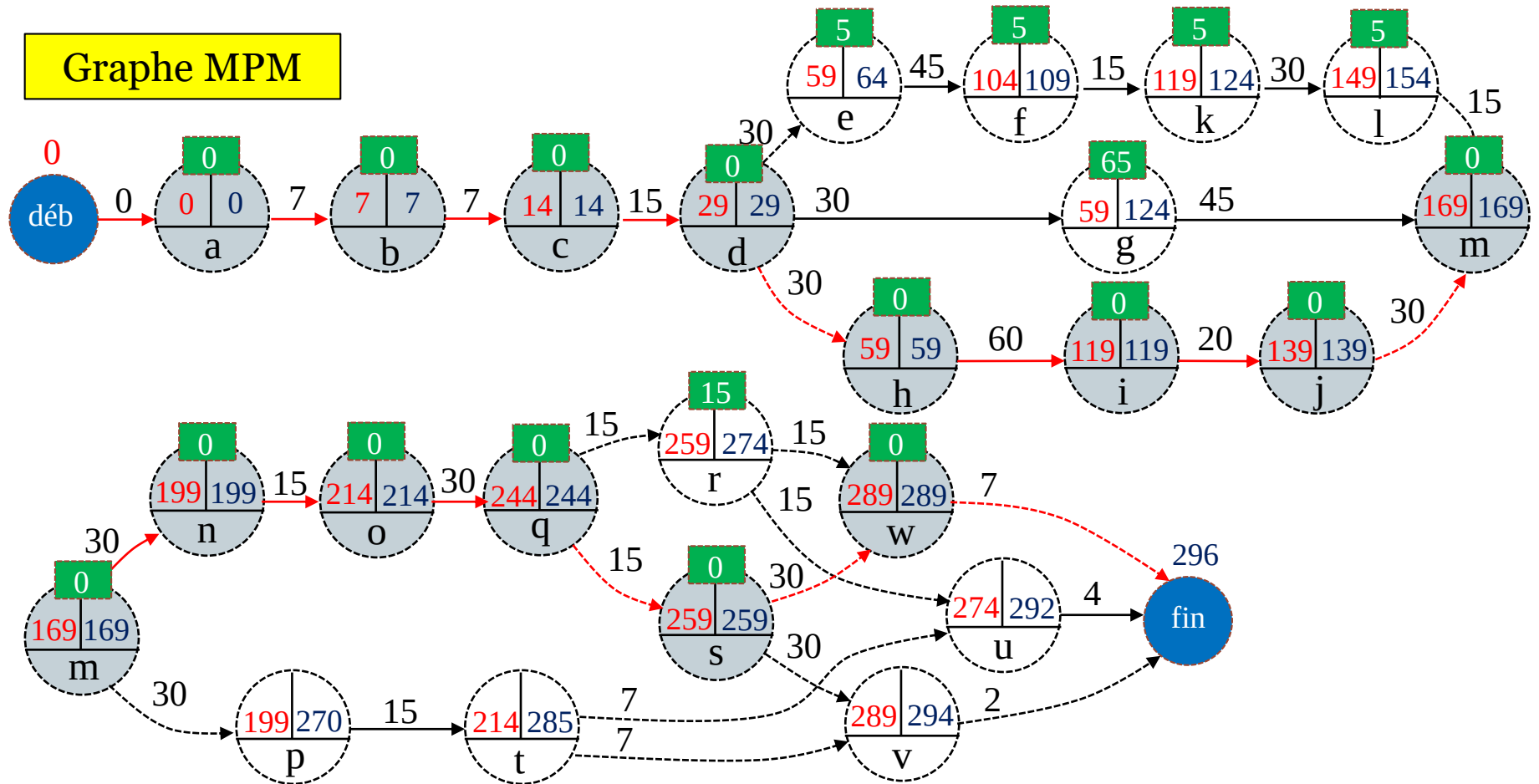
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Graphe MPM



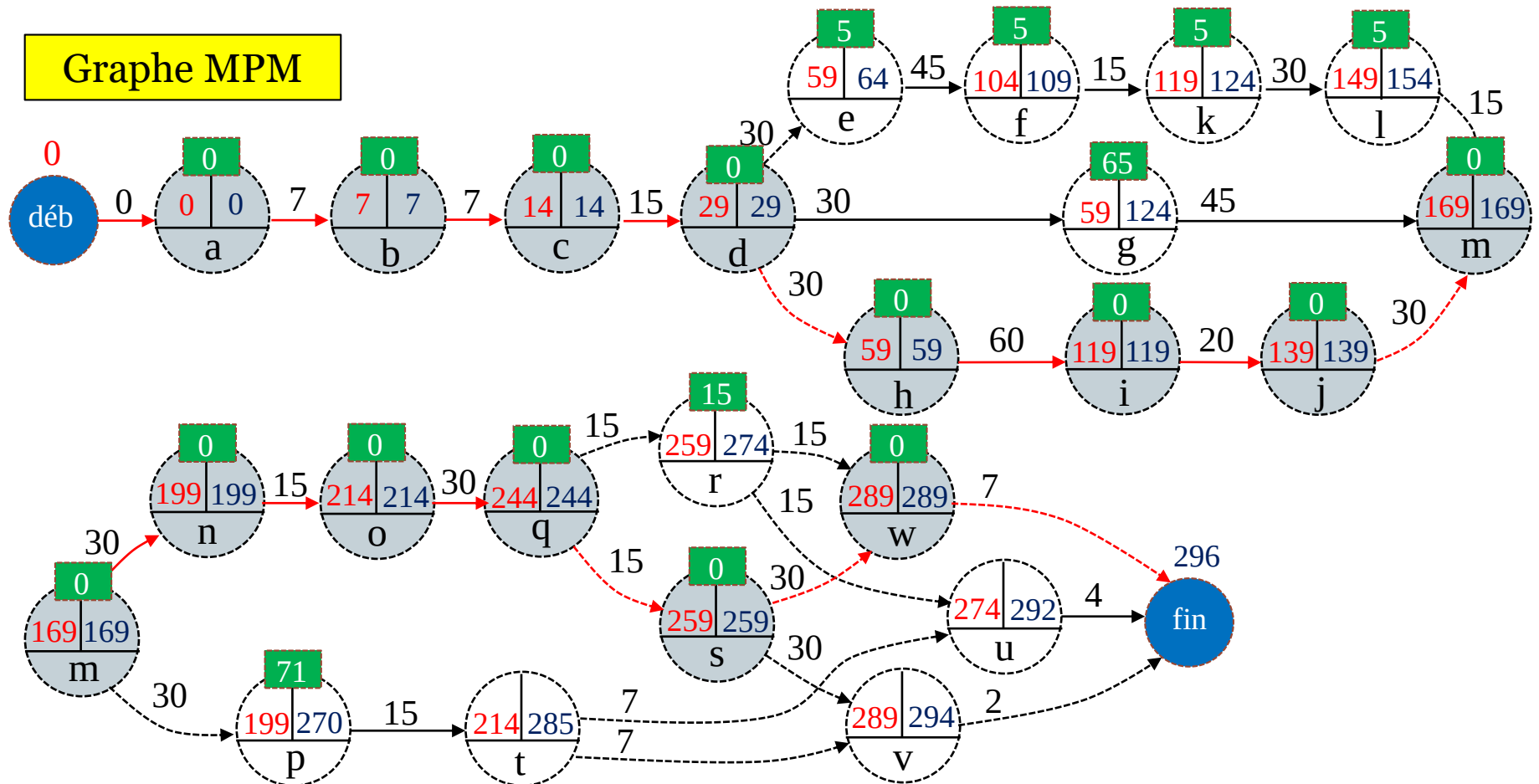
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Graphe MPM



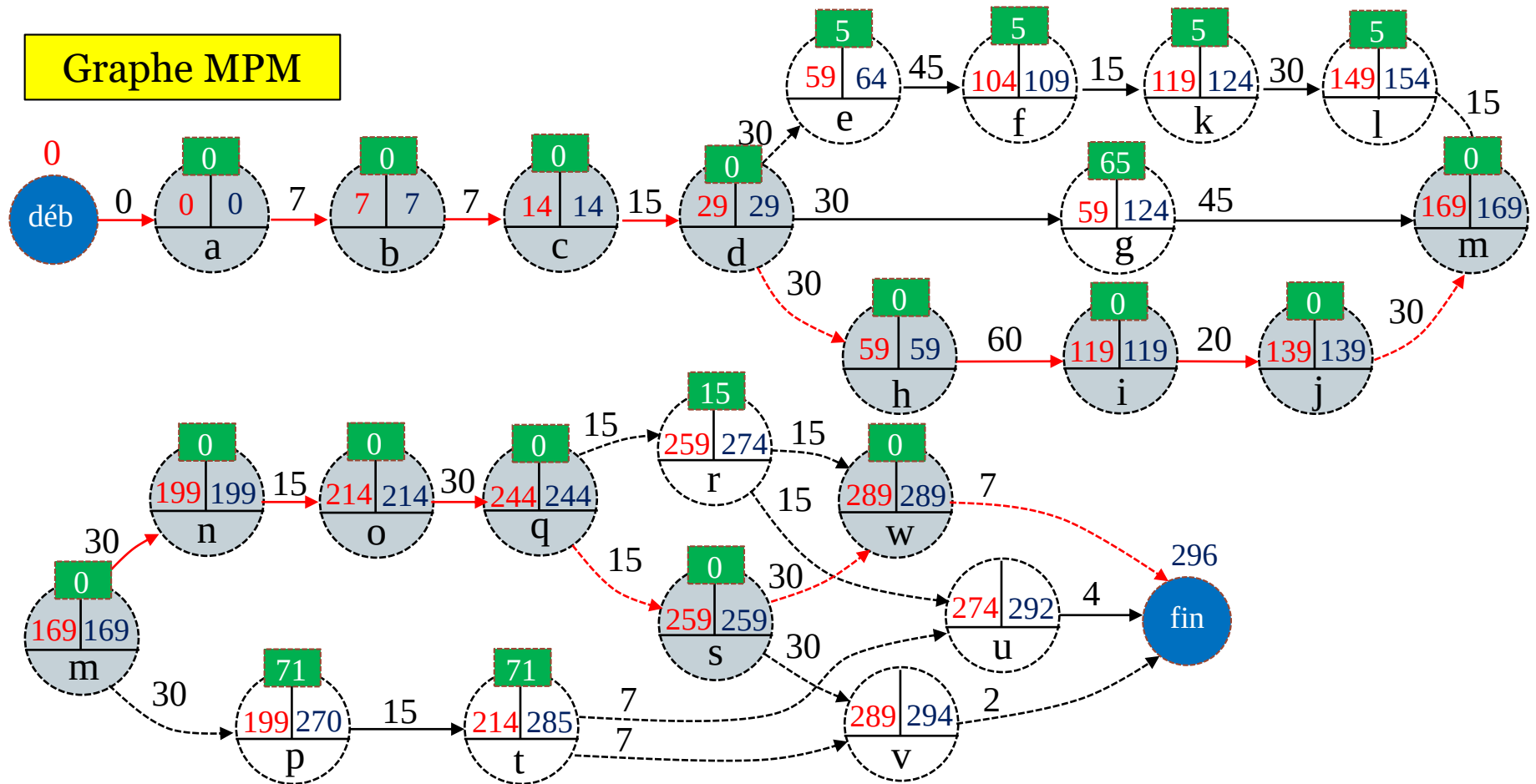
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Graphe MPM



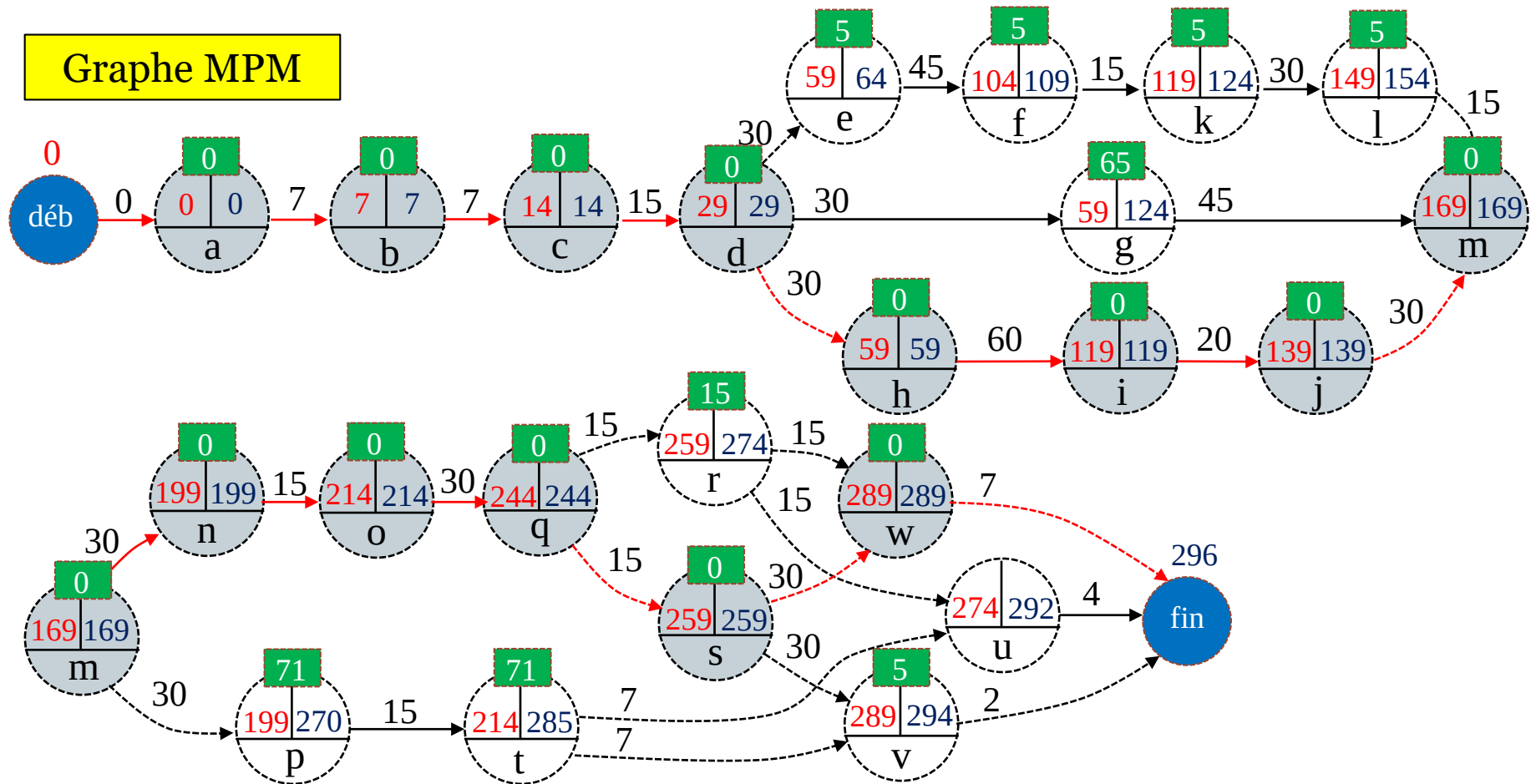
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Graphe MPM



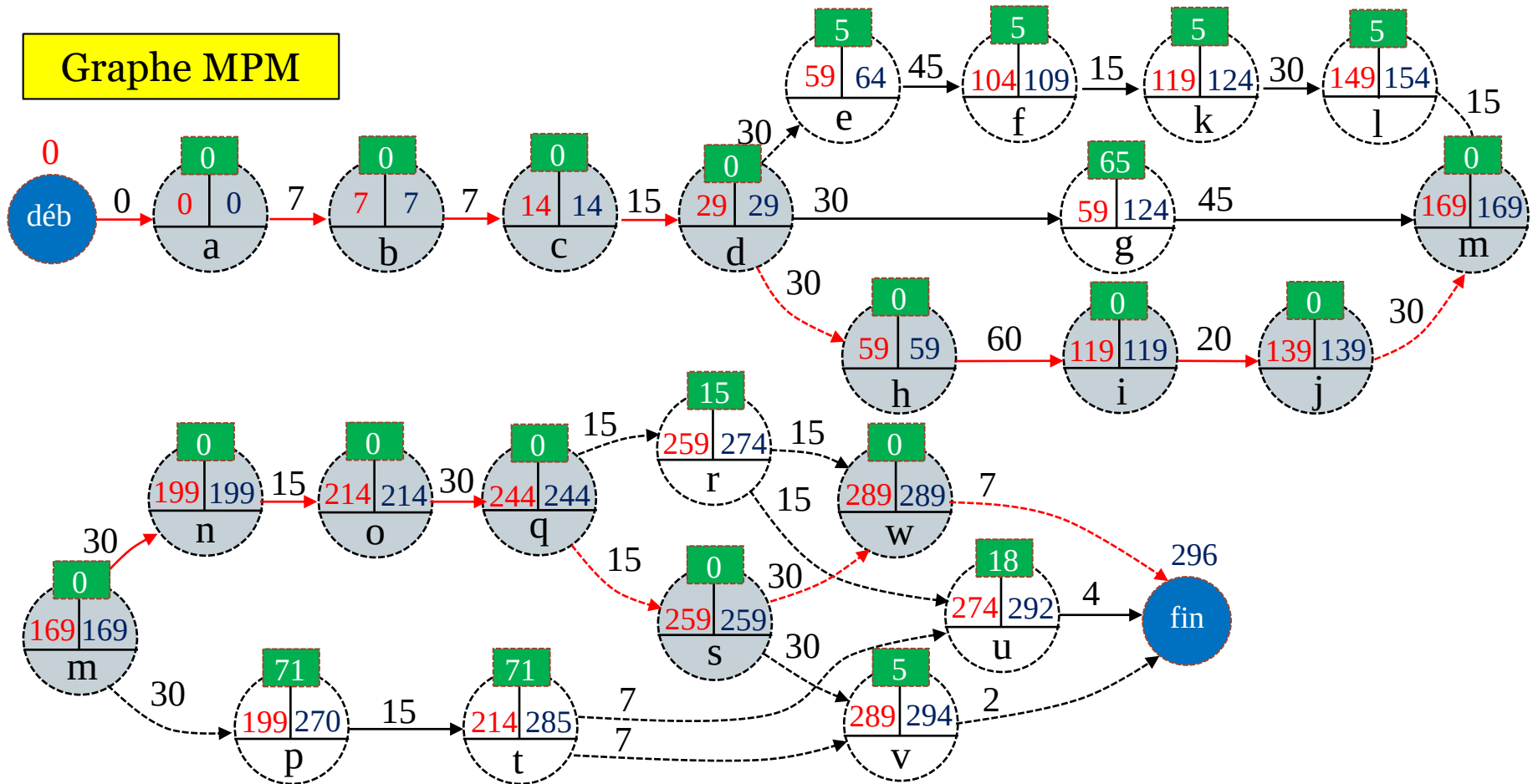
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Graphe MPM



Tâche	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
Durée	7	7	15	30	45	15	45	60	20	30	30	15	30	15	30	15	15	15	30	7	4	2	7

Graphe MPM



Tâche	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
Durée	7	7	15	30	45	15	45	60	20	30	30	15	30	15	30	15	15	15	30	7	4	2	7
T.suc.	b	c	d	e g h	f	k	m	i	j	m	l	m	n p	o	q	t	r s	u w	v w	u v	fin	fin	fin

Graphe CPM

